Project: **Mansourah Massarah Gold Project**

Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **CALCULATION**

3120 SECONDARY CRUSHING

CIVIL DESIGN CALCULATION

AS BUILT

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REFERENCE DOCUMENTS

DOCUMENT NUMBER	DOCUMENT DESCRIPTION
MD-216-1000-EG-ST-CRT-1001	Civil Structural Design Criteria
MD-216-3000-EG-CV-RPT-1008	Geotechnical Report
SBC 301 :2007	Saudi Structural - Loading & Forces Code
SBC 304 :2018	Saudi Concrete Structures Code
ACI 318 :2008	Building Code Requirement of Structural Concrete

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Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
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QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning a

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1. INTENT

The document is prepared for civil design Calculation of secondary cone crusher support structure and its foundation as well as foundation and pedestal for the steel super structure of secondary crusher building. This building have the secondary cone crusher, a surge bin of 162 m3 capacity, vibrating feeder and the feed conveyor from screen building.

2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.
MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.
The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

This document is prepared for design of Concrete Floor supporting the secodary cone crusher at RL 919.5m level, Columns and foundations.
This structure as well as foundation has been kept isolated from the steel structure. Separate staad model has been used for the analysis and design.
Isolated footing design have been performed using MS-Excel and for combined footing separate staad-pro model has been generated.
All the loads applied to the structure for analysis & design followed Civil -Structural Design Criteria and SBC 301:2007
Design of Reinforced Concrete members followed Strength Design as per SBC 304:2018 & ACI 318:2008.
The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

4 CONCRETE STRUCTURE

4.1 Design

Design of concrete structures and foundations shall be in accordance with SBC-304 and ACI 318M.

4.2 Materials

Concrete Strength of 28 days
· Foundation and Structural Members: 35 MPa
· Grade slab and pavement: 30 MPa
· Blinding Concrete below foundation bases: 15 Mpa

Reinforcement Grade
· Deformed Bar : ASTM A 615 / A 615M Grade 60 (fy = 420MPa)
· Welded Wire Fabric : ASTM A 185(Plain) / A 497(Deformed)

4.3 Foundations for Structures

Foundation shall be Isolated and combined Shallow Concrete foundation. The design calculations are prepared as per recommendation of standard SBC-304 Clause 13.3.

4.4 Minimum Concrete Cover

Items	Minimum Cover(mm)
Foundation (Side & Bottom)	75
Foundation (Top)	50
Column & Pedestal	50
Beam	50
Slab	40

4.5 Allowable Soil Bearing Capacity

The net safe bearing capacity of soil for foundations design has been considered as 500 kN/m2 at 1.5m depth below FGL (ie. RL 912.50m) refer Table 4 - BH-19 of Geotechnical Report (MD-216-3000-EG-CV-RPT-1008).

4.6 Reference

Reference is made to codes, standards and specifications as indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.1.

5 LOAD CALCULATONS

5.1 DEAD LOAD

Selfweight of structure is applied through Staad Pro.

Dead load on floor @ 919.50m
Unit Weight of RCC = 24.00 kN/m^3
Slab Thickness = 0.40 m
Total dead load from Slab = 9.60 kN/m^2

Equipment Loads :
Equipments Loads are consist of Static load of Crusher, material load, Dedusting duct load and Cable tray load
For Crusher & other equipment load, Refer Attachment 1 - Mechanical General Arrangement drawings.

Crusher Material Load = 6 x 18 kN = 108 kN
For 4 Nos Support, Load per Suppport = 27 kN

5.2 LIVE LOAD

Live load on floor @ 919.50m = 5.00 kN/m^2

5.3 SUPERIMPOSED DEAD LOAD (For Crusher Dynamic Load)

Dynamic Load Case I (Reversible) : **SDL1_R** = Dynamic Load of Crusher at start/stop : Vertical Load - Downward & Horizontal load in + Z direction
Dynamic Load Case II (Reversible) : **SDL2_R** = Dynamic Load of Crusher at start/stop : Vertical Load - Downward & Horizontal load in + X direction

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5.4 WIND LOAD

Wind Load Parameters (As per design Criteria) :

Basic wind speed, V_b	=	43	m/s	= 155	Km/h
Exposure category	=	C	(For Process Plant Area)		
Building Category	=	II	(Ref. Table 1.6-1 : SBC 301-2007)		
Wind directionablity factor, K_d	=	0.85	(Ref. Table 6.4-1 : SBC 301-2007)		
Topography Factor, K_{zt}	=	1	(As The project is not located in isolated hills or escarpments)		
Importance Factor, I	=	1	(Ref. Table 6.5-1 : SBC 301-2007)		
Velocity Pressure Co-efficient	=	Kz	(Ref. Table 7.2-2 : SBC 301-2007)		

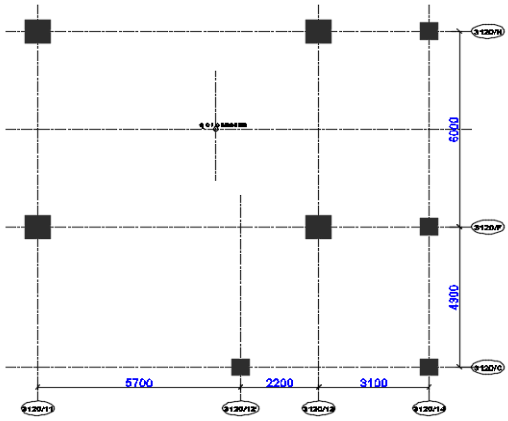
Wind Velocity Pressure, qz = $0.0473 \times 10^{-3} K_z K_{zt} K_d V^2 I$ (KN/m^2) (Ref. Eq. 7.2-12 : SBC 301-200

Level	Height	kz	Pressure	Unit
EL +919.5	6.5 m	1.0	0.966	kN/m^2

Average Pressure on wall, qz = **0.966** kN/m^2

C02 Natural frequency of the structure , n = 8.46 Hz > 1.0 Hz. So, Rigid Structure
(Ref. Doc. No. MD-216-3120-EG-CV-CAL-1007)

Gust Effect Factor, G = 0.85 (For Rigid Structure)
(Ref. Section 7.2.7.1 : SBC 301-2007)



I) WIND LOAD IN +X DIRECTION (From Axis 3120/11 to 3120/14)

a) Wind Load due to Structural Building Structure
Area for Structural Frame work at Axis 3120 / 14

Member	W (m)	L (m)	A (m^2)	Nos	Total Area (m^2)
Column	0.7	5.80	4.06	3	12.18
C02 Column	1	6.00	6.00	1	6.00
Slab	0.4	11.60	4.64	1	4.64
Beam	0.30	11.60	3.48	1	3.48
C02 Wall1	5.00	2.20	11.00	1	11.00
C02 Wall2	7.30	5.50	40.15	1	40.15
			Total Effective Projected Area =	A_f =	**77.45**

Height of Building	=	6.5	m
Frontal Area	=	98.475	m^2
Solidity ratio, Φ	=	0.79	
Force Co-efficient, C_f	=	**1.60**	(Ref. Table 7.2-17 : SBC 301-2007)
Design Wind Load	F =	$q_z G C_f A_f$	(Ref. Eq. 7.2-22 : SBC 301-2007)
Total Wind load	=	101.7	KN

Wind load in column

At Row 3120/H = 101.7 x 3 / (14.3 x 6.5) =	3.28	kN/m
At Row 3120/F = 101.7 x 5.15 / (14.3 x 6.5) =	5.64	kN/m
At Row 3120/C = 101.7 x 4.15 / (14.3 x 6.5) =	4.54	kN/m
At Row 3120/C' = 101.7 x 2.0 / (14.3 x 98.475) =	2.19	kN/m

b) Secondary Crusher

Height	=	4.177 m	
Width	=	3 m	
Area of Straight Portion	=	12.531 m^2	
h/d	=	1.39	
Force Co-efficient, Cf	=	1.30 (Ref. Table 7.2-15 : SBC 301-2007)	
Wind pressue	=	0.966 kN/m^2	
Total Wind Load	=	1.3 x 0.966 x 12.531 =	15.7 kN
Total Wind Load / Support	Hx =	15.736 / 4 =	**3.93** kN
	Vx =	15.7 x 4.177 / 2 / 3.145 / 2	**5.22** kN (Up / Down)

II) WIND LOAD IN +Z DIRECTION (From Axis 3120/J to 3120/C)

a) Wind Load due to Structural Building Structure
Area for Structural Frame work at Axis 3120/ H

Member	W (m)	L (m)	A (m^2)	Nos	Total Area (m^2)
Column	0.7	5.80	4.06	3	12.18
Slab	0.4	12.30	4.92	1	4.92
Beam	0.30	12.30	3.69	1	3.69
C02 Wall	7.20	5.50	39.60	1	39.60
			Total Effective Projected Area =	A_f =	**60.39**

Height of Building	=	6.5	m
Frontal Area	=	74.1	m^2
Solidity ratio, Φ	=	0.81	
Force Co-efficient, C_f	=	**1.60**	(Ref. Table 7.2-17 : SBC 301-2007)
Design Wind Load	F =	$q_z G C_f A_f$	(Ref. Eq. 7.2-22 : SBC 301-2007)
Design Wind Load	F =	$q_z G C_f A_f$	(Ref. Eq. 7.2-22 : SBC 301-2007)
Total Wind load	=	79.3	KN

Wind load in column

At Row 3120/11 = 79.3 x 3.95 / (11 x 6.5) =	4.38	kN/m
At Row 3120/13 = 79.3 x 5.5 / (11 x 6.5) =	6.10	kN/m
At Row 3120/14 = 79.3 x 1.55 / (11 x 6.5) =	1.72	kN/m
At Row 3120/12' = 79.3 x 2.65 / (11 x 6.5) =	2.94	kN/m

b) Secondary Crusher & Motor

Crusher Height	=	4.177 m	Motor Height =	1.70 m
Crusher Width	=	3 m	Motor Width =	2.40 m
Total Area of crusher with motor	=	16.6 m^2		
h/d	=	1.39		
Force Co-efficient, Cf	=	1.30 (Ref. Table 7.2-15 : SBC 301-2007)		
Wind pressue	=	0.966 kN/m^2		
Total Wind Load	=	20.9 kN		
Total Wind Load / Support	Hx =	20.859 / 4 =	**5.21** kN	
	Vx =	20.9 x 4.177 / 2 / 3.29 / 2	**6.62** kN (Up / Down)	

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5.5 EARTHQUAKE LOAD

Seismic Design Category			=	A
Site class			=	B (It is corresponds to Site Class 2 in Staad Pro)
Spectral response acceleration (1-sec)		S1	=	0.02
Spectral response acceleration (.02-sec)		Ss	=	0.1
Importance factor		I	=	1 (From Table 9.5, SBC 301:2007)
Response Reduction factor along X		Rx	=	2.5 (From Table 10.2, SBC 301:2007)
Response Reduction factor along Z		Rz	=	2.5 (From Table 10.2, SBC 301:2007)
Height of building		h	=	6.5 m
Time period along X		Tx	=	0.237 (eqn. 10.9.3.2-1, SBC 301:2007)
Time period along Z		Tz	=	0.237 (eqn. 10.9.3.2-1, SBC 301:2007)
Factor Fa		Fa	=	1
Factor Fv		Fv	=	1
SM1		Fa x S1	=	0.02
SMS		Fa x Ss	=	0.1
Design spectral acceleration (1-sec)	SD1 = 2/3 x SM1		=	0.0133
Design spectral acceleration (0.2-sec)	SDS = 2/3 x SMS		=	0.0667

AG

The above parameters are used in the staad input to find the final earthquake load of the structure as per IBC-2003

5.6 LOAD COMBINATIONS

5.6.1 Load Combination for Base Pressure & Servicibility (As Per Section 2.4 of SBC-301 : 2007)

Load cases	Reference
LOAD 301 (DL + SDL1_R) LOAD 302 (DL + SDL2_R) LOAD 303 (DL - SDL1_R) LOAD 304 (DL - SDL2_R)	(Eq. 2.4.1-1 of SBC 301)
LOAD 305 (DL + SDL2_R) + LL LOAD 306 (DL + SDL1_R) + LL LOAD 307 (DL - SDL2_R) + LL LOAD 308 (DL - SDL1_R) + LL	(Eq. 2.4.1-2 of SBC 301)
LOAD 309 (DL+ SDL1_R) + 0.75 LL + 0.75 WL(+X) LOAD 310 (DL+ SDL1_R) + 0.75 LL + 0.75 WL(-X) LOAD 311 (DL+ SDL1_R) + 0.75 LL + 0.75 WL(+Z) LOAD 312 (DL+ SDL1_R) + 0.75 LL + 0.75 WL(-Z) LOAD 313 (DL+ SDL2_R) + 0.75 LL + 0.75 WL(+X) LOAD 314 (DL+ SDL2_R) + 0.75 LL + 0.75 WL(-X) LOAD 315 (DL+ SDL2_R) + 0.75 LL + 0.75 WL(+Z) LOAD 316 (DL+ SDL2_R) + 0.75 LL + 0.75 WL(-Z) LOAD 317 (DL- SDL1_R) + 0.75 LL + 0.75 WL(+X) LOAD 318 (DL- SDL1_R) + 0.75 LL + 0.75 WL(-X) LOAD 319 (DL- SDL1_R) + 0.75 LL + 0.75 WL(+Z) LOAD 320 (DL- SDL1_R) + 0.75 LL + 0.75 WL(-Z) LOAD 321 (DL- SDL2_R) + 0.75 LL + 0.75 WL(+X) LOAD 322 (DL- SDL2_R) + 0.75 LL + 0.75 WL(-X) LOAD 323 (DL- SDL2_R) + 0.75 LL + 0.75 WL(+Z) LOAD 324 (DL- SDL2_R) + 0.75 LL + 0.75 WL(-Z)	(Eq. 2.4.1-6 of SBC 301)
LOAD 325 (DL+ SDL1_R) + 0.75 LL + 0.525 EL(+X) LOAD 326 (DL+ SDL1_R) + 0.75 LL + 0.525 EL(-X) LOAD 327 (DL+ SDL1_R) + 0.75 LL + 0.525 EL(+Z) LOAD 328 (DL+ SDL1_R) + 0.75 LL + 0.525 EL(-Z) LOAD 329 (DL+ SDL2_R) + 0.75 LL + 0.525 EL(+X) LOAD 330 (DL+ SDL2_R) + 0.75 LL + 0.525 EL(-X) LOAD 331 (DL+ SDL2_R) + 0.75 LL + 0.525 EL(+Z) LOAD 332 (DL+ SDL2_R) + 0.75 LL + 0.525 EL(-Z) LOAD 333 (DL- SDL1_R) + 0.75 LL + 0.525 EL(+X) LOAD 334 (DL- SDL1_R) + 0.75 LL + 0.525 EL(-X) LOAD 335 (DL- SDL1_R) + 0.75 LL + 0.525 EL(+Z) LOAD 336 (DL- SDL1_R) + 0.75 LL + 0.525 EL(-Z) LOAD 337 (DL- SDL2_R) + 0.75 LL + 0.525 EL(+X) LOAD 338 (DL- SDL2_R) + 0.75 LL + 0.525 EL(-X) LOAD 339 (DL- SDL2_R) + 0.75 LL + 0.525 EL(+Z) LOAD 340 (DL- SDL2_R) + 0.75 LL + 0.525 EL(-Z)	(Eq. 2.4.1-6 of SBC 301)

Load cases	Reference
LOAD 341 0.6 (DL + SDL1_R) + WL(+X) LOAD 342 0.6 (DL + SDL1_R) + WL(-X) LOAD 343 0.6 (DL + SDL1_R) + WL(+Z) LOAD 344 0.6 (DL + SDL1_R) + WL(-Z) LOAD 345 0.6 (DL + SDL2_R) + WL(+X) LOAD 346 0.6 (DL + SDL2_R) + WL(-X) LOAD 347 0.6 (DL + SDL2_R) + WL(+Z) LOAD 348 0.6 (DL + SDL2_R) + WL(-Z) LOAD 349 0.6 (DL - SDL1_R) + WL(+X) LOAD 350 0.6 (DL - SDL1_R) + WL(-X) LOAD 351 0.6 (DL - SDL1_R) + WL(+Z) LOAD 352 0.6 (DL - SDL1_R) + WL(-Z) LOAD 353 0.6 (DL - SDL2_R) + WL(+X) LOAD 354 0.6 (DL - SDL2_R) + WL(-X) LOAD 355 0.6 (DL - SDL2_R) + WL(+Z) LOAD 356 0.6 (DL - SDL2_R) + WL(-Z)	(Eq. 2.4.1-7 of SBC 301)
LOAD 357 0.6 (DL + SDL1_R) + 0.7 EL(+X) LOAD 358 0.6 (DL + SDL1_R) + 0.7 EL(-X) LOAD 359 0.6 (DL + SDL1_R) + 0.7 EL(+Z) LOAD 360 0.6 (DL + SDL1_R) + 0.7 EL(-Z) LOAD 361 0.6 (DL + SDL2_R) + 0.7 EL(+X) LOAD 362 0.6 (DL + SDL2_R) + 0.7 EL(-X) LOAD 363 0.6 (DL + SDL2_R) + 0.7 EL(+Z) LOAD 364 0.6 (DL + SDL2_R) + 0.7 EL(-Z) LOAD 365 0.6 (DL + SDL1_R) + 0.7 EL(+X) LOAD 366 0.6 (DL + SDL1_R) + 0.7 EL(-X) LOAD 367 0.6 (DL + SDL1_R) + 0.7 EL(+Z) LOAD 368 0.6 (DL + SDL1_R) + 0.7 EL(-Z) LOAD 369 0.6 (DL + SDL2_R) + 0.7 EL(+X) LOAD 370 0.6 (DL + SDL2_R) + 0.7 EL(-X) LOAD 371 0.6 (DL + SDL2_R) + 0.7 EL(+Z) LOAD 372 0.6 (DL + SDL2_R) + 0.7 EL(-Z)	(Eq. 2.4.1-8 of SBC 301)

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Approved	AG	AG	AG	AG

5.6.2 Load Combination for Concrete design

Load cases	Reference
LOAD 401 1.4 (DL+SDL1_R)	(Eq. 2.3.2-1 of SBC 301)
LOAD 402 1.4 (DL+SDL2_R)	
LOAD 403 1.4 (DL-SDL1_R)	
LOAD 404 1.4 (DL-SDL2_R)	
LOAD 405 1.4 (DL + SDL1_R) + 1.7 LL	(Eq. 2.3.2-2 of SBC 301)
LOAD 406 1.4 (DL + SDL2_R) + 1.7 LL	
LOAD 407 1.4 (DL - SDL1_R) + 1.7 LL	
LOAD 408 1.4 (DL - SDL2_R) + 1.7 LL	
LOAD 409 1.2 (DL + SDL1_R) + 0.5 LL + 1.6 WL(+X)	(Eq. 2.3.2-4 of SBC 301)
LOAD 410 1.2 (DL + SDL1_R) + 0.5 LL + 1.6 WL(-X)	
LOAD 411 1.2 (DL + SDL1_R) + 0.5 LL + 1.6 WL(+Z)	
LOAD 412 1.2 (DL + SDL1_R) + 0.5 LL + 1.6 WL(-Z)	
LOAD 413 1.2 (DL + SDL2_R) + 0.5 LL + 1.6 WL(+X)	
LOAD 414 1.2 (DL + SDL2_R) + 0.5 LL + 1.6 WL(-X)	
LOAD 415 1.2 (DL + SDL2_R) + 0.5 LL + 1.6 WL(+Z)	
LOAD 416 1.2 (DL + SDL2_R) + 0.5 LL + 1.6 WL(-Z)	
LOAD 417 1.2 (DL - SDL1_R) + 0.5 LL + 1.6 WL(+X)	
LOAD 418 1.2 (DL - SDL1_R) + 0.5 LL + 1.6 WL(-X)	
LOAD 419 1.2 (DL - SDL1_R) + 0.5 LL + 1.6 WL(+Z)	
LOAD 420 1.2 (DL - SDL1_R) + 0.5 LL + 1.6 WL(-Z)	
LOAD 421 1.2 (DL - SDL2_R) + 0.5 LL + 1.6 WL(+X)	
LOAD 422 1.2 (DL - SDL2_R) + 0.5 LL + 1.6 WL(-X)	
LOAD 423 1.2 (DL - SDL2_R) + 0.5 LL + 1.6 WL(+Z)	
LOAD 424 1.2 (DL - SDL2_R) + 0.5 LL + 1.6 WL(-Z)	
LOAD 425 1.2 (DL + SDL1_R) + 0.5 LL + 1.0 EL(+X)	(Eq. 2.3.2-5 of SBC 301)
LOAD 426 1.2 (DL + SDL1_R) + 0.5 LL + 1.0 EL(-X)	
LOAD 427 1.2 (DL + SDL1_R) + 0.5 LL + 1.0 EL(+Z)	
LOAD 428 1.2 (DL + SDL1_R) + 0.5 LL + 1.0 EL(-X)	
LOAD 429 1.2 (DL + SDL2_R) + 0.5 LL + 1.0 EL(+X)	
LOAD 430 1.2 (DL + SDL2_R) + 0.5 LL + 1.0 EL(-X)	
LOAD 431 1.2 (DL + SDL2_R) + 0.5 LL + 1.0 EL(+Z)	
LOAD 432 1.2 (DL + SDL2_R) + 0.5 LL + 1.0 EL(-X)	
LOAD 433 1.2 (DL - SDL1_R) + 0.5 LL + 1.0 EL(+X)	
LOAD 434 1.2 (DL - SDL1_R) + 0.5 LL + 1.0 EL(-X)	
LOAD 435 1.2 (DL - SDL1_R) + 0.5 LL + 1.0 EL(+Z)	
LOAD 436 1.2 (DL - SDL1_R) + 0.5 LL + 1.0 EL(-X)	
LOAD 437 1.2 (DL - SDL2_R) + 0.5 LL + 1.0 EL(+X)	
LOAD 438 1.2 (DL - SDL2_R) + 0.5 LL + 1.0 EL(-X)	
LOAD 439 1.2 (DL - SDL2_R) + 0.5 LL + 1.0 EL(+Z)	
LOAD 440 1.2 (DL - SDL2_R) + 0.5 LL + 1.0 EL(-X)	

Load cases	Reference
LOAD 441 0.9 (DL + SDL1_R) + 1.6 WL(+X)	(Eq. 2.3.2-6 of SBC 301)
LOAD 442 0.9 (DL + SDL1_R) + 1.6 WL(-X)	
LOAD 443 0.9 (DL + SDL1_R) + 1.6 WL(+Z)	
LOAD 444 0.9 (DL + SDL1_R) + 1.6 WL(-Z)	
LOAD 445 0.9 (DL + SDL2_R) + 1.6 WL(+X)	
LOAD 446 0.9 (DL + SDL2_R) + 1.6 WL(-X)	
LOAD 447 0.9 (DL + SDL2_R) + 1.6 WL(+Z)	
LOAD 448 0.9 (DL + SDL2_R) + 1.6 WL(-Z)	
LOAD 449 0.9 (DL - SDL1_R) + 1.6 WL(+X)	
LOAD 450 0.9 (DL - SDL1_R) + 1.6 WL(-X)	
LOAD 451 0.9 (DL - SDL1_R) + 1.6 WL(+Z)	
LOAD 452 0.9 (DL - SDL1_R) + 1.6 WL(-Z)	
LOAD 453 0.9 (DL - SDL2_R) + 1.6 WL(+X)	
LOAD 454 0.9 (DL - SDL2_R) + 1.6 WL(-X)	
LOAD 455 0.9 (DL - SDL2_R) + 1.6 WL(+Z)	
LOAD 456 0.9 (DL - SDL2_R) + 1.6 WL(-Z)	
LOAD 457 0.9 (DL + SDL1_R) + 1.0 EL(+X)	(Eq. 2.3.2-7 of SBC 301)
LOAD 458 0.9 (DL + SDL1_R) + 1.0 EL(-X)	
LOAD 459 0.9 (DL + SDL1_R) + 1.0 EL(+Z)	
LOAD 460 0.9 (DL + SDL1_R) + 1.0 EL(-Z)	
LOAD 461 0.9 (DL + SDL2_R) + 1.0 EL(+X)	
LOAD 462 0.9 (DL + SDL2_R) + 1.0 EL(-X)	
LOAD 463 0.9 (DL + SDL2_R) + 1.0 EL(+Z)	
LOAD 464 0.9 (DL + SDL2_R) + 1.0 EL(-Z)	
LOAD 465 0.9 (DL - SDL1_R) + 1.0 EL(+X)	
LOAD 466 0.9 (DL - SDL1_R) + 1.0 EL(-X)	
LOAD 467 0.9 (DL - SDL1_R) + 1.0 EL(+Z)	
LOAD 468 0.9 (DL - SDL1_R) + 1.0 EL(-Z)	
LOAD 469 0.9 (DL - SDL2_R) + 1.0 EL(+X)	
LOAD 470 0.9 (DL - SDL2_R) + 1.0 EL(-X)	
LOAD 471 0.9 (DL - SDL2_R) + 1.0 EL(+Z)	
LOAD 472 0.9 (DL - SDL2_R) + 1.0 EL(-Z)	

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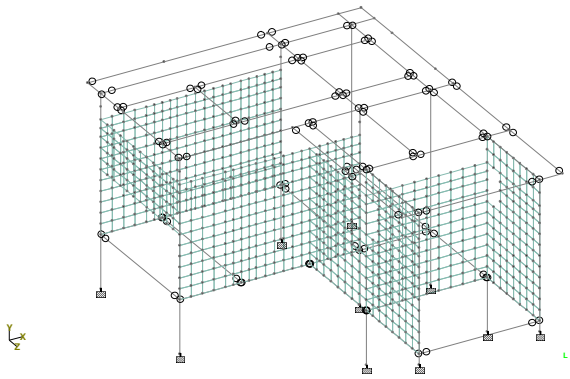
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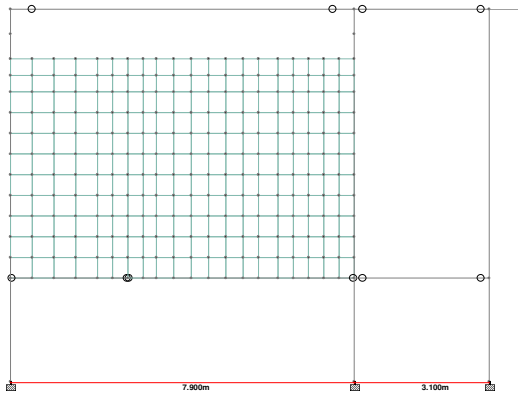
C02

6 GEOMETRY

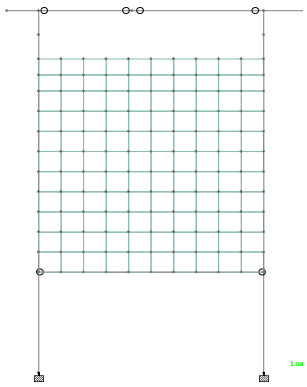
6.1 Geometry of Secondary Crusher Support Structure



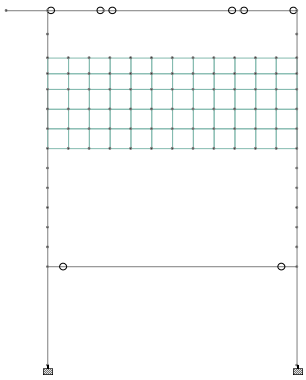
Isometric View of Structure



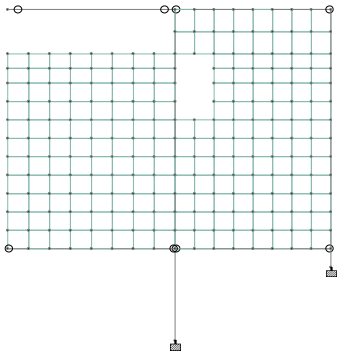
Elevation Along Row - 3120/F and 3120/ H



Elevation Along Row - 3120/C

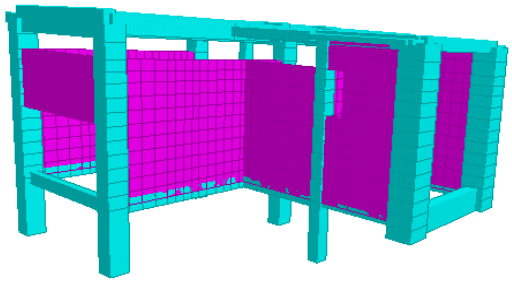


Elevation Along Axis - 3120/11

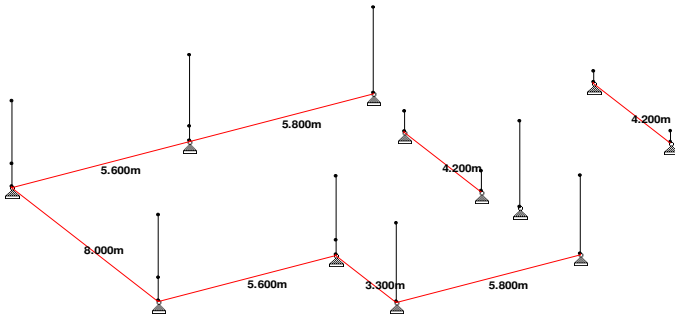


Elevation Along Axis - 3120/12

6.2 Geometry of Secondary Crusher House Column layout



3D-Model of Crusher support Structure



Isometric View of Column Layout of Secondary Crusher House

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Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
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7 DESIGN OF SECONDARY CRUSHER SUPPORT STRUCTURE

7.1 Analysis & Results

C02

7.1.1 Design summary of the existing structure

The design summary of various components of crusher support structure after adding RCC wall and additional foundationis given below -

I) Foundation

a) base Pressure

Foundation marked	Size (LXB) m	Revised Base pressure Max (kN/m^2)	Revised Base pressure Min (kN/m^2)	Earlier Base pressure Min (kN/m^2)	Earlier Base pressure Min (kN/m^2)	Change in maximum Gross Base pressure	Allowable Groos Pressure	Remarks	
F1A	3 x 2	150	57	107.7	25.5	39.2%	536	Safe	
CF2	9 x 2	265	22	221.6	-5.7	19.6%	536	Safe	
CF3_1	6.1 x 3	228	30.8	222.8	10.9	2.3%	536	Safe	
CF3_2	6.1 x 3	240.5	31.7	247.1	14.3	-2.7%	536	Safe	
CF4	7.3 x 4	121.4	20.7	-	-	-	536	Safe	(New Foundation)

b) Reinforcement

All the reinforcement of existing foundations are okay. Refer detail calculation in clause no 7.5.

II) Column

Column size of Existing structure is safe after adding of additonal wall and summary is given below. For design refer staad output file and clause 7.2.

Column Mkd.	Size	Revised Ast required (mm^2)	Earlier Ast required (mm^2)	Longitudinal Reinforcement	Links	Remarks	
C1	500X500	2500	2500	12 Nos - 20 mm Dia	10 mm Dia @ 250 c/c	Safe	
C2	700X700	5243	8124	16 Nos - 32 mm Dia	10 mm Dia @ 250 c/c	Safe	
C3	1000X1000	10000	-	24 Nos - 25 mm Dia	10 mm Dia @ 250 c/c	Safe	(New Column)

III) Slab

Slab of Existing structure is safe as there is no such effect due to adding of RCC wall. For design detail, refer clause 7.3.

IV) Floor Beam & Tie Beam

All Floor Beams of Existing structure is safe after adding of RCC wall & fondation. For design detail, refer clause 7.4.

Beam mkd.	FB1	FB2	FB3	FB4	TB1	TB2	NB1	NB2
Size (B x D)	400x1000	400x900	400x700	300x500	300x550	300x500	500x500	300x500
Top Reinforcement	4-D32 + 4-D20	4-D32	4-D25	3-D25	3-D25+ 3-D16	3-D25	4-D20	3-D20
Bottom Reinforcement	4-D32 + 4-D20	4-D32 + 4-D32	4-D20	3-D20	3-D25	3-D25	4-D20	3-D20
Stirrup	4L-D16 @150	4L-D12 @150	4L-D10 @200	4L-D10 @200	2L-D10 @200	2L-D10@200	2L-D10 @200	2L-D10@200
Remarks	Safe	Safe	Safe	Safe	Safe	Safe	Safe	Safe

V) New RCC Wall

Detail design of new Additional wall (two type) has been performed in clause no 7.6.

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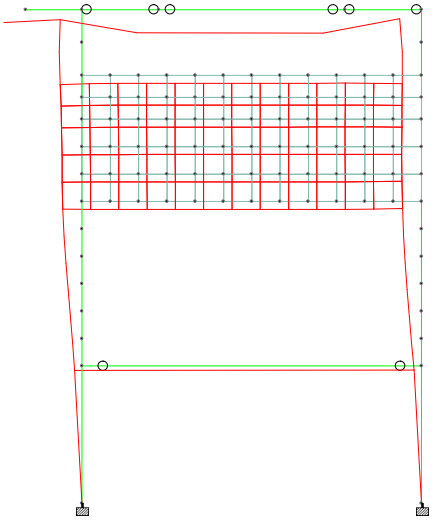
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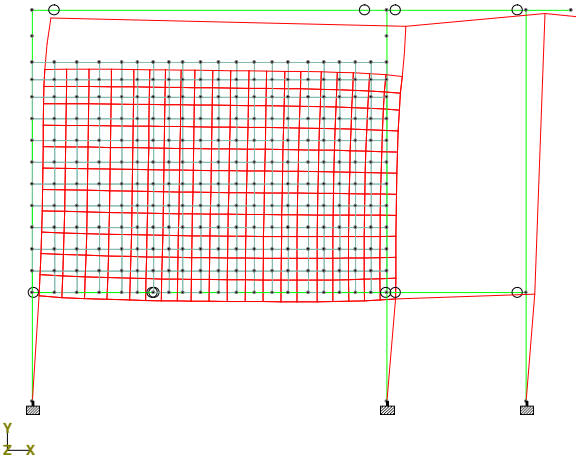
C02

7.1.2 Deflection & Bending moment diagram

Deflection Diagram Along Grid 3120/11 — Load case - 320 — δz max = **0.77** mm
Allowable deflection — H/400 = (6500+2500) / 400 = 22.5 mm — **OK**

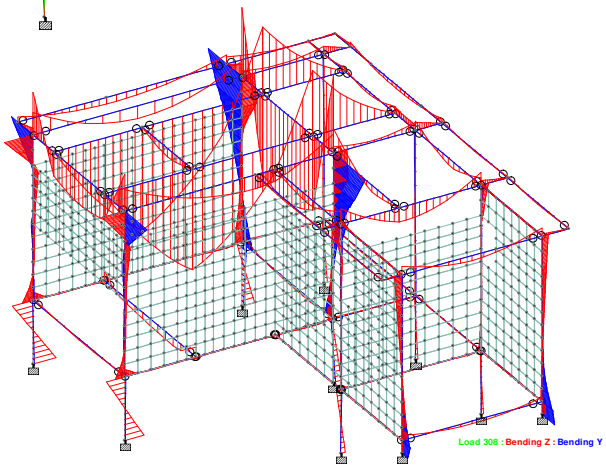


Deflection Diagram Along Grid 3130/H — Load case - 313 — δx max = **0.833** mm
Allowable deflection — H/400 = (6500+2500) / 400 = 22.5 mm — **OK**



Load 313 : Displacement

Bending moment diagram — Load Case- 308



Load 308 : Bending Z : Bending Y

7.1.3 Seismic Base Shear

C02

Design base shear along X	=	0.022 x 5988.58 KN	=	134.43 KN
Design base shear along Z	=	0.022 x 5988.58 KN	=	134.43 KN

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7.1.4 Frequency Check Of Crusher Support Beam

C02

Modal analysis and detail dynamic analysis have been performed in ANSYS Software. For frequency checking along with harmonic analysis report, refer document no. **MD-216-3120-EG-CV-CAL-1007**.

7.2 Design of column

The column of secondary crusher support structure is design using staad (**Crusher Support Structure R2.STD**) and the reinforcements are provided accordingly.

The parameters for design are defined in the staad model.

Column Reinforcement

Member	Width(mm)	depth(mm)	Bar dia 1	Number of bars	Bar dia 2	Number of bars	Total longitudinal reinforcement	Staad Member number	Ast required as per staad (mm2)	Ast provided	Links
			mm								mm
Column C2 -Grid 3120/10-3130/H	700	700	32	16			16-32 Dia	1,2, 729 to 740	4900	12868.0	10 Dia @ 250 c/c
Column C2 -Grid 3120/10-3130/F	700	700	32	16			16-32 Dia	11,12, 412 to 423	4900	12868.0	10 Dia @ 250 c/c
Column C2 -Grid 3120/13-3130/H	700	700	32	16			16-32 Dia	3,4, 741 to 752	5243	12868.0	10 Dia @ 250 c/c
Column C2 -Grid 3120/13-3130/F	700	700	32	16			16-32 Dia	13,14, 424 to 435	4900	12868.0	10 Dia @ 250 c/c
Column C1 -Grid 3120/14-3120/H	500	500	20	12			12-20 Dia	5,6	2500	3769.9	10 Dia @ 250 c/c
Column C1 -Grid 3120/14-3120/F	500	500	20	12			12-20 Dia	15,16	2500	3769.9	10 Dia @ 250 c/c
Column C1 -Grid 3120/12'-3120/C	500	500	20	12			12-20 Dia	-	2500	3769.9	10 Dia @ 250 c/c
Column C1 -Grid 3120/14-3120/C	500	500	20	12			12-20 Dia	-	2500	3769.9	10 Dia @ 250 c/c
Column C3 -Grid 3120/12'-3120/C'	1000	1000	25	24			24-25 Dia	-	10000	11781.0	10 Dia @ 250 c/c
Column C3 -Grid 3120/14-3120/C'	1000	1000	25	24			24-25 Dia	-	10000	11781.0	10 Dia @ 250 c/c

C02

SUMMARY OF COLUMN REINFORCEMENT

Column Mkd.	Size	Longitudinal Reinforcement	Links
C1	500X500	12 Nos - 20 mm Dia	10 mm Dia @ 250 c/c
C2	700X700	16 Nos - 32 mm Dia	10 mm Dia @ 250 c/c
C3	1000X1000	24 Nos - 25 mm Dia	10 mm Dia @ 250 c/c

C02

7.3 Design of Slab

7.3.1 Design of Slab 1- 400mm thick

Slab is designed as simply supported slab along shorter span

Span of slab	=	3.4 m	max span considered	Clear cover	=	40 mm	
Thickness of slab	=	0.4 m		Bar Dia	=	12 mm	
Grade of steel	=	420 N/mm2		Unit weight of concrete	=	24 KN/m3	
Grade of concrete	=	35 N/mm2		Live load on slab	=	5 KN/m2	
Mimimum thickness of one way slab	=	span/20	*Table 7.3.1.1 SBC 304*	β_1	=	0.8	
	=	0.17 m	Hence slab thickness okay				

Load calculation on slab

Dead load from slab	=	0.4 x 24	=	9.6 KN/m2			
Live load from slab	=	5 KN/m2					
Load combination	=	1.4DL + 1.7LL	=	1.4 x 9.6 + 1.7 x 5	=	21.94 KN/m2	

Calculation of bending moment and shear force

BM at midspan	=	wl2/8	=	31.7 KNm			
SF at support	=	wl/2	=	37.3 KN			

Calculation of flexural reinforcement

				Effective Depth Along X	=	354 mm	
	M	=	31.70 KNm				
	M_{lim}	=	$\emptyset\, 0.85\, f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$	For tension controlled section	$\left(\frac{c}{d}\right)_{max}$ =	$\frac{0.003}{0.003+0.005}$ =	0.375
		=	855.61 KNm				
$M_{lim} > M$	Tension controlled section			Min reinforcement	A_{stmin} =	$\frac{0.0018 \times 420}{f_y} bd$	*Table 7.6.1.1 SBC 304*
Therefore	Φ	=	0.9 *Table 21.2.2 SBC 304*	greater of or			
					A_{stmin} =	0.0014bd	
	A_{st}	=	$\frac{0.85\, f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$		A_{stmin} =	637.2 mm^2	
					A_{req} =	237.8 mm^2	
		=	237.8 mm^2		Required Spacing =	177.5 mm	
	Provide	***12mm dia @***	***150 c/c***	***Top & bottom both directions***	A_{prov} =	754.0 mm^2	

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Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

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Shear Check

Lesser of the following

i) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$ — *Table 22.5.5.1 SBC 304*

Reinforcement ratio	ρ_w	=	0.0021		λ	=	1	Normal concrete
Effective Depth	d	=	354	mm	Mu	=	31.70	KNm
Shear Force	V_u	=	37.30	KN	φ	=	0.75	for shear
Allowable shear strength	φVc	=	255.32	KN O.K	φVc	=	255.32	KN

ii) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\right)b_w d$ = 260.93 KN — *Table 22.5.5.1 SBC 304*

iii) $V_c = 0.29\lambda\sqrt{f_c'}b_w d$ = 455.51 KN

7.3.2 Design of Slab 2- 200mm thick

Slab is designed as simply supported slab along shorter span

Span of slab	=	3.9	m	max span considered	Clear cover	=	40	mm
Thickness of slab	=	0.2	m		Bar Dia	=	12	mm
Grade of steel	=	420	N/mm2		Unit weight of concrete	=	24	KN/m3
Grade of concrete	=	35	N/mm2		Live load on slab	=	5	KN/m2
Mimimum thickness of one way slab	=	span/20		*Table 7.3.1.1 SBC 304*	β_1	=	0.8	
	=	0.195	m	Hence slab thickness okay				

Load calculation on slab

Dead load from slab	=	0.2 x 24	=	4.8	KN/m2		
Live load from slab	=	5	KN/m2				
Load combination	=	1.4DL + 1.7LL	=	1.4 x 4.8 + 1.7 x 5	=	15.22	KN/m2

Calculation of bending moment and shear force

BM at midspan	=	wl2/8	=	28.9	KNm
SF at support	=	wl/2	=	29.7	KN

Calculation of flexural reinforcement

					Effective Depth Along X	=	154	mm
	M	=	28.94	KNm				
	M_{lim}	=	$\emptyset\, 0.85\, f'_c \beta_1 \left(\frac{c}{d}\right)_{max}\left(1-\frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)bd^2$		For tension controlled section	$\left(\frac{c}{d}\right)_{max}$ = $\frac{0.003}{0.003+0.005}$	=	0.375
		=	161.92	KNm				

$M_{lim} > M$ — Tension controlled section — Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\, f'_c bd}{f_y}\left(1-\sqrt{1-\frac{2.35M_u}{\emptyset f'_c bd^2}}\right)$ = 508.3 mm^2

Min reinforcement $A_{stmin} = \frac{0.0018 \times 420}{f_y}bd$ *Table 7.6.1.1 SBC 304*

greater of or

A_{stmin} = 0.0014bd

A_{stmin} = 277.2 mm^2

A_{req} = 508.3 mm^2

Required Spacing = 222.5 mm

A_{prov} = 754.0 mm^2

Provide 12mm dia @ 150 c/c Top & bottom both directions

Shear Check

Lesser of the following

i) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$ — *Table 22.5.5.1 SBC 304*

Reinforcement ratio	ρ_w	=	0.0049		λ	=	1	Normal concrete
Effective Depth	d	=	154	mm	Mu	=	28.94	KNm
Shear Force	V_u	=	29.68	KN	φ	=	0.75	for shear
Allowable shear strength	φVc	=	110.85	KN O.K	φVc	=	110.85	KN

ii) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\right)b_w d$ = 118.94 KN — *Table 22.5.5.1 SBC 304*

iii) $V_c = 0.29\lambda\sqrt{f_c'}b_w d$ = 198.16 KN

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7.4 DESIGN OF BEAM

7.4.1 Beam Design Summary

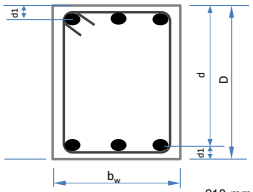
C02

Beam mkd.	FB1	FB2	FB3	FB4	TB1	TB2	NB1	NB2
Size (B x D)	400x1000	400x900	400x700	300x500	300x550	300x500	500x500	300x500
Top Reinforcement	4-D32 + 4-D20	4-D32	4-D25	3-D25	3-D25+ 3-D16	3-D25	4-D20	3-D20
Bottom Reinforcement	4-D32 + 4-D20	4-D32 + 4-D32	4-D20	3-D20	3-D25	3-D25	4-D20	3-D20
Stirrup	4L-D16 @150	4L-D12 @150	4L-D10 @200	4L-D10 @200	2L-D10 @200	2L-D10@200	2L-D10 @200	2L-D10@200

7.4.2.1 Beam marked FB1

Staad beam number - 33 to 39, 52, 54

Width of beam	=	**400**	mm
Depth of beam	=	**1000**	mm
Grade of steel	=	420	MPa
Grade of concrete	=	35	MPa
Cover	=	50	mm
Stress reduction factor for flexure	=	0.9	*Table 21.2.2 SBC 304*
Stress reduction factor for shear	=	0.75	*Table 21.2.1 SBC 304*
Stress reduction factor for torsion	=	0.75	*Table 21.2.1 SBC 304*
Effective Depth	=	918	mm
Dia of longitudinal bar	=	**32**	mm
Dia of stirrup bar	=	**16**	mm



d	=	918	mm
d'	=	82	mm
h	=	132	mm

ANALYSIS RESULTS

C02

Analysis results as obtained from staad

B.M at support	Mz	L/C -	405	Beam No =	38	=	**920.1**	KNm
B.M at span	Mz	L/C -	405	Beam No =	37	=	**1035.5**	KNm
Shear Force	Fy	L/C -	405	Beam No =	38	=	**1372.0**	KN
Torsion	Mx					=	**0**	KNm

DESIGN FOR FLEXURE

Support Moment

Mu lim = $\emptyset\, 0.85\, f_c'\beta_1\left(\frac{c}{d}\right)_{max} bd^2\left(1-\frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assuptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 400 x 918 x 918 (1 - 0.8/2 x 0.375)
= 2301.5 KNm
Singly reinforced beam

$\frac{c}{d}$ max = $\frac{0.003}{0.003+0.005}$ = 0.375

b_1 = 0.80 *Table 22.2.2.4.3 SBC 304*

Mu = 920.055 KNm

Mu_2 = Mu - Mu_{lim} = 920.055 - 2301.6
= 0.00 KNm

A_{st1} = $\frac{0.85\, f_c' bd}{f_y}\left(1-\sqrt{1-\frac{2.35\, M_u}{\emptyset\, f_c' bd^2}}\right)$
= 0.85 x 35x 400 x 918 x (1-√(1-(2.35 x 920.1 x 10^6)/(0.9 x 35 x 400 x 918^2)))/420
= 2798.679648 mm^2

A_{st2} = $\frac{M_u - M_{u\,lim}}{\emptyset\, f_s'(d-d')}$
= 0 x 10^6/(0.9 x 420 x (918 - 82))
= 0 mm^2

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y} b_w d$ or $\frac{1.4}{f_y} b_w d$

e_{sc} = 0.0023
Asc = 0.0000 mm^2

= 1293.08601 mm^2 or 1224 mm^2 = 1293.08601 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}
= 2798.7 + 0.00 + 0
= 2798.7 mm^2

Top layer1	**4**	*nos*	**32**	*mm*	*dia*	+	**0**	*nos*	**0**	*mm*	*dia*
Top layer2	**4**	*nos*	**20**	*mm*	*dia*	+		*nos*		*mm*	*dia*

Ast provided = 4473.63 mm^2 Safe

ρ_w = 0.0121831

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Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
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Span Moment

Mu lim = $\emptyset\, 0.85\, f_c'\beta_1\left(\frac{c}{d}\right)_{max} bd^2\left(1-\frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assuptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 400 x 918 x 918 (1 - 0.8/2 x 0.375)

= 2301.5 KNm

Singly reinforced beam

$\frac{c}{d}$ max = $\frac{0.003}{0.003+0.005}$ = 0.375

b_1 = 0.80

Mu = 1035.47 KNm

Mu_2 = Mu - Mu_{lim} = 1035.47 - 2301.6

= 0 KNm

A_{st} = $\frac{0.85\, f_c' bd}{f_y}\left(1-\sqrt{1-\frac{2.35\, M_u}{\emptyset\, f_c' bd^2}}\right)$

= 0.85 x 35x 400 x 918 x (1-√(1-(2.35 x 1035.5 x 10^6)/(0.9 x 35 x 400 x 918^2)))/420

= 3174.0 mm^2

A_{st2} = $\frac{M_u - M_{u\,lim}}{\emptyset\, f_s'(d-d')}$

= 0 x 10^6/(0.9 x 420 x (918 - 82))

= 0 mm^2

e_{sc} = 0.002285403

Asc = 0.0000 mm^2

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y} b_w d$ or $\frac{1.4}{f_y} b_w d$

= 1293.1 mm^2 or 1224 mm^2 = 1293.08601 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 3174 + 0 + 0

= 3174 mm^2

bottom layer1	4	nos	32	mm	dia	+	0	nos	0	mm	dia
bottom layer2	4	nos	20	mm	dia	+		nos		mm	dia

Ast provided = 4473.6 mm^2 Safe

ρ_w = 0.012183

DESIGN FOR SHEAR

Allowable shear force = Minimum of the following Vc

Correspo

V_c = $\left(0.16\,\lambda\sqrt{f_c'}+17\rho_w\left(\frac{V_u d}{M_u}\right)\right) b_w d$ = 451.7 KN

λ = 1 Normal concrete

V_c = $\left(0.16\,\lambda\sqrt{f_c'}+17\rho_w\right) b_w d$ = 423.6 KN

V_c = $0.29\,\lambda\sqrt{f_c'}\, b_w d$ = 630.0 KN

fV_c = 0.75 x 423.7 = 317.72 KN < 1372 KN *Shear Reinforcement required*

Vs = $V_U - \emptyset V_C$ = $\frac{\emptyset\, A_v f_y d}{s}$

$\frac{A_v}{S}$ = $\frac{V_U - \emptyset V_C}{\emptyset\, f_y d}$ = ((1372-317.8) x 10^3)/(0.75 x 420 x 918) = 3.6 mm^2/mm

No. of legs = 4 Spacing provided = 150 mm

Provide 4L -16 mm stirrups @ 150 C/C

Spacing of stirrup is minimum of the following

i) Shear Reinforcement	s =	221 mm	*If Vu > φVc*
ii) Min Asv = $0.062\sqrt{f_c'}\frac{b_w S}{f_t}$	s =	2302 mm	*Corresponding to table 9.6.3.3 of SBC 304*
or Min Asv = $0.65\frac{b_w s}{f_{yt}}$	s =	1299 mm	*Corresponding to table 9.6.3.3 of SBC 304*
iii) effective depth/2	s =	459 mm	*Corresponding to table 9.7.6.2.2 of SBC 304*
iv) Combined shear & torsion	s =	0 mm	

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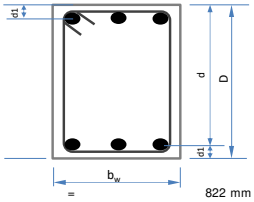
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REFERENCE DOCUMENTS

DOCUMENT NUMBER	DOCUMENT DESCRIPTION
MD-216-1000-EG-ST-CRT-1001	Civil Structural Design Criteria

7.4.2.2 Beam marked FB2 Staad beam number - 41 to 44, 53, 55



Width of beam	=	**400** mm	
Depth of beam	=	**900** mm	
Grade of steel	=	420 MPa	
Grade of concrete	=	35 MPa	
Cover	=	50 mm	
Stress reduction factor for flexure	=	0.9	*Table 21.2.2 SBC 304*
Stress reduction factor for shear	=	0.75	*Table 21.2.1 SBC 304*
Stress reduction factor for torsion	=	0.75	*Table 21.2.1 SBC 304*
Effective Depth	=	822 mm	
Dia of longitudinal bar	=	**32** mm	
Dia of stirrup bar	=	**12** mm	

d	=	822	mm
d'	=	78	mm
h	=	124	mm

ANALYSIS RESULTS

Analysis results as obtained from staad

C02

B.M at support	Mz	L/C -	405	Beam No =	44	=	**794.7**	KNm
B.M at span	Mz	L/C -	406	Beam No =	44	=	**1242.3**	KNm
Shear Force	Fy	L/C -	405	Beam No =	44	=	**844.6**	KN
Torsion	Mx					=	**0**	KNm

DESIGN FOR FLEXURE

Support Moment

Mu lim = $\emptyset\, 0.85\, f_c'\beta_1 \left(\frac{c}{d}\right)_{max} bd^2 \left(1-\frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assuptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 400 x 822 x 822 (1 - 0.8/2 x 0.375)
= 1845.3 KNm
Singly reinforced beam

$\frac{c}{d}$ max = $\frac{0.003}{0.003+0.005}$ = 0.375

b_1 = 0.80 *Table 22.2.2.4.3 SBC 304*

Mu = 794.715 KNm

Mu_2 = Mu - Mu_{lim} = 794.715 - 1845.4
= 0.00 KNm

A_{st1} = $\frac{0.85\, f_c' bd}{f_y}\left(1-\sqrt{1-\frac{2.35\, M_u}{\emptyset\, f_c' bd^2}}\right)$

A_{st2} = $\frac{M_u - M_{u\,lim}}{\emptyset\, f_s'(d-d')}$

= 0.85 x 35x 400 x 822 x (1-√(1-(2.35 x 794.8 x 10^6)/(0.9 x 35 x 400 x 822^2)))/420
= 2712.44 mm^2

= 0 x 10^6/(0.9 x 420 x (822 - 78))
= 0 mm^2

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y} b_w d$ or $\frac{1.4}{f_y} b_w d$

e_{sc} = 0.0022
Asc = 0.0000 mm^2

= 1157.9 mm^2 or 1096 mm^2 = 1157.9 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}
= 2712.5 + 0.00 + 0
= 2712.5 mm^2

Top layer1	***10***	*nos*	***25***	*mm*	*dia*	+	***0***	*nos*	***0***	*mm*	*dia*
Top layer2		*nos*		*mm*	*dia*	+		*nos*		*mm*	*dia*

Ast provided = 4908.7 mm^2 Safe

ρ_w = 0.0149293

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Span Moment

Mu lim = $\emptyset\, 0.85\, f_c'\beta_1\left(\frac{c}{d}\right)_{max} bd^2\left(1-\frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assuptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 400 x 822 x 822 (1 - 0.8/2 x 0.375)

= 1845.3 KNm

Singly reinforced beam

$\frac{c}{d}$ max = $\frac{0.003}{0.003+0.005}$ = 0.375

b_1 = 0.80

Mu = 1242.333 KNm

Mu_2 = Mu - Mu_{lim} = 1242.333 - 1845.4

= 0 KNm

A_{st} = $\frac{0.85\, f_c' bd}{f_y}\left(1-\sqrt{1-\frac{2.35\, M_u}{\emptyset\, f_c' bd^2}}\right)$

= 0.85 x 35x 400 x 822 x (1-√(1-(2.35 x 1242.4 x 10^6)/(0.9 x 35 x 400 x 822^2)))/420

= 4411.0 mm^2

A_{st2} = $\frac{M_u - M_{u\,lim}}{\emptyset\, f_s'(d-d')}$

= 0 x 10^6/(0.9 x 420 x (822 - 78))

= 0 mm^2

e_{sc} = 0.002240876

Asc = 0.0000 mm^2

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y} b_w d$ or $\frac{1.4}{f_y} b_w d$

= 1157.861329 mm^2 or 1096 mm^2 = 1157.86133 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 4411.1 + 0 + 0

= 4411.1 mm^2

bottom layer1	**4**	*nos*	**32**	*mm*	*dia*	+	**0**	*nos*	**0**	*mm*	*dia*
bottom layer2	**4**	*nos*	**32**	*mm*	*dia*	+		*nos*		*mm*	*dia*

Ast provided = 6433.982 mm^2 Safe

ρ_w = 0.019568071

DESIGN FOR SHEAR

Allowable shear force = Minimum of the following Vc

Correspo

V_c = $\left(0.16\,\lambda\sqrt{f_c'}+17\rho_w\left(\frac{V_u d}{M_u}\right)\right) b_w d$ = 384.13 KN λ = 1 Normal concrete

V_c = $\left(0.16\,\lambda\sqrt{f_c'}+17\rho_w\right) b_w d$ = 394.68 KN

V_c = $0.29\,\lambda\sqrt{f_c'}\, b_w d$ = 564.11 KN

fV_c = 0.75 x 384.2 = 288.1001934 KN < 844.6 KN *Shear Reinforcement required*

Vs = $V_U - \emptyset V_C$ = $\frac{\emptyset\, A_v f_y d}{s}$

$\frac{A_v}{S}$ = $\frac{V_U - \emptyset V_C}{\emptyset\, f_y d}$ = ((844.6-288.2) x 10^3)/(0.75 x 420 x 822) = 2.15 mm^2/mm

No. of legs = **4** Spacing provided = **150** mm

Provide 4L -12 mm stirrups @ 150 C/C

Spacing of stirrup is minimum of the following

i) Shear Reinforcement	s =	210 mm	*If Vu > φVc*	
ii) Min Asv = $0.062\sqrt{f_c'}\frac{b_w S}{f_t}$	s =	1295 mm	*Corresponding to table 9.6.3.3 of SBC 304*	
or Min Asv = $0.65\frac{b_w s}{f_{yt}}$	s =	731 mm	*Corresponding to table 9.6.3.3 of SBC 304*	
iii) effective depth/2	s =	411 mm	*Corresponding to table 9.7.6.2.2 of SBC 304*	
iv) Combined shear & torsion	s =	0 mm		

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Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
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Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

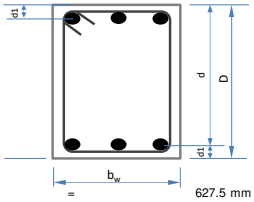
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REFERENCE DOCUMENTS

DOCUMENT NUMBER	DOCUMENT DESCRIPTION
MD-216-1000-EG-ST-CRT-1001	Civil Structural Design Criteria

7.4.2.3 Beam marked FB3 Staad beam number - 9, 19, 26, 46, 48, 50, 51, 71



Width of beam	=	**400** mm		
Depth of beam	=	**700** mm		
Grade of steel	=	420 MPa		
Grade of concrete	=	35 MPa		
Cover	=	50 mm		
Stress reduction factor for flexure	=	0.9	*Table 21.2.2 SBC 304*	
Stress reduction factor for shear	=	0.75	*Table 21.2.1 SBC 304*	
Stress reduction factor for torsion	=	0.75	*Table 21.2.1 SBC 304*	
Effective Depth	=	627.5 mm		
Dia of longitudinal bar	=	**25** mm		
Dia of stirrup bar	=	**10** mm		

d	=	627.5	mm
d'	=	72.5	mm
h	=	120	mm

ANALYSIS RESULTS

Analysis results as obtained from staad

C02

B.M at support	Mz	L/C -	406	Beam No =	46	=	**205.2**	KNm
B.M at span	Mz	L/C -	408	Beam No =	19	=	**119.8**	KNm
Shear Force	Fy	L/C -	406	Beam No =	46	=	**164.5**	KN
Torsion	Mx					=	**0**	KNm

DESIGN FOR FLEXURE

Support Moment

Mu lim = $\emptyset\, 0.85\, f_c'\beta_1 \left(\frac{c}{d}\right)_{max} bd^2 \left(1-\frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assuptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 400 x 627.5 x 627.5 (1 - 0.8/2 x 0.375)
= 1075.4 KNm
Singly reinforced beam

$\frac{c}{d}$ max = $\frac{0.003}{0.003+0.005}$ = 0.375

b_1 = 0.80 *Table 22.2.2.4.3 SBC 304*

Mu = 205.201 KNm

Mu_2 = Mu - Mu_{lim} = 205.201 - 1075.4
= 0.00 KNm

A_{st1} = $\frac{0.85\, f_c' bd}{f_y}\left(1-\sqrt{1-\frac{2.35\, M_u}{\emptyset\, f_c' bd^2}}\right)$

= 0.85 x 35x 400 x 627.5 x (1-√(1-(2.35 x 205.3 x 10^6)/(0.9 x 35 x 400 x 627.5^2)))/420
= 886.12 mm^2

A_{st2} = $\frac{M_u - M_{u\,lim}}{\emptyset\, f_s'(d-d')}$
= 0 x 10^6/(0.9 x 420 x (627.5 - 72.5))
= 0 mm^2

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y} b_w d$ or $\frac{1.4}{f_y} b_w d$

e_{sc} = 0.0021
Asc = 0.0000 mm^2

= 883.9 mm^2 or 836.6666667 mm^2 = 883.9 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}
= 886.2 + 0.00 + 0
= 886.2 mm^2

Top layer1	**4**	*nos*	**25**	*mm*	*dia*	+		*nos*		*mm*	*dia*
Top layer2		*nos*		*mm*	*dia*	+		*nos*		*mm*	*dia*

Ast provided = 1963.5 mm^2 Safe

ρ_w = 0.0078227

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Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	C02	D01
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Approved	AG	AG	AG	AG

Span Moment

Mu lim = $\emptyset\, 0.85\, f_c'\beta_1\left(\frac{c}{d}\right)_{max} bd^2\left(1-\frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assuptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 400 x 627.5 x 627.5 (1 - 0.8/2 x 0.375)

$\frac{c}{d}$ max = $\frac{0.003}{0.003+0.005}$ = 0.375

= 1075.4 KNm

Singly reinforced beam

b_1 = 0.80

Mu = 119.831 KNm

Mu_2 = Mu - Mu_{lim} = 119.831 - 1075.4

= 0 KNm

A_{st} = $\frac{0.85\, f_c' bd}{f_y}\left(1-\sqrt{1-\frac{2.35\, M_u}{\emptyset\, f_c' bd^2}}\right)$

A_{st2} = $\frac{M_u - M_{u\,lim}}{\emptyset\, f_s'(d-d')}$

= 0.85 x 35x 400 x 627.5 x (1-√(1-(2.35 x 119.9 x 10^6)/(0.9 x 35 x 400 x 627.5^2)))/420

= 0 x 10^6/(0.9 x 420 x (627.5 - 72.5))

= 511.94 mm^2

= 0 mm^2

e_{sc} = 0.002075697

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y} b_w d$ or $\frac{1.4}{f_y} b_w d$

Asc = 0.0000 mm^2

= 883.9 mm^2 or 836.67 mm^2 = 883.9 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 883.9 + 0 + 0

= 883.9 mm^2

bottom layer1	**4**	*nos*	**20**	*mm*	*dia*	+		*nos*		*mm*	*dia*
bottom layer2		*nos*		*mm*	*dia*	+		*nos*		*mm*	*dia*

Ast provided = 1256.64 mm^2 Safe

ρ_w = 0.005007

DESIGN FOR SHEAR

Allowable shear force = Minimum of the following Vc *Correspo*

V_c = $\left(0.16\,\lambda\sqrt{f_c'}+17\rho_w\left(\frac{V_u d}{M_u}\right)\right) b_w d$ = 254.4 KN

λ = 1 Normal concrete

V_c = $\left(0.16\,\lambda\sqrt{f_c'}+17\rho_w\right) b_w d$ = 271.0 KN

V_c = $0.29\,\lambda\sqrt{f_c'}\, b_w d$ = 430.6 KN

fV_c = 0.75 x 254.4 = 190.79 KN > 164.5 KN *Shear Reinforcement not required*

Vs = $V_U - \emptyset V_C = \frac{\emptyset A_v f_y d}{s}$

$\frac{A_v}{S} = \frac{V_U - \emptyset V_C}{\emptyset f_y d}$ = ((164.5-190.8) x 10^3)/(0.75 x 420 x 627.5) = 0.000 mm^2/mm

No. of legs = **2** Spacing provided = **200** mm

Provide 2L -10 mm stirrups @ 200 C/C

Spacing of stirrup is minimum of the following

i) Shear Reinforcement s = 0 mm *If Vu > φVc*

ii) Min Asv = $0.062\sqrt{f_c'}\frac{b_w S}{f_t}$ s = 450 mm *Corresponding to table 9.6.3.3 of SBC 304*

or

Min Asv = $0.65\frac{b_w s}{f_{yt}}$ s = 254 mm *Corresponding to table 9.6.3.3 of SBC 304*

iii) effective depth/2 s = 314 mm *Corresponding to table 9.7.6.2.2 of SBC 304*

iv) Combined shear & torsion s = 0 mm

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Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

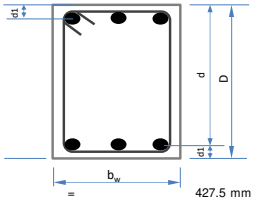
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Approved	AG	AG	AG	AG

REFERENCE DOCUMENTS

DOCUMENT NUMBER	DOCUMENT DESCRIPTION
MD-216-1000-EG-ST-CRT-1001	Civil Structural Design Criteria

7.4.2.4 Beam marked FB4 Staad beam number - 40, 45, 47, 49



Width of beam	=	**300** mm		
Depth of beam	=	**500** mm		
Grade of steel	=	420 MPa		
Grade of concrete	=	35 MPa		
Cover	=	50 mm		
Stress reduction factor for flexure	=	0.9	*Table 21.2.2 SBC 304*	
Stress reduction factor for shear	=	0.75	*Table 21.2.1 SBC 304*	
Stress reduction factor for torsion	=	0.75	*Table 21.2.1 SBC 304*	
Effective Depth	=	427.5 mm		
Dia of longitudinal bar	=	**25** mm		
Dia of stirrup bar	=	**10** mm		

d	=	427.5	mm
d'	=	72.5	mm
h	=	120	mm

ANALYSIS RESULTS

Analysis results as obtained from staad

C02

B.M at support	Mz	L/C -	406	Beam No =	49	=	**240.8**	KNm
B.M at span	Mz	L/C -	406	Beam No =	45	=	**72.2**	KNm
Shear Force	Fy	L/C -	406	Beam No =	49	=	**152.3**	KN
Torsion	Mx					=	**0**	KNm

DESIGN FOR FLEXURE

Support Moment

Mu lim = $\emptyset\, 0.85\, f_c'\beta_1\left(\frac{c}{d}\right)_{max} bd^2\left(1-\frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assuptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 300 x 427.5 x 427.5 (1 - 0.8/2 x 0.375)
= 374.3 KNm
Singly reinforced beam

$\frac{c}{d}$ max = $\frac{0.003}{0.003+0.005}$ = 0.375

b_1 = 0.80 *Table 22.2.2.4.3 SBC 304*

Mu = 240.753 KNm

Mu_2 = Mu - Mu_{lim} = 240.753 - 374.4
= 0.00 KNm

A_{st1} = $\frac{0.85\, f_c' bd}{f_y}\left(1-\sqrt{1-\frac{2.35\, M_u}{\emptyset\, f_c' bd^2}}\right)$
= 0.85 x 35x 300 x 427.5 x (1-√(1-(2.35 x 240.8 x 10^6)/(0.9 x 35 x 300 x 427.5^2)))/420
= 1635.2 mm^2

A_{st2} = $\frac{M_u - M_{u\,lim}}{\emptyset\, f_s'(d-d')}$
= 0 x 10^6/(0.9 x 420 x (427.5 - 72.5))
= 0 mm^2

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y} b_w d$ or $\frac{1.4}{f_y} b_w d$

e_{sc} = 0.0016
Asc = 0.0000 mm^2

= 451.629 mm^2 or 427.5 mm^2 = 451.629305 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}
= 1635.2 + 0.00 + 0
= 1635.2 mm^2

Top layer1	**3**	*nos*	**25**	*mm*	*dia*	+	**3**	*nos*	**20**	*mm*	*dia*
Top layer2		*nos*		*mm*	*dia*	+		*nos*		*mm*	*dia*

Ast provided = 2415.099 mm^2 Safe

ρ_w = 0.0188312

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Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

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Span Moment

Mu lim = $\emptyset\, 0.85\, f_c'\beta_1 \left(\frac{c}{d}\right)_{max} bd^2 \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assuptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 300 x 427.5 x 427.5 (1 - 0.8/2 x 0.375)

$\frac{c}{d}$ max = $\frac{0.003}{0.003 + 0.005}$ = 0.375

= 374.3 KNm

Singly reinforced beam

b_1 = 0.80

Mu = 72.214 KNm

Mu_c = Mu - Mu_{lim} = 72.214 - 374.4

= 0 KNm

A_{st} = $\frac{0.85\, f_c' bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35\, M_u}{\emptyset\, f_c' bd^2}}\right)$

A_{st2} = $\frac{M_u - M_{u\,lim}}{\emptyset\, f_s'(d - d')}$

= 0.85 x 35x 300 x 427.5 x (1-√(1-(2.35 x 72.3 x 10^6)/(0.9 x 35 x 300 x 427.5^2)))/420

= 0 x 10^6/(0.9 x 420 x (427.5 - 72.5))

= 457.9 mm^2

= 0 mm^2

e_{sc} = 0.001643275

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y} b_w d$ or $\frac{1.4}{f_y} b_w d$

Asc = 0.0000 mm^2

= 451.6 mm^2 or 427.5 mm^2 = 451.629305 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 457.9 + 0 + 0

= 457.9 mm^2

bottom layer1	**3**	*nos*	**20**	*mm*	*dia*	+		*nos*		*mm*	*dia*
bottom layer2		*nos*		*mm*	*dia*	+		*nos*		*mm*	*dia*

Ast provided = 942.5 mm^2 Safe

ρ_w = 0.007348755

DESIGN FOR SHEAR

Allowable shear force = Minimum of the following Vc *Correspo*

V_c = $\left(0.16\,\lambda\sqrt{f_c'} + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right) b_w d$ = 132.5 KN

λ = 1 Normal concrete

V_c = $\left(0.16\,\lambda\sqrt{f_c'} + 17\rho_w\right) b_w d$ = 162.5 KN

V_c = $0.29\,\lambda\sqrt{f_c'}\, b_w d$ = 220.0 KN

fV_c = 0.75 x 132.6 = 99.3759 KN < 152.3 KN *Shear Reinforcement required*

Vs = $V_U - \emptyset V_C$ = $\frac{\emptyset\, A_v f_y d}{s}$

$\frac{A_v}{S}$ = $\frac{V_U - \emptyset V_C}{\emptyset\, f_y d}$ = ((152.3-99.4) x 10^3)/(0.75 x 420 x 427.5) = 0.39 mm^2/mm

No. of legs = **2** Spacing provided = **200** mm

Provide 2L -10 mm stirrups @ 200 C/C

Spacing of stirrup is minimum of the following

i) Shear Reinforcement	s =	400 mm	*If Vu > φVc*	
ii) Min Asv = $0.062\sqrt{f_c'}\frac{b_w S}{f_t}$ or	s =	600 mm	*Corresponding to table 9.6.3.3 of SBC 304*	
Min Asv = $0.65\frac{b_w s}{f_{yt}}$	s =	338 mm	*Corresponding to table 9.6.3.3 of SBC 304*	
iii) effective depth/2	s =	214 mm	*Corresponding to table 9.7.6.2.2 of SBC 304*	
iv) Combined shear & torsion	s =	0 mm		

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Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

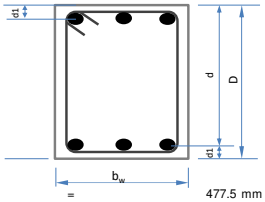
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REFERENCE DOCUMENTS

DOCUMENT NUMBER	DOCUMENT DESCRIPTION
MD-216-1000-EG-ST-CRT-1001	Civil Structural Design Criteria

7.4.2.5 Beam marked TB1 Staad beam number - 7, 8, 17, 18, 31



Width of beam	=	**300**	mm	
Depth of beam	=	**550**	mm	
Grade of steel	=	420	MPa	
Grade of concrete	=	35	MPa	
Cover	=	50	mm	
Stress reduction factor for flexure	=	0.9		*Table 21.2.2 SBC 304*
Stress reduction factor for shear	=	0.75		*Table 21.2.1 SBC 304*
Stress reduction factor for torsion	=	0.75		*Table 21.2.1 SBC 304*
Effective Depth	=	477.5	mm	
Dia of longitudinal bar	=	**25**	mm	
Dia of stirrup bar	=	**10**	mm	

d	=	477.5	mm
d'	=	72.5	mm
h	=	120	mm

ANALYSIS RESULTS

C02

Analysis results as obtained from staad

B.M at support	Mz	L/C -	413	Beam No =	452	=	**83.3**	KNm
B.M at span	Mz	L/C -	413	Beam No =	18	=	**37.6**	KNm
Shear Force	Fy	L/C -	406	Beam No =	452	=	**222.9**	KN
Torsion	Mx					=	**0**	KNm

DESIGN FOR FLEXURE

Support Moment

Mu lim = $\emptyset\, 0.85\, f_c'\beta_1\left(\frac{c}{d}\right)_{max} bd^2\left(1-\frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assuptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 300 x 477.5 x 477.5 (1 - 0.8/2 x 0.375)

= 467.0 KNm

Singly reinforced beam

$\frac{c}{d}$ max = $\frac{0.003}{0.003+0.005}$ = 0.375

b_1 = 0.80 *Table 22.2.2.4.3 SBC 304*

Mu = 83.295 KNm

Mu_2 = Mu - Mu_{lim} = 83.295 - 467.1

= 0.00 KNm

A_{st1} = $\frac{0.85\, f_c' bd}{f_y}\left(1-\sqrt{1-\frac{2.35\, M_u}{\emptyset\, f_c' bd^2}}\right)$

A_{st2} = $\frac{M_u - M_{u\,lim}}{\emptyset\, f_s'(d-d')}$

= 0.85 x 35x 300 x 477.5 x (1-√(1-(2.35 x 83.3 x 10^6)/(0.9 x 35 x 300 x 477.5^2)))/420

= 471.9 mm^2

= 0 x 10^6/(0.9 x 420 x (477.5 - 72.5))

= 0 mm^2

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y} b_w d$ or $\frac{1.4}{f_y} b_w d$

e_{sc} = 0.0018

Asc = 0.0000 mm^2

= 504.451 mm^2 or 477.5 mm^2 = 504.5 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 504.5 + 0.00 + 0

= 504.5 mm^2

Top layer1	**3**	*nos*	**25**	*mm*	*dia*	+		*nos*		*mm* *dia*
Top layer2	**3**	*nos*	**16**	*mm*	*dia*	+		*nos*		*mm* *dia*

Ast provided = 2075.807 mm^2 Safe

ρ_w = 0.0144908

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	C02	D01
Date	1/27/2020	3/12/2020	4/25/2022	12/10/2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS DT. 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER SUPPORT STRUCTURE	AS BUILT
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Checked	SBY	SBY	SBY	SBY
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Outotec / L&T Revision No.:	0	1	2	3
Date	1/27/2020	3/12/2020	4/25/2022	12/10/2023
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Approved	AG	AG	AG	AG

Span Moment

Mu lim = $\emptyset\, 0.85\, f_c'\beta_1 \left(\frac{c}{d}\right)_{max} bd^2 \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assuptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 300 x 477.5 x 477.5 (1 - 0.8/2 x 0.375)

= 467.0 KNm

$\frac{c}{d}$ max = $\frac{0.003}{0.003 + 0.005}$ = 0.375

Singly reinforced beam

b_1 = 0.80

Mu = 37.646 KNm

Mu_2 = Mu - Mu_{lim} = 37.646 - 467.1

= 0 KNm

A_{st} = $\frac{0.85\, f_c' bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35\, M_u}{\emptyset\, f_c' bd^2}}\right)$

= 0.85 x 35x 300 x 477.5 x (1-√(1-(2.35 x 37.7 x 10^6)/(0.9 x 35 x 300 x 477.5^2)))/420

= 210.5 mm^2

A_{st2} = $\frac{M_u - M_{u\,lim}}{\emptyset\, f_s'(d - d')}$

= 0 x 10^6/(0.9 x 420 x (477.5 - 72.5))

= 0 mm^2

e_{sc} = 0.00178534

Asc = 0.0000 mm^2

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y} b_w d$ or $\frac{1.4}{f_y} b_w d$

= 504.5 mm^2 or 477.5 mm^2 = 504.5 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 504.5 + 0 + 0

= 504.5 mm^2

bottom layer1	3	nos	25	mm	dia	+		nos		mm dia
bottom layer2		nos		mm	dia	+		nos		mm dia

Ast provided = 1472.6 mm^2 Safe

ρ_w = 0.010280081

DESIGN FOR SHEAR

Allowable shear force = Minimum of the following Vc

Corresponding to table 22.5.5.1 of SBC 304

V_c = $\left(0.16\lambda\sqrt{f_c'} + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right) b_w d$ = 180.7 KN

λ = 1 Normal concrete

V_c = $\left(0.16\lambda\sqrt{f_c'} + 17\rho_w\right) b_w d$ = 170.9 KN

V_c = $0.29\lambda\sqrt{f_c'}\, b_w d$ = 245.8 KN

fV_c = 0.75 x 170.9 = 128.2 KN < 222.9 KN *Shear Reinforcement required*

Vs = $V_U - \emptyset V_C$ = $\frac{\emptyset\, A_v f_y d}{s}$

$\frac{A_v}{S}$ = $\frac{V_U - \emptyset V_C}{\emptyset\, f_y d}$ = ((222.9-128.2) x 10^3)/(0.75 x 420 x 477.5) = 0.63 mm^2/mm

No. of legs = 2 Spacing provided = 200 mm

Provide 2L -10 mm stirrups @ 200 C/C

Spacing of stirrup is minimum of the following

i) Shear Reinforcement s = 249 mm *if Vu > φVc*

ii) Min Asv = $0.062\sqrt{f_c'}\frac{b_w s}{f_t}$ s = 600 mm *Corresponding to table 9.6.3.3 of SBC 304*

or

Min Asv = $0.65\frac{b_w s}{f_{yt}}$ s = 338 mm *Corresponding to table 9.6.3.3 of SBC 304*

iii) effective depth/2 s = 239 mm *Corresponding to table 9.7.6.2.2 of SBC 304*

iv) Combined shear & torsion s = 0 mm

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

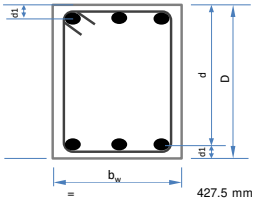
Ma'aden Revision No:	A01	C01	C02	D01			Outotec / L&T Revision No.:	0	1	2	3	
Date	1/27/2020	3/12/2020	4/25/2022	12/10/2023			Date	1/27/2020	3/12/2020	4/25/2022	12/10/2023	
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Prepared By	APR	APR	APR	APR			Prepared By	APR	APR	APR	APR	
Checked	SBY	SBY	SBY	SBY			Checked	SBY	SBY	SBY	SBY	
Approved	AG	AG	AG	AG			Approved	AG	AG	AG	AG	

REFERENCE DOCUMENTS

DOCUMENT NUMBER	DOCUMENT DESCRIPTION
MD-216-1000-EG-ST-CRT-1001	Civil Structural Design Criteria

7.4.2.6 Beam marked TB2

Staad beam number - 7, 8, 17, 18, 31



Width of beam	=	**300** mm	
Depth of beam	=	**500** mm	
Grade of steel	=	420 MPa	
Grade of concrete	=	35 MPa	
Cover	=	50 mm	
Stress reduction factor for flexure	=	0.9	*Table 21.2.2 SBC 304*
Stress reduction factor for shear	=	0.75	*Table 21.2.1 SBC 304*
Stress reduction factor for torsion	=	0.75	*Table 21.2.1 SBC 304*
Effective Depth	=	427.5 mm	
Dia of longitudinal bar	=	**25** mm	
Dia of stirrup bar	=	**10** mm	

d	=	427.5	mm
d'	=	72.5	mm
h	=	120	mm

ANALYSIS RESULTS

C02

Analysis results as obtained from staad

B.M at support	Mz	L/C -	406	Beam No =	28	=	**99.1**	KNm
B.M at span	Mz	L/C -	407	Beam No =	70	=	**69.5**	KNm
Shear Force	Fy	L/C -	407	Beam No =	70	=	**163.2**	KN
Torsion	Mx					=	**0**	KNm

DESIGN FOR FLEXURE

Support Moment

Mu lim = $\emptyset\, 0.85\, f_c'\beta_1 \left(\frac{c}{d}\right)_{max} bd^2 \left(1-\frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assuptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 300 x 427.5 x 427.5 (1 - 0.8/2 x 0.375)

= 374.3 KNm

Singly reinforced beam

$\frac{c}{d}$ max = $\frac{0.003}{0.003+0.005}$ = 0.375

b_1 = 0.80 *Table 22.2.2.4.3 SBC 304*

Mu = 99.059 KNm

Mu_2 = Mu - Mu_{lim} = 99.059 - 374.4

= 0.00 KNm

A_{st1} = $\frac{0.85\, f_c' bd}{f_y}\left(1-\sqrt{1-\frac{2.35\, M_u}{\emptyset\, f_c' bd^2}}\right)$

= 0.85 x 35x 300 x 427.5 x (1-√(1-(2.35 x 99.1 x 10^6)/(0.9 x 35 x 300 x 427.5^2)))/420

= 634.39 mm^2

A_{st2} = $\frac{M_u - M_{u\,lim}}{\emptyset\, f_s'(d-d')}$

= 0 x 10^6/(0.9 x 420 x (427.5 - 72.5))

= 0 mm^2

e_{sc} = 0.0016

Asc = 0.0000 mm^2

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y} b_w d$ or $\frac{1.4}{f_y} b_w d$

= 451.629 mm^2 or 427.5 mm^2 = 451.6 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 634.4 + 0.00 + 0

= 634.4 mm^2

Top layer1	**3**	*nos*	**25**	*mm*	*dia*	+		*nos*		*mm*	*dia*
Top layer2		*nos*		*mm*	*dia*	+		*nos*		*mm*	*dia*

Ast provided = 1472.6 mm^2 Safe

ρ_w = 0.0114824

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Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

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Checked	SBY	SBY	SBY	SBY			Checked	SBY	SBY	SBY	SBY	
Approved	AG	AG	AG	AG			Approved	AG	AG	AG	AG	

Span Moment

Mu lim = $\emptyset\, 0.85\, f_c'\beta_1 \left(\frac{c}{d}\right)_{max} bd^2 \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assuptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 300 x 427.5 x 427.5 (1 - 0.8/2 x 0.375)

= 374.3 KNm

$\frac{c}{d}$ max = $\frac{0.003}{0.003 + 0.005}$ = 0.375

Singly reinforced beam

b_1 = 0.80

Mu = 69.501 KNm

Mu_2 = Mu - Mu_{lim} = 69.501 - 374.4

= 0 KNm

A_{st} = $\frac{0.85\, f_c' bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35\, M_u}{\emptyset\, f_c' bd^2}}\right)$

= 0.85 x 35x 300 x 427.5 x (1-√(1-(2.35 x 69.6 x 10^6)/(0.9 x 35 x 300 x 427.5^2)))/420

= 440.2 mm^2

A_{st2} = $\frac{M_u - M_{u\,lim}}{\emptyset\, f_s'(d - d')}$

= 0 x 10^6/(0.9 x 420 x (427.5 - 72.5))

= 0 mm^2

e_{sc} = 0.001643275

Asc = 0.0000 mm^2

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y} b_w d$ or $\frac{1.4}{f_y} b_w d$

= 451.63 mm^2 or 427.5 mm^2 = 451.6 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 451.7 + 0 + 0

= 451.7 mm^2

bottom layer1	**3**	*nos*	**25**	*mm*	*dia*	+		*nos*		*mm*	*dia*
bottom layer2		*nos*		*mm*	*dia*	+		*nos*		*mm*	*dia*

Ast provided = 1472.6 mm^2 Safe

ρ_w = 0.011482429

DESIGN FOR SHEAR

Allowable shear force = Minimum of the following Vc *Correspo*

V_c = $\left(0.16\,\lambda\sqrt{f_c'} + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right) b_w d$ = 139.0 KN

λ = 1 Normal concrete

V_c = $\left(0.16\,\lambda\sqrt{f_c'} + 17\rho_w\right) b_w d$ = 146.4 KN

V_c = $0.29\,\lambda\sqrt{f_c'}\, b_w d$ = 220.0 KN

fV_c = 0.75 x 139.1 = 104.27 KN < 163.2 KN *Shear Reinforcement required*

Vs = $V_U - \emptyset V_C$ = $\frac{\emptyset\, A_v f_y d}{s}$

$\frac{A_v}{S}$ = $\frac{V_U - \emptyset V_C}{\emptyset\, f_y d}$ = ((163.2-104.3) x 10^3)/(0.75 x 420 x 427.5) = 0.44 mm^2/mm

No. of legs = **2** Spacing provided = **200** mm

Provide 2L -10 mm stirrups @ 200 C/C

Spacing of stirrup is minimum of the following

i) Shear Reinforcement s = 359 mm *if Vu > φVc*

ii) Min Asv = $0.062\sqrt{f_c'}\frac{b_w s}{f_t}$ s = 600 mm *Corresponding to table 9.6.3.3 of SBC 304*

or

Min Asv = $0.65\frac{b_w s}{f_{yt}}$ s = 338 mm *Corresponding to table 9.6.3.3 of SBC 304*

iii) effective depth/2 s = 214 mm *Corresponding to table 9.7.6.2.2 of SBC 304*

iv) Combined shear & torsion s = 0 mm

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

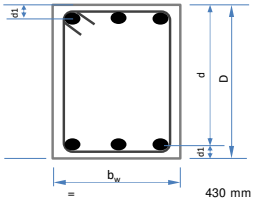
Ma'aden Revision No:	A01	C01	C02	D01			Outotec / L&T Revision No.:	0	1	2	3		
Date	1/27/2020	3/12/2020	4/25/2022	12/10/2023			Date	1/27/2020	3/12/2020	4/25/2022	12/10/2023		
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS DT. 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER SUPPORT STRUCTURE	AS BUILT			Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS DT. 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER SUPPORT STRUCTURE	AS BUILT		
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Checked	SBY	SBY	SBY	SBY			Checked	SBY	SBY	SBY	SBY		
Approved	AG	AG	AG	AG			Approved	AG	AG	AG	AG		

REFERENCE DOCUMENTS

DOCUMENT NUMBER	DOCUMENT DESCRIPTION
MD-216-1000-EG-ST-CRT-1001	Civil Structural Design Criteria

7.4.2.7 Beam marked NB1

Staad beam number - 40, 45, 47, 49



Width of beam	=	**500**	mm
Depth of beam	=	**500**	mm
Grade of steel	=	420	MPa
Grade of concrete	=	35	MPa
Cover	=	50	mm
Stress reduction factor for flexure	=	0.9	*Table 21.2.2 SBC 304*
Stress reduction factor for shear	=	0.75	*Table 21.2.1 SBC 304*
Stress reduction factor for torsion	=	0.75	*Table 21.2.1 SBC 304*
Effective Depth	=	430	mm
Dia of longitudinal bar	=	**20**	mm
Dia of stirrup bar	=	**10**	mm

d	=	430	mm
d'	=	70	mm
h	=	120	mm

ANALYSIS RESULTS

Analysis results as obtained from staad

B.M at support	Mz	L/C -	406	Beam No =	1038	=	**60.1** KNm
B.M at span	Mz	L/C -	408	Beam No =	1038	=	**30.4** KNm
Shear Force	Fy	L/C -	406	Beam No =	1038	=	**59.5** KN
Torsion	Mx					=	**0** KNm

DESIGN FOR FLEXURE

Support Moment

Mu lim = $\emptyset\, 0.85\, f_c' \beta_1 \left(\frac{c}{d}\right)_{max} bd^2 \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assuptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 500 x 430 x 430 (1 - 0.8/2 x 0.375)
= 631.2 KNm
Singly reinforced beam

$\frac{c}{d}$ max = $\frac{0.003}{0.003 + 0.005}$ = 0.375

b_1 = 0.80 *Table 22.2.2.4.3 SBC 304*

Mu = 60.096 KNm

Mu_2 = Mu - Mu_{lim} = 60.096 - 631.3
= 0.00 KNm

A_{st1} = $\frac{0.85\, f_c' bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35\, M_u}{\emptyset\, f_c' bd^2}}\right)$

A_{st2} = $\frac{M_u - M_{u\,lim}}{\emptyset\, f_s'(d - d')}$

= 0.85 x 35x 500 x 430 x (1-√(1-(2.35 x 60.1 x 10^6)/(0.9 x 35 x 500 x 430^2)))/420
= 373.9 mm^2

= 0 x 10^6/(0.9 x 420 x (430 - 70))
= 0 mm^2

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y} b_w d$ or $\frac{1.4}{f_y} b_w d$

e_{sc} = 0.0017
Asc = 0.0000 mm^2

= 757.117 mm^2 or 716.6666667 mm^2 = 757.117353 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}
= 757.2 + 0.00 + 0
= 757.2 mm^2

Top layer1	**4**	*nos*	**20**	*mm*	*dia*	+		*nos*		*mm*	*dia*
Top layer2		*nos*		*mm*	*dia*	+		*nos*		*mm*	*dia*

Ast provided = 1256.637 mm^2 Safe

ρ_w = 0.0058448

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Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	C02	D01
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Span Moment

Mu lim = $\emptyset\, 0.85\, f_c'\beta_1\left(\frac{c}{d}\right)_{max} bd^2\left(1-\frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assuptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 500 x 430 x 430 (1 - 0.8/2 x 0.375)

= 631.2 KNm

Singly reinforced beam

$\frac{c}{d}$ max = $\frac{0.003}{0.003+0.005}$ = 0.375

b_1 = 0.80

Mu = 30.387 KNm

Mu_2 = Mu - Mu_{lim} = 30.387 - 631.3

= 0 KNm

A_{st} = $\frac{0.85\, f_c' bd}{f_y}\left(1-\sqrt{1-\frac{2.35\, M_u}{\emptyset\, f_c' bd^2}}\right)$

= 0.85 x 35x 500 x 430 x (1-√(1-(2.35 x 30.4 x 10^6)/(0.9 x 35 x 500 x 430^2)))/420

= 187.9 mm^2

A_{st2} = $\frac{M_u - M_{u\,lim}}{\emptyset\, f_s'(d-d')}$

= 0 x 10^6/(0.9 x 420 x (430 - 70))

= 0 mm^2

e_{sc} = 0.001697674

Asc = 0.0000 mm^2

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y} b_w d$ or $\frac{1.4}{f_y} b_w d$

= 757.1 mm^2 or 716.6666667 mm^2 = 757.117353 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 757.2 + 0 + 0

= 757.2 mm^2

bottom layer1	**4**	*nos*	**20**	*mm*	*dia*	+		*nos*		*mm*	*dia*
bottom layer2		*nos*		*mm*	*dia*	+		*nos*		*mm*	*dia*

Ast provided = 1256.6 mm^2 Safe

ρ_w = 0.005844824

DESIGN FOR SHEAR

Allowable shear force = Minimum of the following Vc *Correspo*

V_c = $\left(0.16\,\lambda\sqrt{f_c'}+17\rho_w\left(\frac{V_u d}{M_u}\right)\right) b_w d$ = 212.6 KN

λ = 1 Normal concrete

V_c = $\left(0.16\,\lambda\sqrt{f_c'}+17\rho_w\right) b_w d$ = 224.9 KN

V_c = $0.29\,\lambda\sqrt{f_c'}\, b_w d$ = 368.9 KN

fV_c = 0.75 x 212.7 = 159.4560 KN > 59.5 KN *Shear Reinforcement not required*

Vs = $V_U - \emptyset V_C$ = $\frac{\emptyset\, A_v f_y d}{s}$

$\frac{A_v}{S}$ = $\frac{V_U - \emptyset V_C}{\emptyset\, f_y d}$ = ((59.5-159.5) x 10^3)/(0.75 x 420 x 430) = 0.00 mm^2/mm

No. of legs = **2** Spacing provided = **200** mm

Provide 2L -10 mm stirrups @ 200 C/C

Spacing of stirrup is minimum of the following

i) Shear Reinforcement	s =	0 mm	*If Vu > φVc*	
ii) Min Asv = $0.062\sqrt{f_c'}\frac{b_w S}{f_t}$	s =	360 mm	*Corresponding to table 9.6.3.3 of SBC 304*	
or Min Asv = $0.65\frac{b_w s}{f_{yt}}$	s =	203 mm	*Corresponding to table 9.6.3.3 of SBC 304*	
iii) effective depth/2	s =	215 mm	*Corresponding to table 9.7.6.2.2 of SBC 304*	
iv) Combined shear & torsion	s =	0 mm		

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Project Country	**KINGDOM OF SAUDI ARABIA**

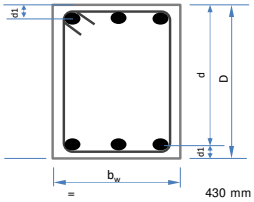
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Approved	AG	AG	AG	AG			Approved	AG	AG	AG	AG		

REFERENCE DOCUMENTS

DOCUMENT NUMBER	DOCUMENT DESCRIPTION
MD-216-1000-EG-ST-CRT-1001	Civil Structural Design Criteria

7.4.2.8 Beam marked NB2

Staad beam number - 40, 45, 47, 49



Width of beam	=	**300** mm	
Depth of beam	=	**500** mm	
Grade of steel	=	420 MPa	
Grade of concrete	=	35 MPa	
Cover	=	50 mm	
Stress reduction factor for flexure	=	0.9	*Table 21.2.2 SBC 304*
Stress reduction factor for shear	=	0.75	*Table 21.2.1 SBC 304*
Stress reduction factor for torsion	=	0.75	*Table 21.2.1 SBC 304*
Effective Depth	=	430 mm	
Dia of longitudinal bar	=	**20** mm	
Dia of stirrup bar	=	**10** mm	

d	=	430	mm
d'	=	70	mm
h	=	120	mm

ANALYSIS RESULTS

Analysis results as obtained from staad

B.M at support	Mz	L/C -	406	Beam No =	1683	=	**72.5**	KNm
B.M at span	Mz	L/C -	408	Beam No =	1683	=	**49.5**	KNm
Shear Force	Fy	L/C -	406	Beam No =	1683	=	**79.1**	KN
Torsion	Mx					=	**0**	KNm

DESIGN FOR FLEXURE

Support Moment

Mu lim = $\emptyset\, 0.85\, f_c'\beta_1 \left(\frac{c}{d}\right)_{max} bd^2 \left(1-\frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assuptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 300 x 430 x 430 (1 - 0.8/2 x 0.375)
= 378.7 KNm
Singly reinforced beam

$\frac{c}{d}$ max = $\frac{0.003}{0.003+0.005}$ = 0.375

b_1 = 0.80 *Table 22.2.2.4.3 SBC 304*

Mu = 72.53 KNm

Mu_2 = Mu - Mu_{lim} = 72.53 - 378.8
= 0.00 KNm

A_{st1} = $\frac{0.85\, f_c' bd}{f_y}\left(1-\sqrt{1-\frac{2.35\, M_u}{\emptyset\, f_c' bd^2}}\right)$
= 0.85 x 35x 300 x 430 x (1-√(1-(2.35 x 72.6 x 10^6)/(0.9 x 35 x 300 x 430^2)))/420
= 457.1 mm^2

A_{st2} = $\frac{M_u - M_{u\,lim}}{\emptyset\, f_s'(d-d')}$
= 0 x 10^6/(0.9 x 420 x (430 - 70))
= 0 mm^2

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y} b_w d$ or $\frac{1.4}{f_y} b_w d$

e_{sc} = 0.0017
Asc = 0.0000 mm^2

= 454.270 mm^2 or 430 mm^2 = 454.270412 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}
= 457.2 + 0.00 + 0
= 457.2 mm^2

Top layer1	**3**	*nos*	**20**	*mm*	*dia*	+		*nos*		*mm*	*dia*
Top layer2		*nos*		*mm*	*dia*	+		*nos*		*mm*	*dia*

Ast provided = 942.478 mm^2 Safe

ρ_w = 0.0073060

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Span Moment

Mu lim = $\emptyset\, 0.85\, f_c'\beta_1 \left(\frac{c}{d}\right)_{max} bd^2 \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assuptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 300 x 430 x 430 (1 - 0.8/2 x 0.375)

= 378.7 KNm

$\frac{c}{d}$ max = $\frac{0.003}{0.003 + 0.005}$ = 0.375

Singly reinforced beam

b_1 = 0.80

Mu = 49.45 KNm

Mu_2 = Mu - Mu_{lim} = 49.45 - 378.8

= 0 KNm

A_{st} = $\frac{0.85\, f_c' bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35\, M_u}{\emptyset\, f_c' bd^2}}\right)$

A_{st2} = $\frac{M_u - M_{u\,lim}}{\emptyset\, f_s'(d - d')}$

= 0.85 x 35x 300 x 430 x (1-√(1-(2.35 x 49.5 x 10^6)/(0.9 x 35 x 300 x 430^2)))/420

= 0 x 10^6/(0.9 x 420 x (430 - 70))

= 309.1 mm^2

= 0 mm^2

e_{sc} = 0.001697674

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y} b_w d$ or $\frac{1.4}{f_y} b_w d$

Asc = 0.0000 mm^2

= 454.3 mm^2 or 430 mm^2 = 454.270412 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 454.3 + 0 + 0

= 454.3 mm^2

bottom layer1	**3**	*nos*	**20**	*mm*	*dia*	+		*nos*		*mm*	*dia*
bottom layer2		*nos*		*mm*	*dia*	+		*nos*		*mm*	*dia*

Ast provided = 942.5 mm^2 Safe

ρ_w = 0.007306029

DESIGN FOR SHEAR

Allowable shear force = Minimum of the following Vc *Correspo*

V_c = $\left(0.16\,\lambda\sqrt{f_c'} + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right) b_w d$ = 129.6 KN

λ = 1 Normal concrete

V_c = $\left(0.16\,\lambda\sqrt{f_c'} + 17\rho_w\right) b_w d$ = 138.1 KN

V_c = $0.29\,\lambda\sqrt{f_c'}\, b_w d$ = 221.3 KN

fV_c = 0.75 x 129.7 = 97.2161 KN > 79.1 KN *Shear Reinforcement not required*

Vs = $V_U - \emptyset V_C$ = $\frac{\emptyset\, A_v f_y d}{s}$

$\frac{A_v}{S}$ = $\frac{V_U - \emptyset V_C}{\emptyset\, f_y d}$ = ((79.1-97.3) x 10^3)/(0.75 x 420 x 430) = 0.00 mm^2/mm

No. of legs = **2** Spacing provided = **200** mm

Provide 2L -10 mm stirrups @ 200 C/C

Spacing of stirrup is minimum of the following

i) Shear Reinforcement s = 0 mm *If Vu > φVc*

ii) Min Asv = $0.062\sqrt{f_c'}\frac{b_w S}{f_t}$ s = 600 mm *Corresponding to table 9.6.3.3 of SBC 304*

or

Min Asv = $0.65\frac{b_w s}{f_{yt}}$ s = 338 mm *Corresponding to table 9.6.3.3 of SBC 304*

iii) effective depth/2 s = 215 mm *Corresponding to table 9.7.6.2.2 of SBC 304*

iv) Combined shear & torsion s = 0 mm

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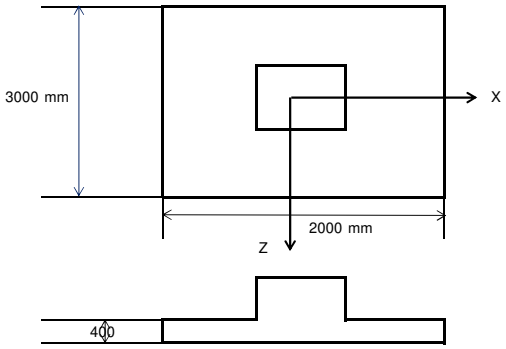
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7.5 FOUNDATION DESIGN

7.5.1.1 DESIGN OF ISOLATED FOUNDATION FOR COLUMN AT 3120/12'_2130/C AND MARKED AS F1A

Finished Ground level	F.G.L	=	-0.500	m
Bottom of foundation	B.O.F	=	-2.500	m
Top of footing (staad support)	T.O.F	=	-2.500	m
Ground Water Level	G.W.L	=	-10.000	m
Surcharge	Ws	=	0.000	kN/m^2
Length of footing	Lz	=	3.00	m
Width of footing	Lx	=	2.00	m
Thickness of footing	D	=	0.40	m
Depth of pedestal	Bx	=	0.80	m
Width of pedestal	Bz	=	0.80	m
Height of Load point from bottom of footing	H	=	0.00	m
Section Modulus of Pedestal	Zxx	=	3.00	m^3
Section Modulus of Pedestal	Zzz	=	2.00	m^3
Allowable Net Bearing Capacity		=	500	kN/m^2
Gross Bearing Capacity		=	536	kN/m^2
Grade of concrete		=	35	N/mm^2
Grade of steel		=	420	N/mm^2
Clear cover		=	50	mm
Top Reinfocement		=	12	mm
Bottom Reinforcement		=	16	mm
Co-efficient of friction between soil & concrete		=	0.3	
Unit wt of soil		=	18	kN/m^3
Unit wt of concrete		=	24	kN/m^3

Weight of soil	=	154.37	kN
Weight of foundation	=	57.60	kN
Uplift	=	0.00	kN
Surcharge	=	0	kN
Pressure on top for tension	=	35.3	kN/m^2



$$\beta_1 = 0.85 - \frac{0.05(f'_c - 28)}{7} \qquad \beta_1 = 0.8 \quad \textit{Table 22.2.2.4.3 SBC 304}$$

CHECK FOR BEARING PRESSURE CAPACITY

L/C	Fx	Fy	Fz	Mx	My	Mz	Sum Fy	Pmax	Pmin	B.C Factor	F.O.S B.P
301	4.4	422.6	10.7	11.74	0.5	-7.6	634.6	113.48	98.05	1	0.21
302	16.2	499.4	4.5	3.07	-0.31	-23.7	711.4	131.45	105.67	1	0.25
303	-8.6	558.2	-1.7	-4.90	0.299	9.4	770.2	134.71	122.01	1	0.25
304	-20.3	481.4	4.6	3.77	1.135	25.6	693.4	129.60	101.52	1	0.24
305	4.5	481.7	11.1	11.91	0.509	-7.9	693.7	123.56	107.67	1	0.23
306	16.3	558.5	4.8	3.24	-0.3	-24.1	770.5	141.53	115.29	1	0.26
307	-8.5	617.2	-1.3	-4.73	0.282	9.1	829.2	144.32	132.07	1	0.27
308	-20.3	540.4	4.9	3.95	1.118	25.2	752.4	139.32	111.47	1	0.26
309	18.9	454.8	11.1	12.02	0.0	-27.5	666.8	128.87	93.38	1	0.24
310	-11.4	481.3	10.8	11.70	1.186	13.8	693.3	126.34	104.75	1	0.24
311	6.4	432.3	13.6	15.51	0.624	-10.3	644.3	117.72	97.04	1	0.22
312	3.4	483.5	9.6	9.99	0.469	-6.4	695.5	122.45	109.37	1	0.23
313	30.6	531.6	4.9	3.35	-0.87	-43.6	743.6	146.85	101.01	1	0.27
314	0.4	558.1	4.6	3.03	0.4	-2.3	770.1	130.53	126.16	1	0.24
315	18.2	509.1	7.4	6.83	-0.212	-26.5	721.1	135.69	104.66	1	0.25
316	15.1	560.3	3.4	1.32	-0.367	-22.6	772.3	140.43	117.00	1	0.26
317	5.9	590.3	-1.3	-4.62	-0.261	-10.4	802.3	140.47	126.95	1	0.26
318	-24.4	616.8	-1.5	-4.94	0.959	30.8	828.8	155.19	121.07	1	0.29
319	-6.6	567.8	1.2	-1.13	0.4	6.7	779.8	133.69	126.23	1	0.25
320	-9.7	619.1	-2.8	-6.65	0.242	10.6	831.1	146.04	130.99	1	0.27
321	-5.9	513.5	5.0	4.05	0.575	5.7	725.5	125.11	116.71	1	0.23
322	-36.2	540	4.7	3.73	1.8	47.0	752.0	150.05	100.61	1	0.28
323	-18.4	491	7.5	7.54	1.233	22.8	703.0	131.09	103.23	1	0.24
324	-21.4	542.3	3.5	2.03	1.078	26.8	754.3	139.76	111.66	1	0.26
325	4.5	466.9	11.0	11.87	0.513	-7.9	678.9	121.03	105.26	1	0.23
326	4.5	466.9	11.0	11.87	0.513	-7.9	678.9	121.03	105.26	1	0.23
327	4.5	466.9	11.0	11.87	0.5	-7.9	678.9	121.03	105.26	1	0.23
328	4.5	466.9	11.0	11.87	0.513	-7.9	678.9	121.03	105.26	1	0.23
329	16.2	543.7	4.7	3.20	-0.323	-24.0	755.7	139.01	112.88	1	0.26
330	16.2	543.7	4.7	3.20	-0.323	-24.0	755.7	139.01	112.88	1	0.26
331	16.2	543.7	4.7	3.20	-0.323	-24.0	755.7	139.01	112.88	1	0.26
332	16.2	543.7	4.7	3.20	-0.3	-24.0	755.7	139.01	112.88	1	0.26
333	-8.5	602.4	-1.4	-4.77	0.286	9.2	814.4	141.91	129.55	1	0.26
334	-8.5	602.4	-1.4	-4.77	0.286	9.2	814.4	141.91	129.55	1	0.26
335	-8.5	602.4	-1.4	-4.77	0.286	9.2	814.4	141.91	129.55	1	0.26
336	-8.5	602.4	-1.4	-4.77	0.3	9.2	814.4	141.91	129.55	1	0.26
337	-20.3	525.6	4.9	3.90	1.122	25.3	737.6	136.89	108.97	1	0.26
338	-20.3	525.6	4.9	3.90	1.122	25.3	737.6	136.89	108.97	1	0.26
339	-20.3	525.6	4.9	3.90	1.122	25.3	737.6	136.89	108.97	1	0.26
340	-20.3	525.6	4.9	3.90	1.122	25.3	737.6	136.89	108.97	1	0.26
341	21.9	237.4	6.6	7.24	-0.413	-30.7	449.4	92.67	57.12	1	0.17
342	-18.5	272.7	6.2	6.81	1.213	24.3	484.7	95.20	66.36	1	0.18
343	5.2	207.5	10.0	11.89	0.5	-7.9	419.5	77.81	62.01	1	0.15

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344	1.2	275.8	4.6	4.54	0.256	-2.6	487.8	84.13	78.46	1	0.16
345	28.9	283.5	2.8	2.04	-0.915	-40.4	495.5	103.45	61.70	1	0.19
346	-11.5	318.8	2.5	1.61	0.711	14.6	530.8	96.31	80.61	1	0.18
347	12.3	253.5	6.2	6.69	0.0	-17.5	465.5	88.58	66.58	1	0.17
348	8.3	321.8	0.8	-0.66	-0.245	-12.3	533.8	95.34	82.58	1	0.18
349	14.1	318.7	-0.8	-2.75	-0.55	-20.5	530.7	99.61	77.28	1	0.19
350	-26.3	354.1	-1.2	-3.17	1.076	34.5	566.1	112.66	76.03	1	0.21
351	-2.6	288.8	2.5	1.91	0.327	2.4	500.8	85.28	81.65	1	0.16
352	-6.6	357.1	-2.8	-5.44	0.12	7.6	569.1	100.45	89.24	1	0.19
353	7.0	272.7	2.9	2.46	-0.048	-10.8	484.7	87.00	74.55	1	0.16
354	-33.4	308	2.6	2.03	1.6	44.2	520.0	109.44	63.88	1	0.20
355	-9.6	242.7	6.3	7.11	0.829	12.0	454.7	84.17	67.39	1	0.16
356	-13.7	311	0.9	-0.24	0.622	17.3	523.0	95.87	78.45	1	0.18
357	2.7	253.6	6.4	7.04	0.315	-4.6	465.6	82.22	72.97	1	0.15
358	2.7	253.6	6.4	7.04	0.3	-4.6	465.6	82.22	72.97	1	0.15
359	2.7	253.6	6.4	7.04	0.315	-4.6	465.6	82.22	72.97	1	0.15
360	2.7	253.6	6.4	7.04	0.315	-4.6	465.6	82.22	72.97	1	0.15
361	9.7	299.7	2.7	1.84	-0.186	-14.2	511.7	93.01	77.54	1	0.17
362	9.7	299.7	2.7	1.84	-0.2	-14.2	511.7	93.01	77.54	1	0.17
363	9.7	299.7	2.7	1.84	-0.186	-14.2	511.7	93.01	77.54	1	0.17
364	9.7	299.7	2.7	1.84	-0.186	-14.2	511.7	93.01	77.54	1	0.17
365	-5.1	334.9	-1.0	-2.94	0.179	5.7	546.9	94.96	87.33	1	0.18
366	-5.1	334.9	-1.0	-2.94	0.2	5.7	546.9	94.96	87.33	1	0.18
367	-5.1	334.9	-1.0	-2.94	0.179	5.7	546.9	94.96	87.33	1	0.18
368	-5.1	334.9	-1.0	-2.94	0.179	5.7	546.9	94.96	87.33	1	0.18
369	-12.2	288.8	2.7	2.26	0.681	15.3	500.8	91.89	75.04	1	0.17
370	-12.2	288.8	2.7	2.26	0.681	15.3	500.8	91.89	75.04	1	0.17
371	-12.2	288.8	2.7	2.26	0.7	15.3	500.8	91.89	75.04	1	0.17
372	-12.2	288.8	2.7	2.26	0.681	15.3	500.8	91.89	75.04	1	0.17

C02

Max P_{max} = 150.05 kN/m^2 Load Case 322 Resultant F.O.S = 0.28

Max P_{min} = 57.12 kN/m^2 Load Case 341 Resultant F.O.S

BEARING PRESSURE CALCULATION FOR BENDING MOMENT AND SHEAR CHECK

L/C	Fx	Fy	Fz	Mx	My	Mz	Pmax	Pmin
401	6.2	591.7	15.0	16.44	0.7	-10.64	109.42	87.82
402	22.7	699.2	6.2	4.29	-0.434	-33.23	134.58	98.49
403	-12.0	781.4	-2.3	-6.86	0.418	13.22	139.13	121.34
404	-28.5	673.9	6.4	5.28	1.589	35.80	131.98	92.66
405	6.3	692.0	15.6	16.73	0.707	-11.23	126.53	104.14
406	22.8	799.5	6.9	4.59	-0.463	-33.82	151.69	114.81
407	-11.9	881.8	-1.7	-6.56	0.389	12.63	155.47	138.47
408	-28.4	774.2	7.0	5.58	1.56	35.21	148.50	109.57
409	36.1	510.8	13.3	14.49	-0.5	-51.15	115.54	54.73
410	-28.5	567.3	12.7	13.81	2.058	36.88	117.59	71.51
411	9.5	462.9	18.7	21.93	0.859	-14.59	91.75	62.55
412	3.0	572.2	10.1	10.17	0.528	-6.22	101.87	88.87
413	50.2	603.0	5.8	4.08	-1.547	-70.50	137.11	63.89
414	-14.4	659.5	5.2	3.40	1.055	17.52	119.81	100.02
415	23.6	555.1	11.2	11.52	-0.144	-33.94	113.33	71.70
416	17.1	664.3	2.6	-0.24	-0.475	-25.58	123.58	97.85
417	20.5	673.5	-1.6	-5.48	-0.8	-30.70	129.43	95.08
418	-44.2	730.0	-2.1	-6.16	1.785	57.33	152.39	90.95
419	-6.1	625.5	3.8	1.96	0.586	5.87	107.84	100.66
420	-12.6	734.8	-4.8	-9.80	0.255	14.23	132.85	112.09
421	6.3	581.3	5.9	4.93	0.187	-11.34	104.19	89.57
422	-58.3	637.8	5.4	4.25	2.788	76.69	146.06	66.54
423	-20.3	533.4	11.3	12.37	1.59	25.22	105.63	72.17
424	-26.7	642.6	2.8	0.61	1.259	33.59	124.10	90.10
425	5.4	536.7	13.0	14.18	0.623	-9.30	98.82	80.08
426	5.4	536.7	13.0	14.18	0.623	-9.30	98.82	80.08
427	5.4	536.7	13.0	14.18	0.623	-9.30	98.82	80.08
428	5.4	536.7	13.0	14.18	0.623	-9.30	98.82	80.08
429	19.5	628.8	5.5	3.77	-0.381	-28.65	120.38	89.22
430	19.5	628.8	5.5	3.77	-0.381	-28.65	120.38	89.22
431	19.5	628.8	5.5	3.77	-0.381	-28.65	120.38	89.22
432	19.5	628.8	5.5	3.77	-0.381	-28.65	120.38	89.22
433	-10.3	699.3	-1.8	-5.79	0.35	11.15	124.06	109.04
434	-10.3	699.3	-1.8	-5.79	0.35	11.15	124.06	109.04
435	-10.3	699.3	-1.8	-5.79	0.35	11.15	124.06	109.04
436	-10.3	699.3	-1.8	-5.79	0.35	11.15	124.06	109.04
437	-24.4	607.1	5.7	4.61	1.353	30.51	117.98	84.39
438	-24.4	607.1	5.7	4.61	1.353	30.51	117.98	84.39
439	-24.4	607.1	5.7	4.61	1.353	30.51	117.98	84.39
440	-24.4	607.1	5.7	4.61	1.353	30.51	117.98	84.39
441	34.7	354.5	9.9	10.88	-0.7	-48.69	87.06	31.11
442	-29.9	411.0	9.3	10.20	1.908	39.34	91.57	45.43
443	8.1	306.6	15.3	18.32	0.71	-12.13	63.27	38.93

Global Project ID	**900707**
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Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	C02	D01			Outotec / L&T	0	1	2	3		
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Approved	AG	AG	AG	AG			Approved	AG	AG	AG	AG		

444	1.6	415.8	6.7	6.56	0.379	-3.77	73.37	65.23
445	45.3	423.7	4.3	3.07	-1.445	-63.21	103.25	37.99
446	-19.3	480.2	3.7	2.39	1.156	24.82	93.24	66.83
447	18.7	375.7	9.7	10.52	-0.043	-26.65	79.45	45.79
448	12.2	485.0	1.1	-1.24	-0.374	-18.28	90.39	71.28
449	23.0	476.5	-1.2	-4.10	-0.9	-33.36	97.46	61.37
450	-41.6	533.0	-1.8	-4.78	1.704	54.67	117.76	59.90
451	-3.6	428.6	4.1	3.35	0.505	3.21	74.15	68.72
452	-10.1	537.8	-4.4	-8.42	0.174	11.57	98.22	81.04
453	12.4	407.4	4.4	3.71	-0.145	-18.84	78.55	57.25
454	-52.2	463.9	3.8	3.03	2.456	69.19	112.92	41.71
455	-14.2	359.4	9.8	11.15	1.258	17.73	72.48	47.32
456	-20.7	468.7	1.2	-0.61	0.927	26.09	91.37	64.87
457	4.0	380.4	9.6	10.57	0.473	-6.84	70.34	56.46
458	4.0	380.4	9.6	10.57	0.473	-6.84	70.34	56.46
459	4.0	380.4	9.6	10.57	0.473	-6.84	70.34	56.46
460	4.0	380.4	9.6	10.57	0.473	-6.84	70.34	56.46
461	14.6	449.5	4.0	2.76	-0.279	-21.36	86.52	63.32
462	14.6	449.5	4.0	2.76	-0.279	-21.36	86.52	63.32
463	14.6	449.5	4.0	2.76	-0.279	-21.36	86.52	63.32
464	14.6	449.5	4.0	2.76	-0.279	-21.36	86.52	63.32
465	-7.7	502.3	-1.5	-4.41	0.269	8.50	89.43	78.00
466	-7.7	502.3	-1.5	-4.41	0.269	8.50	89.43	78.00
467	-7.7	502.3	-1.5	-4.41	0.269	8.50	89.43	78.00
468	-7.7	502.3	-1.5	-4.41	0.269	8.50	89.43	78.00
469	-18.3	433.2	4.1	3.40	1.021	23.01	84.84	59.56
470	-18.3	433.2	4.1	3.40	1.021	23.01	84.84	59.56
471	-18.3	433.2	4.1	3.40	1.021	23.01	84.84	59.56
472	-18.3	433.2	4.1	3.40	1.021	23.01	84.84	59.56

C02

Max Net factored B.P = 155.47 kN/m² Pmin = 31.11 kN/m²

DETERMINATION OF REINFORCEMENT

Bottom Reinforcement

Length of cantilever along X = 0.6 m

Moment about Z Muz = 27.98 kNm

$M_{uzlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

= 798.59 kNm

As $M_{uzlim} > M_{uz}$ Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

= 217.2 mm²

Effective Depth Along X = 342 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement $A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*

greater of or $A_{stmin} = 0.0014bd$

A_{stmin} = 615.6 mm²

A_{req} = 217.2 mm²

Required Spacing = 326.6 mm

Length of cantilever along Z = 1.1 m

Moment about X Mux = 94.06 kNm

$M_{uxlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

= 725.61 kNm

As $M_{uxlim} > M_{ux}$ Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

= 775.3 mm²

Effective Depth Along X = 326 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement $A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*

greater of or $A_{stmin} = 0.0014bd$

A_{stmin} = 586.8 mm²

A_{req} = 775.3 mm²

Required Spacing = 259.3 mm

Provide Bottom Reinforcement = 16 mm dia @ 150 c/c

Ast prov = 1340.4 mm²

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Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	C02	D01			Outotec / L&T	0	1	2	3		
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Top Reinforcement

Length of cantilever along X = 0.6 m
Moment about Z Muz = 0.00 kNm

$M_{uzlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$
= 807.95 kNm

As $M_{uzlim} > M_{uz}$ Tension controlled section
Therefore φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$
= 0.0 mm^2

Effective Depth Along X = 344 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement greater of
A_{stmin} = 50% of $\frac{0.0018 \times 420}{f_y} bd$ since 50% or more prov in
A_{stmin} = 50% of 0.0014bd bottom
Table 24.4.3.2 SBC 304
A_{stmin} = 309.6 mm^2
A_{req} = 0.0 mm^2
Required Spacing = 365.3 mm

Length of cantilever along Z = 1.1 m
Moment about X Mux = 0.00 kNm

$M_{uxlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$
= 752.57 kNm

As $M_{uxlim} > M_{ux}$ Tension controlled section
Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$
= 0.0 mm^2

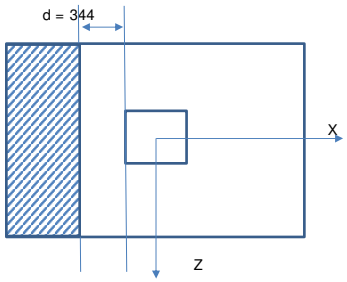
Effective Depth Along Z = 332 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement greater of
A_{stmin} = 50% of $\frac{0.0018 \times 420}{f_y} bd$ since 50% or more prov in
A_{stmin} = 50% of 0.0014bd bottom
Table 24.4.3.2 SBC 304
A_{stmin} = 298.8 mm^2
A_{req} = 0.0 mm^2
Required Spacing = 378.5 mm

Provide top Reinforcement = 12 mm dia @ 150 c/c Ast prov = 754.0 mm^2

ONE WAY SHEAR CKECK



Along X Direction Cantilever L = 0.256 m
Reinforcement ratio ρ_w = 0.0039
Effective Depth Along X d = 344 mm
Shear Force at d distance V_u = 39.80 kN
Allowable shear strength φVc = 244.22 kN O.K

Least of

i) $V_c = \left(0.16\lambda\sqrt{f'_c} + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right) b_w d$
Table 22.5.5.1 SBC 304
λ = 1 Normal concrete
Mu = 5.09 kNm
φ = 0.75 for shear
φVc = 290.145858 kN

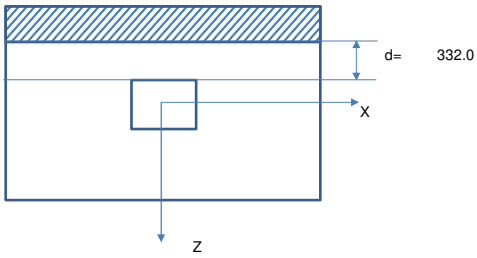
ii) $V_c = \left(0.16\lambda\sqrt{f'_c} + 17\rho_w\right) b_w d$
φVc = 261.306037 kN *Table 22.5.5.1 SBC 304*

iii) $V_c = 0.29\lambda\sqrt{f'_c} b_w d$
= 244.215773 kN

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Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	C02	D01			Outotec / L&T	0	1	2	3		
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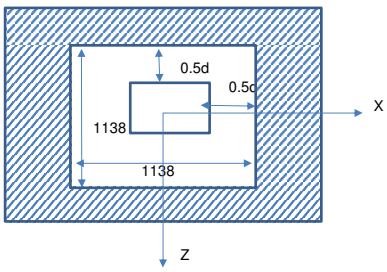
ONE WAY SHEAR CKECK



Along Z Direction Cantilever	L	=	0.768	m		Least of	i)	V_c	=	$\left(0.16\lambda\sqrt{f_c'}+17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$
Reinforcement ratio	ρ_w	=	0.0040							*Table 22.5.5.1 SBC 304*
Effective Depth Along Z	d	=	332	mm				λ	=	1 Normal concrete
Shear Force at d distance	V_u	=	119.40	kN				Mu	=	45.85 kNm
Allowable shear strength	φVc	=	235.70	kN	O.K			φ	=	0.75 for shear
								φVc	=	250.47 kN
							ii)	V_c	=	$\left(0.16\lambda\sqrt{f_c'}+17\rho_w\right)b_w d$
								φVc	=	252.786883 kN
							iii)	V_c	=	$0.29\lambda\sqrt{f_c'}b_w d$
								φVc	=	235.70 kN

TWO WAY SHEAR CKECK

Punching at 0.5d from face of column



Total area		=	6	m2
Punching area		=	1.30	m2
	λ	=	1	
	β	=	1	
	αs	=	40	
	b0	=	4552.0	mm

Allowable shear stress — least of — *Table 22.6.5.2 SBC 304*

v_c	$0.33\lambda\sqrt{f_c'}$	=	1.95 N/mm²
v_c	$0.17\left(1+\frac{2}{\beta}\right)\lambda\sqrt{f_c'}$	=	3.02 N/mm²
v_c	$0.083\left(2+\frac{\alpha_s d}{b_0}\right)\lambda\sqrt{f_c'}$	=	2.44 N/mm²

Design vc = 1.95 N/mm²

Check at 0.5d from column face

Perimeter at 0.5d from column face	u	=	4.55	m		Vc	=	$v_c b_0 d$	
Depth at section	d	=	338	mm			=	3003.77	kN
Shear Force at section	Vu	=	731.5	kN		φ	=	0.75	for shear
Allowable shear strength	φVc	=	2252.83	kN	O.K	φV_n	=	φVc	
						φVc	=	2252.8	kN

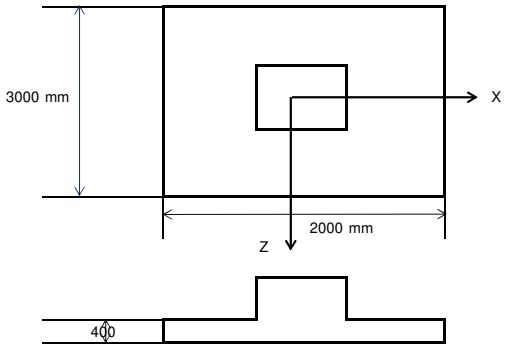
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Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
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7.5.1.2 DESIGN OF ISOLATED FOUNDATION FOR COLUMN AT 3120/14_2130/C AND MARKED AS F1A

Finished Ground level	F.G.L	=	-0.500	m
Bottom of foundation	B.O.F	=	-2.500	m
Top of footing (staad support)	T.O.F	=	-2.500	m
Ground Water Level	G.W.L	=	-10.000	m
Surcharge	Ws	=	0.000	kN/m^2
Length of footing	Lz	=	3.00	m
Width of footing	Lx	=	2.00	m
Thickness of footing	D	=	0.40	m
Depth of pedestal	Bx	=	0.80	m
Width of pedestal	Bz	=	0.80	m
Height of Load point from bottom of footing	H	=	0.000	m
Section Modulus of Pedestal	Zxx	=	3.00	m^3
Section Modulus of Pedestal	Zzz	=	2.00	m^3
Allowable Net Bearing Capacity		=	500	kN/m^2
Gross Bearing Capacity		=	536	kN/m^2
Grade of concrete		=	35	N/mm^2
Grade of steel		=	420	N/mm^2
Clear cover		=	50	mm
Top Reinfocement		=	12	mm
Bottom Reinforcement		=	16	mm
Co-efficient of friction between soil & concrete		=	0.3	
Unit wt of soil		=	18	kN/m^3
Unit wt of concrete		=	24	kN/m^3

Weight of soil	=	154.37	kN
Weight of foundation	=	57.60	kN
Uplift	=	0.00	kN
Surcharge	=	0	kN
Pressure on top for tension	=	35.3	kN/m^2



$$\beta_1 = 0.85 - \frac{0.05(f'_c - 28)}{7}$$ β_1 = 0.8 *Table 22.2.2.4.3 SBC 304*

CHECK FOR BEARING PRESSURE CAPACITY

L/C	Fx	Fy	Fz	Mx	My	Mz	Sum Fy	Pmax	Pmin	B.C Factor	F.O.S B.P
301	13.3	391.3	0.1	-0.04	-0.4	-15.9	603.3	108.53	92.56	1	0.20
302	26.4	447.5	0.0	-0.27	-1.444	-33.3	659.5	126.63	93.19	1	0.24
303	3.3	508.4	-2.1	-3.22	-0.036	-1.6	720.4	121.94	118.18	1	0.23
304	-9.7	452.2	-2.0	-2.99	1.054	15.7	664.2	119.54	101.85	1	0.22
305	14.7	458.3	-0.1	-0.31	-0.327	-17.5	670.3	120.58	102.85	1	0.22
306	27.7	514.6	-0.2	-0.54	-1.4	-34.8	726.6	138.69	103.50	1	0.26
307	4.7	575.4	-2.3	-3.50	-0.009	-3.2	787.4	133.98	128.47	1	0.25
308	-8.3	519.2	-2.2	-3.27	1.081	14.1	731.2	130.01	113.71	1	0.24
309	28.8	471.2	0.0	-0.22	-1.2	-36.8	683.2	132.33	95.40	1	0.25
310	-1.6	406	-0.1	-0.25	0.679	4.6	618.0	105.37	100.62	1	0.20
311	15.6	410.3	1.3	1.58	-0.4	-19.0	622.3	113.72	93.70	1	0.21
312	13.6	453.5	-0.6	-0.92	-0.305	-16.0	665.5	119.20	102.62	1	0.22
313	41.8	527.4	-0.1	-0.45	-2.333	-54.1	739.4	150.42	96.03	1	0.28
314	11.5	462.3	-0.2	-0.48	-0.4	-12.7	674.3	118.90	105.86	1	0.22
315	28.6	466.6	1.2	1.35	-1.49	-36.3	678.6	131.69	94.50	1	0.25
316	26.6	509.7	-0.7	-1.14	-1.395	-33.3	721.7	137.29	103.26	1	0.26
317	18.8	588.3	-2.2	-3.40	-0.925	-22.4	800.3	145.74	121.02	1	0.27
318	-11.6	523.2	-2.3	-3.43	0.997	18.9	735.2	133.13	111.92	1	0.25
319	5.6	527.5	-0.9	-1.60	-0.1	-4.6	739.5	126.10	120.39	1	0.24
320	3.5	570.6	-2.8	-4.10	0.013	-1.6	782.6	132.61	128.25	1	0.25
321	5.7	532.1	-2.1	-3.17	0.165	-5.1	744.1	127.64	120.38	1	0.24
322	-24.6	466.9	-2.2	-3.20	2.1	36.2	678.9	132.33	93.96	1	0.25
323	-7.5	471.2	-0.8	-1.37	1.008	12.7	683.2	120.65	107.07	1	0.23
324	-9.5	514.4	-2.7	-3.87	1.103	15.7	726.4	130.19	111.93	1	0.24
325	14.4	441.5	-0.1	-0.25	-0.334	-17.1	653.5	117.56	100.27	1	0.22
326	14.4	441.5	-0.1	-0.25	-0.334	-17.1	653.5	117.56	100.27	1	0.22
327	14.4	441.5	-0.1	-0.25	-0.3	-17.1	653.5	117.56	100.27	1	0.22
328	14.4	441.5	-0.1	-0.25	-0.334	-17.1	653.5	117.56	100.27	1	0.22
329	27.4	497.8	-0.2	-0.47	-1.424	-34.4	709.8	135.67	100.92	1	0.25
330	27.4	497.8	-0.2	-0.47	-1.424	-34.4	709.8	135.67	100.92	1	0.25
331	27.4	497.8	-0.2	-0.47	-1.424	-34.4	709.8	135.67	100.92	1	0.25
332	27.4	497.8	-0.2	-0.47	-1.4	-34.4	709.8	135.67	100.92	1	0.25
333	4.3	558.7	-2.3	-3.43	-0.016	-2.8	770.7	130.98	125.91	1	0.24
334	4.3	558.7	-2.3	-3.43	-0.016	-2.8	770.7	130.98	125.91	1	0.24
335	4.3	558.7	-2.3	-3.43	-0.016	-2.8	770.7	130.98	125.91	1	0.24
336	4.3	558.7	-2.3	-3.43	0.0	-2.8	770.7	130.98	125.91	1	0.24
337	-8.7	502.4	-2.2	-3.20	1.074	14.5	714.4	127.39	110.74	1	0.24
338	-8.7	502.4	-2.2	-3.20	1.074	14.5	714.4	127.39	110.74	1	0.24
339	-8.7	502.4	-2.2	-3.20	1.074	14.5	714.4	127.39	110.74	1	0.24
340	-8.7	502.4	-2.2	-3.20	1.074	14.5	714.4	127.39	110.74	1	0.24
341	27.2	274.3	0.1	0.01	-1.425	-35.8	486.3	98.94	63.15	1	0.18
342	-13.2	187.4	0.0	-0.03	1.138	19.4	399.4	76.26	56.86	1	0.14
343	9.6	193.2	1.9	2.41	-0.3	-12.0	405.2	74.35	60.71	1	0.14

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	C02	D01			Outotec / L&T	0	1	2	3		
Date	27-01-2020	12-03-2020	25-04-2022	12/10/2023			Date	27-01-2020	12-03-2020	25-04-2022	12/10/2023		
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS DT. 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER SUPPORT STRUCTURE	AS BUILT			Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS DT. 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER SUPPORT STRUCTURE	AS BUILT		
Prepared By	APR	APR	APR	APR			Prepared By	APR	APR	APR	APR		
Checked	SBY	SBY	SBY	SBY			Checked	SBY	SBY	SBY	SBY		
Approved	AG	AG	AG	AG			Approved	AG	AG	AG	AG		

344	6.9	250.7	-0.6	-0.92	-0.174	-8.0	462.7	81.43	72.80	1	0.15
345	35.1	308	0.1	-0.13	-2.079	-46.2	520.0	109.79	63.54	1	0.20
346	-5.4	221.2	-0.1	-0.16	0.484	9.0	433.2	76.75	67.64	1	0.14
347	17.5	226.9	1.9	2.27	-1.0	-22.4	438.9	85.11	61.18	1	0.16
348	14.8	284.4	-0.7	-1.05	-0.828	-18.4	496.4	92.28	73.18	1	0.17
349	21.2	344.6	-1.2	-1.90	-1.234	-27.2	556.6	106.98	78.54	1	0.20
350	-19.3	257.7	-1.3	-1.94	1.329	28.0	469.7	92.91	63.64	1	0.17
351	3.6	263.5	0.6	0.50	-0.11	-3.4	475.5	81.13	77.36	1	0.15
352	0.9	321	-2.0	-2.82	0.017	0.6	533.0	90.06	87.59	1	0.17
353	13.4	310.8	-1.1	-1.76	-0.58	-16.8	522.8	96.11	78.15	1	0.18
354	-27.1	224	-1.3	-1.80	2.0	38.4	436.0	92.44	52.88	1	0.17
355	-4.2	229.7	0.7	0.64	0.544	7.0	441.7	77.30	69.92	1	0.14
356	-6.9	287.2	-1.9	-2.69	0.671	11.0	499.2	89.58	76.81	1	0.17
357	8.0	234.8	0.0	-0.02	-0.212	-9.6	446.8	79.25	69.67	1	0.15
358	8.0	234.8	0.0	-0.02	-0.2	-9.6	446.8	79.25	69.67	1	0.15
359	8.0	234.8	0.0	-0.02	-0.212	-9.6	446.8	79.25	69.67	1	0.15
360	8.0	234.8	0.0	-0.02	-0.212	-9.6	446.8	79.25	69.67	1	0.15
361	15.8	268.5	0.0	-0.16	-0.866	-20.0	480.5	90.11	70.05	1	0.17
362	15.8	268.5	0.0	-0.16	-0.9	-20.0	480.5	90.11	70.05	1	0.17
363	15.8	268.5	0.0	-0.16	-0.866	-20.0	480.5	90.11	70.05	1	0.17
364	15.8	268.5	0.0	-0.16	-0.866	-20.0	480.5	90.11	70.05	1	0.17
365	2.0	305.1	-1.3	-1.93	-0.021	-1.0	517.1	87.31	85.05	1	0.16
366	2.0	305.1	-1.3	-1.93	0.0	-1.0	517.1	87.31	85.05	1	0.16
367	2.0	305.1	-1.3	-1.93	-0.021	-1.0	517.1	87.31	85.05	1	0.16
368	2.0	305.1	-1.3	-1.93	-0.021	-1.0	517.1	87.31	85.05	1	0.16
369	-5.8	271.3	-1.2	-1.79	0.633	9.4	483.3	85.85	75.24	1	0.16
370	-5.8	271.3	-1.2	-1.79	0.633	9.4	483.3	85.85	75.24	1	0.16
371	-5.8	271.3	-1.2	-1.79	0.6	9.4	483.3	85.85	75.24	1	0.16
372	-5.8	271.3	-1.2	-1.79	0.633	9.4	483.3	85.85	75.24	1	0.16

C02

Max P_{max} = 135.67 kN/m^2 Load Case 329 Resultant F.O.S = 0.25

Max P_{min} = 52.88 kN/m^2 Load Case 354

BEARING PRESSURE CALCULATION FOR BENDING MOMENT AND SHEAR CHECK

L/C	Fx	Fy	Fz	Mx	My	Mz	Pmax	Pmin
401	18.7	547.8	0.1	-0.05	-0.5	-22.33	102.48	80.12
402	36.9	626.6	0.0	-0.37	-2.021	-46.56	127.84	81.03
403	4.6	711.8	-3.0	-4.51	-0.05	-2.26	121.26	116.00
404	-13.6	633.0	-2.8	-4.19	1.476	21.97	117.88	93.12
405	21.0	661.7	-0.3	-0.52	-0.45	-25.00	122.96	97.61
406	39.3	740.5	-0.4	-0.84	-1.976	-49.23	148.31	98.52
407	7.0	825.7	-3.3	-4.98	-0.005	-4.93	141.74	133.49
408	-11.3	746.9	-3.2	-4.66	1.521	19.30	135.69	113.28
409	47.5	566.3	0.1	-0.13	-2.4	-61.86	125.36	63.41
410	-17.3	427.3	-0.1	-0.19	1.749	26.40	84.48	57.96
411	19.3	436.5	3.0	3.71	-0.553	-23.86	85.92	59.58
412	15.0	528.5	-1.1	-1.61	-0.35	-17.44	97.34	78.83
413	63.1	633.8	0.0	-0.41	-3.659	-82.63	147.08	64.18
414	-1.7	494.8	-0.2	-0.47	0.441	5.63	85.44	79.50
415	35.0	504.0	2.9	3.44	-1.861	-44.63	107.46	60.54
416	30.6	596.0	-1.2	-1.89	-1.658	-38.21	119.07	79.60
417	35.4	706.9	-2.5	-3.95	-2.0	-44.65	141.46	94.17
418	-29.4	567.9	-2.8	-4.01	2.131	43.60	117.79	71.51
419	7.3	577.1	0.4	-0.11	-0.172	-6.66	99.55	92.82
420	2.9	669.1	-3.8	-5.43	0.031	-0.24	113.44	109.59
421	19.8	639.3	-2.4	-3.68	-0.661	-23.88	119.72	93.38
422	-45.0	500.4	-2.6	-3.73	3.439	64.37	116.83	49.97
423	-8.4	509.5	0.5	0.17	1.137	14.11	92.03	77.80
424	-12.7	601.6	-3.6	-5.15	1.34	20.53	112.25	88.28
425	16.7	503.0	0.0	-0.18	-0.411	-19.92	93.86	73.81
426	16.7	503.0	0.0	-0.18	-0.411	-19.92	93.86	73.81
427	16.7	503.0	0.0	-0.18	-0.411	-19.92	93.86	73.81
428	16.7	503.0	0.0	-0.18	-0.411	-19.92	93.86	73.81
429	32.3	570.6	-0.1	-0.46	-1.719	-40.69	115.60	74.60
430	32.3	570.6	-0.1	-0.46	-1.719	-40.69	115.60	74.60
431	32.3	570.6	-0.1	-0.46	-1.719	-40.69	115.60	74.60
432	32.3	570.6	-0.1	-0.46	-1.719	-40.69	115.60	74.60
433	4.6	643.6	-2.7	-4.00	-0.03	-2.72	109.96	104.57
434	4.6	643.6	-2.7	-4.00	-0.03	-2.72	109.96	104.57
435	4.6	643.6	-2.7	-4.00	-0.03	-2.72	109.96	104.57
436	4.6	643.6	-2.7	-4.00	-0.03	-2.72	109.96	104.57
437	-11.0	576.1	-2.5	-3.73	1.279	18.05	106.28	85.75
438	-11.0	576.1	-2.5	-3.73	1.279	18.05	106.28	85.75
439	-11.0	576.1	-2.5	-3.73	1.279	18.05	106.28	85.75
440	-11.0	576.1	-2.5	-3.73	1.279	18.05	106.28	85.75
441	42.8	415.4	0.2	0.02	-2.3	-56.29	97.38	41.08
442	-22.0	276.4	0.0	-0.04	1.842	31.97	62.06	30.07
443	14.6	285.6	3.1	3.86	-0.46	-18.29	58.03	37.17

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	C02	D01			Outotec / L&T	0	1	2	3		
Date	27-01-2020	12-03-2020	25-04-2022	12/10/2023			Date	27-01-2020	12-03-2020	25-04-2022	12/10/2023		
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS DT. 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER SUPPORT STRUCTURE	AS BUILT			Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS DT. 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER SUPPORT STRUCTURE	AS BUILT		
Prepared By	APR	APR	APR	APR			Prepared By	APR	APR	APR	APR		
Checked	SBY	SBY	SBY	SBY			Checked	SBY	SBY	SBY	SBY		
Approved	AG	AG	AG	AG			Approved	AG	AG	AG	AG		

444	10.3	377.6	-1.0	-1.46	-0.257	-11.87	69.36	56.51
445	54.5	466.0	0.1	-0.19	-3.239	-71.86	113.66	41.67
446	-10.3	327.1	-0.1	-0.25	0.861	16.39	62.79	46.24
447	26.4	336.2	3.0	3.66	-1.441	-33.87	74.19	37.88
448	22.0	428.3	-1.1	-1.67	-1.238	-27.45	85.66	57.10
449	33.7	520.8	-1.8	-2.85	-2.0	-43.38	109.44	64.16
450	-31.0	381.9	-2.0	-2.90	2.128	44.87	87.05	40.25
451	5.6	391.0	1.1	1.00	-0.174	-5.39	68.19	62.14
452	1.3	483.1	-3.0	-4.33	0.029	1.03	82.47	78.56
453	22.0	470.2	-1.7	-2.64	-0.991	-27.81	93.15	63.58
454	-42.8	331.2	-1.9	-2.70	3.109	60.45	86.32	24.08
455	-6.1	340.4	1.2	1.20	0.807	10.19	62.23	51.24
456	-10.5	432.4	-2.9	-4.12	1.01	16.61	81.74	62.39
457	12.0	352.1	0.1	-0.03	-0.318	-14.35	65.87	51.50
458	12.0	352.1	0.1	-0.03	-0.318	-14.35	65.87	51.50
459	12.0	352.1	0.1	-0.03	-0.318	-14.35	65.87	51.50
460	12.0	352.1	0.1	-0.03	-0.318	-14.35	65.87	51.50
461	23.7	402.8	0.0	-0.24	-1.299	-29.93	82.18	52.09
462	23.7	402.8	0.0	-0.24	-1.299	-29.93	82.18	52.09
463	23.7	402.8	0.0	-0.24	-1.299	-29.93	82.18	52.09
464	23.7	402.8	0.0	-0.24	-1.299	-29.93	82.18	52.09
465	3.0	457.6	-1.9	-2.90	-0.032	-1.45	77.96	74.58
466	3.0	457.6	-1.9	-2.90	-0.032	-1.45	77.96	74.58
467	3.0	457.6	-1.9	-2.90	-0.032	-1.45	77.96	74.58
468	3.0	457.6	-1.9	-2.90	-0.032	-1.45	77.96	74.58
469	-8.8	406.9	-1.8	-2.69	0.949	14.13	75.78	59.86
470	-8.8	406.9	-1.8	-2.69	0.949	14.13	75.78	59.86
471	-8.8	406.9	-1.8	-2.69	0.949	14.13	75.78	59.86
472	-8.8	406.9	-1.8	-2.69	0.949	14.13	75.78	59.86

C02

Max Net factored B.P = 148.31 kN/m² Pmin = 24.08 kN/m²

DETERMINATION OF REINFORCEMENT

Bottom Reinforcement

Length of cantilever along X = 0.6 m

Moment about Z Muz = 26.70 kNm

$M_{uzlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

= 798.59 kNm

As $M_{uzlim} > M_{uz}$ Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

= 207.1 mm²

Effective Depth Along X = 342 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement greater of

$A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*

or

$A_{stmin} = 0.0014bd$

A_{stmin} = 615.6 mm²

A_{req} = 207.1 mm²

Required Spacing = 326.6 mm

Length of cantilever along Z = 1.1 m

Moment about X Mux = 89.73 kNm

$M_{uxlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

= 725.61 KNm

As $M_{uxlim} > M_{ux}$ Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

= 739.1 mm²

Effective Depth Along X = 326 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement greater of

$A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*

or

$A_{stmin} = 0.0014bd$

A_{stmin} = 586.8 mm²

A_{req} = 739.1 mm²

Required Spacing = 272.0 mm

Provide Bottom Reinforcement = 16 mm dia @ 150 c/c Ast prov = 1340.4 mm²

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	C02	D01		
Date	27-01-2020	12-03-2020	25-04-2022	12/10/2023		
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS DT. 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER SUPPORT STRUCTURE	AS BUILT		
Prepared By	APR	APR	APR	APR		
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Approved	AG	AG	AG	AG		

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Top Reinforcement

Length of cantilever along X = 0.6 m

Moment about Z: M_{uz} = 0.00 kNm

$$M_{uzlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$$

M_{uzlim} = 807.95 kNm

As $M_{uzlim} > M_{uz}$ Therefore: Tension controlled section

φ = 0.9 *Table 21.2.2 SBC 304*

$$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$$

A_{st} = 0.0 mm^2

Effective Depth Along X = 344 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement greater of:

$A_{stmin} = 50\%\ of\ \frac{0.0018 \times 420}{f_y} bd$ since 50% or more prov in bottom

$A_{stmin} = 50\%\ of\ 0.0014bd$ *Table 24.4.3.2 SBC 304*

A_{stmin} = 309.6 mm^2

A_{req} = 0.0 mm^2

Required Spacing = 365.3 mm

Length of cantilever along Z = 1.1 m

Moment about X: M_{ux} = 0.00 kNm

$$M_{uxlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$$

M_{uxlim} = 752.57 kNm

As $M_{uxlim} > M_{ux}$ Therefore: Tension controlled section

Φ = 0.9 *Table 21.2.2 SBC 304*

$$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$$

A_{st} = 0.0 mm^2

Effective Depth Along Z = 332 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement greater of:

$A_{stmin} = 50\%\ of\ \frac{0.0018 \times 420}{f_y} bd$ since 50% or more prov in bottom

$A_{stmin} = 50\%\ of\ 0.0014bd$ *Table 24.4.3.2 SBC 304*

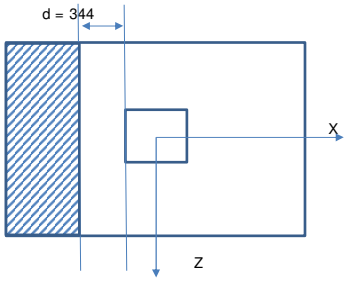
A_{stmin} = 298.8 mm^2

A_{req} = 0.0 mm^2

Required Spacing = 378.5 mm

Provide top Reinforcement = 12 mm dia @ 150 c/c Ast prov = 754.0 mm^2

ONE WAY SHEAR CKECK



Along X Direction Cantilever: L = 0.256 m

Reinforcement ratio: ρ_w = 0.0039

Effective Depth Along X: d = 344 mm

Shear Force at d distance: V_u = 37.97 kN

Allowable shear strength: φVc = 244.22 kN O.K

Least of

i) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right) b_w d$ *Table 22.5.5.1 SBC 304*

λ = 1 Normal concrete

Mu = 4.86 kNm

φ = 0.75 for shear

φVc = 290.145858 kN

ii) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\right) b_w d$

φVc = 261.306037 KN *Table 22.5.5.1 SBC 304*

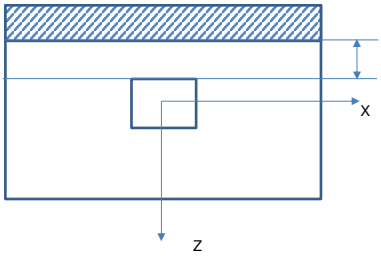
iii) $V_c = 0.29\lambda\sqrt{f_c'} b_w d$

= 244.215773 KN

Global Project ID	**900707**
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Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	C02	D01			Outotec / L&T	0	1	2	3		
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Approved	AG	AG	AG	AG			Approved	AG	AG	AG	AG		

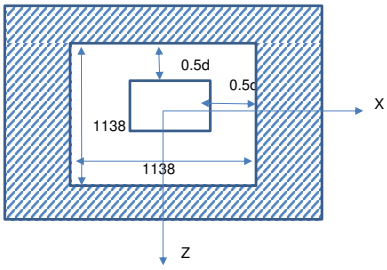
ONE WAY SHEAR CKECK



Along Z Direction Cantilever	L	=	0.768	m		Least of	i)	V_c	=	$\left(0.16\lambda\sqrt{f_c'}+17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$
Reinforcement ratio	ρ_w	=	0.0040							*Table 22.5.5.1 SBC 304*
Effective Depth Along Z	d	=	332	mm				λ	=	1 Normal concrete
Shear Force at d distance	V_u	=	113.90	kN				Mu	=	43.74 kNm
Allowable shear strength	φVc	=	235.70	kN	O.K			φ	=	0.75 for shear
								φVc	=	250.47 kN
							ii)	V_c	=	$\left(0.16\lambda\sqrt{f_c'}+17\rho_w\right)b_w d$
								φVc	=	252.786883 kN
							iii)	V_c	=	$0.29\lambda\sqrt{f_c'}b_w d$
								φVc	=	235.70 kN

TWO WAY SHEAR CKECK

Punching at 0.5d from face of column



Total area		=	6	m2	
Punching area		=	1.30	m2	
	λ	=	1		
	β	=	1		
	αs	=	40		
	b0	=	4552.0	mm	

Allowable shear stress least of *Table 22.6.5.2 SBC 304*

v_c	$0.33\lambda\sqrt{f_c'}$	=	1.95 N/mm²
v_c	$0.17\left(1+\frac{2}{\beta}\right)\lambda\sqrt{f_c'}$	=	3.02 N/mm²
v_c	$0.083\left(2+\frac{\alpha_s d}{b_0}\right)\lambda\sqrt{f_c'}$	=	2.44 N/mm²

Design vc = 1.95 N/mm²

Check at 0.5d from column face

Perimeter at 0.5d from column face	u	=	4.55	m		Vc	=	$v_c b_0 d$	
Depth at section	d	=	338	mm			=	3003.77	kN
Shear Force at section	Vu	=	697.8	kN		φ	=	0.75	for shear
Allowable shear strength	φVc	=	2252.83	kN	O.K	φV_n	=	φVc	
						φVc	=	2252.8	kN

Global Project ID	900707
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Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

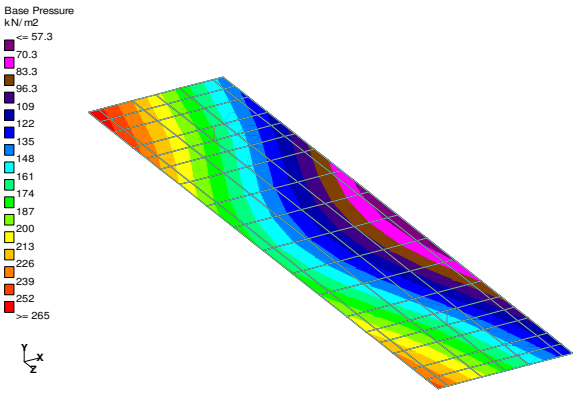
Ma'aden Revision No:	A01	C01	C02	D01			Outotec / L&T Revision No.:	0	1	2	3	
Date	27-01-2020	12-03-2020	25-04-2022	12/10/2023			Date	27-01-2020	12-03-2020	25-04-2022	12/10/2023	
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Checked	SBY	SBY	SBY	SBY			Checked	SBY	SBY	SBY	SBY	
Approved	AG	AG	AG	AG			Approved	AG	AG	AG	AG	

7.5.2.1 DESIGN OF COMBINED FOUNDATION FOR COLUMN AT 3120/H_3120/11 & 3120/F_3120/11 AND FOUNDATION MKD. AS CF2

For Calculation of Base pressure and design forces for column & foundation Raft, Refer STAAD File **CF2.STD**

Staad Supoort Node for column at **3120/H_3120/11** **1**
3120/F_3120/11 **4**

Finished Floor level	F.F.L	=	0.0	m	(Corresponds to RL +313.0m)
Finished Ground level	F.G.L	=	-0.5	m	
Bottom of foundation	B.O.F	=	-2.5	m	
Top of pedestal (staad support)	T.O.F	=	-2.3	m	
Ground Water Level	G.W.L	=	-10	m	
Length of footing	Lz	=	9.0	m	
Width of footing	Lx	=	2.0	m	
Thickness of footing	D	=	0.5	m	
Depth of pedestal	Bx	=	0.7	m	
Width of pedestal	Bz	=	0.7	m	
Unit wt of soil		=	18.0	KN/m^3	
Allowable Net Bearing Capacity		=	500	KN/m^2	
Gross Bearing Capacity		=	536	KN/m^2	



1) Check for Gross Base Pressure :

Maximum Base Pressure,	=	265.0	KN/m^2	Load Case	322
Minimum Base Pressure	=	22.0	KN/m^2	Load Case	354

2) Design of foundation Raft :

Slab Thickness =	500 mm	Yield Strength of Main Reinforcement , fy =	420 N/mm^2
Width =	1000 mm	Yield Strength of Shear Reinforcement , fyt =	420 N/mm^2
Clear Cover =	50 mm	Strength Reduction Factor of Main Reinforcement , Φ =	0.9
Compressive Strength of Concrete , fc' =	35 N/mm^2	Strength Reduction Factor of Shear Reinforcement , Φt =	0.75

STAAD OUTPUT FROM FACTORED LOAD COMBINATION :

(Staad Group : RAFT_MOMENT)

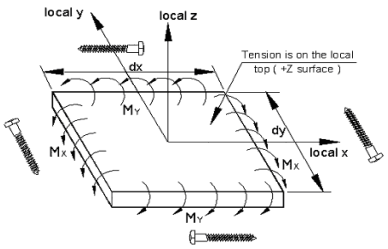
MOMENT	VALUE	UNIT	TORSIONAL MOMENT	VALUE	UNIT	
Max. Mx	314.191	kN-m	Corresponding Mxy	4.549	kN-m	*(Plate No. - 52 , L/C No. - 408)*
Min. Mx	-190.331	kN-m	Corresponding Mxy	25.040	kN-m	*(Plate No. - 17 , L/C No. - 408)*
Max. My	9.214	kN-m	Corresponding Mxy	-23.232	kN-m	*(Plate No. - 97 , L/C No. - 454)*
Min. My	-99.737	kN-m	Corresponding Mxy	-20.957	kN-m	*(Plate No. - 22 , L/C No. - 408)*

FOR TOP FACE REINFORCEMRNT :

Mx1 =	Mx + abs(Mxy) =	314.191 + 4.549 =	318.74 kN-m
My1 =	My + abs(Mxy) =	9.214 + 23.232 =	32.446 kN-m
Mx2 =	Mx+abs(Mxy^2/My) =	314.191 + 2.246 =	316.437 kN-m
My2 =	My+abs(Mxy^2/Mx) =	9.214 + 1.718 =	10.932 kN-m

DESIGN MOMENT CALCULATION FOR TOP FACE REINFORCEMENT :

Mxd = 318.74 kN-m
Myd = 32.446 kN-m



ALONG LOCAL X - DIRECTION :

Mu = 318.74 kN-m
Φ*Kn = Mu/bd^2
Mu/bd^2 = 318.74X10^6/(1000X440^2) = 1.646 N/mm^2 deff. = 440 mm < 6.9 N/mm^2
Pt = 0.0045 Reqd. Ast = 0.0045X1000X440/100 = 19.8 cm^2
Provided **20** mm @ **150** C/C (Provide Ast = 20.94 cm2) at Horizontal
Provided **0** mm @ **150** C/C (Provide Ast = 0 cm2) at Horizontal
Total Provide Ast = 20.94 cm2 at Horizontal
Total Provide Pt = 0.0048

ALONG LOCAL Y - DIRECTION :

Mu = 32.446 kN-m
Φ*Kn = Mu/bd^2
Mu/bd^2 = 32.446X10^6/(1000X420^2) = 0.168 N/mm^2 deff. = 420 mm < 6.9 N/mm^2
Pt = 0.0004 Reqd. Ast = 0.0004X1000X440/100 = 1.68 cm^2
Provided **20** mm @ **150** C/C (Provide Ast = 20.94 cm2) at Vertical
Provided **0** mm @ **150** C/C (Provide Ast = 0 cm2) at Vertical
Total Provide Ast = 20.94 cm2 at Vertical **O.K.**
Total Provide Pt = 0.005

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Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	C02	D01			Outotec / L&T Revision No.:	0	1	2	3	
Date	27-01-2020	12-03-2020	25-04-2022	12/10/2023			Date	27-01-2020	12-03-2020	25-04-2022	12/10/2023	
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Checked	SBY	SBY	SBY	SBY			Checked	SBY	SBY	SBY	SBY	
Approved	AG	AG	AG	AG			Approved	AG	AG	AG	AG	

FOR BOTTOM FACE REINFORCEMRNT :

Mx1 = Mx - abs(Mxy) = -190.331 - 25.04 = -215.371 kN-m
My1 = My - abs(Mxy) = -99.737 - 20.957 = -120.694 kN-m
Mx2 = Mx-abs(Mxy^2/My) = -190.331 - 6.287 = -196.618 kN-m
My2 = My-abs(Mxy^2/Mx) = -99.737 - 2.308 = -97.429 kN-m

DESIGN MOMENT CALCULATION FOR BOTTOM FACE REINFORCEMENT :

Mxd = -215.371 kN-m
Myd = -120.694 kN-m

ALONG LOCAL X - DIRECTION :

Mu = 215.371 kN-m
Φ*Kn = Mu/bd^2 deff. = 442 mm
Mu/bd^2 = 215.371X10^6/(1000X442^2) = 1.102 N/mm^2 < 6.9 N/mm^2
Pt = 0.0030 Reqd. Ast = 0.003X1000X442/100 = 13.26 cm^2
Provided **16** mm @ **150** C/C (Provide Ast = 13.4 cm2) at Horizontal
Provided **0** mm @ **150** C/C (Provide Ast = 0 cm2) at Horizontal
Total Provide Ast = 13.4 cm2) at Horizontal **O.K.**
Total Provide Pt = 0.003

ALONG LOCAL Y - DIRECTION :

Mu = 120.694 kN-m
Φ*Kn = Mu/bd^2 deff. = 426 mm
Mu/bd^2 = 120.694X10^6/(1000X426^2) = 0.665 N/mm^2 < 6.9 N/mm^2
Pt = 0.0018 Reqd. Ast = 0.0018X1000X426/100 = 7.67 cm^2
Provided **16** mm @ **150** C/C (Provide Ast = 13.4 cm2) at Vertical
Provided **0** mm @ **150** C/C (Provide Ast = 0 cm2) at Vertical
Total Provide Ast = 13.4 cm2) at Vertical **O.K.**
Total Provide Pt = 0.0031

SHEAR CHECK

(Staad Group : RAFT_SHEAR)

(FROM FACTORED LOAD COMBINATION)

ALONG LOCAL X - DIRECTION :

SQx = 0.701 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 50 , L/C No. - 408)*
Vu = 309.84 kN

Provided Pt = 0.003
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.003X309.842X10^3X442/215.371/10^6)X1000X442/10^3
= 432.719 kN
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.003)X1000X442/10^3
= 440.927 kN
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X5.92X1000X442/10^3
= 758.323 kN
HENCE, Vc = 432.719 kN
Φt*Vc = 0.75X432.719 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 324.539 kN **> 309.842T , Hence O.K.**

ALONG LOCAL Y - DIRECTION :

SQy = -0.1 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 14 , L/C No. - 408)*
Vu = 42.6 kN

Pt provided = 0.0031
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0031X42.6X10^3X426/120.694/10^3)X1000X426/10^3
= 406.616 kN
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0031)X1000X426/10^3
= 425.69 KN
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X5.92X1000X426/10^3
= 730.872 kN
HENCE, Vc = 406.616 kN
Φt*Vc = 0.75X406.616 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 304.962 kN **> 42.6T , Hence O.K.**

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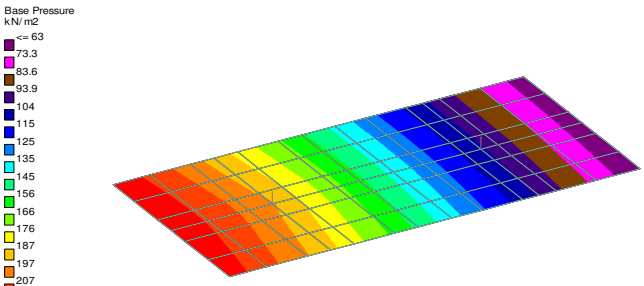
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Checked	SBY	SBY	SBY	SBY			Checked	SBY	SBY	SBY	SBY	
Approved	AG	AG	AG	AG			Approved	AG	AG	AG	AG	

7.5.3.1 DESIGN OF COMBINED FOUNDATION FOR COLUMN AT 3120/H_3120/13 & 3120/H_3120/14 AND FOUNDATION MKD. AS CF3

For Calculation of Base pressure and design forces for column & foundation Raft, Refer STAAD File **CF3_1.STD**

Staad Supoort Node for column at **3120/H_3120/13** **2**
3120/H_3120/14 **3**

Finished Floor level	F.F.L	=	0.0	m	(Corresponds to RL +313.0m)
Finished Ground level	F.G.L	=	-0.5	m	
Bottom of foundation	B.O.F	=	-2.5	m	
Top of pedestal (staad support)	T.O.F	=	-2.3	m	
Ground Water Level	G.W.L	=	-10	m	
Length of footing	Lz	=	6.1	m	
Width of footing	Lx	=	3.0	m	
Thickness of footing	D	=	0.6	m	Surcharge Load =
Depth of pedestal	Bx	=	0.7	m	50 kN/m^2
Width of pedestal	Bz	=	0.7	m	
Unit wt of soil		=	18.0	KN/m^3	
Allowable Net Bearing Capacity		=	500	KN/m^2	
Gross Bearing Capacity		=	536	KN/m^2	



1) Check for Gross Base Pressure :

Maximum Base Pressure,	=	228.0	KN/m^2	Load Case	306
Minimum Base Pressure	=	30.8	KN/m^2	Load Case	342

2) Design of foundation Raft :

Slab Thickness =	600 mm	Yield Strength of Main Reinforcement , fy =	420 N/mm^2
Width =	1000 mm	Yield Strength of Shear Reinforcement , fyt =	420 N/mm^2
Clear Cover =	50 mm	Strength Reduction Factor of Main Reinforcement , Φ =	0.9
Compressive Strength of Concrete , fc' =	35 N/mm^2	Strength Reduction Factor of Shear Reinforcement , Φt =	0.75

STAAD OUTPUT FROM FACTORED LOAD COMBINATION :

(Staad Group : RAFT_MOMENT)

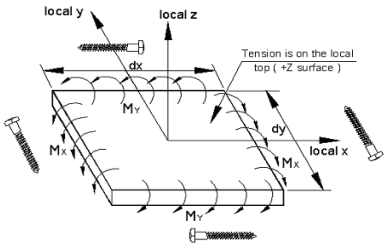
MOMENT	VALUE	UNIT	TORSIONAL MOMENT	VALUE	UNIT	
Max. Mx	76.660	kN-m	Corresponding Mxy	3.392	kN-m	*(Plate No. - 112 , L/C No. - 405)*
Min. Mx	-262.346	kN-m	Corresponding Mxy	28.218	kN-m	*(Plate No. - 88 , L/C No. - 406)*
Max. My	-0.930	kN-m	Corresponding Mxy	-4.182	kN-m	*(Plate No. - 52 , L/C No. - 454)*
Min. My	-217.896	kN-m	Corresponding Mxy	-29.297	kN-m	*(Plate No. - 69 , L/C No. - 406)*

FOR TOP FACE REINFORCEMRNT :

Mx1 =	Mx + abs(Mxy) =	76.66 + 3.392 =	80.052 kN-m
My1 =	My + abs(Mxy) =	-0.93 + 4.182 =	3.252 kN-m
Mx2 =	Mx+abs(Mxy^2/My) =	76.66 + 12.372 =	89.032 kN-m
My2 =	My+abs(Mxy^2/Mx) =	-0.93 + 0.228 =	-0.702 kN-m

DESIGN MOMENT CALCULATION FOR TOP FACE REINFORCEMENT :

Mxd = 80.052 kN-m
Myd = 3.252 kN-m



ALONG LOCAL X - DIRECTION :

Mu =	80.052 kN-m							
Φ*Kn =	Mu/bd^2					deff. =	544	mm
Mu/bd^2 =	80.052X10^6/(1000X544^2) =	0.271	N/mm^2			<	6.9	N/mm^2
Pt =	0.0007	Reqd. Ast =	0.0007X1000X544/100 =	3.81	cm^2			
Provided	12	mm @	150	C/C	(Provide Ast =	7.54	cm2) at Horizontal	
Provided	0	mm @	150	C/C	(Provide Ast =	0	cm2) at Horizontal	
					Total Provide Ast =	7.54	cm2 at Horizontal	
					Total Provide Pt =	0.0014		

ALONG LOCAL Y - DIRECTION :

Mu =	3.252 kN-m							
Φ*Kn =	Mu/bd^2					deff. =	532	mm
Mu/bd^2 =	3.252X10^6/(1000X532^2) =	0.011	N/mm^2			<	6.9	N/mm^2
Pt =	0.0000	Reqd. Ast =	0X1000X544/100 =	0	cm^2			
Provided	12	mm @	150	C/C	(Provide Ast =	7.54	cm2) at Vertical	
Provided	0	mm @	150	C/C	(Provide Ast =	0	cm2) at Vertical	
					Total Provide Ast =	7.54	cm2 at Vertical	O.K.
					Total Provide Pt =	0.0014		

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	C02	D01			Outotec / L&T Revision No.:	0	1	2	3	
Date	27-01-2020	12-03-2020	25-04-2022	12/10/2023			Date	27-01-2020	12-03-2020	25-04-2022	12/10/2023	
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS DT. 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER SUPPORT STRUCTURE	AS BUILT			Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS DT. 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER SUPPORT	AS BUILT	
Prepared By	APR	APR	APR	APR			Prepared By	APR	APR	APR	APR	
Checked	SBY	SBY	SBY	SBY			Checked	SBY	SBY	SBY	SBY	
Approved	AG	AG	AG	AG			Approved	AG	AG	AG	AG	

FOR BOTTOM FACE REINFORCEMRNT :

Mx1 = Mx - abs(Mxy) = -262.346 - 28.218 = -290.564 kN-m
My1 = My - abs(Mxy) = -217.896 - 29.297 = -247.193 kN-m
Mx2 = Mx-abs(Mxy^2/My) = -262.346 - 3.654 = -266 kN-m
My2 = My-abs(Mxy^2/Mx) = -217.896 - 3.272 = -214.624 kN-m

DESIGN MOMENT CALCULATION FOR BOTTOM FACE REINFORCEMENT :

Mxd = -290.564 kN-m
Myd = -247.193 kN-m

ALONG LOCAL X - DIRECTION :

Mu = 290.564 kN-m
Φ*Kn = Mu/bd^2
Mu/bd^2 = 290.564X10^6/(1000X540^2) = 0.996 N/mm^2 deff. = 540 mm
< 6.9 N/mm^2
Pt = 0.0027 Reqd. Ast = 0.0027X1000X540/100 = 14.58 cm^2
Provided 20 mm @ 150 C/C (Provide Ast = 20.94 cm2) at Horizontal
Provided 0 mm @ 150 C/C (Provide Ast = 0 cm2) at Horizontal
Total Provide Ast = 20.94 cm2) at Horizontal **O.K.**
Total Provide Pt = 0.0039

ALONG LOCAL Y - DIRECTION :

Mu = 247.193 kN-m
Φ*Kn = Mu/bd^2
Mu/bd^2 = 247.193X10^6/(1000X520^2) = 0.914 N/mm^2 deff. = 520 mm
< 6.9 N/mm^2
Pt = 0.0025 Reqd. Ast = 0.0025X1000X520/100 = 13 cm^2
Provided 20 mm @ 150 C/C (Provide Ast = 20.94 cm2) at Vertical
Provided 0 mm @ 150 C/C (Provide Ast = 0 cm2) at Vertical
Total Provide Ast = 20.94 cm2) at Vertical **O.K.**
Total Provide Pt = 0.004

SHEAR CHECK

(Staad Group : RAFT_SHEAR)
(FROM FACTORED LOAD COMBINATION)

ALONG LOCAL X - DIRECTION :

SQx = 0.448 N/mm^2 *(At d-distance from face of the column)* *(Plate No. - 80 , L/C No. - 406)*
Vu = 241.92 kN

Provided Pt = 0.0039
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0039X241.92X10^3X540/290.564/10^6)X1000X540/10^3
= 527.246 kN
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0039)X1000X540/10^3
= 546.951 kN
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X5.92X1000X540/10^3
= 926.458 kN
HENCE, Vc = 527.246 kN
Φt*Vc = 0.75X527.246 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 395.435 kN **> 241.92T , Hence O.K.**

ALONG LOCAL Y - DIRECTION :

SQy = 0.166 N/mm^2 *(At d-distance from face of the column)* *(Plate No. - 98 , L/C No. - 406)*
Vu = 86.32 kN

Pt provided = 0.004
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.004X86.32X10^3X520/247.193/10^3)X1000X520/10^3
= 498.639 kN
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.004)X1000X520/10^3
= 527.578 KN
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X5.92X1000X520/10^3
= 892.145 kN
HENCE, Vc = 498.639 kN
Φt*Vc = 0.75X498.639 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 373.979 kN **> 86.32T , Hence O.K.**

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

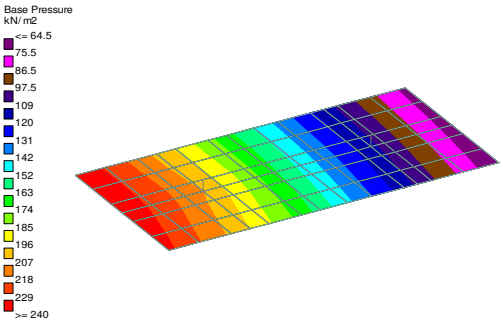
Ma'aden Revision No:	A01	C01	C02	D01			Outotec / L&T Revision No.:	0	1	2	3	
Date	27-01-2020	12-03-2020	25-04-2022	12/10/2023			Date	27-01-2020	12-03-2020	25-04-2022	12/10/2023	
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS DT. 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER SUPPORT STRUCTURE	AS BUILT			Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS DT. 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER SUPPORT	AS BUILT	
Prepared By	APR	APR	APR	APR			Prepared By	APR	APR	APR	APR	
Checked	SBY	SBY	SBY	SBY			Checked	SBY	SBY	SBY	SBY	
Approved	AG	AG	AG	AG			Approved	AG	AG	AG	AG	

7.5.3.2 DESIGN OF COMBINED FOUNDATION FOR COLUMN AT 3120/F_3120/13 & 3120/F_3120/14 AND FOUNDATION MKD. AS CF3

For Calculation of Base pressure and design forces for column & foundation Raft, Refer STAAD File **CF3_2.STD**

Staad Supoort Node for column at **3120/F_3120/13** **5**
3120/F_3120/14 **6**

Finished Floor level	F.F.L	=	0.0	m	(Corresponds to RL +313.0m)
Finished Ground level	F.G.L	=	-0.5	m	
Bottom of foundation	B.O.F	=	-2.5	m	
Top of pedestal (staad support)	T.O.F	=	-2.3	m	
Ground Water Level	G.W.L	=	-10	m	
Length of footing	Lz	=	6.1	m	
Width of footing	Lx	=	3.0	m	Surcharge Load =
Thickness of footing	D	=	0.6	m	50 kN/m^2
Depth of pedestal	Bx	=	0.7	m	
Width of pedestal	Bz	=	0.7	m	
Unit wt of soil		=	18.0	KN/m^3	
Allowable Net Bearing Capacity		=	500	KN/m^2	
Gross Bearing Capacity		=	536	KN/m^2	



1) Check for Gross Base Pressure :

Maximum Base Pressure,	=	240.5	KN/m^2	Load Case 306
Minimum Base Pressure	=	31.7	KN/m^2	Load Case 342

2) Design of foundation Raft :

Slab Thickness =	600 mm	Yield Strength of Main Reinforcement , fy =	420 N/mm^2
Width =	1000 mm	Yield Strength of Shear Reinforcement , fyt =	420 N/mm^2
Clear Cover =	50 mm	Strength Reduction Factor of Main Reinforcement , Φ =	0.9
Compressive Strength of Concrete , fc' =	35 N/mm^2	Strength Reduction Factor of Shear Reinforcement , Φt =	0.75

STAAD OUTPUT FROM FACTORED LOAD COMBINATION :

(Staad Group : RAFT_MOMENT)

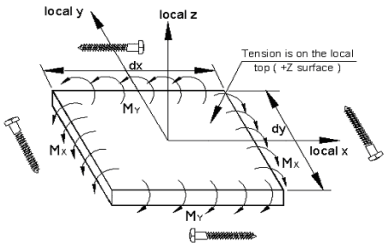
MOMENT	VALUE	UNIT	TORSIONAL MOMENT	VALUE	UNIT	
Max. Mx	74.392	kN-m	Corresponding Mxy	-0.543	kN-m	*(Plate No. - 112 , L/C No. - 405)*
Min. Mx	-275.772	kN-m	Corresponding Mxy	27.719	kN-m	*(Plate No. - 88 , L/C No. - 406)*
Max. My	-1.206	kN-m	Corresponding Mxy	-4.815	kN-m	*(Plate No. - 52 , L/C No. - 454)*
Min. My	-228.297	kN-m	Corresponding Mxy	-35.451	kN-m	*(Plate No. - 69 , L/C No. - 406)*

FOR TOP FACE REINFORCEMRNT :

Mx1 =	Mx + abs(Mxy) =	74.392 + 0.543 =	74.935 kN-m
My1 =	My + abs(Mxy) =	-1.206 + 4.815 =	3.609 kN-m
Mx2 =	Mx+abs(Mxy^2/My) =	74.392 + 0.244 =	74.636 kN-m
My2 =	My+abs(Mxy^2/Mx) =	-1.206 + 0.312 =	-0.894 kN-m

DESIGN MOMENT CALCULATION FOR TOP FACE REINFORCEMENT :

Mxd = 74.935 kN-m
Myd = 3.609 kN-m



ALONG LOCAL X - DIRECTION :

Mu =	74.935 kN-m							
Φ*Kn =	Mu/bd^2					deff. =	544 mm	
Mu/bd^2 =	74.935X10^6/(1000X544^2) =	0.253	N/mm^2	<	6.9	N/mm^2		
Pt =	0.0007	Reqd. Ast =	0.0007X1000X544/100 =	3.81	cm^2			
Provided	12	mm @	150	C/C	(Provide Ast =	7.54	cm2) at Horizontal	
Provided	0	mm @	150	C/C	(Provide Ast =	0	cm2) at Horizontal	
					Total Provide Ast =	7.54	cm2 at Horizontal	
					Total Provide Pt =	0.0014		

ALONG LOCAL Y - DIRECTION :

Mu =	3.609 kN-m							
Φ*Kn =	Mu/bd^2					deff. =	532 mm	
Mu/bd^2 =	3.609X10^6/(1000X532^2) =	0.012	N/mm^2	<	6.9	N/mm^2		
Pt =	0.0000	Reqd. Ast =	0X1000X544/100 =	0	cm^2			
Provided	12	mm @	150	C/C	(Provide Ast =	7.54	cm2) at Vertical	
Provided	0	mm @	150	C/C	(Provide Ast =	0	cm2) at Vertical	
					Total Provide Ast =	7.54	cm2 at Vertical	O.K.
					Total Provide Pt =	0.0014		

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	C02	D01		
Date	27-01-2020	12-03-2020	25-04-2022	12/10/2023		
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS DT. 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER SUPPORT STRUCTURE	AS BUILT		
Prepared By	APR	APR	APR	APR		
Checked	SBY	SBY	SBY	SBY		
Approved	AG	AG	AG	AG		

Outotec / L&T Revision No.:	0	1	2	3	
Date	27-01-2020	12-03-2020	25-04-2022	12/10/2023	
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS DT. 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER SUPPORT	AS BUILT	
Prepared By	APR	APR	APR	APR	
Checked	SBY	SBY	SBY	SBY	
Approved	AG	AG	AG	AG	

FOR BOTTOM FACE REINFORCEMRNT :

Mx1 = Mx - abs(Mxy) = -275.772 - 27.719 = -303.491 kN-m
My1 = My - abs(Mxy) = -228.297 - 35.451 = -263.748 kN-m
Mx2 = Mx-abs(Mxy^2/My) = -275.772 - 3.366 = -279.138 kN-m
My2 = My-abs(Mxy^2/Mx) = -228.297 - 4.557 = -223.74 kN-m

DESIGN MOMENT CALCULATION FOR BOTTOM FACE REINFORCEMENT :

Mxd = -303.491 kN-m
Myd = -263.748 kN-m

ALONG LOCAL X - DIRECTION :

Mu = 303.491 kN-m
Φ*Kn = Mu/bd^2
Mu/bd^2 = 303.491X10^6/(1000X540^2) = 1.041 N/mm^2 deff. = 540 mm
< 6.9 N/mm^2
Pt = 0.0028 Reqd. Ast = 0.0028X1000X540/100 = 15.12 cm^2
Provided 20 mm @ 150 C/C (Provide Ast = 20.94 cm2) at Horizontal
Provided 0 mm @ 150 C/C (Provide Ast = 0 cm2) at Horizontal
Total Provide Ast = 20.94 cm2) at Horizontal **O.K.**
Total Provide Pt = 0.0039

ALONG LOCAL Y - DIRECTION :

Mu = 263.748 kN-m
Φ*Kn = Mu/bd^2
Mu/bd^2 = 263.748X10^6/(1000X520^2) = 0.975 N/mm^2 deff. = 520 mm
< 6.9 N/mm^2
Pt = 0.0026 Reqd. Ast = 0.0026X1000X520/100 = 13.52 cm^2
Provided 20 mm @ 150 C/C (Provide Ast = 20.94 cm2) at Vertical
Provided 0 mm @ 150 C/C (Provide Ast = 0 cm2) at Vertical
Total Provide Ast = 20.94 cm2) at Vertical **O.K.**
Total Provide Pt = 0.004

SHEAR CHECK

(Staad Group : RAFT_SHEAR)
(FROM FACTORED LOAD COMBINATION)

ALONG LOCAL X - DIRECTION :

SQx = 0.461 N/mm^2 *(At d-distance from face of the column)* *(Plate No. - 80 , L/C No. - 406)*
Vu = 248.94 kN

Provided Pt = 0.0039
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0039X248.94X10^3X540/303.491/10^6)X1000X540/10^3
= 527.007 kN
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0039)X1000X540/10^3
= 546.951 kN
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X5.92X1000X540/10^3
= 926.458 kN
HENCE, Vc = 527.007 kN
Φt*Vc = 0.75X527.007 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 395.255 kN **> 248.94T , Hence O.K.**

ALONG LOCAL Y - DIRECTION :

SQy = 0.179 N/mm^2 *(At d-distance from face of the column)* *(Plate No. - 98 , L/C No. - 406)*
Vu = 93.08 kN

Pt provided = 0.004
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.004X93.08X10^3X520/263.748/10^3)X1000X520/10^3
= 498.707 kN
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.004)X1000X520/10^3
= 527.578 KN
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X5.92X1000X520/10^3
= 892.145 kN
HENCE, Vc = 498.707 kN
Φt*Vc = 0.75X498.707 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 374.03 kN **> 93.08T , Hence O.K.**

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

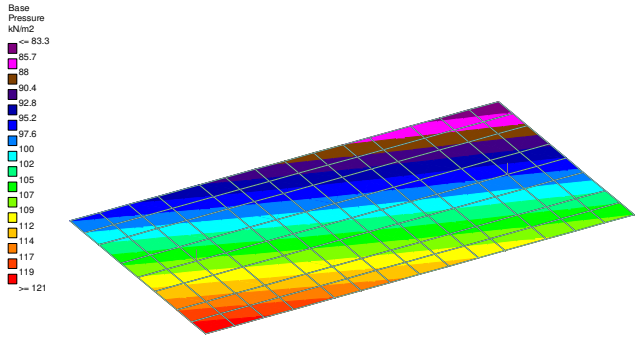
Ma'aden Revision No:	A01	C01	C02	D01			Outotec / L&T Revision No.:	0	1	2	3	
Date	27-01-2020	12-03-2020	25-04-2022	12/10/2023			Date	27-01-2020	12-03-2020	25-04-2022	12/10/2023	
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS DT. 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER SUPPORT STRUCTURE	AS BUILT			Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS DT. 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER SUPPORT STRUCTURE	AS BUILT	
Prepared By	APR	APR	APR	APR			Prepared By	APR	APR	APR	APR	
Checked	SBY	SBY	SBY	SBY			Checked	SBY	SBY	SBY	SBY	
Approved	AG	AG	AG	AG			Approved	AG	AG	AG	AG	

7.5.3.3 DESIGN OF COMBINED FOUNDATION FOR COLUMN AT 3120/12' & 3120/14 AND FOUNDATION MKD. AS CF4

For Calculation of Base pressure and design forces for column & foundation Raft, Refer STAAD File **CF3_1.STD**

Staad Supoort Node for column at **3120/12' 1002**
3120/14 1003

Finished Floor level	F.F.L	=	0.0	m	(Corresponds to RL +313.0m)
Finished Ground level	F.G.L	=	-0.5	m	
Bottom of foundation	B.O.F	=	-2.5	m	
Top of pedestal (staad support)	T.O.F	=	-2.3	m	
Ground Water Level	G.W.L	=	-10	m	
Length of footing	Lz	=	7.3	m	
Width of footing	Lx	=	4.0	m	
Thickness of footing	D	=	2.4	m	Surcharge Load =
Depth of pedestal	Bx	=	1.0	m	50 kN/m^2
Width of pedestal	Bz	=	1.0	m	
Unit wt of soil		=	18.0	KN/m^3	
Allowable Net Bearing Capacity		=	500	KN/m^2	
Gross Bearing Capacity		=	536	KN/m^2	



1) Check for Gross Base Pressure :

Maximum Base Pressure,	=	121.4	KN/m^2	Load Case	311
Minimum Base Pressure	=	20.7	KN/m^2	Load Case	352

2) Design of foundation Raft :

Slab Thickness =	2350 mm	Yield Strength of Main Reinforcement , fy =	420 N/mm^2
Width =	1000 mm	Yield Strength of Shear Reinforcement , fyt =	420 N/mm^2
Clear Cover =	50 mm	Strength Reduction Factor of Main Reinforcement , Φ =	0.9
Compressive Strength of Concrete , fc' =	35 N/mm^2	Strength Reduction Factor of Shear Reinforcement , Φt =	0.75

STAAD OUTPUT FROM FACTORED LOAD COMBINATION :

(Staad Group : RAFT_MOMENT)

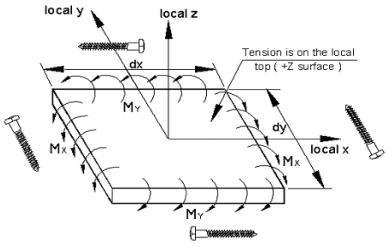
MOMENT	VALUE	UNIT	TORSIONAL MOMENT	VALUE	UNIT	
Max. Mx	171.860	kN-m	Corresponding Mxy	15.523	kN-m	*(Plate No. - 165 , L/C No. - 405)*
Min. Mx	-41.995	kN-m	Corresponding Mxy	-4.425	kN-m	*(Plate No. - 26 , L/C No. - 411)*
Max. My	59.818	kN-m	Corresponding Mxy	2.880	kN-m	*(Plate No. - 17 , L/C No. - 420)*
Min. My	-153.418	kN-m	Corresponding Mxy	-5.078	kN-m	*(Plate No. - 22 , L/C No. - 411)*

FOR TOP FACE REINFORCEMRNT :

Mx1 =	Mx + abs(Mxy) =	171.86 + 15.523 =	187.383	kN-m
My1 =	My + abs(Mxy) =	59.818 + 2.88 =	62.698	kN-m
Mx2 =	Mx+abs(Mxy^2/My) =	171.86 + 4.028 =	175.888	kN-m
My2 =	My+abs(Mxy^2/Mx) =	59.818 + 0.048 =	59.866	kN-m

DESIGN MOMENT CALCULATION FOR TOP FACE REINFORCEMENT :

Mxd = 187.383 kN-m
Myd = 62.698 kN-m



ALONG LOCAL X - DIRECTION :

deff. = 2290 mm

Mu =	187.383 kN-m						
Φ*Kn =	Mu/bd^2				Ast min =	20.61	cm2
Mu/bd^2 =	187.383X10^6/(1000X2290^2) =	0.036	N/mm^2	<	6.9	N/mm^2	
Pt =	0.0001	Reqd. Ast =	0.0001X1000X2290/100 =	2.29	cm^2		
Provided	20 mm @	150	C/C	(Provide Ast =	20.94	cm2) at Horizontal	
Provided	0 mm @	150	C/C	(Provide Ast =	0	cm2) at Horizontal	
				Total Provide Ast =	20.94	cm2 at Horizontal	
				Total Provide Pt =	0.0009		

ALONG LOCAL Y - DIRECTION :

deff. = 2270 mm

Mu =	62.698 kN-m						
Φ*Kn =	Mu/bd^2						
Mu/bd^2 =	62.698X10^6/(1000X2270^2) =	0.012	N/mm^2	<	6.9	N/mm^2	
Pt =	0.0000	Reqd. Ast =	0X1000X2290/100 =		0	cm^2	
Provided	20 mm @	150	C/C	(Provide Ast =	20.94	cm2) at Vertical	
Provided	0 mm @	150	C/C	(Provide Ast =	0	cm2) at Vertical	
				Total Provide Ast =	20.94	cm2 at Vertical	O.K.
				Total Provide Pt =	0.0009		

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	C02	D01		
Date	27-01-2020	12-03-2020	25-04-2022	12/10/2023		
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS DT. 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER SUPPORT STRUCTURE	AS BUILT		
Prepared By	APR	APR	APR	APR		
Checked	SBY	SBY	SBY	SBY		
Approved	AG	AG	AG	AG		

Outotec / L&T Revision No.:	0	1	2	3	
Date	27-01-2020	12-03-2020	25-04-2022	12/10/2023	
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS DT. 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER SUPPORT STRUCTURE	AS BUILT	
Prepared By	APR	APR	APR	APR	
Checked	SBY	SBY	SBY	SBY	
Approved	AG	AG	AG	AG	

FOR BOTTOM FACE REINFORCEMRNT :

Mx1 = Mx - abs(Mxy) = -41.995 - 4.425 = -46.42 kN-m
My1 = My - abs(Mxy) = -153.418 - 5.078 = -158.496 kN-m
Mx2 = Mx-abs(Mxy^2/My) = -41.995 - 0.128 = -42.123 kN-m
My2 = My-abs(Mxy^2/Mx) = -153.418 - 0.614 = -152.804 kN-m

DESIGN MOMENT CALCULATION FOR BOTTOM FACE REINFORCEMENT :

Mxd = -46.42 kN-m
Myd = -158.496 kN-m

ALONG LOCAL X - DIRECTION :

Mu = 46.42 kN-m
Φ*Kn = Mu/bd^2
Mu/bd^2 = 46.42X10^6/(1000X2290^2) = 0.009 N/mm^2 < 6.9 N/mm^2 deff. = 2290 mm
Pt = 0.0000 Reqd. Ast = 0X1000X2290/100 = 0 cm^2
Provided **20** mm @ **150** C/C (Provide Ast = 20.94 cm2) at Horizontal
Provided **0** mm @ **150** C/C (Provide Ast = 0 cm2) at Horizontal
Total Provide Ast = 20.94 cm2) at Horizontal **O.K.**
Total Provide Pt = 0.0009

ALONG LOCAL Y - DIRECTION :

Mu = 158.496 kN-m
Φ*Kn = Mu/bd^2
Mu/bd^2 = 158.496X10^6/(1000X2270^2) = 0.031 N/mm^2 < 6.9 N/mm^2 deff. = 2270 mm
Pt = 0.0001 Reqd. Ast = 0.0001X1000X2270/100 = 2.27 cm^2
Provided **20** mm @ **150** C/C (Provide Ast = 20.94 cm2) at Vertical
Provided **0** mm @ **150** C/C (Provide Ast = 0 cm2) at Vertical
Total Provide Ast = 20.94 cm2) at Vertical **O.K.**
Total Provide Pt = 0.0009

SHEAR CHECK

(Staad Group : RAFT_SHEAR)
(FROM FACTORED LOAD COMBINATION)

ALONG LOCAL X - DIRECTION :

SQx = 0.043 N/mm^2 *(At d-distance from face of the column)* *(Plate No. - 70 , L/C No. - 411)*
Vu = 98.47 kN

Provided Pt = 0.0009
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0009X98.47X10^3X2290/46.42/10^6)X1000X2290/10^3
= 2337.852 kN
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0009)X1000X2290/10^3
= 2202.689 kN
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X5.92X1000X2290/10^3
= 3928.869 kN
HENCE, Vc = 2202.689 kN
Φt*Vc = 0.75X2202.689 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 1652.017 kN **> 98.47T , Hence O.K.**

ALONG LOCAL Y - DIRECTION :

SQy = -0.038 N/mm^2 *(At d-distance from face of the column)* *(Plate No. - 14 , L/C No. - 411)*
Vu = 86.26 kN

Pt provided = 0.0009
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0009X86.26X10^3X2270/158.496/10^3)X1000X2270/10^3
= 2191.628 kN
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0009)X1000X2270/10^3
= 2183.451 KN
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X5.92X1000X2270/10^3
= 3894.555 kN
HENCE, Vc = 2183.451 kN
Φt*Vc = 0.75X2183.451 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 1637.588 kN **> 86.26T , Hence O.K.**

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No	A01	C01	C02	D01
Date	1/27/2020	3/12/2020	4/25/2022	12/10/2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS DT. 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER SUPPORT STRUCTURE	AS BUILT
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Checked	SBY	SBY	SBY	SBY
Approved	AG	AG	AG	AG

Outotec / L&T Revision No.:	0	1	2	3
Date	1/27/2020	3/12/2020	4/25/2022	12/10/2023
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Prepared By	APR	APR	APR	APR
Checked	SBY	SBY	SBY	SBY
Approved	AG	AG	AG	AG

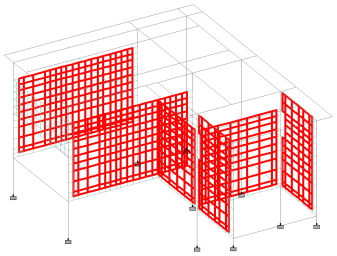
C02

7.6 DESIGN OF WALL :

(STAAD File : Crusher Support Structure R2)

7.6.1 DESIGN OF 300MM WALL :

Wall Thickness =	300	mm
Width =	1000	mm
Clear Cover =	40	mm
Compressive Strength of Concrete , fc' =	45	N/mm^2
Yield Strength of Main Reinforcement , fy =	420	N/mm^2
Yield Strength of Shear Reinforcement , fyt =	420	N/mm^2
Strength Reduction Factor of Main Reinforcement , Φ =	0.9	
Strength Reduction Factor of Shear Reinforcement , Φt =	0.75	



STAAD Group - 300MM_WALL_B

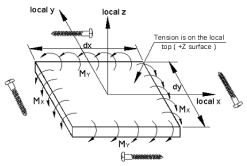
STAAD OUTPUT FROM FACTORED LOAD COMBINATION :

(Staad Group : 300MM_WALL_B)

MOMENT	VALUE	UNIT	TORSIONAL MOMENT	VALUE	UNIT	
Max. Mx	3.1	Mton-m	Corresponding Mxy	0.9	Mton-m	*(Plate No. - 687 , L/C No. - 405)*
Min. Mx	-3.4	Mton-m	Corresponding Mxy	-0.3	Mton-m	*(Plate No. - 643 , L/C No. - 405)*
Max. My	4.2	Mton-m	Corresponding Mxy	-0.1	Mton-m	*(Plate No. - 814 , L/C No. - 406)*
Min. My	-4	Mton-m	Corresponding Mxy	-0.2	Mton-m	*(Plate No. - 495 , L/C No. - 405)*

REINFORCEMENT FOR OUTER FACE (TOP +Z):

Mx1 =	Mx + abs(Mxy) =	3.1 + 0.9 =	4	Mton-m
My1 =	My + abs(Mxy) =	4.2 + 0.1 =	4.3	Mton-m
Mx2 =	Mx+abs(Mxy^2/My) =	3.1 + 0.193 =	3.293	Mton-m
My2 =	My+abs(Mxy^2/Mx) =	4.2 + 0.003 =	4.203	Mton-m



DESIGN MOMENT CALCULATION FOR OUTER FACE REINFORCEMENT :

Mxd =	4	Mton-m			
Myd =	4.3	Mton-m			

ALONG X - DIRECTION :

Mu =	4	Mton-m					
Φ*Kn =	Mu/bd^2				deff. =	254	mm
Mu/bd^2 =			4X10^7/(1000X254^2) =	0.62 N/mm^2	<	8.5	N/mm^2
Pt =	0.0017		Reqd. Ast =	0.0017X1000X254/100 =		4.32	cm^2
Provided	12	mm @	200	C/C (Provide As:	5.65	cm2)	
Provided	0	mm @	200	C/C (Provide As:	0	cm2)	
				Total Provide Ast =	5.65	cm2	O.K.
				Total Provide Pt =	0.002		

ALONG Y - DIRECTION :

Mu =	4.3	Mton-m					
Φ*Kn =	Mu/bd^2				deff. =	242	mm
Mu/bd^2 =			4.3X10^7/(1000X242^2) =	0.667 N/mm^2	<	8.5	N/mm^2
Pt =	0.0018		Reqd. Ast =	0.0018X1000X254/100 =		4.36	cm^2
Provided	12	mm @	200	C/C (Provide As:	5.65	cm2)	
Provided	0	mm @	200	C/C (Provide As:	0	cm2)	
				Total Provide Ast =	5.65	cm2	O.K.
				Total Provide Pt =	0.002		

REINFORCEMENT FOR INNER FACE (BOTTOM +Z):

Mx1 =	Mx - abs(Mxy) =	-3.4 - 0.3 =	-3.7	Mton-m
My1 =	My - abs(Mxy) =	-4 - 0.2 =	-4.2	Mton-m
Mx2 =	Mx-abs(Mxy^2/My) =	-3.4 - 0.023 =	-3.42	Mton-m
My2 =	My-abs(Mxy^2/Mx) =	-4 - 0.012 =	-3.99	Mton-m

DESIGN MOMENT CALCULATION FOR INNER FACE REINFORCEMENT :

Mxd =	-3.7	Mton-m
Myd =	-4.2	Mton-m

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Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No	A01	C01	C02	D01
Date	1/27/2020	3/12/2020	4/25/2022	12/10/2023
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Date	1/27/2020	3/12/2020	4/25/2022	12/10/2023
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ALONG X - DIRECTION :

Mu = 3.7 Mton-m

Φ*Kn = Mu/bd^2 = 3.7X10^7/(1000X254^2) = 0.574 N/mm^2 < deff. = 254 mm

Mu/bd^2 = 8.5 N/mm^2

Pt = 0.0015 Reqd. Ast = 0.0015X1000X254/100 = 3.81 cm^2

Provided 12 mm @ 200 C/C (Provide As: 5.65 cm2)

Provided 0 mm @ 200 C/C (Provide As: 0 cm2)

Total Provide Ast = 5.65 cm2) **O.K.**

Total Provide Pt = 0.002

ALONG Y - DIRECTION :

Mu = 4.2 Mton-m

Φ*Kn = Mu/bd^2 = 4.2X10^7/(1000X242^2) = 0.717 N/mm^2 < deff. = 242 mm

Mu/bd^2 = 8.5 N/mm^2

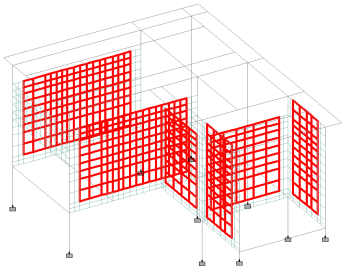
Pt = 0.0019 AG Reqd. Ast = 0.0019X1000X242/100 = 4.6 cm^2

Provided 12 mm @ 200 C/C (Provide As: 5.65 cm2)

Provided 0 mm @ 200 C/C (Provide As: 0 cm2)

Total Provide Ast = 5.65 cm2) **O.K.**

Total Provide Pt = 0.002



STAAD Group - 300MM_WALL_SF

SHEAR CHECK

(Staad Group : 300MM_WALL_SF)

(FROM FACTORED LOAD COMBINATION)

ALONG X - DIRECTION :

SQx = 0.2 N/mm^2 *(At d-distance from face of the Column)* *(Plate No. - 666 , L/C No. - 405)*

Vu = 6 Mton

Provided Pt = 0.0022

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*

[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1

= (0.16X1X45^0.5+17X0.0022X6X10^4X254/3.7/10^7)X1000X254/10^4

= 27.653 Mton

[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff

= (0.16X1X45^0.5+17X0.0022)X1000X254/10^4

= 28.212 Mton

[c] = 0.29*λ*fc'^0.5*b*d.eff

= 0.29X1X6.71X1000X254/10^4

= 49.413 Mton

HENCE, Vc = 27.653 Mton

Φt*Vc = 0.75X27.653 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*

= 20.74 Mton **> 6Mton , Hence O.K.**

ALONG Y - DIRECTION :

SQy = -0.2 N/mm^2 *(At d-distance from face of the Column)* *(Plate No. - 1180 , L/C No. - 408)*

Vu = 6 Mton

Pt provided = 0.0023

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*

[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1

= (0.16X1X45^0.5+17X0.0023X6X10^4X242/4.2/10^7)X1000X242/10^4

= 26.301 Mton

[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff

= (0.16X1X45^0.5+17X0.0023)X1000X242/10^4

= 26.92 Mton

[c] = 0.29*λ*fc'^0.5*b*d.eff

= 0.29X1X6.71X1000X242/10^4

= 47.078 Mton

HENCE, Vc = 26.301 Mton

Φt*Vc = 0.75X26.301 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*

= 19.726 Mton **> 6Mton , Hence O.K.**

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

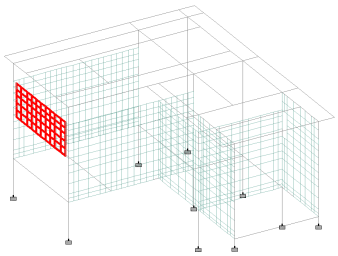
Ma'aden Revision No	A01	C01	C02	D01
Date	1/27/2020	3/12/2020	4/25/2022	12/10/2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS DT. 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER SUPPORT STRUCTURE	AS BUILT
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Prepared By	APR	APR	APR	APR

(STAAD File : Crusher Support Structure R2)

7.6.2 DESIGN OF 500MM WALL :

Wall Thickness =	500	mm
Width =	1000	mm
Clear Cover =	40	mm
Compressive Strength of Concrete , fc' =	45	N/mm^2
Yield Strength of Main Reinforcement , fy =	420	N/mm^2
Yield Strength of Shear Reinforcement , fyt =	420	N/mm^2
Strength Reduction Factor of Main Reinforcement , Φ =	0.9	
Strength Reduction Factor of Shear Reinforcement , Φt =	0.75	



STAAD Group - 500MM_WALL_B

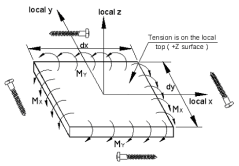
STAAD OUTPUT FROM FACTORED LOAD COMBINATION :

(Staad Group : 500MM_WALL_B)

MOMENT	VALUE	UNIT	TORSIONAL MOMENT	VALUE	UNIT	
Max. Mx	0.4	Mton-m	Corresponding Mxy	0.4	Mton-m	*(Plate No. - 1681 , L/C No. - 410)*
Min. Mx	-0.9	Mton-m	Corresponding Mxy	0.1	Mton-m	*(Plate No. - 1624 , L/C No. - 411)*
Max. My	1.2	Mton-m	Corresponding Mxy	0.7	Mton-m	*(Plate No. - 1669 , L/C No. - 422)*
Min. My	-1	Mton-m	Corresponding Mxy	-0.2	Mton-m	*(Plate No. - 1669 , L/C No. - 413)*

REINFORCEMENT FOR OUTER FACE (TOP +Z):

Mx1 =	Mx + abs(Mxy) =	0.4 + 0.4 =	0.8	Mton-m
My1 =	My + abs(Mxy) =	1.2 + 0.7 =	1.55	Mton-m
Mx2 =	Mx+abs(Mxy^2/My) =	0.4 + 0.133 =	0.533	Mton-m
My2 =	My+abs(Mxy^2/Mx) =	1.2 + 1.225 =	2.425	Mton-m



DESIGN MOMENT CALCULATION FOR OUTER FACE REINFORCEMENT :

Mxd = 0.8 Mton-m
Myd = 1.55 Mton-m

ALONG X - DIRECTION :

Mu = 0.8 Mton-m
Φ*Kn = Mu/bd^2 deff. = 452 mm
Mu/bd^2 = 0.8X10^7/(1000X452^2) = 0.039 N/mm^2 < 8.5 N/mm^2
Pt = 0.0001 Reqd. Ast = 0.0001X1000X452/100 = 0.45 cm^2
Provided **16** mm @ **200** C/C (Provide A: 10.05 cm2) Minimum r/f 0.12% of Gross area (on both faces)
Provided **0** mm @ **200** C/C (Provide A: 0 cm2) required = 3 cm2
Total Provide Ast = 10.05 cm2 **O.K.**
Total Provide Pt = 0.002

ALONG Y - DIRECTION :

Mu = 1.55 Mton-m
Φ*Kn = Mu/bd^2 deff. = 436 mm
Mu/bd^2 = 1.55X10^7/(1000X436^2) = 0.076 N/mm^2 < 8.5 N/mm^2
Pt = 0.0002 Reqd. Ast = 0.0002X1000X452/100 = 0.87 cm^2
Provided **16** mm @ **200** C/C (Provide A: 10.05 cm2) Minimum r/f 0.12% of Gross area (on both faces)
Provided **0** mm @ **200** C/C (Provide A: 0 cm2) required = 3 cm2
Total Provide Ast = 10.05 cm2 **O.K.**
Total Provide Pt = 0.002

REINFORCEMENT FOR INNER FACE (BOTTOM +Z):

Mx1 =	Mx - abs(Mxy) =	-0.9 - 0.1 =	-1	Mton-m
My1 =	My - abs(Mxy) =	-1 - 0.2 =	-1.2	Mton-m
Mx2 =	Mx-abs(Mxy^2/My) =	-0.9 - 0.01 =	-0.91	Mton-m
My2 =	My-abs(Mxy^2/Mx) =	-1 - 0.044 =	-0.96	Mton-m

DESIGN MOMENT CALCULATION FOR INNER FACE REINFORCEMENT :

Mxd = -1 Mton-m
Myd = -1.2 Mton-m

ALONG X - DIRECTION :

Mu = 1 Mton-m
Φ*Kn = Mu/bd^2 deff. = 452 mm
Mu/bd^2 = 1X10^7/(1000X452^2) = 0.049 N/mm^2 < 8.5 N/mm^2
Pt = 0.0001 Reqd. Ast = 0.0001X1000X452/100 = 0.45 cm^2
Provided **16** mm @ **200** C/C (Provide A: 10.05 cm2) Minimum r/f 0.12% of Gross area (on both faces)
Provided **0** mm @ **200** C/C (Provide A: 0 cm2) required = 3 cm2
Total Provide Ast = 10.05 cm2 **O.K.**
Total Provide Pt = 0.002

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No	A01	C01	C02	D01
Date	1/27/2020	3/12/2020	4/25/2022	12/10/2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS DT. 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER SUPPORT STRUCTURE	AS BUILT
Prepared By	APR	APR	APR	APR

Outotec / L&T Revision No.:	0	1	2	3
Date	1/27/2020	3/12/2020	4/25/2022	12/10/2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS DT. 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER SUPPORT STRUCTURE	AS BUILT
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ALONG Y - DIRECTION :

Mu = 1.2 Mton-m

Φ*Kn = Mu/bd^2

Mu/bd^2 = 1.2X10^7/(1000X436^2) = 0.063 N/mm^2 < deff. = 436 mm, 8.5 N/mm^2

Pt = 0.0002 Reqd. Ast = 0.0002X1000X436/100 = 0.87 cm^2

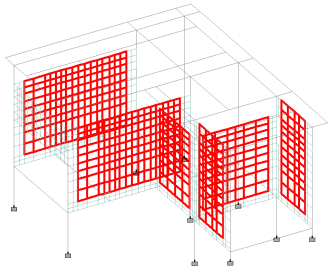
Minimum r/f 0.12% of Gross area (on both faces)

Provided **16** mm @ **200** C/C (Provide As 10.05 cm2)

Provided **0** mm @ **200** C/C (Provide As 0 cm2) required = 3 cm2

Total Provide Ast = 10.05 cm2) **O.K.**

Total Provide Pt = 0.002



STAAD Group - 500MM_WALL_SF

SHEAR CHECK

(Staad Group : 500MM_WALL_SF)

(FROM FACTORED LOAD COMBINATION)

ALONG X - DIRECTION :

SQx = 0 N/mm^2 *(At d-distance from face of the Column)* *(Plate No. - 1625 , L/C No. - 422)*

Vu = 0 Mton

Provided Pt = 0.0022

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*

[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1

= (0.16X1X45^0.5+17X0.0022X0X10^4X452/1/10^7)X1000X452/10^4

= 48.514 Mton

[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff

= (0.16X1X45^0.5+17X0.0022)X1000X452/10^4

= 50.204 Mton

[c] = 0.29*λ*fc'^0.5*b*d.eff

= 0.29X1X6.71X1000X452/10^4

= 87.931 Mton

HENCE, Vc = 48.514 Mton

Φt*Vc = 0.75X48.514 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*

= 36.386 Mton **> 0Mton , Hence O.K.**

ALONG Y - DIRECTION :

SQy = 0 N/mm^2 *(At d-distance from face of the Column)* *(Plate No. - 1644 , L/C No. - 408)*

Vu = 0 Mton

Pt provided = 0.0023

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*

[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1

= (0.16X1X45^0.5+17X0.0023X0X10^4X436/1.2/10^7)X1000X436/10^4

= 46.796 Mton

[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff

= (0.16X1X45^0.5+17X0.0023)X1000X436/10^4

= 48.501 Mton

[c] = 0.29*λ*fc'^0.5*b*d.eff

= 0.29X1X6.71X1000X436/10^4

= 84.819 Mton

HENCE, Vc = 46.796 Mton

Φt*Vc = 0.75X46.796 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*

= 35.097 Mton **> 0Mton , Hence O.K.**

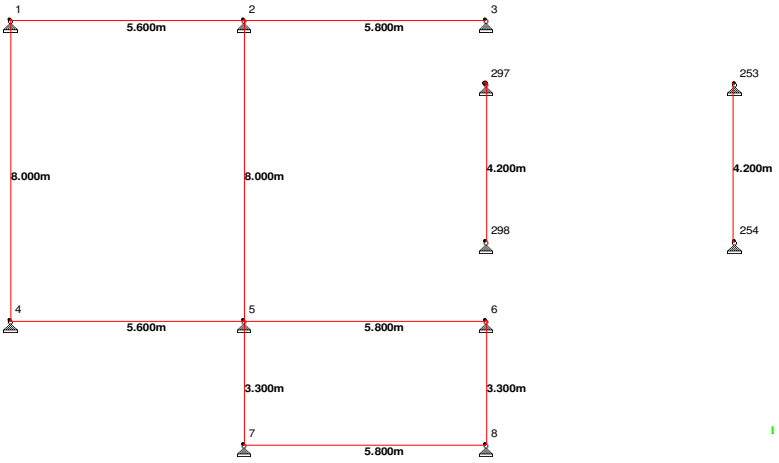
Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	C02	D01			Outotec / L&T Revision No.:	0	1	2	3		
Date	27-01-2020	12-03-2020	25-04-2022	12/10/2023			Date	27-01-2020	12-03-2020	25-04-2022	12/10/2023		
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS DT. 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER SUPPORT STRUCTURE	AS BUILT			`	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS DT. 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER SUPPORT STRUCTURE	AS BUILT		
Prepared By	APR	APR	APR	APR			Prepared By	APR	APR	APR	APR		
Checked	SBY	SBY	SBY	SBY			Checked	SBY	SBY	SBY	SBY		
Approved	AG	AG	AG	AG			Approved	AG	AG	AG	AG		

8 DESIGN OF SECONDARY CRUSHER HOUSE FOUNDATION

Basic Load cases for column foundation of Crusher house

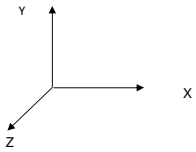
LOAD 1 DL	Dead Load
LOAD 2 LL	Live Load
LOAD 3 SDL1	Superimposed Dead Load1 - Chute & Bunker empty condition
LOAD 4 SDL2	Superimposed Dead Load2 - Chute & Bunker filled condition
LOAD 5 WL(+X)	Wind Load in (+) X Direction
LOAD 6 WL(-X)	Wind Load in (-) X Direction
LOAD 7 WL(+Z)	Wind Load in (+) Z Direction
LOAD 8 WL(-Z)	Wind Load in (-) Z Direction
LOAD 9 TL1	Temperature Load 1 for (+) 25^0
LOAD 10 TL2	Temperature Load 2 for (-) 25^0
LOAD 11 EL(+X)	Earthquake Load in (+X) Direction
LOAD 12 EL (+Z)	Earthquake Load in (+Z) Direction



8.1 Basic load data

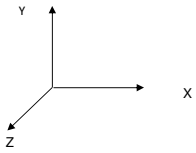
Staad support node - 1 Column support at grid 3120/J-3120/9

Load case	Force-X kN	Force-Y kN	Force-Z kN	Moment-X kN-m	Moment-Y kN-m	Moment-Z kN-m
1 DL	0.0	-87.5	-6.4	0	0	0
2 LL	0.0	-95.4	-13.8	0	0	0
3 SDL 1	0.1	-114.6	-21.0	0	0	0
4 SDL 2	0.1	-115.0	-27.4	0	0	0
5 WL(+X)	1.8	22.7	1.2	0	0	0
6 WL(-X)	-1.8	-23.6	-1.3	0	0	0
7 WL(+Z)	0.0	95.5	31.1	0	0	0
8 WL(-Z)	0.0	-97.7	-32.1	0	0	0
9 TL 1	-1.2	-4.0	-51.8	0	0	0
10 TL 2	1.2	4.0	51.8	0	0	0
11 EL (+X)	0.0	2.6	0.1	0	0	0
12 EL (+Z)	0.0	4.3	1.0	0	0	0



Staad support node - 2 Column support at grid 3120/J-3120/10

Load case	Force-X kN	Force-Y kN	Force-Z kN	Moment-X kN-m	Moment-Y kN-m	Moment-Z kN-m
1 DL	-1.6	-181.1	-0.2	0	0	0
2 LL	-0.5	-212.2	-1.0	0	0	0
3 SDL 1	-40.9	-505.3	-0.2	0	0	0
4 SDL 2	-21.6	-1405.3	1.2	0	0	0
5 WL(+X)	84.0	422.3	-0.3	0	0	0
6 WL(-X)	-83.1	-412.5	0.2	0	0	0
7 WL(+Z)	-14.2	301.0	-14.5	0	0	0
8 WL(-Z)	13.9	-318.7	15.0	0	0	0
9 TL 1	-6.0	-33.3	-233.4	0	0	0
10 TL 2	6.0	33.3	233.4	0	0	0
11 EL (+X)	4.8	32.7	-0.1	0	0	0
12 EL (+Z)	-0.1	20.9	-0.7	0	0	0



Sensitivity: LNT Construction Internal Use

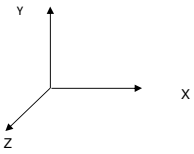
Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	C02	D01		
Date	27-01-2020	12-03-2020	25-04-2022	12/10/2023		
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Prepared By	APR	APR	APR	APR		
Checked	SBY	SBY	SBY	SBY		
Approved	AG	AG	AG	AG		

Outotec / L&T Revision No.:	0	1	2	3
Date	27-01-2020	12-03-2020	25-04-2022	12/10/2023
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Approved	AG	AG	AG	AG

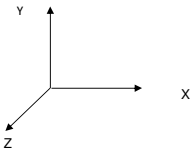
Staad support node - 3 Column support at grid 3120/J-3120/12

Load case	Force-X kN	Force-Y kN	Force-Z kN	Moment-X kN-m	Moment-Y kN-m	Moment-Z kN-m
1 DL	0.4	-149.2	-0.4	0	0	0
2 LL	-0.6	-161.5	-1.8	0	0	0
3 SDL 1	-41.3	98.5	2.3	0	0	0
4 SDL 2	-22.1	-1636.8	5.0	0	0	0
5 WL(+X)	84.2	-452.3	0.2	0	0	0
6 WL(-X)	-83.2	444.4	-0.2	0	0	0
7 WL(+Z)	-14.2	210.9	5.8	0	0	0
8 WL(-Z)	13.9	-224.4	-5.8	0	0	0
9 TL 1	-1.4	29.7	-1.0	0	0	0
10 TL 2	1.4	-29.7	1.0	0	0	0
11 EL (+X)	4.8	-36.3	0.0	0	0	0
12 EL (+Z)	-0.1	14.7	0.0	0	0	0



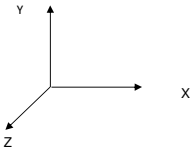
Staad support node - 4 Column support at grid 3120/F-3120/9

Load case	Force-X kN	Force-Y kN	Force-Z kN	Moment-X kN-m	Moment-Y kN-m	Moment-Z kN-m
1 DL	0.0	-114.8	4.7	0	0	0
2 LL	0.0	-128.1	12.0	0	0	0
3 SDL 1	0.1	-158.3	20.7	0	0	0
4 SDL 2	0.1	-143.2	24.2	0	0	0
5 WL(+X)	1.8	12.8	1.3	0	0	0
6 WL(-X)	-1.8	-13.5	-1.4	0	0	0
7 WL(+Z)	0.0	-96.5	31.2	0	0	0
8 WL(-Z)	0.0	98.9	-32.1	0	0	0
9 TL 1	-1.2	14.7	48.4	0	0	0
10 TL 2	1.2	-14.7	-48.4	0	0	0
11 EL (+X)	0.0	2.3	0.1	0	0	0
12 EL (+Z)	0.0	-4.5	1.0	0	0	0



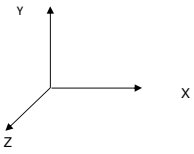
Staad support node - 5 Column support at grid 3120/F-3120/10

Load case	Force-X kN	Force-Y kN	Force-Z kN	Moment-X kN-m	Moment-Y kN-m	Moment-Z kN-m
1 DL	-0.4	-202.9	-6.3	0	0	0
2 LL	0.7	-285.4	-8.7	0	0	0
3 SDL 1	-35.7	-521.9	0.1	0	0	0
4 SDL 2	-14.0	-1496.9	-16.1	0	0	0
5 WL(+X)	98.4	475.4	12.0	0	0	0
6 WL(-X)	-98.1	-474.1	-11.3	0	0	0
7 WL(+Z)	14.2	-34.0	123.0	0	0	0
8 WL(-Z)	-14.0	44.0	-126.0	0	0	0
9 TL 1	-1.3	19.6	164.3	0	0	0
10 TL 2	1.3	-19.6	-164.3	0	0	0
11 EL (+X)	5.3	34.5	0.8	0	0	0
12 EL (+Z)	0.1	-10.5	4.2	0	0	0



Staad support node - 6 Column support at grid 3120/F-3120/12

Load case	Force-X kN	Force-Y kN	Force-Z kN	Moment-X kN-m	Moment-Y kN-m	Moment-Z kN-m
1 DL	1.7	-155.8	-3.9	0	0	0
2 LL	0.6	-181.3	-4.4	0	0	0
3 SDL 1	-35.7	183.3	8.7	0	0	0
4 SDL 2	-14.2	-1195.4	-12.8	0	0	0
5 WL(+X)	98.4	-463.2	-8.1	0	0	0
6 WL(-X)	-98.1	459.4	7.7	0	0	0
7 WL(+Z)	14.1	168.4	54.6	0	0	0
8 WL(-Z)	-13.9	-186.6	-58.3	0	0	0
9 TL 1	3.2	-5.4	-27.6	0	0	0
10 TL 2	-3.2	5.4	27.6	0	0	0
11 EL (+X)	5.3	-33.5	-0.5	0	0	0
12 EL (+Z)	0.1	16.0	3.7	0	0	0



Sensitivity: LNT Construction Internal Use

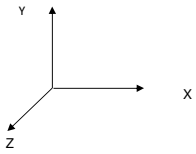
Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	C02	D01		
Date	27-01-2020	12-03-2020	25-04-2022	12/10/2023		
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Prepared By	APR	APR	APR	APR		
Checked	SBY	SBY	SBY	SBY		
Approved	AG	AG	AG	AG		

Outotec / L&T Revision No.:	0	1	2	3
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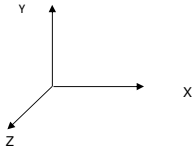
Staad support node - 7 Column support at grid 3120/C-3120/10

Load case	Force-X kN	Force-Y kN	Force-Z kN	Moment-X kN-m	Moment-Y kN-m	Moment-Z kN-m
1 DL	-0.9	-47.0	8.2	0	0	0
2 LL	0.0	-66.6	11.8	0	0	0
3 SDL 1	-0.1	-13.6	19.3	0	0	0
4 SDL 2	-0.1	-46.8	58.9	0	0	0
5 WL(+X)	16.0	72.5	-13.9	0	0	0
6 WL(-X)	-16.0	-73.9	14.6	0	0	0
7 WL(+Z)	0.0	-265.0	154.1	0	0	0
8 WL(-Z)	0.0	272.3	-158.5	0	0	0
9 TL 1	-1.2	-7.5	69.2	0	0	0
10 TL 2	1.2	7.5	-69.2	0	0	0
11 EL (+X)	0.2	1.5	-0.8	0	0	0
12 EL (+Z)	0.0	-10.0	5.9	0	0	0



Staad support node - 8 Column support at grid 3120/C-3120/12

Load case	Force-X kN	Force-Y kN	Force-Z kN	Moment-X kN-m	Moment-Y kN-m	Moment-Z kN-m
1 DL	0.9	-57.5	4.4	0	0	0
2 LL	0.0	-75.6	6.0	0	0	0
3 SDL 1	0.1	-3.7	4.2	0	0	0
4 SDL 2	0.1	-239.3	33.4	0	0	0
5 WL(+X)	16.1	-90.2	7.7	0	0	0
6 WL(-X)	-16.1	93.9	-8.3	0	0	0
7 WL(+Z)	0.0	-380.3	77.3	0	0	0
8 WL(-Z)	0.0	412.1	-83.0	0	0	0
9 TL 1	1.2	-12.5	31.8	0	0	0
10 TL 2	-1.2	12.5	-31.8	0	0	0
11 EL (+X)	0.2	-3.8	0.5	0	0	0
12 EL (+Z)	0.0	-30.9	5.6	0	0	0



Staad support node - 253 Conveyor walkway support post

Load case	Force-X kN	Force-Y kN	Force-Z kN	Moment-X kN-m	Moment-Y kN-m	Moment-Z kN-m
1 DL	0.0	-4.5	-5.2	0	0	0
2 LL	0.0	-12.6	-12.8	0	0	0
3 SDL 1	0.0	-15.0	-26.8	0	0	0
9 TL 1	0.0	1.3	-32.0	0	0	0
10 TL 2	0.0	-1.3	32.0	0	0	0

Staad support node - 254 Conveyor walkway support post

Load case	Force-X kN	Force-Y kN	Force-Z kN	Moment-X kN-m	Moment-Y kN-m	Moment-Z kN-m
1 DL	0.0	-4.5	5.2	0	0	0
2 LL	0.0	-11.3	12.8	0	0	0
3 SDL 1	0.0	-15.5	29.0	0	0	0
9 TL 1	0.0	1.2	32.0	0	0	0
10 TL 2	0.0	-1.2	-32.0	0	0	0

Staad support node - 297 Conveyor walkway support post

Load case	Force-X kN	Force-Y kN	Force-Z kN	Moment-X kN-m	Moment-Y kN-m	Moment-Z kN-m
1 DL	0.0	-12.1	-9.7	0	0	0
2 LL	0.0	-32.2	-24.0	0	0	0
3 SDL 1	0.1	-61.7	-66.2	0	0	0
9 TL 1	-0.2	-0.4	-9.3	0	0	0
10 TL 2	0.2	0.4	9.3	0	0	0

Staad support node - 298 Conveyor walkway support post

Load case	Force-X kN	Force-Y kN	Force-Z kN	Moment-X kN-m	Moment-Y kN-m	Moment-Z kN-m
1 DL	0.0	-12.1	9.6	0	0	0
2 LL	0.0	-32.2	24.0	0	0	0
3 SDL 1	0.1	-63.6	72.3	0	0	0
9 TL 1	0.1	-62.0	70.5	0	0	0
10 TL 2	-0.2	-0.4	9.2	0	0	0

Sensitivity: LNT Construction Internal Use

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Ma'aden Revision No:	A01	C01	C02	D01
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Approved	AG	AG	AG	AG

8.2 LOAD COMBINATION

8.2.1 Load Combination for Base pressure calculation

(As per Section 2.4 of SBC 301 -2007)

Load Cases	Reference
LOAD 301 (DL + SDL1)	(Eq. 2.4.1-1 of SBC 301)
LOAD 302 (DL + SDL2)	
LOAD 303 (DL + SDL1 + TL1) + LL	(Eq. 2.4.1-2 of SBC 301)
LOAD 304 (DL + SDL2 + TL1) + LL	
LOAD 305 (DL + SDL1 + TL2) + LL	
LOAD 306 (DL + SDL2 + TL2) + LL	
LOAD 307 (DL+ SDL1) + 0.75 LL + 0.75 WL(+X)	(Eq. 2.4.1-6 of SBC 301)
LOAD 308 (DL+ SDL1) + 0.75 LL + 0.75 WL(-X)	
LOAD 309 (DL+ SDL1) + 0.75 LL + 0.75 WL(+Z)	
LOAD 310 (DL+ SDL1) + 0.75 LL + 0.75 WL(-Z)	
LOAD 311 (DL+ SDL2) + 0.75 LL + 0.75 WL(+X)	
LOAD 312 (DL+ SDL2) + 0.75 LL + 0.75 WL(-X)	
LOAD 313 (DL+ SDL2) + 0.75 LL + 0.75 WL(+Z)	
LOAD 314 (DL+ SDL2) + 0.75 LL + 0.75 WL(-Z)	
LOAD 315 (DL+ SDL1) + 0.75 LL + 0.525 EL(+X)	(Eq. 2.4.1-6 of SBC 301)
LOAD 316 (DL+ SDL1) + 0.75 LL + 0.525 EL(-X)	
LOAD 317 (DL+ SDL1) + 0.75 LL + 0.525 EL(+Z)	
LOAD 318 (DL+ SDL1) + 0.75 LL + 0.525 EL(-Z)	
LOAD 319 (DL+ SDL2) + 0.75 LL + 0.525 EL(+X)	
LOAD 320 (DL+ SDL2) + 0.75 LL + 0.525 EL(-X)	
LOAD 321 (DL+ SDL2) + 0.75 LL + 0.525 EL(+Z)	
LOAD 322 (DL+ SDL2) + 0.75 LL + 0.525 EL(-Z)	

Load Cases	Reference
LOAD 323 0.6 (DL+SDL1) + 1.0 WL(+X)	(Eq. 2.4.1-7 of SBC 301)
LOAD 324 0.6 (DL+SDL1) + 1.0 WL(-X)	
LOAD 325 0.6 (DL+SDL1) + 1.0 WL(+Z)	
LOAD 326 0.6 (DL+SDL1) + 1.0 WL(-Z)	
LOAD 327 0.6 (DL+SDL2) + 1.0 WL(+X)	
LOAD 328 0.6 (DL+SDL2) + 1.0 WL(-X)	
LOAD 329 0.6 (DL+SDL2) + 1.0 WL(+Z)	
LOAD 330 0.6 (DL+SDL2) + 1.0 WL(-Z)	
LOAD 331 0.6 (DL+SDL1) + 0.7 EL(+X)	(Eq. 2.4.1-8 of SBC 301)
LOAD 332 0.6 (DL+SDL1) + 0.7 EL(-X)	
LOAD 333 0.6 (DL+SDL1) + 0.7 EL(+Z)	
LOAD 334 0.6 (DL+SDL1) + 0.7 EL(-Z)	
LOAD 335 0.6 (DL+SDL2) + 0.7 EL(+X)	
LOAD 336 0.6 (DL+SDL2) + 0.7 EL(-X)	
LOAD 337 0.6 (DL+SDL2) + 0.7 EL(+Z)	
LOAD 338 0.6 (DL+SDL2) + 0.7 EL(-Z)	

8.2.2 Load Combination for concrete design

Load Case	Reference
LOAD 401 1.4 (DL + SDL1)	(Eq. 2.3.2-1 of SBC 301)
LOAD 402 1.4 (DL + SDL2)	
LOAD 403 1.4 (DL + SDL1 + TL1) + 1.7 LL	(Eq. 2.3.2-2 of SBC 301)
LOAD 404 1.4 (DL + SDL2 + TL1) + 1.7 LL	
LOAD 405 1.4 (DL + SDL1 + TL2) + 1.7 LL	
LOAD 406 1.4 (DL + SDL2 + TL2) + 1.7 LL	
LOAD 407 1.2 (DL+ SDL1) + 0.5 LL + 1.6 WL(+X)	(Eq. 2.3.2-4 of SBC 301)
LOAD 408 1.2 (DL+ SDL1) + 0.5 LL + 1.6 WL(-X)	
LOAD 409 1.2 (DL+ SDL1) + 0.5 LL + 1.6 WL(+Z)	
LOAD 410 1.2 (DL+ SDL1)+ 0.5 LL + 1.6 WL(-Z)	
LOAD 411 1.2 (DL+ SDL2) + 0.5 LL + 1.6 WL(+X)	
LOAD 412 1.2 (DL+ SDL2) + 0.5 LL + 1.6 WL(-X)	
LOAD 413 1.2 (DL+ SDL2) + 0.5 LL + 1.6 WL(+Z)	
LOAD 414 1.2 (DL+ SDL2)+ 0.5 LL + 1.6 WL(-Z)	
LOAD 415 1.2 (DL+ SDL1) + 0.5 LL + 1.0 EL(+X)	(Eq. 2.3.2-5 of SBC 301)
LOAD 416 1.2 (DL+ SDL1) + 0.5 LL + 1.0 EL(-X)	
LOAD 417 1.2 (DL+ SDL1) + 0.5 LL + 1.0 EL(+Z)	
LOAD 418 1.2 (DL+ SDL1)+ 0.5 LL + 1.0 EL(-Z)	
LOAD 419 1.2 (DL+ SDL2) + 0.5 LL + 1.0 EL(+X)	
LOAD 420 1.2 (DL+ SDL2) + 0.5 LL + 1.0 EL(-X)	
LOAD 421 1.2 (DL+ SDL2) + 0.5 LL + 1.0 EL(+Z)	
LOAD 422 1.2 (DL+ SDL2) + 0.5 LL + 1.0 EL(-Z)	

LOAD CASE	Reference
LOAD 423 0.9 (DL+ SDL1) + 1.6 WL(+X)	(Eq. 2.3.2-6 of SBC 301)
LOAD 424 0.9 (DL+ SDL1) + 1.6 WL(-X)	
LOAD 425 0.9 (DL+ SDL1) + 1.6 WL(+Z)	
LOAD 426 0.9 (DL+ SDL1) + 1.6 WL(-Z)	
LOAD 427 0.9 (DL+ SDL2) + 1.6 WL(+X)	
LOAD 428 0.9 (DL+ SDL2) + 1.6 WL(-X)	
LOAD 429 0.9 (DL+ SDL2) + 1.6 WL(+Z)	
LOAD 430 0.9 (DL+ SDL2) + 1.6 WL(-Z)	
LOAD 431 0.9 (DL+ SDL1) + 1.0 EL(+X)	(Eq. 2.3.2-7 of SBC 301)
LOAD 432 0.9 (DL+ SDL1) + 1.0 EL(-X)	
LOAD 433 0.9 (DL+ SDL1) + 1.0 EL(+Z)	
LOAD 434 0.9 (DL+ SDL1) + 1.0 EL(-Z)	
LOAD 435 0.9 (DL+ SDL2) + 1.0 EL(+X)	
LOAD 436 0.9 (DL+ SDL2) + 1.0 EL(-X)	
LOAD 437 0.9 (DL+ SDL2) + 1.0 EL(+Z)	
LOAD 438 0.9 (DL+ SDL2) + 1.0 EL(-Z)	

Sensitivity: LNT Construction Internal Use

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

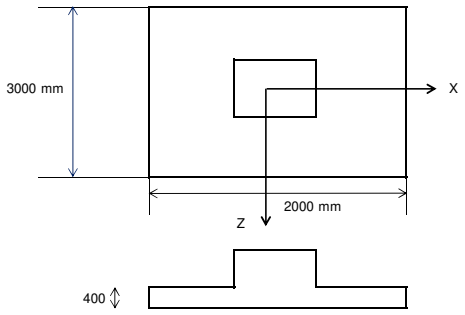
Ma'aden Revision No:	A01	C01	C02	D01			Outotec / L&T Revision No.:	0	1	2	3	
Date	27-01-2020	12-03-2020	25-04-2022	12/10/2023			Date	27-01-2020	12-03-2020	25-04-2022	12/10/2023	
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS DT. 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER SUPPORT STRUCTURE	AS BUILT			Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS DT. 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER SUPPORT STRUCTURE	AS BUILT	
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Checked	SBY	SBY	SBY	SBY			Checked	SBY	SBY	SBY	SBY	
Approved	AG	AG	AG	AG			Approved	AG	AG	AG	AG	

8.3 DESIGN OF FOUNDATION

8.3.1.1 DESIGN OF ISOLATED FOUNDATION FOR COLUMN AT 3120/9_2130/J AND MARKED AS F1

Finished Ground level	F.G.L	=	-0.500	m
Bottom of foundation	B.O.F	=	-2.500	m
Top of footing (staad support)	T.O.F	=	0.500	m
Ground Water Level	G.W.L	=	-10.000	m
Surcharge	Ws	=	0.000	kN/m^2
Length of footing	Lz	=	3.00	m
Width of footing	Lx	=	2.00	m
Thickness of footing	D	=	0.40	m
Depth of pedestal	Bx	=	0.80	m
Width of pedestal	Bz	=	0.80	m
Height of Load point from bottom of footing	H	=	3.000	m
Section Modulus of Pedestal	Zxx	=	3.00	m^3
Section Modulus of Pedestal	Zzz	=	2.00	m^3
Allowable Net Bearing Capacity		=	500	kN/m^2
Gross Bearing Capacity		=	536	kN/m^2
Grade of concrete		=	35	N/mm^2
Grade of steel		=	420	N/mm^2
Clear cover		=	50	mm
Top Reinfocement		=	12	mm
Bottom Reinforcement		=	16	mm
Co-efficient of friction between soil & concrete		=	0.3	
Unit wt of soil		=	18	kN/m^3
Unit wt of concrete		=	24	kN/m^3

Weight of soil	=	154.37 KN
Weight of foundation	=	97.54 KN
Uplift	=	0.00 KN
Surcharge	=	0 KN
Pressure on top for tension	=	42.0 KN/m^2



$$\beta_1 = 0.85 - \frac{0.05(f'_c - 28)}{7}$$ β_1 = 0.8 *Table 22.2.2.4.3 SBC 304*

CHECK FOR BEARING PRESSURE CAPACITY

L/C	Fx	Fy	Fz	Mx	My	Mz	Sum Fy	Pmax	Pmin	B.C Factor	F.O.S B.P
301	0.1	202.118	-27.4	-82.34	0.0	-0.4	454.0	103.31	48.03	1	0.19
302	0.1	202.479	-33.8	-101.36	0	-0.3	454.4	109.67	41.79	1	0.20
303	-1.1	301.506	-93.1	-279.18	0	3.2	553.4	186.90	-2.43	1	0.35
304	-1.1	301.866	-99.4	-298.21	0	3.3	553.8	193.35	-8.76	1	0.36
305	1.4	293.473	10.5	31.43	0	-4.1	545.4	103.43	78.36	1	0.19
306	1.3	293.833	4.1	12.41	0.0	-4.0	545.7	97.11	84.80	1	0.18
307	1.5	256.591	-36.9	-110.72	0	-4.4	508.5	123.86	45.63	1	0.23
308	-1.2	291.333	-38.8	-116.51	0	3.5	543.2	131.15	49.93	1	0.24
309	0.1	202.014	-14.5	-43.41	0.0	-0.4	453.9	90.31	61.00	1	0.17
310	0.2	346.943	-61.9	-185.62	0	-0.5	598.8	161.94	37.68	1	0.30
311	1.4	256.952	-43.2	-129.75	0	-4.3	508.9	130.22	39.39	1	0.24
312	-1.2	291.693	-45.2	-135.54	0	3.6	543.6	137.59	43.61	1	0.26
313	0.1	202.374	-20.8	-62.44	0	-0.3	454.3	96.67	54.76	1	0.18
314	0.1	347.303	-68.2	-204.65	0.0	-0.4	599.2	168.30	31.44	1	0.31
315	0.1	273.647	-37.8	-113.49	0	-0.4	525.6	125.64	49.54	1	0.23
316	0.1	273.647	-37.8	-113.49	0	-0.4	525.6	125.64	49.54	1	0.23
317	0.1	273.647	-37.8	-113.49	0	-0.4	525.6	125.64	49.54	1	0.23
318	0.1	273.647	-37.8	-113.49	0	-0.4	525.6	125.64	49.54	1	0.23
319	0.1	274.007	-44.2	-132.52	0.0	-0.4	525.9	132.00	43.30	1	0.25
320	0.1	274.007	-44.2	-132.52	0	-0.4	525.9	132.00	43.30	1	0.25
321	0.1	274.007	-44.2	-132.52	0	-0.4	525.9	132.00	43.30	1	0.25
322	0.1	274.007	-44.2	-132.52	0.0	-0.4	525.9	132.00	43.30	1	0.25
323	1.8	98.531	-15.2	-45.71	0	-5.5	350.4	76.41	40.40	1	0.14
324	-1.7	144.853	-17.8	-53.43	0	5.1	396.8	86.47	45.78	1	0.16
325	0.0	25.761	14.7	44.04	0	-0.1	277.7	61.03	31.52	1	0.11
326	0.1	218.999	-48.5	-145.58	0	-0.3	470.9	127.18	29.79	1	0.24
327	1.8	98.747	-19.0	-57.13	0.0	-5.5	350.7	80.23	36.65	1	0.15
328	-1.7	145.069	-21.6	-64.85	0	5.1	397.0	90.34	41.99	1	0.17
329	0.0	25.977	10.9	32.63	0	-0.1	277.9	57.24	35.39	1	0.11
330	0.1	219.215	-52.3	-157.00	0	-0.3	471.1	130.99	26.05	1	0.24
331	0.1	121.271	-16.5	-49.40	0	-0.2	373.2	78.78	45.61	1	0.15
332	0.1	121.271	-16.5	-49.40	0.0	-0.2	373.2	78.78	45.61	1	0.15
333	0.1	121.271	-16.5	-49.40	0	-0.2	373.2	78.78	45.61	1	0.15
334	0.1	121.271	-16.5	-49.40	0	-0.2	373.2	78.78	45.61	1	0.15
335	0.1	121.487	-20.3	-60.82	0	-0.2	373.4	82.60	41.87	1	0.15
336	0.1	121.487	-20.3	-60.82	0.0	-0.2	373.4	82.60	41.87	1	0.15
337	0.1	121.487	-20.3	-60.82	0	-0.2	373.4	82.60	41.87	1	0.15
338	0.1	121.487	-20.3	-60.82	0	-0.2	373.4	82.60	41.87	1	0.15

Max P_{max}	=	193.35 kN/m^2	Load Case	304	Resultant F.O.S	=	0.25
Max P_{min}	=	-8.76 kN/m^2	Load Case	304			

Sensitivity: LNT Construction Internal Use

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Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	C02	D01
Date	27-01-2020	12-03-2020	25-04-2022	12/10/2023
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BEARING PRESSURE CALCULATION FOR BENDING MOMENT AND SHEAR CHECK

L/C	Fx	Fy	Fz	Mx	My	Mz	Pmax	Pmin
401	0.2	283.0	-38.4	-115.27	0.0	-0.56	85.86	8.46
402	0.1	283.5	-47.3	-141.91	0	-0.44	94.77	-0.28
403	-1.5	450.7	-134.4	-403.31	0	4.49	211.80	-61.56
404	1.9	439.5	10.5	31.55	0	-5.78	86.65	59.84
405	-1.5	451.2	-143.3	-429.95	0	4.61	220.82	-70.42
406	1.9	440.0	1.6	4.91	0	-5.66	77.79	68.86
407	3.0	253.8	-37.9	-113.67	0	-8.99	84.69	-0.08
408	-2.7	328.0	-42.0	-126.02	0	8.00	100.66	8.66
409	0.1	137.4	10.0	29.93	0.0	-0.36	33.06	12.75
410	0.2	446.6	-91.2	-273.46	0	-0.66	165.91	-17.05
411	3.0	254.3	-45.5	-136.50	0	-8.89	92.32	-7.57
412	-2.7	328.4	-49.6	-148.85	0	8.09	108.39	1.07
413	0.1	137.8	2.4	7.10	0	-0.26	25.47	20.48
414	0.2	447.0	-98.8	-296.29	0	-0.56	173.55	-24.54
415	0.2	290.2	-39.9	-119.57	0	-0.50	88.48	8.26
416	0.2	290.2	-39.9	-119.57	0	-0.50	88.48	8.26
417	0.2	290.2	-39.9	-119.57	0.0	-0.50	88.48	8.26
418	0.2	290.2	-39.9	-119.57	0	-0.50	88.48	8.26
419	0.1	290.7	-47.5	-142.40	0	-0.40	96.11	0.78
420	0.1	290.7	-47.5	-142.40	0	-0.40	96.11	0.78
421	0.1	290.7	-47.5	-142.40	0	-0.40	96.11	0.78
422	0.1	290.7	-47.5	-142.40	0	-0.40	96.11	0.78
423	1.9	159.2	-23.5	-70.41	0	-5.66	52.83	0.23
424	-1.7	205.5	-26.0	-78.13	0	4.95	62.77	5.73
425	0.1	86.4	6.4	19.34	0	-0.27	20.98	7.82
426	0.2	279.6	-56.8	-170.28	0	-0.46	103.59	-10.38
427	1.9	159.5	-29.2	-87.53	0	-5.59	58.55	-5.39
428	-1.7	205.8	-31.8	-95.25	0	5.03	68.57	0.04
429	0.1	86.7	0.7	2.22	0	-0.19	15.29	13.62
430	0.1	280.0	-62.5	-187.40	0	-0.38	109.32	-16.00
431	0.1	181.9	-24.7	-74.10	0	-0.36	55.20	5.44
432	0.1	181.9	-24.7	-74.10	0	-0.36	55.20	5.44
433	0.1	181.9	-24.7	-74.10	0	-0.36	55.20	5.44
434	0.1	181.9	-24.7	-74.10	0	-0.36	55.20	5.44
435	0.1	182.2	-30.4	-91.22	0	-0.28	60.92	-0.18
436	0.1	182.2	-30.4	-91.22	0	-0.28	60.92	-0.18
437	0.1	182.2	-30.4	-91.22	0	-0.28	60.92	-0.18
438	0.1	182.2	-30.4	-91.22	0	-0.28	60.92	-0.18

Max Net factored B.P = 220.82 kN/m^2 Pmin = -70.42 kN/m^2

DETERMINATION OF REINFORCEMENT

Bottom Reinforcement

Length of cantilever along X = 0.6 m

Moment about Z Muz = 39.75 kNm

$M_{uzlim} = \emptyset\, 0.85\, f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

= 798.59 kNm

As $M_{uzlim} > M_{uz}$ Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\, f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

= 309.1 mm^2

Effective Depth Along X = 342 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement $A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*

or greater of

$A_{stmin} = 0.0014bd$

A_{stmin} = 615.6 mm^2

A_{req} = 309.1 mm^2

Required Spacing = 326.6 mm

Sensitivity: LNT Construction Internal Use

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Project Country	**KINGDOM OF SAUDI ARABIA**

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Length of cantilever along Z = 1.1 m

Moment about X: Mux = 133.60 KNm

$M_{uxlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

= 725.61 kNm

As $M_{uxlim} > M_{ux}$ Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBY 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

= 1109.5 mm^2

Provide Bottom Reinforcement = 16 mm dia @ 150 c/c

Effective Depth Along X = 326 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005} = 0.375$

Min reinforcement greater of

$A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*

or $A_{stmin} = 0.0014bd$

A_{stmin} = 586.8 mm^2

A_{req} = 1109.5 mm^2

Required Spacing = 181.2 mm

Ast prov = 1340.4 mm^2

Top Reinforcement

Length of cantilever along X = 0.6 m

Moment about Z: Muz = 7.56 kNm

$M_{uzlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

= 807.95 kNm

As $M_{uzlim} > M_{uz}$ Tension controlled section

Therefore φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

= 58.1 mm^2

Effective Depth Along X = 344 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005} = 0.375$

Min reinforcement greater of

A_{stmin} = 50% of $\frac{0.0018 \times 420}{f_y} bd$ since 50% or more prov in bottom

or A_{stmin} = 50% of 0.0014bd *Table 24.4.3.2 SBC 304*

A_{stmin} = 309.6 mm^2

A_{req} = 58.1 mm^2

Required Spacing = 365.3 mm

Length of cantilever along Z = 1.1 m

Moment about X: Mux = 25.40 kNm

$M_{uxlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

= 752.57 kNm

As $M_{uxlim} > M_{ux}$ Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

= 203.0 mm^2

Effective Depth Along Z = 332 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005} = 0.375$

Min reinforcement greater of

A_{stmin} = 50% of $\frac{0.0018 \times 420}{f_y} bd$ since 50% or more prov in bottom

or A_{stmin} = 50% of 0.0014bd *Table 24.4.3.2 SBC 304*

A_{stmin} = 298.8 mm^2

A_{req} = 203.0 mm^2

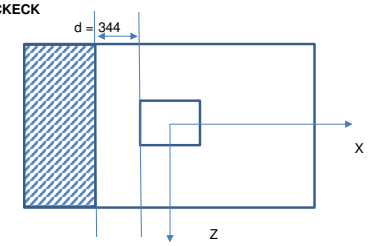
Required Spacing = 378.5 mm

Provide top Reinforcement = 12 mm dia @ 150 c/c

Ast prov = 754.0 mm^2

ONE WAY SHEAR CKECK

Along X Direction



Cantilever L = 0.256 m

Reinforcement ratio ρ_w = 0.00390

Effective Depth Along X d = 344 mm

Shear Force at d distance V_u = 56.53 kN

Allowable shear strength φVc = 244.22 kN O.K

Least of

i) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right) b_w d$ *Table 22.5.5.1 SBC 304*

λ = 1 Normal concrete

Mu = 7.24 KNm

φ = 0.75 for shear

φVc = 290.15 KN

ii) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\right) b_w d$

φVc = 261.306037 kN *Table 22.5.5.1 SBC 304*

iii) $V_c = 0.29\lambda\sqrt{f_c'} b_w d$

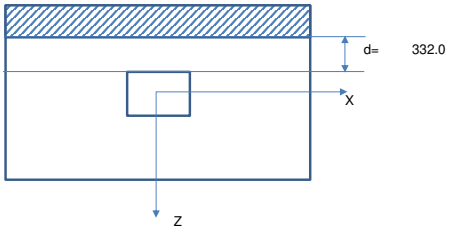
= 244.2 kN

Sensitivity: LNT Construction Internal Use

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	C02	D01			Outotec / L&T Revision No.:	0	1	2	3		
Date	27-01-2020	12-03-2020	25-04-2022	12/10/2023			Date	27-01-2020	12-03-2020	25-04-2022	12/10/2023		
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS DT. 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER SUPPORT STRUCTURE	AS BUILT			Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS DT. 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER SUPPORT STRUCTURE	AS BUILT		
Prepared By	APR	APR	APR	APR			Prepared By	APR	APR	APR	APR		
Checked	SBY	SBY	SBY	SBY			Checked	SBY	SBY	SBY	SBY		
Approved	AG	AG	AG	AG			Approved	AG	AG	AG	AG		

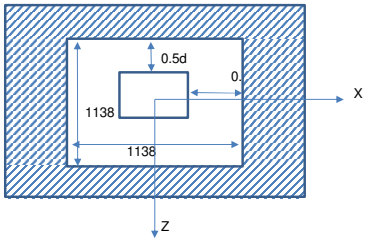
Along Z Direction



Cantilever	L	=	0.768 m		Least of	i)	V_c	=	$\left(0.16\lambda\sqrt{f_c'}+17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$	
Reinforcement ratio	ρ_w	=	0.00							*Table 22.5.5.1 SBC 304*
Effective Depth Along Z	d	=	332 mm				λ	=	1 Normal concrete	
Shear Force at d distance	V_u	=	169.59 kN				Mu	=	65.12 KNm	
Allowable shear strength	φVc	=	235.70 kN	O.K			φ	=	0.75 for shear	
							φVc	=	250.47 kN	
						ii)	V_c	=	$\left(0.16\lambda\sqrt{f_c'}+17\rho_w\right)b_w d$	
							φVc	=	252.786883 kN	
						iii)	V_c	=	$0.29\lambda\sqrt{f_c'}b_w d$	
							φVc	=	235.70 kN	

TWO WAY SHEAR CKECK

Punching at 0.5d from face of column



Total area		=	6 m2				
Punching area		=	1.30 m2				
	λ	=	1				
	β	=	1				
	αs	=	40				
	b0	=	4552.0 mm				
Allowable shear stress		least of		*Table 22.6.5.2 SBC 304*			
v_c	$0.33\lambda\sqrt{f_c'}$	=			1.95 N/mm²		
v_c	$0.17\left(1+\frac{2}{\beta}\right)\lambda\sqrt{f_c'}$	=			3.02 N/mm²		
v_c	$0.083\left(2+\frac{\alpha_s d}{b_0}\right)\lambda\sqrt{f_c'}$	=			2.44 N/mm²		
Design vc		=	1.95 N/mm²				

Check at 0.5d from column face

Perimeter at 0.5d from column face	u	=	4.55 m		Vc	=	$v_c b_0 d$	
Depth at section	d	=	338 mm			=	3003.77 kN	
Shear Force at section	Vu	=	1039.0 kN		φ	=	0.75 for shear	
Allowable shear strength	φVc	=	2252.83 kN	O.K	φV_n	=	φVc	
					φVc	=	2252.8 kN	

Sensitivity: LNT Construction Internal Use

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

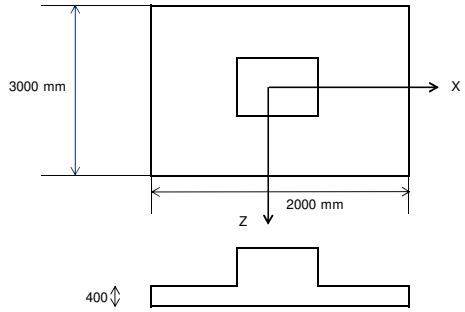
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Approved	AG	AG	AG	AG

Outotec / L&T Revision No.:	0	1	2	3
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8.3.1.2 DESIGN OF ISOLATED FOUNDATION FOR COLUMN AT 3120/9_2130/E AND MARKED AS F1

Finished Ground level	F.G.L	=	-0.500	m
Bottom of foundation	B.O.F	=	-2.500	m
Top of footing (staad support)	T.O.F	=	0.500	m
Ground Water Level	G.W.L	=	-10.000	m
Surcharge	Ws	=	0.000	kN/m^2
Length of footing	Lz	=	3.00	m
Width of footing	Lx	=	2.00	m
Thickness of footing	D	=	0.40	m
Depth of pedestal	Bx	=	0.80	m
Width of pedestal	Bz	=	0.80	m
Height of Load point from bottom of footing	H	=	3.000	m
Section Modulus of Pedestal	Zxx	=	3.00	m^3
Section Modulus of Pedestal	Zzz	=	2.00	m^3
Allowable Net Bearing Capacity		=	500	kN/m^2
Gross Bearing Capacity		=	536	kN/m^2
Grade of concrete		=	35	N/mm^2
Grade of steel		=	420	N/mm^2
Clear cover		=	50	mm
Top Reinfocement		=	12	mm
Bottom Reinforcement		=	16	mm
Co-efficient of friction between soil & concrete		=	0.3	
Unit wt of soil		=	18	kN/m^3
Unit wt of concrete		=	24	kN/m^3

Weight of soil	=	154.37	KN
Weight of foundation	=	97.54	KN
Uplift	=	0.00	KN
Surcharge	=	0	KN
Pressure on top for tension	=	42.0	KN/m^2



$$\beta_1 = 0.85 - \frac{0.05(f'_c - 28)}{7}$$ β_1 = 0.8 *Table 22.2.2.4.3 SBC 304*

CHECK FOR BEARING PRESSURE CAPACITY

L/C	Fx	Fy	Fz	Mx	My	Mz	Sum Fy	Pmax	Pmin	B.C Factor	F.O.S B.P
301	0.1	273.128	25.4	76.14	0.0	-0.4	525.0	113.08	61.93	1	0.21
302	0.1	258.016	28.9	86.67	0	-0.3	509.9	114.04	55.94	1	0.21
303	-1.1	386.586	85.8	257.33	0	3.3	638.5	193.82	19.01	1	0.36
304	-1.1	371.474	89.3	267.86	0	3.3	623.4	194.84	12.95	1	0.36
305	1.4	415.967	-11.1	-33.33	0	-4.2	667.9	124.50	98.12	1	0.23
306	1.4	400.855	-7.6	-22.81	0.0	-4.1	652.8	118.44	99.15	1	0.22
307	1.5	359.673	35.3	105.86	0	-4.4	611.6	139.43	64.43	1	0.26
308	-1.2	379.399	33.3	99.94	0	3.5	631.3	140.30	70.13	1	0.26
309	0.2	441.642	57.7	173.12	0.0	-0.5	693.5	173.56	57.62	1	0.32
310	0.1	295.03	10.3	30.89	0	-0.3	546.9	101.63	80.68	1	0.19
311	1.5	344.561	38.8	116.39	0	-4.4	596.5	140.38	58.44	1	0.26
312	-1.2	364.287	36.8	110.47	0	3.6	616.2	141.33	64.07	1	0.26
313	0.2	426.53	61.2	183.65	0	-0.5	678.4	174.52	51.63	1	0.33
314	0.1	279.918	13.8	41.42	0.0	-0.3	531.8	102.58	74.69	1	0.19
315	0.1	369.24	34.3	103.03	0	-0.4	621.1	138.09	68.96	1	0.26
316	0.1	369.24	34.3	103.03	0	-0.4	621.1	138.09	68.96	1	0.26
317	0.1	369.24	34.3	103.03	0	-0.4	621.1	138.09	68.96	1	0.26
318	0.1	369.24	34.3	103.03	0	-0.4	621.1	138.09	68.96	1	0.26
319	0.1	354.128	37.9	113.56	0.0	-0.4	606.0	139.04	62.97	1	0.26
320	0.1	354.128	37.9	113.56	0	-0.4	606.0	139.04	62.97	1	0.26
321	0.1	354.128	37.9	113.56	0	-0.4	606.0	139.04	62.97	1	0.26
322	0.1	354.128	37.9	113.56	0.0	-0.4	606.0	139.04	62.97	1	0.26
323	1.8	151.121	16.5	49.45	0	-5.5	403.0	86.43	47.92	1	0.16
324	-1.7	177.422	13.9	41.56	0	5.1	429.3	87.94	55.17	1	0.16
325	0.1	260.413	46.4	139.14	0	-0.4	512.3	131.94	38.83	1	0.25
326	0.0	64.931	-16.8	-50.51	0	-0.1	316.8	69.70	35.91	1	0.13
327	1.8	142.054	18.6	55.77	0.0	-5.5	394.0	87.00	44.32	1	0.16
328	-1.7	168.355	16.0	47.88	0	5.1	420.3	88.56	51.53	1	0.17
329	0.1	251.346	48.5	145.46	0	-0.3	503.3	132.51	35.24	1	0.25
330	0.0	55.864	-14.7	-44.19	0	-0.1	307.8	66.06	36.53	1	0.12
331	0.1	163.877	15.2	45.68	0	-0.2	415.8	84.64	53.95	1	0.16
332	0.1	163.877	15.2	45.68	0.0	-0.2	415.8	84.64	53.95	1	0.16
333	0.1	163.877	15.2	45.68	0	-0.2	415.8	84.64	53.95	1	0.16
334	0.1	163.877	15.2	45.68	0	-0.2	415.8	84.64	53.95	1	0.16
335	0.1	154.81	17.3	52.00	0	-0.2	406.7	85.22	50.36	1	0.16
336	0.1	154.81	17.3	52.00	0.0	-0.2	406.7	85.22	50.36	1	0.16
337	0.1	154.81	17.3	52.00	0	-0.2	406.7	85.22	50.36	1	0.16
338	0.1	154.81	17.3	52.00	0	-0.2	406.7	85.22	50.36	1	0.16

Max P_{max}	=	194.84 kN/m^2	Load Case	304	Resultant F.O.S	=	0.26
Max P_{min}	=	12.95 kN/m^2	Load Case	304			

Sensitivity: LNT Construction Internal Use

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	C02	D01
Date	27-01-2020	12-03-2020	25-04-2022	12/10/2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS DT. 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER SUPPORT STRUCTURE	AS BUILT
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Checked	SBY	SBY	SBY	SBY
Approved	AG	AG	AG	AG

Outotec / L&T Revision No.:	0	1	2	3
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BEARING PRESSURE CALCULATION FOR BENDING MOMENT AND SHEAR CHECK

L/C	Fx	Fy	Fz	Mx	My	Mz	Pmax	Pmin
401	0.2	382.4	35.5	106.59	0.0	-0.56	99.54	27.92
402	0.2	361.2	40.4	121.33	0	-0.46	100.88	19.53
403	-1.5	579.7	123.7	371.02	0	4.53	222.55	-29.33
404	1.9	620.8	-12.0	-35.91	0	-5.84	118.36	88.57
405	-1.5	558.5	128.6	385.76	0	4.63	223.99	-37.82
406	1.9	599.6	-7.1	-21.17	0	-5.75	109.87	90.01
407	3.0	371.4	38.4	115.33	0	-8.99	104.84	18.96
408	-2.7	413.5	34.2	102.70	0	7.99	107.14	30.69
409	0.2	546.3	86.3	258.82	0.0	-0.69	177.67	4.43
410	0.1	233.5	-14.9	-44.61	0	-0.31	53.94	23.89
411	3.0	353.3	42.7	127.96	0	-8.91	105.99	11.77
412	-2.7	395.4	38.4	115.34	0	8.07	108.37	23.41
413	0.2	528.2	90.5	271.46	0	-0.60	178.81	-2.76
414	0.1	215.4	-10.7	-31.98	0	-0.23	46.67	25.13
415	0.2	391.8	36.4	109.30	0	-0.50	101.99	28.62
416	0.2	391.8	36.4	109.30	0	-0.50	101.99	28.62
417	0.2	391.8	36.4	109.30	0.0	-0.50	101.99	28.62
418	0.2	391.8	36.4	109.30	0	-0.50	101.99	28.62
419	0.1	373.7	40.6	121.93	0	-0.42	103.14	21.43
420	0.1	373.7	40.6	121.93	0	-0.42	103.14	21.43
421	0.1	373.7	40.6	121.93	0	-0.42	103.14	21.43
422	0.1	373.7	40.6	121.93	0	-0.42	103.14	21.43
423	1.9	233.1	24.1	72.29	0	-5.66	65.77	11.91
424	-1.7	259.4	21.5	64.40	0	4.95	67.17	19.28
425	0.2	342.4	54.0	161.98	0	-0.47	111.29	2.83
426	0.1	146.9	-9.2	-27.67	0	-0.23	33.82	15.14
427	1.9	219.5	27.3	81.77	0	-5.60	66.63	6.52
428	-1.7	245.8	24.6	73.88	0	5.02	68.09	13.83
429	0.1	328.8	57.2	171.45	0	-0.41	112.15	-2.56
430	0.1	133.3	-6.1	-18.19	0	-0.17	28.36	16.06
431	0.1	245.8	22.8	68.52	0	-0.36	63.99	17.95
432	0.1	245.8	22.8	68.52	0	-0.36	63.99	17.95
433	0.1	245.8	22.8	68.52	0	-0.36	63.99	17.95
434	0.1	245.8	22.8	68.52	0	-0.36	63.99	17.95
435	0.1	232.2	26.0	78.00	0	-0.29	64.85	12.56
436	0.1	232.2	26.0	78.00	0	-0.29	64.85	12.56
437	0.1	232.2	26.0	78.00	0	-0.29	64.85	12.56
438	0.1	232.2	26.0	78.00	0	-0.29	64.85	12.56

Max Net factored B.P = 223.99 kN/m^2 Pmin = -37.82 kN/m^2

DETERMINATION OF REINFORCEMENT

Bottom Reinforcement

Length of cantilever along X = 0.6 m

Moment about Z Muz = 40.32 kNm

$M_{uzlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

= 798.59 kNm

As $M_{uzlim} > M_{uz}$ Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

= 313.5 mm^2

Effective Depth Along X = 342 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement $A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*

greater of or A_{stmin} = 0.0014bd

A_{stmin} = 615.6 mm^2

A_{req} = 313.5 mm^2

Required Spacing = 326.6 mm

Sensitivity: LNT Construction Internal Use

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

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Length of cantilever along Z = 1.1 m
Moment about X: Mux = 135.51 kNm

$M_{uxlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$
= 725.61 kNm

As $M_{uxlim} > M_{ux}$ Tension controlled section
Therefore Φ = 0.9 *Table 21.2.2 SBY 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$
= 1125.8 mm^2

Provide Bottom Reinforcement = 16 mm dia @ 150 c/c

Effective Depth Along X = 326 mm
For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement greater of
$A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*
or
A_{stmin} = 0.0014bd
A_{stmin} = 586.8 mm^2
A_{req} = 1125.8 mm^2
Required Spacing = 178.6 mm

Ast prov = 1340.4 mm^2

Top Reinforcement
Length of cantilever along X = 0.6 m
Moment about Z: Muz = 7.56 KNm

$M_{uzlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$
= 807.95 kNm

As $M_{uzlim} > M_{uz}$ Tension controlled section
Therefore φ = 0.9 *Table 21.2.2 SBY 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$
= 58.1 mm^2

Effective Depth Along X = 344 mm
For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement greater of
A_{stmin} = 50% of $\frac{0.0018 \times 420}{f_y} bd$ since 50% or more prov in bottom
or
A_{stmin} = 50% of 0.0014bd *Table 24.4.3.2 SBC 304*
A_{stmin} = 309.6 mm^2
A_{req} = 58.1 mm^2
Required Spacing = 365.3 mm

Length of cantilever along Z = 1.1 m
Moment about X: Mux = 25.40 kNm

$M_{uxlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$
= 752.57 kNm

As $M_{uxlim} > M_{ux}$ Tension controlled section
Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$
= 203.0 mm^2

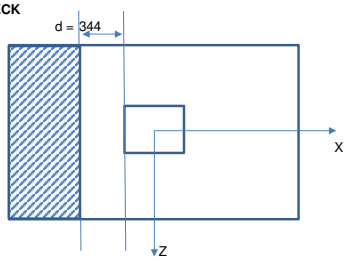
Effective Depth Along Z = 332 mm
For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement greater of
A_{stmin} = 50% of $\frac{0.0018 \times 420}{f_y} bd$ since 50% or more prov in bottom
or
A_{stmin} = 50% of 0.0014bd *Table 24.4.3.2 SBC 304*
A_{stmin} = 298.8 mm^2
A_{req} = 203.0 mm^2
Required Spacing = 378.5 mm

Provide top Reinforcement = 12 mm dia @ 150 c/c

Ast prov = 754.0 mm^2

ONE WAY SHEAR CKECK



Along X Direction Cantilever: L = 0.256 m
Reinforcement ratio: ρ_w = 0.0039
Effective Depth Along X: d = 344 mm
Shear Force at d distance: V_u = 57.34 kN
Allowable shear strength: φVc = 244.22 kN O.K

Least of
i) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right) b_w d$ *Table 22.5.5.1 SBC 304*
λ = 1 Normal concrete
Mu = 7.34 kNm
φ = 0.75 for shear
φVc = 290.145858 kN

ii) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\right) b_w d$
φVc = 261.31 kN *Table 22.5.5.1 SBC 304*

iii) $V_c = 0.29\lambda\sqrt{f_c'} b_w d$
= 244.22 kN

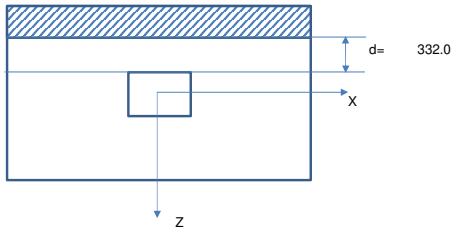
Sensitivity: LNT Construction Internal Use

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	C02	D01
Date	27-01-2020	12-03-2020	25-04-2022	12/10/2023
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Checked	SBY	SBY	SBY	SBY
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Outotec / L&T Revision No.:	0	1	2	3
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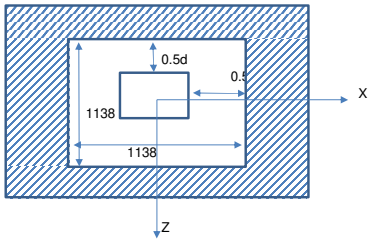
Along Z Direction



Cantilever	L	=	0.768 m		Least of	i)	V_c	=	$\left(0.16\lambda\sqrt{f_c'}+17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$
Reinforcement ratio	ρ_w	=	0.0040						*Table 22.5.5.1 SBC 304*
Effective Depth Along Z	d	=	332 mm				λ	=	1 Normal concrete
Shear Force at d distance	V_u	=	172.02 kN				Mu	=	66.06 kNm
Allowable shear strength	φVc	=	235.70 kN	O.K			φ	=	0.75 for shear
							φVc	=	250.47 kN
						ii)	V_c	=	$\left(0.16\lambda\sqrt{f_c'}+17\rho_w\right)b_w d$
							φVc	=	252.786883 kN
						iii)	V_c	=	$0.29\lambda\sqrt{f_c'}b_w d$
							φVc	=	235.70 kN

TWO WAY SHEAR CKECK

Punching at 0.5d from face of column



Total area	=	6	m2
Punching area	=	1.30	m2
λ	=	1	
β	=	1	
αs	=	40	
b0	=	4552.0	mm

Allowable shear stress least of *Table 22.6.5.2 SBC 304*

$v_c = 0.33\lambda\sqrt{f_c'}$	=	1.95	N/mm²
$v_c = 0.17\left(1+\frac{2}{\beta}\right)\lambda\sqrt{f_c'}$	=	3.02	N/mm²
$v_c = 0.083\left(2+\frac{\alpha_s d}{b_0}\right)\lambda\sqrt{f_c'}$	=	2.44	N/mm²

Design vc = 1.95 N/mm²

Check at 0.5d from column face

Perimeter at 0.5d from column face	u	=	4.55 m		Vc	=	$v_c b_0 d$
Depth at section	d	=	338 mm			=	3003.77 kN
Shear Force at section	Vu	=	1053.9 kN		φ	=	0.75 for shear
Allowable shear strength	φVc	=	2252.83 kN	O.K	φV_n	=	φVc
					φVc	=	2252.8 kN

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Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

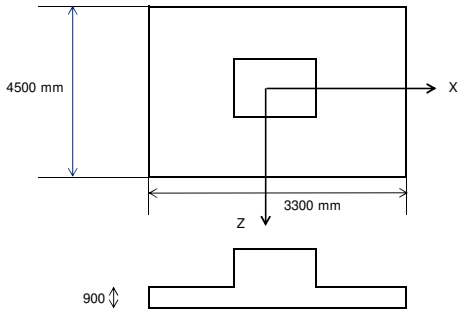
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8.3.2.1 DESIGN OF ISOLATED FOUNDATION FOR COLUMN AT 3120/10_2130/J AND MARKED AS F2

Finished Ground level	F.G.L	=	-0.500	m
Bottom of foundation	B.O.F	=	-2.500	m
Top of footing (staad support)	T.O.F	=	0.500	m
Ground Water Level	G.W.L	=	-10.000	m
Surcharge	Ws	=	0.000	kN/m^2
Length of footing	Lz	=	4.50	m
Width of footing	Lx	=	3.30	m
Thickness of footing	D	=	0.90	m
Depth of pedestal	Bx	=	0.80	m
Width of pedestal	Bz	=	0.80	m
Height of Load point from bottom of footing	H	=	3.000	m
Section Modulus of Pedestal	Zxx	=	11.14	m^3
Section Modulus of Pedestal	Zzz	=	8.17	m^3
Allowable Net Bearing Capacity		=	500	kN/m^2
Gross Bearing Capacity		=	536	kN/m^2
Grade of concrete		=	35	N/mm^2
Grade of steel		=	420	N/mm^2
Clear cover		=	50	mm
Top Reinfocement		=	12	mm
Bottom Reinforcement		=	20	mm
Co-efficient of friction between soil & concrete		=	0.3	
Unit wt of soil		=	18	kN/m^3
Unit wt of concrete		=	24	kN/m^3

Weight of soil	=	281.36	KN
Weight of foundation	=	353.02	KN
Uplift	=	0.00	KN
Surcharge	=	0	KN
Pressure on top for tension	=	42.7	KN/m^2



$$\beta_1 = 0.85 - \frac{0.05(f'_c - 28)}{7} \qquad \beta_1 = 0.8$$ *Table 22.2.2.4.3 SBC 304*

CHECK FOR BEARING PRESSURE CAPACITY

L/C	Fx	Fy	Fz	Mx	My	Mz	Sum Fy	Pmax	Pmin	B.C Factor	F.O.S B.P
301	-42.6	686.363	-0.5	-1.36	0.0	127.7	1320.7	104.69	73.19	1	0.20
302	-23.2	1586.39	1.0	3.09	0	69.7	2220.8	158.35	140.74	1	0.30
303	-49.1	931.912	-234.9	-704.55	0	147.2	1566.3	186.75	24.20	1	0.35
304	-29.7	1831.939	-233.4	-700.10	0	89.2	2466.3	239.86	92.31	1	0.45
305	-37.1	865.254	231.8	695.54	0	111.3	1499.6	177.07	24.90	1	0.33
306	-17.8	1765.281	233.3	700.00	0.0	53.3	2399.7	230.97	92.21	1	0.43
307	20.1	528.814	-1.5	-4.48	0	-60.2	1163.2	86.11	70.55	1	0.16
308	-105.3	1154.927	-1.1	-3.23	0	315.8	1789.3	159.45	81.53	1	0.30
309	-53.6	619.801	-12.1	-36.25	0.0	160.7	1254.2	107.39	61.53	1	0.20
310	-32.5	1084.579	10.0	30.04	0	97.5	1719.0	130.39	101.12	1	0.24
311	39.4	1428.841	0.0	-0.02	0	-118.2	2063.2	153.42	124.46	1	0.29
312	-85.9	2054.954	0.4	1.23	0	257.8	2689.3	212.78	149.42	1	0.40
313	-34.2	1519.828	-10.6	-31.79	0	102.7	2154.2	160.49	129.64	1	0.30
314	-13.2	1984.606	11.5	34.49	0.0	39.5	2619.0	184.30	168.43	1	0.34
315	-43.0	845.528	-1.2	-3.72	0	128.9	1479.9	115.77	83.55	1	0.22
316	-43.0	845.528	-1.2	-3.72	0	128.9	1479.9	115.77	83.55	1	0.22
317	-43.0	845.528	-1.2	-3.72	0	128.9	1479.9	115.77	83.55	1	0.22
318	-43.0	845.528	-1.2	-3.72	0	128.9	1479.9	115.77	83.55	1	0.22
319	-23.6	1745.555	0.2	0.74	0.0	70.8	2379.9	169.00	151.52	1	0.32
320	-23.6	1745.555	0.2	0.74	0	70.8	2379.9	169.00	151.52	1	0.32
321	-23.6	1745.555	0.2	0.74	0	70.8	2379.9	169.00	151.52	1	0.32
322	-23.6	1745.555	0.2	0.74	0.0	70.8	2379.9	169.00	151.52	1	0.32
323	58.5	-10.467	-0.6	-1.83	0	-175.5	623.9	63.67	20.36	1	0.12
324	-108.6	824.35	-0.1	-0.16	0	325.9	1458.7	138.15	58.31	1	0.26
325	-39.7	110.848	-14.7	-44.19	0	119.1	745.2	68.73	31.64	1	0.13
326	-11.6	730.553	14.7	44.20	0	34.8	1364.9	100.15	83.68	1	0.19
327	70.1	529.549	0.3	0.84	0.0	-210.3	1163.9	104.21	52.55	1	0.19
328	-97.0	1364.366	0.8	2.51	0	291.1	1998.7	170.46	98.73	1	0.32
329	-28.1	650.864	-13.8	-41.52	0	84.3	1285.2	100.59	72.50	1	0.19
330	0.0	1270.569	15.6	46.87	0	0.0	1904.9	132.49	124.07	1	0.25
331	-25.5	411.818	-0.3	-0.82	0	76.6	1046.2	79.90	61.00	1	0.15
332	-25.5	411.818	-0.3	-0.82	0.0	76.6	1046.2	79.90	61.00	1	0.15
333	-25.5	411.818	-0.3	-0.82	0	76.6	1046.2	79.90	61.00	1	0.15
334	-25.5	411.818	-0.3	-0.82	0	76.6	1046.2	79.90	61.00	1	0.15
335	-13.9	951.834	0.6	1.86	0	41.8	1586.2	112.10	101.53	1	0.21
336	-13.9	951.834	0.6	1.86	0.0	41.8	1586.2	112.10	101.53	1	0.21
337	-13.9	951.834	0.6	1.86	0	41.8	1586.2	112.10	101.53	1	0.21
338	-13.9	951.834	0.6	1.86	0	41.8	1586.2	112.10	101.53	1	0.21

Max P_{max}	=	239.86	kN/m^2	Load Case	304	Resultant F.O.S =	0.32
Max P_{min}	=	20.36	kN/m^2	Load Case	323		

Sensitivity: LNT Construction Internal Use

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Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

APR SBY AG

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BEARING PRESSURE CALCULATION FOR BENDING MOMENT AND SHEAR CHECK

L/C	Fx	Fy	Fz	Mx	My	Mz	Pmax	Pmin
401	-59.6	960.9	-0.6	-1.91	0.0	178.73	86.76	42.65
402	-32.5	2220.9	1.4	4.33	0	97.52	161.89	137.23
403	-68.8	1368.3	-329.1	-987.32	0	206.51	206.08	-21.79
404	-52.1	1275.0	324.3	972.82	0	156.34	192.35	-20.63
405	-41.8	2628.4	-327.0	-981.08	0	125.30	280.42	73.57
406	-25.0	2535.1	326.4	979.05	0	75.12	267.81	73.61
407	83.1	254.1	-1.6	-4.83	0	-249.39	48.08	-13.86
408	-184.3	1589.8	-0.7	-2.15	0	552.91	174.95	39.17
409	-74.0	448.2	-24.2	-72.60	0.0	221.93	63.87	-3.51
410	-29.1	1439.7	22.9	68.81	0	87.17	113.80	80.10
411	106.3	1334.1	0.2	0.52	0	-319.01	128.94	50.74
412	-161.1	2669.8	1.1	3.19	0	483.29	239.25	120.33
413	-50.8	1528.2	-22.4	-67.26	0	152.32	127.60	78.22
414	-5.9	2519.8	24.7	74.16	0	17.56	178.49	160.87
415	-51.3	929.7	-1.1	-3.20	0	153.99	81.75	43.47
416	-51.3	929.7	-1.1	-3.20	0	153.99	81.75	43.47
417	-51.3	929.7	-1.1	-3.20	0.0	153.99	81.75	43.47
418	-51.3	929.7	-1.1	-3.20	0	153.99	81.75	43.47
419	-28.1	2009.8	0.7	2.14	0	84.38	145.86	124.82
420	-28.1	2009.8	0.7	2.14	0	84.38	145.86	124.82
421	-28.1	2009.8	0.7	2.14	0	84.38	145.86	124.82
422	-28.1	2009.8	0.7	2.14	0	84.38	145.86	124.82
423	45.7	195.4	-0.7	-2.24	0	-137.22	30.16	-3.84
424	-121.4	1030.3	-0.2	-0.57	0	364.22	114.02	24.73
425	-52.5	316.8	-14.9	-44.60	0	157.36	44.60	-1.94
426	-24.4	936.5	14.6	43.79	0	73.13	75.95	50.18
427	63.1	1005.5	0.6	1.77	0	-189.43	91.06	44.36
428	-104.0	1840.3	1.1	3.44	0	312.01	162.44	85.41
429	-35.1	1126.8	-13.5	-40.59	0	105.15	92.40	59.36
430	-7.0	1746.5	15.9	47.80	0	20.93	124.46	110.76
431	-38.3	617.7	-0.4	-1.22	0	114.90	55.78	27.42
432	-38.3	617.7	-0.4	-1.22	0	114.90	55.78	27.42
433	-38.3	617.7	-0.4	-1.22	0	114.90	55.78	27.42
434	-38.3	617.7	-0.4	-1.22	0	114.90	55.78	27.42
435	-20.9	1427.8	0.9	2.78	0	62.69	104.07	88.22
436	-20.9	1427.8	0.9	2.78	0	62.69	104.07	88.22
437	-20.9	1427.8	0.9	2.78	0	62.69	104.07	88.22
438	-20.9	1427.8	0.9	2.78	0	62.69	104.07	88.22

Max Net factored B.P = 280.42 kN/m^2 Pmin = -21.79 kN/m^2

DETERMINATION OF REINFORCEMENT

Bottom Reinforcement

Length of cantilever along X = 1.25 m

Moment about Z Muz = 219.08 kNm

$M_{uzlim} = \emptyset\, 0.85\, f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

= 4817.57 kNm

As $M_{uzlim} > M_{uz}$ Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\, f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

= 693.2 mm^2

Effective Depth Along X = 840 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement $A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*

greater of or A_{stmin} = 0.0014bd

A_{stmin} = 1512.0 mm^2

A_{req} = 693.2 mm^2

Required Spacing = 207.8 mm

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Approved	AG	AG	AG	AG

Outotec / L&T Revision No.:	0	1	2	3
Date	27-01-2020	12-03-2020	25-04-2022	12/10/2023
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Length of cantilever along Z = 1.85 m
Moment about X: Mux = 479.88 kNm

$M_{uxlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$ = 4590.90 kNm

As $M_{uxlim} > M_{ux}$ Tension controlled section
Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$ = 1567.4 mm^2

Provide Bottom Reinforcement = 20 mm dia @ 150 c/c

Effective Depth Along X = 820 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement greater of
$A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*
or
A_{stmin} = 0.0014bd
A_{stmin} = 1476.0 mm^2
A_{req} = 1567.4 mm^2
Required Spacing = 200.4 mm

Ast prov = 2094.4 mm^2

Top Reinforcement
Length of cantilever along X = 1.25 m
Moment about Z: Muz = 33.37 KNm

$M_{uzlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$ = 4863.56 KNm

As $M_{uzlim} > M_{uz}$ Tension controlled section
Therefore φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$ = 104.6 mm^2

Effective Depth Along X = 844 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement greater of
A_{stmin} = 50% of $\frac{0.0018 \times 420}{f_y} bd$ since 50% or more prov in bottom
or
A_{stmin} = 50% of 0.0014bd *Table 24.4.3.2 SBC 304*
A_{stmin} = 759.6 mm^2
A_{req} = 104.6 mm^2
Required Spacing = 148.9 mm

Length of cantilever along Z = 1.85 m
Moment about X: Mux = 73.10 KNm

$M_{uxlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$ = 4726.25 KNm

As $M_{uxlim} > M_{ux}$ Tension controlled section
Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$ = 232.6 mm^2

Effective Depth Along Z = 832 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

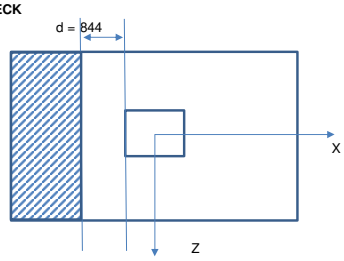
Min reinforcement greater of
A_{stmin} = 50% of $\frac{0.0018 \times 420}{f_y} bd$ since 50% or more prov in bottom
or
A_{stmin} = 50% of 0.0014bd *Table 24.4.3.2 SBC 304*
A_{stmin} = 748.8 mm^2
A_{req} = 232.6 mm^2
Required Spacing = 151.0 mm

Provide top Reinforcement = 12 mm dia @ 150 c/c

Ast prov = 754.0 mm^2

ONE WAY SHEAR CKECK

Along X Direction



Cantilever L = 0.406 m
Reinforcement ratio ρ_w = 0.0025
Effective Depth Along X d = 844 mm
Shear Force at d distance V_u = 113.85 kN
Allowable shear strength φVc = 599.18 kN O.K

Least of

i) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right) b_w d$ *Table 22.5.5.1 SBC 304*
λ = 1 Normal concrete
Mu = 23.11 kNm
φ = 0.75 for shear
φVc = 710.204 kN

ii) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\right) b_w d$
φVc = 625.884098 KN *Table 22.5.5.1 SBC 304*

iii) $V_c = 0.29\lambda\sqrt{f_c'} b_w d$
= 599.18056 kN

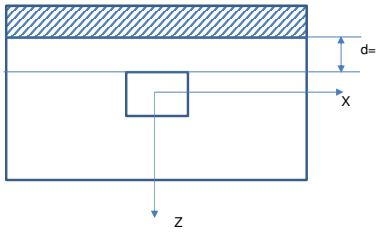
Sensitivity: LNT Construction Internal Use

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

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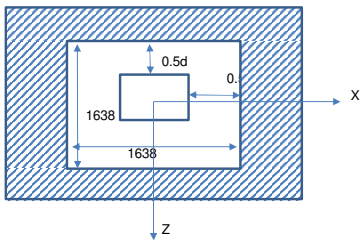
Along Z Direction



Along Z Direction Cantilever	L	=	1.018 m		Least of	i)	V_c	=	$\left(0.16\lambda\sqrt{f_c'}+17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$
Reinforcement ratio	ρ_w	=	0.0025						*Table 22.5.5.1 SBC 304*
Effective Depth Along Z	d	=	832 mm				λ	=	1 Normal concrete
Shear Force at d distance	V_u	=	285.47 kN				Mu	=	145.31 kNm
Allowable shear strength	φVc	=	590.66 kN	O.K			φ	=	0.75 for shear
							φVc	=	634.31 kN
						ii)	V_c	=	$\left(0.16\lambda\sqrt{f_c'}+17\rho_w\right)b_w d$
							φVc	=	617.364943 kN
						iii)	V_c	=	$0.29\lambda\sqrt{f_c'}b_w d$
							φVc	=	590.66 kN

TWO WAY SHEAR CKECK

Punching at 0.5d from face of column



Total area		=	14.85 m2			
Punching area		=	2.68 m2			
	λ	=	1			
	β	=	1			
	αs	=	40			
	b0	=	6552.0 mm			
Allowable shear stress	least of		*Table 22.6.5.2 SBC 304*			
v_c	$0.33\lambda\sqrt{f_c'}$	=	1.95 N/mm²			
v_c	$0.17\left(1+\frac{2}{\beta}\right)\lambda\sqrt{f_c'}$	=	3.02 N/mm²			
v_c	$0.083\left(2+\frac{\alpha_s d}{b_0}\right)\lambda\sqrt{f_c'}$	=	3.49 N/mm²			
Design vc		=	1.95 N/mm²			

Check at 0.5d from column face								
Perimeter at 0.5d from column face	u	=	6.55 m		Vc	=	$v_c b_0 d$	
Depth at section	d	=	838 mm			=	10719.29 kN	
Shear Force at section	Vu	=	3411.9 kN		φ	=	0.75 for shear	
Allowable shear strength	φVc	=	8039.46 kN	O.K	φV_n	=	φVc	
					φVc	=	8039.5 kN	

Sensitivity: LNT Construction Internal Use

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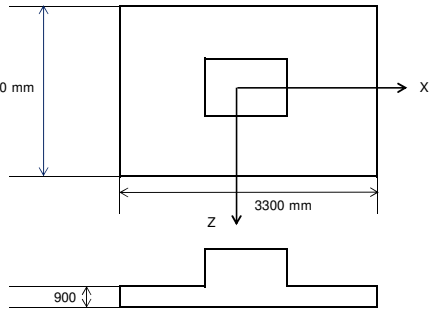
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8.3.2.2 DESIGN OF ISOLATED FOUNDATION FOR COLUMN AT 3120/12_2130/J AND MARKED AS F2

Finished Ground level	F.G.L	=	-0.500	m
Bottom of foundation	B.O.F	=	-2.500	m
Top of footing (staad support)	T.O.F	=	0.500	m
Ground Water Level	G.W.L	=	-10.000	m
Surcharge	Ws	=	0.000	kN/m^2
Length of footing	Lz	=	4.50	m
Width of footing	Lx	=	3.30	m
Thickness of footing	D	=	0.90	m
Depth of pedestal	Bx	=	0.80	m
Width of pedestal	Bz	=	0.80	m
Height of Load point from bottom of footing	H	=	3.000	m
Section Modulus of Pedestal	Zxx	=	11.14	m^3
Section Modulus of Pedestal	Zzz	=	8.17	m^3
Allowable Net Bearing Capacity		=	500	kN/m^2
Gross Bearing Capacity		=	536	kN/m^2
Grade of concrete		=	35	N/mm^2
Grade of steel		=	420	N/mm^2
Clear cover		=	50	mm
Top Reinfocement		=	12	mm
Bottom Reinforcement		=	20	mm
Co-efficient of friction between soil & concrete		=	0.3	
Unit wt of soil		=	18	kN/m^3
Unit wt of concrete		=	24	kN/m^3

Weight of soil	=	281.36	KN
Weight of foundation	=	353.02	KN
Uplift	=	0.00	KN
Surcharge	=	0	KN
Pressure on top for tension	=	42.7	KN/m^2



$\beta_1 = 0.85 - \frac{0.05(f'_c - 28)}{7}$ β_1 = 0.8 *Table 22.2.2.4.3 SBC 304*

CHECK FOR BEARING PRESSURE CAPACITY

L/C	Fx	Fy	Fz	Mx	My	Mz	Sum Fy	Pmax	Pmin	B.C Factor	F.O.S B.P
301	-40.8	50.714	1.8	5.48	0.0	122.5	685.1	61.63	30.64	1	0.11
302	-21.6	1785.949	4.6	13.73	0	64.9	2420.3	172.17	153.80	1	0.32
303	-42.9	182.519	-0.9	-2.70	0	128.6	816.9	70.99	39.02	1	0.13
304	-23.7	1917.754	1.8	5.54	0	71.0	2552.1	181.05	162.68	1	0.34
305	-40.0	241.948	1.0	3.11	0	120.1	876.3	73.99	44.03	1	0.14
306	-20.8	1977.183	3.8	11.35	0.0	62.5	2611.6	184.53	167.19	1	0.34
307	21.8	511.112	0.7	1.99	0	-65.5	1145.5	85.33	68.94	1	0.16
308	-103.7	-161.435	0.4	1.08	0	311.1	472.9	70.04	-6.34	1	0.13
309	-51.9	13.663	4.9	14.68	0.0	155.7	648.0	64.02	23.25	1	0.12
310	-30.9	340.126	-3.9	-11.57	0	92.6	974.5	77.99	53.25	1	0.15
311	41.0	2246.347	3.4	10.22	0	-123.1	2880.7	209.98	178.00	1	0.39
312	-84.5	1573.8	3.1	9.32	0	253.5	2208.2	180.58	116.82	1	0.34
313	-32.7	1748.898	7.6	22.91	0	98.1	2383.3	174.56	146.42	1	0.33
314	-11.6	2075.361	-1.1	-3.33	0.0	34.9	2709.7	187.05	177.90	1	0.35
315	-41.3	171.854	0.5	1.53	0	123.9	806.2	69.60	38.99	1	0.13
316	-41.3	171.854	0.5	1.53	0	123.9	806.2	69.60	38.99	1	0.13
317	-41.3	171.854	0.5	1.53	0	123.9	806.2	69.60	38.99	1	0.13
318	-41.3	171.854	0.5	1.53	0	123.9	806.2	69.60	38.99	1	0.13
319	-22.1	1907.089	3.3	9.77	0.0	66.3	2541.5	180.13	162.15	1	0.34
320	-22.1	1907.089	3.3	9.77	0	66.3	2541.5	180.13	162.15	1	0.34
321	-22.1	1907.089	3.3	9.77	0	66.3	2541.5	180.13	162.15	1	0.34
322	-22.1	1907.089	3.3	9.77	0.0	66.3	2541.5	180.13	162.15	1	0.34
323	59.7	482.773	1.3	3.90	0	-179.0	1117.1	97.49	52.97	1	0.18
324	-107.7	-413.956	0.9	2.70	0	323.2	220.4	54.66	-24.97	1	0.10
325	-38.7	-180.493	6.9	20.82	0	116.0	453.9	46.64	14.49	1	0.09
326	-10.6	254.791	-4.7	-14.17	0	31.8	889.2	65.04	54.72	1	0.12
327	71.2	1523.914	2.9	8.85	0.0	-213.5	2158.3	172.28	118.40	1	0.32
328	-96.2	627.185	2.5	7.64	0	288.6	1261.6	120.98	48.93	1	0.23
329	-27.1	860.648	8.6	25.77	0	81.4	1495.0	112.96	88.39	1	0.21
330	0.9	1295.932	-3.1	-9.23	0	-2.8	1930.3	131.16	128.81	1	0.24
331	-24.5	30.428	1.1	3.29	0	73.5	664.8	54.06	35.47	1	0.10
332	-24.5	30.428	1.1	3.29	0.0	73.5	664.8	54.06	35.47	1	0.10
333	-24.5	30.428	1.1	3.29	0	73.5	664.8	54.06	35.47	1	0.10
334	-24.5	30.428	1.1	3.29	0	73.5	664.8	54.06	35.47	1	0.10
335	-13.0	1071.569	2.7	8.24	0	39.0	1705.9	120.39	109.37	1	0.22
336	-13.0	1071.569	2.7	8.24	0.0	39.0	1705.9	120.39	109.37	1	0.22
337	-13.0	1071.569	2.7	8.24	0	39.0	1705.9	120.39	109.37	1	0.22
338	-13.0	1071.569	2.7	8.24	0	39.0	1705.9	120.39	109.37	1	0.22

Max P_{max}	=	209.98	kN/m^2	Load Case	311	Resultant F.O.S	= 0.34
Max P_{min}	=	-24.97	kN/m^2	Load Case	324		

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BEARING PRESSURE CALCULATION FOR BENDING MOMENT AND SHEAR CHECK

L/C	Fx	Fy	Fz	Mx	My	Mz	Pmax	Pmin
401	-57.2	71.0	2.6	7.68	0.0	171.55	26.47	-16.91
402	-30.3	2500.3	6.4	19.22	0	90.89	181.23	155.52
403	-60.2	304.0	-1.8	-5.36	0	180.55	43.06	-2.12
404	-56.2	387.2	0.9	2.77	0	168.66	46.97	5.17
405	-33.3	2733.3	2.1	6.18	0	99.88	196.84	171.28
406	-29.3	2816.5	4.8	14.30	0	88.00	201.72	177.61
407	85.3	865.4	1.6	4.92	0	-256.02	90.06	26.49
408	-182.5	-569.4	1.0	2.99	0	547.42	28.95	-105.64
409	-72.0	-195.9	10.7	31.99	0.0	215.90	16.12	-42.50
410	-27.0	500.6	-8.0	-23.99	0	81.11	45.80	21.63
411	108.4	2947.7	4.9	14.81	0	-325.16	239.64	157.35
412	-159.4	1512.9	4.3	12.88	0	478.28	161.59	42.16
413	-48.9	1886.4	14.0	41.88	0	146.76	148.76	105.30
414	-4.0	2582.9	-4.7	-14.10	0	11.97	176.66	171.20
415	-49.3	141.6	1.3	3.94	0	147.94	28.00	-8.93
416	-49.3	141.6	1.3	3.94	0	147.94	28.00	-8.93
417	-49.3	141.6	1.3	3.94	0.0	147.94	28.00	-8.93
418	-49.3	141.6	1.3	3.94	0	147.94	28.00	-8.93
419	-26.3	2223.9	4.6	13.83	0	78.80	160.65	138.87
420	-26.3	2223.9	4.6	13.83	0	78.80	160.65	138.87
421	-26.3	2223.9	4.6	13.83	0	78.80	160.65	138.87
422	-26.3	2223.9	4.6	13.83	0	78.80	160.65	138.87
423	47.4	498.0	1.8	5.55	0	-142.19	51.44	15.63
424	-120.0	-398.7	1.4	4.34	0	359.96	17.61	-71.31
425	-50.9	-165.3	7.5	22.47	0	152.76	9.59	-31.85
426	-22.8	270.0	-4.2	-12.52	0	68.51	27.70	8.67
427	64.7	2059.7	4.3	12.96	0	-194.05	163.62	113.78
428	-102.7	1163.0	3.9	11.76	0	308.10	117.09	39.54
429	-33.6	1396.4	10.0	29.88	0	100.90	109.07	79.00
430	-5.6	1831.7	-1.7	-5.11	0	16.66	125.85	120.85
431	-36.8	45.6	1.6	4.94	0	110.28	17.02	-10.87
432	-36.8	45.6	1.6	4.94	0	110.28	17.02	-10.87
433	-36.8	45.6	1.6	4.94	0	110.28	17.02	-10.87
434	-36.8	45.6	1.6	4.94	0	110.28	17.02	-10.87
435	23.0	1550.6	-3.4	-10.19	0	-69.01	113.78	95.05
436	23.0	1550.6	-3.4	-10.19	0	-69.01	113.78	95.05
437	23.0	1550.6	-3.4	-10.19	0	-69.01	113.78	95.05
438	23.0	1550.6	-3.4	-10.19	0	-69.01	113.78	95.05

Max Net factored B.P = 239.64 KN/m^2 Pmin = -105.64 kN/m^2

DETERMINATION OF REINFORCEMENT

Bottom Reinforcement

Length of cantilever along X = 1.25 m

Moment about Z Muz = 187.22 KNm

$M_{uzlim} = \emptyset\, 0.85\, f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

= 4817.57 KNm

As $M_{uzlim} > M_{uz}$ Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\, f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

= 591.8 mm^2

Effective Depth Along X = 840 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement greater of $A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*

or $A_{stmin} = 0.0014bd$

A_{stmin} = 1512.0 mm^2

A_{req} = 591.8 mm^2

Required Spacing = 207.8 mm

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	C02	D01
Date	27-01-2020	12-03-2020	25-04-2022	12/10/2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS DT. 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER SUPPORT STRUCTURE	AS BUILT
Prepared By	APR	APR	APR	APR
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Outotec / L&T Revision No.:	0	1	2	3
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Length of cantilever along Z = 1.85 m
Moment about X: Mux = 410.08 KNm

$M_{uxlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$ = 4590.90 KNm

As $M_{uxlim} > M_{ux}$ Tension controlled section
Therefore Φ = 0.9 *Table 21.2.2 SBY 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$ = 1336.7 mm^2

Provide Bottom Reinforcement = 20 mm dia @ 150 c/c

Effective Depth Along X = 820 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement greater of
$A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*
or
A_{stmin} = 0.0014bd
A_{stmin} = 1476.0 mm^2
A_{req} = 1336.7 mm^2
Required Spacing = 212.8 mm

Ast prov = 2094.4 mm^2

Top Reinforcement
Length of cantilever along X = 1.25 m
Moment about Z: Muz = 33.37 KNm

$M_{uzlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$ = 4863.56 KNm

As $M_{uzlim} > M_{uz}$ Tension controlled section
Therefore φ = 0.9 *Table 21.2.2 SBY 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$ = 104.6 mm^2

Effective Depth Along X = 844 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement greater of
A_{stmin} = 50% of $\frac{0.0018 \times 420}{f_y} bd$ since 50% or more prov in bottom
or
A_{stmin} = 50% of 0.0014bd *Table 24.4.3.2 SBC 304*
A_{stmin} = 759.6 mm^2
A_{req} = 104.6 mm^2
Required Spacing = 148.9 mm

Length of cantilever along Z = 1.85 m
Moment about X: Mux = 73.10 KNm

$M_{uxlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$ = 4726.25 KNm

As $M_{uxlim} > M_{ux}$ Tension controlled section
Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$ = 232.6 mm^2

Effective Depth Along Z = 832 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

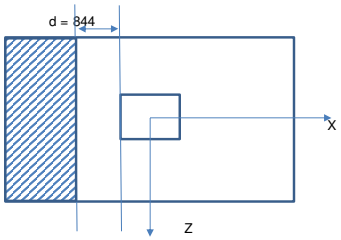
Min reinforcement greater of
A_{stmin} = 50% of $\frac{0.0018 \times 420}{f_y} bd$ since 50% or more prov in bottom
or
A_{stmin} = 50% of 0.0014bd *Table 24.4.3.2 SBC 304*
A_{stmin} = 748.8 mm^2
A_{req} = 232.6 mm^2
Required Spacing = 151.0 mm

Provide top Reinforcement = 12 mm dia @ 150 c/c

Ast prov = 754.0 mm^2

ONE WAY SHEAR CKECK

Along X Direction



Cantilever L = 0.406 m
Reinforcement ratio ρ_w = 0.0025
Effective Depth Along X d = 844 mm
Shear Force at d distance V_u = 97.29 KN
Allowable shear strength φVc = 599.18 KN O.K

Least of

i) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right) b_w d$ *Table 22.5.5.1 SBC 304*
λ = 1 Normal concrete
Mu = 19.75 KNm
φ = 0.75 for shear
φVc = 710.204135 KN

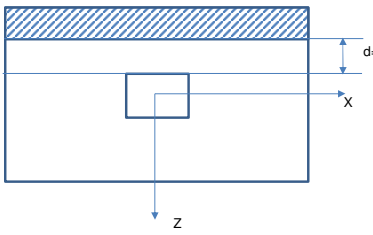
ii) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\right) b_w d$
φVc = 625.884098 KN *Table 22.5.5.1 SBC 304*

iii) $V_c = 0.29\lambda\sqrt{f_c'} b_w d$ = 599.18056 KN

Global Project ID	**900707**
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Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	C02	D01			Outotec / L&T Revision No.:	0	1	2	3		
Date	27-01-2020	12-03-2020	25-04-2022	12/10/2023			Date	27-01-2020	12-03-2020	25-04-2022	12/10/2023		
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Approved	AG	AG	AG	AG			Approved	AG	AG	AG	AG		

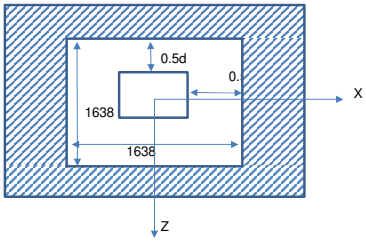
Along Z Direction



Cantilever	L	=	1.018 m		Least of	i)	V_c	=	$\left(0.16\lambda\sqrt{f_c'}+17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$	
Reinforcement ratio	ρ_w	=	0.00							*Table 22.5.5.1 SBC 304*
Effective Depth Along Z	d	=	832 mm				λ	=	1 Normal concrete	
Shear Force at d distance	V_u	=	243.95 kN				Mu	=	124.17 kNm	
Allowable shear strength	φVc	=	590.66 kN	O.K			φ	=	0.75 for shear	
							φVc	=	634.31 kN	
						ii)	V_c	=	$\left(0.16\lambda\sqrt{f_c'}+17\rho_w\right)b_w d$	
							φVc	=	617.364943 kN	
						iii)	V_c	=	$0.29\lambda\sqrt{f_c'}b_w d$	
							φVc	=	590.66 kN	

TWO WAY SHEAR CKECK

Punching at 0.5d from face of column



Total area		=	14.85 m2	
Punching area		=	2.68 m2	
	λ	=	1	
	β	=	1	
	αs	=	40	
	b0	=	6552.0 mm	

Allowable shear stress — least of — *Table 22.6.5.2 SBC 304*

v_c	= $0.33\lambda\sqrt{f_c'}$	=	1.95 N/mm²
v_c	= $0.17\left(1+\frac{2}{\beta}\right)\lambda\sqrt{f_c'}$	=	3.02 N/mm²
v_c	= $0.083\left(2+\frac{\alpha_s d}{b_0}\right)\lambda\sqrt{f_c'}$	=	3.49 N/mm²

Design vc = 1.95 N/mm²

Check at 0.5d from column face

Perimeter at 0.5d from column face	u	=	6.55 m		Vc	=	$v_c b_0 d$	
Depth at section	d	=	838 mm			=	10719.29 kN	
Shear Force at section	Vu	=	2915.6 kN		φ	=	0.75 for shear	
Allowable shear strength	φVc	=	8039.46 kN	O.K	φV_n	=	φVc	
					φVc	=	8039.5 kN	

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Project Country	KINGDOM OF SAUDI ARABIA

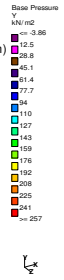
Ma'aden Revision No:	A01	C01	C02	D01			Outotec / L&T Revision No.:	0	1	2	3		
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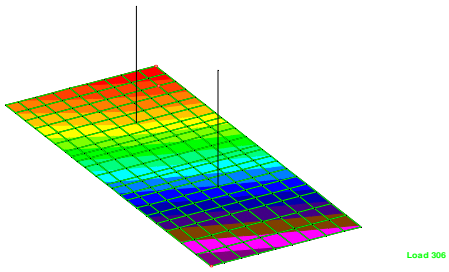
8.3.3.1 DESIGN OF COMBINED FOUNDATION FOR COLUMN AT 3120/E_3120/9 & 3120/C_3120/9 AND FOUNDATION MKD. AS CF1

For Calculation of Base pressure and design forces for column & foundation Raft, Refer STAAD File **CF1_1.STD**

Staad Supoort Node for column at **3120/E_3120/9** **5**
3120/C_3120/9 **7**

Finished Floor level	F.F.L	=	0.0	m	(Corresponds to RL +313.0m)
Finished Ground level	F.G.L	=	-0.5	m	
Bottom of foundation	B.O.F	=	-2.5	m	
Top of pedestal (staad support)	T.O.F	=	0.5	m	
Ground Water Level	G.W.L	=	-10	m	
Length of footing	Lz	=	8.3	m	
Width of footing	Lx	=	3.5	m	
Thickness of footing	D	=	0.8	m	
Depth of pedestal	Bx	=	0.8	m	
Width of pedestal	Bz	=	0.8	m	
Unit wt of soil		=	18.0	KN/m^3	
Allowable Net Bearing Capacity		=	500	KN/m^2	
Gross Bearing Capacity		=	536	KN/m^2	





1) Check for Gross Base Pressure :

Maximum Base Pressure,	=	257.1	KN/m^2	Load Case 306
Minimum Base Pressure,	=	-18.6	KN/m^2	Load Case 330

2) Design of foundation Raft :

Slab Thickness =	800 mm	Yield Strength of Main Reinforcement , fy =	420 N/mm^2
Width =	1000 mm	Yield Strength of Shear Reinforcement , fyt =	420 N/mm^2
Clear Cover =	50 mm	Strength Reduction Factor of Main Reinforcement , Φ =	0.9
Compressive Strength of Concrete , fc' =	35 N/mm^2	Strength Reduction Factor of Shear Reinforcement , Φt =	0.75

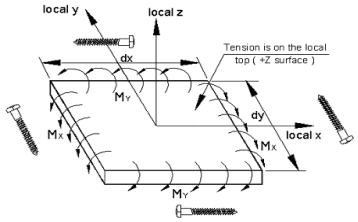
STAAD OUTPUT FROM FACTORED LOAD COMBINATION :

(Staad Group : RAFT_MOMENT)

MOMENT	VALUE	UNIT	TORSIONAL MOMENT	VALUE	UNIT	
Max. Mx	217.497	kN-m	Corresponding Mxy	4.497	kN-m	*(Plate No. - 40 , L/C No. - 430)*
Min. Mx	-452.100	kN-m	Corresponding Mxy	-61.658	kN-m	*(Plate No. - 26 , L/C No. - 405)*
Max. My	64.468	kN-m	Corresponding Mxy	-14.794	kN-m	*(Plate No. - 37 , L/C No. - 426)*
Min. My	-352.070	kN-m	Corresponding Mxy	7.122	kN-m	*(Plate No. - 19 , L/C No. - 406)*

FOR TOP FACE REINFORCEMRNT :

Mx1 =	Mx + abs(Mxy) =	217.497 + 4.497 =	221.994 kN-m
My1 =	My + abs(Mxy) =	64.468 + 14.794 =	79.262 kN-m
Mx2 =	Mx+abs(Mxy^2/My) =	217.497 + 0.314 =	217.811 kN-m
My2 =	My+abs(Mxy^2/Mx) =	64.468 + 1.006 =	65.474 kN-m



DESIGN MOMENT CALCULATION FOR TOP FACE REINFORCEMENT :

Mxd = 221.994 kN-m
Myd = 79.262 kN-m

ALONG LOCAL X - DIRECTION :

Mu = 221.994 kN-m
Φ*Kn = Mu/bd^2 deff. = 742 mm
Mu/bd^2 = 221.994X10^6/(1000X742^2) = 0.403 N/mm^2 < 6.9 N/mm^2
Pt = 0.0011 Reqd. Ast = 0.0011X1000X742/100 = 8.16 cm^2
Provided **16** mm @ **150** C/C (Provide Ast = 13.4 cm2) at Horizontal
Provided **0** mm @ **150** C/C (Provide Ast = 0 cm2) at Horizontal
Total Provide Ast = 13.4 cm2 at Horizontal **O.K.**
Total Provide Pt = 0.0018

ALONG LOCAL Y - DIRECTION :

Mu = 79.262 kN-m
Φ*Kn = Mu/bd^2 deff. = 726 mm
Mu/bd^2 = 79.262X10^6/(1000X726^2) = 0.144 N/mm^2 < 6.9 N/mm^2
Pt = 0.0004 Reqd. Ast = 0.0004X1000X742/100 = 2.9 cm^2
Provided **16** mm @ **150** C/C (Provide Ast = 13.4 cm2) at Vertical
Provided **0** mm @ **150** C/C (Provide Ast = 0 cm2) at Vertical
Total Provide Ast = 13.4 cm2 at Vertical **O.K.**
Total Provide Pt = 0.0018

Sensitivity: LNT Construction Internal Use

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	C02	D01			Outotec / L&T Revision No.:	0	1	2	3		
Date	27-01-2020	12-03-2020	25-04-2022	13/10/2023			Date	27-01-2020	12-03-2020	25-04-2022	13/10/2023		
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Checked	SBY	SBY	SBY	SBY			Checked	SBY	SBY	SBY	SBY		
Approved	AG	AG	AG	AG			Approved	AG	AG	AG	AG		

FOR BOTTOM FACE REINFORCEMRNT :

Mx1 = Mx - abs(Mxy) = -452.1 - 61.658 = -513.758 kN-m
My1 = My - abs(Mxy) = -352.07 - 7.122 = -359.192 kN-m
Mx2 = Mx-abs(Mxy^2/My) = -452.1 - 10.798 = -462.898 kN-m
My2 = My-abs(Mxy^2/Mx) = -352.07 - 0.112 = -351.958 kN-m

DESIGN MOMENT CALCULATION FOR BOTTOM FACE REINFORCEMENT :

Mxd = -513.758 kN-m
Myd = -359.192 kN-m

ALONG LOCAL X - DIRECTION :

Mu = 513.758 kN-m
Φ*Kn = Mu/bd^2 deff. = 740 mm
Mu/bd^2 = 513.758X10^6/(1000X740^2) = 0.938 N/mm^2 < 6.9 N/mm^2
Pt = 0.0025 Reqd. Ast = 0.0025X1000X740/100 = 18.5 cm^2
Provided 20 mm @ 150 C/C (Provide Ast = 20.94 cm2) at Horizontal
Provided 0 mm @ 150 C/C (Provide Ast = 0 cm2) at Horizontal
Total Provide Ast = 20.94 cm2) at Horizontal **O.K.**
Total Provide Pt = 0.0028

ALONG LOCAL Y - DIRECTION :

Mu = 359.192 kN-m
Φ*Kn = Mu/bd^2 deff. = 720 mm
Mu/bd^2 = 359.192X10^6/(1000X720^2) = 0.693 N/mm^2 < 6.9 N/mm^2
Pt = 0.0019 Reqd. Ast = 0.0019X1000X720/100 = 13.68 cm^2
Provided 20 mm @ 150 C/C (Provide Ast = 20.94 cm2) at Vertical
Provided 0 mm @ 150 C/C (Provide Ast = 0 cm2) at Vertical
Total Provide Ast = 20.94 cm2) at Vertical **O.K.**
Total Provide Pt = 0.0029

SHEAR CHECK

(Staad Group : RAFT_SHEAR)
(FROM FACTORED LOAD COMBINATION)

ALONG LOCAL X - DIRECTION :

SQx = 0.403 N/mm^2 *(At d-distance from face of the column)* *(Plate No. - 82 , L/C No. - 405)*
Vu = 298.22 kN

Provided Pt = 0.0028
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0028X298.22X10^3X740/513.758/10^6)X1000X740/10^3
= 715.594 kN
[b] = (0.16*λ*fc^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0028)X1000X740/10^3
= 735.688 kN
[c] = 0.29*λ*fc^0.5*b*d.eff
= 0.29X1X5.92X1000X740/10^3
= 1269.591 kN
HENCE, Vc = 715.594 kN
Φt*Vc = 0.75X715.594 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 536.696 kN **> 298.22T , Hence O.K.**

ALONG LOCAL Y - DIRECTION :

SQy = -0.229 N/mm^2 *(At d-distance from face of the column)* *(Plate No. - 128 , L/C No. - 412)*
Vu = 164.88 kN

Pt provided = 0.0029
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0029X164.88X10^3X720/359.192/10^3)X1000X720/10^3
= 693.264 kN
[b] = (0.16*λ*fc^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0029)X1000X720/10^3
= 717.028 KN
[c] = 0.29*λ*fc^0.5*b*d.eff
= 0.29X1X5.92X1000X720/10^3
= 1235.277 kN
HENCE, Vc = 693.264 kN
Φt*Vc = 0.75X693.264 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 519.948 kN **> 164.88T , Hence O.K.**

Sensitivity: LNT Construction Internal Use

Global Project ID	900707
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Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	C02	D01
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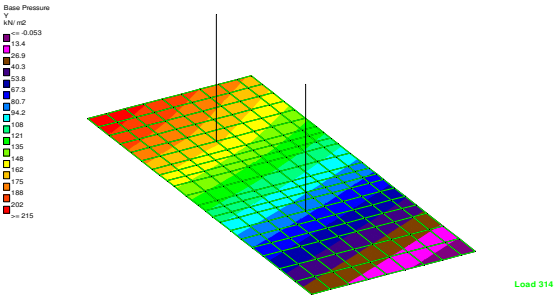
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8.3.3.2 DESIGN OF COMBINED FOUNDATION FOR COLUMN AT 3120/E_3120/10 & 3120/C_3120/10 AND FOUNDATION MKD. AS CF1

For Calculation of Base pressure and design forces for column & foundation Raft, Refer STAAD File **CF1_2.STD**

Staad Supoort Node for column at **3120/E_3120/10** **6**
3120/C_3120/10 **8**

Finished Floor level	F.F.L	=	0.0	m	(Corresponds to RL +313.0m)
Finished Ground level	F.G.L	=	-0.5	m	
Bottom of foundation	B.O.F	=	-2.5	m	
Top of pedestal (staad support)	T.O.F	=	0.5	m	
Ground Water Level	G.W.L	=	-10	m	
Length of footing	Lz	=	8.3	m	
Width of footing	Lx	=	3.5	m	
Thickness of footing	D	=	0.8	m	
Depth of pedestal	Bx	=	0.8	m	
Width of pedestal	Bz	=	0.8	m	
Unit wt of soil		=	18.0	KN/m^3	
Allowable Net Bearing Capacity		=	500	KN/m^2	
Gross Bearing Capacity		=	536	KN/m^2	



1) Check for Gross Base Pressure :

Maximum Base Pressure,	=	215.4	KN/m^2	Load Case	314
Minimum Base Pressure,	=	-30.2	KN/m^2	Load Case	324

2) Design of foundation Raft :

Slab Thickness =	800 mm	Yield Strength of Main Reinforcement , fy =	420 N/mm^2
Width =	1000 mm	Yield Strength of Shear Reinforcement , fyt =	420 N/mm^2
Clear Cover =	50 mm	Strength Reduction Factor of Main Reinforcement , Φ =	0.9
Compressive Strength of Concrete , fc' =	35 N/mm^2	Strength Reduction Factor of Shear Reinforcement , Φt =	0.75

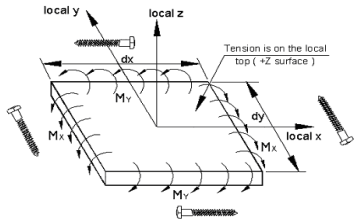
STAAD OUTPUT FROM FACTORED LOAD COMBINATION :

(Staad Group : RAFT_MOMENT)

MOMENT	VALUE	UNIT	TORSIONAL MOMENT	VALUE	UNIT	
Max. Mx	173.395	kN-m	Corresponding Mxy	8.919	kN-m	*(Plate No. - 37 , L/C No. - 426)*
Min. Mx	-371.757	kN-m	Corresponding Mxy	28.701	kN-m	*(Plate No. - 88 , L/C No. - 411)*
Max. My	150.089	kN-m	Corresponding Mxy	4.371	kN-m	*(Plate No. - 26 , L/C No. - 408)*
Min. My	-288.749	kN-m	Corresponding Mxy	-60.637	kN-m	*(Plate No. - 69 , L/C No. - 411)*

FOR TOP FACE REINFORCEMRNT :

Mx1 =	Mx + abs(Mxy) =	173.395 + 8.919 =	182.314 kN-m
My1 =	My + abs(Mxy) =	150.089 + 4.371 =	154.46 kN-m
Mx2 =	Mx+abs(Mxy^2/My) =	173.395 + 0.53 =	173.925 kN-m
My2 =	My+abs(Mxy^2/Mx) =	150.089 + 0.11 =	150.199 kN-m



DESIGN MOMENT CALCULATION FOR TOP FACE REINFORCEMENT :

Mxd = 182.314 kN-m
Myd = 154.46 kN-m

ALONG LOCAL X - DIRECTION :

Mu = 182.314 kN-m
Φ*Kn = Mu/bd^2 deff. = 742 mm
Mu/bd^2 = 182.314X10^6/(1000X742^2) = 0.331 N/mm^2 < 6.9 N/mm^2
Pt = 0.0009 Reqd. Ast = 0.0009X1000X742/100 = 6.68 cm^2
Provided **16** mm @ **150** C/C (Provide Ast = 13.4 cm2) at Horizontal
Provided **0** mm @ **150** C/C (Provide Ast = 0 cm2) at Horizontal
Total Provide Ast = 13.4 cm2 at Horizontal **O.K.**
Total Provide Pt = 0.0018

ALONG LOCAL Y - DIRECTION :

Mu = 154.46 kN-m
Φ*Kn = Mu/bd^2 deff. = 726 mm
Mu/bd^2 = 154.46X10^6/(1000X726^2) = 0.281 N/mm^2 < 6.9 N/mm^2
Pt = 0.0007 Reqd. Ast = 0.0007X1000X742/100 = 5.08 cm^2
Provided **16** mm @ **150** C/C (Provide Ast = 13.4 cm2) at Vertical
Provided **0** mm @ **150** C/C (Provide Ast = 0 cm2) at Vertical
Total Provide Ast = 13.4 cm2 at Vertical **O.K.**
Total Provide Pt = 0.0018

Sensitivity: LNT Construction Internal Use

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	C02	D01			Outotec / L&T Revision No.:	0	1	2	3		
Date	27-01-2020	12-03-2020	25-04-2022	13/10/2023			Date	27-01-2020	12-03-2020	25-04-2022	13/10/2023		
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS DT. 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER SUPPORT STRUCTURE	AS BUILT			Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS DT. 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER SUPPORT STRUCTURE	AS BUILT		
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Checked	SBY	SBY	SBY	SBY			Checked	SBY	SBY	SBY	SBY		
Approved	AG	AG	AG	AG			Approved	AG	AG	AG	AG		

FOR BOTTOM FACE REINFORCEMRNT :

Mx1 = Mx - abs(Mxy) = -371.757 - 28.701 = -400.458 kN-m
My1 = My - abs(Mxy) = -288.749 - 60.637 = -349.386 kN-m
Mx2 = Mx-abs(Mxy^2/My) = -371.757 - 2.853 = -374.61 kN-m
My2 = My-abs(Mxy^2/Mx) = -288.749 - 9.89 = -278.859 kN-m

DESIGN MOMENT CALCULATION FOR BOTTOM FACE REINFORCEMENT :

Mxd = -400.458 kN-m
Myd = -349.386 kN-m

ALONG LOCAL X - DIRECTION :

Mu = 400.458 kN-m
Φ*Kn = Mu/bd^2 deff. = 740 mm
Mu/bd^2 = 400.458X10^6/(1000X740^2) = 0.731 N/mm^2 < 6.9 N/mm^2
Pt = 0.0020 Reqd. Ast = 0.002X1000X740/100 = 14.8 cm^2
Provided **20** mm @ **150** C/C (Provide Ast = 20.94 cm2) at Horizontal
Provided **0** mm @ **150** C/C (Provide Ast = 0 cm2) at Horizontal
Total Provide Ast = 20.94 cm2) at Horizontal **O.K.**
Total Provide Pt = 0.0028

ALONG LOCAL Y - DIRECTION :

Mu = 349.386 kN-m
Φ*Kn = Mu/bd^2 deff. = 720 mm
Mu/bd^2 = 349.386X10^6/(1000X720^2) = 0.674 N/mm^2 < 6.9 N/mm^2
Pt = 0.0018 Reqd. Ast = 0.0018X1000X720/100 = 12.96 cm^2
Provided **20** mm @ **150** C/C (Provide Ast = 20.94 cm2) at Vertical
Provided **0** mm @ **150** C/C (Provide Ast = 0 cm2) at Vertical
Total Provide Ast = 20.94 cm2) at Vertical **O.K.**
Total Provide Pt = 0.0029

SHEAR CHECK
(Staad Group : RAFT_SHEAR)
(FROM FACTORED LOAD COMBINATION)

ALONG LOCAL X - DIRECTION :

SQx = 0.232 N/mm^2 *(At d-distance from face of the column)* *(Plate No. - 82 , L/C No. - 411)*
Vu = 171.68 kN

Provided Pt = 0.0028
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0028X171.68X10^3X740/400.458/10^6)X1000X740/10^3
= 711.638 kN
[b] = (0.16*λ*fc^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0028)X1000X740/10^3
= 735.688 kN
[c] = 0.29*λ*fc^0.5*b*d.eff
= 0.29X1X5.92X1000X740/10^3
= 1269.591 kN
HENCE, Vc = 711.638 kN
Φt*Vc = 0.75X711.638 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 533.729 kN **> 171.68T , Hence O.K.**

ALONG LOCAL Y - DIRECTION :

SQy = 0.191 N/mm^2 *(At d-distance from face of the column)* *(Plate No. - 128 , L/C No. - 411)*
Vu = 137.52 kN

Pt provided = 0.0029
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0029X137.52X10^3X720/349.386/10^3)X1000X720/10^3
= 691.592 kN
[b] = (0.16*λ*fc^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0029)X1000X720/10^3
= 717.028 KN
[c] = 0.29*λ*fc^0.5*b*d.eff
= 0.29X1X5.92X1000X720/10^3
= 1235.277 kN
HENCE, Vc = 691.592 kN
Φt*Vc = 0.75X691.592 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 518.694 kN **> 137.52T , Hence O.K.**

Sensitivity: LNT Construction Internal Use

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	C02	D01
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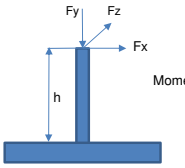
REFERENCE DOCUMENTS

DOCUMENT NUMBER	DOCUMENT DESCRIPTION
MD-216-1000-EG-ST-CRT-1001	Civil Structural Design Criteria

8.4 Design of pedestal

8.4.1 Design of pedestal P1 at grid 3120/J-3130/9

Underside of base plate	=	0.500 m
Bottom of foundation	=	-2.500 m
Thickness of footing	=	0.4 m
Height of column	=	2.600 m



Moment at pedestal Bottom,
Mx = Fz *h
Mz = Fx *h

Calculation of load in pedestal for all load combinations

L/C	Fx	Fy	Fz	Mx	My	Mz
401	0.2	283.0	-38.4	-99.9	0.0	0.5
402	0.1	283.5	-47.3	-123.0	0.0	0.4
403	-1.5	450.7	-134.4	-349.5	0.0	-3.9
404	1.9	439.5	10.5	27.3	0.0	5.0
405	-1.5	451.2	-143.3	-372.6	0.0	-4.0
406	1.9	440.0	1.6	4.3	0.0	4.9
407	3.0	253.8	-37.9	-98.5	0.0	7.8
408	-2.7	328.0	-42.0	-109.2	0.0	-6.9
409	0.1	137.4	10.0	25.9	0.0	0.3
410	0.2	446.6	-91.2	-237.0	0.0	0.6
411	3.0	254.3	-45.5	-118.3	0.0	7.7
412	-2.7	328.4	-49.6	-129.0	0.0	-7.0
413	0.1	137.8	2.4	6.2	0.0	0.2
414	0.2	447.0	-98.8	-256.8	0.0	0.5
415	0.2	290.2	-39.9	-103.6	0.0	0.4
416	0.2	290.2	-39.9	-103.6	0.0	0.4
417	0.2	290.2	-39.9	-103.6	0.0	0.4
418	0.2	290.2	-39.9	-103.6	0.0	0.4
419	0.1	290.7	-47.5	-123.4	0.0	0.3
420	0.1	290.7	-47.5	-123.4	0.0	0.3
421	0.1	290.7	-47.5	-123.4	0.0	0.3
422	0.1	290.7	-47.5	-123.4	0.0	0.3
423	1.9	159.2	-23.5	-61.0	0.0	4.9
424	-1.7	205.5	-26.0	-67.7	0.0	-4.3
425	0.1	86.4	6.4	16.8	0.0	0.2
426	0.2	279.6	-56.8	-147.6	0.0	0.4
427	1.9	159.5	-29.2	-75.9	0.0	4.8
428	-1.7	205.8	-31.8	-82.6	0.0	-4.4
429	0.1	86.7	0.7	1.9	0.0	0.2
430	0.1	280.0	-62.5	-162.4	0.0	0.3
431	0.1	181.9	-24.7	-64.2	0.0	0.3
432	0.1	181.9	-24.7	-64.2	0.0	0.3
433	0.1	181.9	-24.7	-64.2	0.0	0.3
434	0.1	181.9	-24.7	-64.2	0.0	0.3
435	0.1	182.2	-30.4	-79.1	0.0	0.2
436	0.1	182.2	-30.4	-79.1	0.0	0.2
437	0.1	182.2	-30.4	-79.1	0.0	0.2
438	0.1	182.2	-30.4	-79.1	0.0	0.2

Summary of load in pedestal

	L/C	Fy(KN)	Mx(KNm)	Mz(KNm)
FY max	405	451.2	-372.6	-3.99
Fy min	425	86.4	16.8	0.23
Mx max	404	439.5	27.3	5.01
Mx min	405	451.2	-372.6	5.01
Mz max	407	253.8	-98.5	7.79
Mz min	412	328.4	-129.0	-7.01

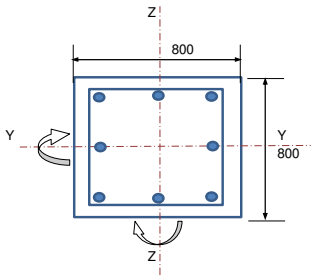
Sensitivity: LNT Construction Internal Use

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	C02	D01
Date	27-01-2020	12-03-2020	25-04-2022	13/10/2023
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Dimension of column along Z	Dz	=	800 mm
Dimenson of column along Y	Dy	=	800 mm
Unsupported length of column along Z	Lz	=	2600 mm
Unsupported length of column along Y	Ly	=	2600 mm
Grade of steel	f_y	=	420 MPa
Grade of concrete	f_c'	=	35 MPa
Effective length factor along Z	Kz	=	2
Effective length factor along Y	Ky	=	2
Effective Depth in Z dir		=	655 mm
Effective Depth in Y dir		=	655 mm
Cover		=	50 mm
Effective cover		=	72.5 mm
Dia of longitudinal bar		=	25 mm
Dia of link		=	10 mm
Percentage of steel reinforcement		=	1 %



ANALYSIS RESULTS

Analysis result as obtained from staad analysis

Axial load	Fy	=	P_u	=	502.1 KN	Load Case No.
B.M about Z	Mz	=	M_{uz}	=	4.26 KNm	406 1.4 DL + 1.4 TL1 + 1.7 LL + 1.7 EQL2
B.M about Y	Mx	=	M_{uy}	=	351 KNm	

CHECK FOR SLENDERNESS

Allowable slenderness for sway column $\frac{kl}{r} \le 22$ where r = 0.3h *Clause 6.2.5 SBC 304*

Slenderness check along Z axis $\frac{k_z l_z}{r_z}$ = 2 x 2600 / 240 r_z = 240

= 21.7 *Non-slender along Z*

Slenderness check along Y axis $\frac{k_y l_y}{r_y}$ = 2 x 2600 / 240 r_z = 240

= 21.7 *Non-slender along Y*

DESIGN FOR CAPACITY OF COLUMN

Moment Capacity about Z axis

P_u = 502.1 KN A_g = 800 x 800 = 640000 mm^2

Assume Strength reduction factor φ = 0.65 *Table 21.2.2 SBC 304* f_c' = 35 N/mm^2

ρ_s = 0.01

$P_n = \frac{P_u}{\phi}$ = 772.46 KN γ_z = 0.81875

$K_n = \frac{P_n}{A_g f_c'}$ = 0.034

Corresponding to the values of K_n ρ_s γ_z f_y f_c' the value of $\frac{M_{nz}}{A_g f_c' D_z}$ is interpolated from Interaction diagr in SP-17M(09)

We get $\frac{M_{nz}}{A_g f_c' D_z}$ = 0.13 *As per graph* *3.3.3* *SP-17M(09)*

M_{nz} = 0.13x640000x35x800/10^6 = 2329.6 KNm

Max moment capacity M_{uz} = ϕM_{nz} = 1514.24 KNm > 4.26 KNm *safe*

Utility ratio = ***0.003***

Moment Capacity about Y axis

P_u = 502.1 KN A_g = 800 x 800 = 640000 mm^2

Assume Strength reduction factor φ = 0.65 *Table 21.2.2 SBC 304* f_c' = 35 N/mm^2

ρ_s = 0.01

$P_n = \frac{P_u}{\phi}$ = 772.46 KN γ_y = 0.81875

$K_n = \frac{P_n}{A_g f_c'}$ = 0.034

Corresponding to the values of K_n ρ_s γ_y f_y f_c' the value of $\frac{M_{ny}}{A_g f_c' D_y}$ is interpolated from Interaction diagr in SP-17M(09)

We get $\frac{M_{ny}}{A_g f_c' D_y}$ = 0.13 *As per graph* *3.3.2* *SP-17M(09)*

M_{ny} = 0.13x640000x35x800/10^6 = 2329.6 KNm

Max moment capacity M_{uy} = ϕM_{ny} = 1514.24 KNm > 351 KNm *safe*

Utility ratio = ***0.232***

Sensitivity: LNT Construction Internal Use

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	C02	D01
Date	27-01-2020	12-03-2020	25-04-2022	13/10/2023
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Combined moment capacity check

M_{noz} = 2329.6 KNm M_{nz} = M_{uz}/φ = 6.553846154 KNm

M_{noy} = 2329.6 KNm M_{ny} = M_{uy}/φ = 540 KNm

Assume β = 0.65 log β = -0.187087

log 0.5 = -0.30103

Check $\left(\frac{M_{nz}}{M_{noz}}\right)^{\log 0.5/\log \beta} + \left(\frac{M_{ny}}{M_{noy}}\right)^{\log 0.5/\log \beta}$ < 1 *Equation 3-15 SP-17M(09)*

= 0.00 + 0.10 = 0.0952 < 1 *Safe*

Longitudinal reinforcement		16	nos	20	mm
Spread equally on all four side		4	nos	25	mm
		Total Ast	=	6990.0437	mm^2 O.K
Minimim Reinforcement	=	0.01Ag	=	6400	mm^2

Required Ast = 6400 mm^2

Transverse Reinforcement

Minimum diameter of tie = 10 mm

Maximum c/c dist.	a)	16db	=	400 mm
	b)	48db	=	480 mm
	c)	Smaller column dimension	=	800 mm

Provide lateral ties: 10 mm @ 250 c/c

Check for Tension

Tension Forces P_u = 86.40 KN M_{uz} = 0.2314 KNm M_{uy} = 16.8 KNm

Assume Strength reduction factor φ = 0.9 Tension controlled *Table 21.2.2 SBC 304*

$P_n = \frac{P_u}{\phi}$ = 96.00 KN A_g = 800 x 800 = 640000 mm^2

f_c' = 35 N/mm^2

$K_n = \frac{P_n}{A_g f_c'}$ = 0.004 ρ_s = 0.01

γ_z = 0.81875

γ_y = 0.81875

From the tension curve corresponding to the values of γ_y f_y f_c' ρ_s K_n as per interaction diagram generatec using clause 3.2.2 of SP-17M(09)

We get,

$\frac{M_{nz}}{A_g f_c' D_z}$ = 0.05 $\frac{M_{ny}}{A_g f_c' D_y}$ = 0.05

M_{nz} = 896 KNm M_{ny} = 896 KNm

$\emptyset M_{nz}$ = 806.4 KNm >0.2314 $\emptyset M_{ny}$ = 806.4 KNm >16.7622

safe **safe**

Utility ratio = ***0.00*** *Utility ratio =* ***0.02***

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Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

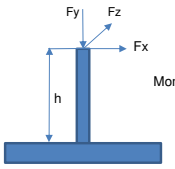
Ma'aden Revision No:	A01	C01	C02	D01
Date	27-01-2020	12-03-2020	25-04-2022	13/10/2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS DT. 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER SUPPORT STRUCTURE	AS BUILT
Prepared By	APR	APR	APR	APR
Checked	SBY	SBY	SBY	SBY
Approved	AG	AG	AG	AG

Outotec / L&T Revision No.:	0	1	2	3
Date	27-01-2020	12-03-2020	25-04-2022	13/10/2023
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REFERENCE DOCUMENTS

DOCUMENT NUMBER	DOCUMENT DESCRIPTION
MD-216-1000-EG-ST-CRT-1001	Civil Structural Design Criteria

8.4.2 Design of pedestal P1 at grid 3120/J-3130/10



Moment at pedestal Bottom,
Mx = Fz *h
Mz = Fx *h

Underside of base plate	=	0.500 m
Bottom of foundation	=	-2.500 m
Thickness of footing	=	0.9 m
Height of column	=	2.100 m

Calculation of load in pedestal for all load combinations

L/C	Fx	Fy	Fz	Mx	My	Mz
401	-59.6	960.9	-0.6	-1.3	0.0	-125.1
402	-32.5	2220.9	1.4	3.0	0.0	-68.3
403	-68.8	1368.3	-329.1	-691.1	0.0	-144.6
404	-52.1	1275.0	324.3	681.0	0.0	-109.4
405	-41.8	2628.4	-327.0	-686.8	0.0	-87.7
406	-25.0	2535.1	326.4	685.3	0.0	-52.6
407	83.1	254.1	-1.6	-3.4	0.0	174.6
408	-184.3	1589.8	-0.7	-1.5	0.0	-387.0
409	-74.0	448.2	-24.2	-50.8	0.0	-155.4
410	-29.1	1439.7	22.9	48.2	0.0	-61.0
411	106.3	1334.1	0.2	0.4	0.0	223.3
412	-161.1	2669.8	1.1	2.2	0.0	-338.3
413	-50.8	1528.2	-22.4	-47.1	0.0	-106.6
414	-5.9	2519.8	24.7	51.9	0.0	-12.3
415	-51.3	929.7	-1.1	-2.2	0.0	-107.8
416	-51.3	929.7	-1.1	-2.2	0.0	-107.8
417	-51.3	929.7	-1.1	-2.2	0.0	-107.8
418	-51.3	929.7	-1.1	-2.2	0.0	-107.8
419	-28.1	2009.8	0.7	1.5	0.0	-59.1
420	-28.1	2009.8	0.7	1.5	0.0	-59.1
421	-28.1	2009.8	0.7	1.5	0.0	-59.1
422	-28.1	2009.8	0.7	1.5	0.0	-59.1
423	45.7	195.4	-0.7	-1.6	0.0	96.1
424	-121.4	1030.3	-0.2	-0.4	0.0	-255.0
425	-52.5	316.8	-14.9	-31.2	0.0	-110.2
426	-24.4	936.5	14.6	30.7	0.0	-51.2
427	63.1	1005.5	0.6	1.2	0.0	132.6
428	-104.0	1840.3	1.1	2.4	0.0	-218.4
429	-35.1	1126.8	-13.5	-28.4	0.0	-73.6
430	-7.0	1746.5	15.9	33.5	0.0	-14.6
431	-38.3	617.7	-0.4	-0.9	0.0	-80.4
432	-38.3	617.7	-0.4	-0.9	0.0	-80.4
433	-38.3	617.7	-0.4	-0.9	0.0	-80.4
434	-38.3	617.7	-0.4	-0.9	0.0	-80.4
435	-20.9	1427.8	0.9	1.9	0.0	-43.9
436	-20.9	1427.8	0.9	1.9	0.0	-43.9
437	-20.9	1427.8	0.9	1.9	0.0	-43.9
438	-20.9	1427.8	0.9	1.9	0.0	-43.9

Summary of load in pedestal

	L/C	Fy(KN)	Mx(KNm)	Mz(KNm)
FY max	412	2669.8	2.2	-338.31
Fy min	423	195.4	-1.6	96.05
Mx max	406	2535.1	685.3	-52.59
Mx min	403	1368.3	-691.1	-52.59
Mz max	411	1334.1	0.4	223.31
Mz min	408	1589.8	-1.5	-387.04

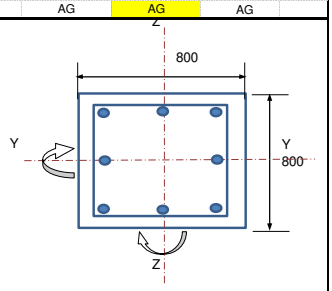
Sensitivity: LNT Construction Internal Use

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	C02	D01
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Outotec / L&T Revision No.:	0	1	2	3
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Dimension of column along Z	Dz	=	800 mm
Dimenson of column along Y	Dy	=	800 mm
Unsupported length of column along Z	Lz	=	2100 mm
Unsupported length of column along Y	Ly	=	2100 mm
Grade of steel	f_y	=	420 MPa
Grade of concrete	f_c'	=	35 MPa
Effective length factor along Z	Kz	=	2
Effective length factor along Y	Ky	=	2
Effective Depth in Z dir		=	655 mm
Effective Depth in Y dir		=	655 mm
Cover		=	50 mm
Effective cover		=	72.5 mm
Dia of longitudinal bar		=	25 mm
Dia of link		=	10 mm
Percentage of steel reinforcement		=	1 %



ANALYSIS RESULTS

Analysis result as obtained from staad analysis

Axial load	Fy	=	P_u	=	2535.1 KN	Load Case No.
B.M about Z	Mz	=	M_{uz}	=	52.6 KNm	409 1.2 (DL + SDL1) + 0.5 LL + 1.6 WL(+Z)
B.M about Y	Mx	=	M_{uy}	=	685.3 KNm	

CHECK FOR SLENDERNESS

Allowable slenderness for sway column: $\frac{kl}{r} \leq 22$ where r = 0.3h *Clause 6.2.5 SBC 304*

Slenderness check along Z axis: $\frac{k_z l_z}{r_z}$ = 2 x 2100 / 240 = 17.5 ; r_z = 240 — *Non-slender along Z*

Slenderness check along Y axis: $\frac{k_y l_y}{r_y}$ = 2 x 2100 / 240 = 17.5 ; r_z = 240 — *Non-slender along Y*

DESIGN FOR CAPACITY OF COLUMN

Moment Capacity about Z axis

P_u = 2535.059 KN ; A_g = 800 x 800 = 640000 mm^2

Assume Strength reduction factor φ = 0.65 *Table 21.2.2 SBC 304* ; f_c' = 35 N/mm^2

ρ_s = 0.01

$P_n = \frac{P_u}{\phi}$ = 3900.09 KN ; γ_z = 0.81875

$K_n = \frac{P_n}{A_g f_c'}$ = 0.174

Corresponding to the values of K_n ρ_s γ_z f_y f_c' the value of $\frac{M_{nz}}{A_g f_c' D_z}$ is interpolated from Interaction diagr. in SP-17M(09)

We get $\frac{M_{nz}}{A_g f_c' D_z}$ = 0.115 *As per graph* *3.3.3* *SP-17M(09)*

M_{nz} = 0.115x640000x35x800/10^6 = 2060.8 KNm

Max moment capacity M_{uz} = ϕM_{nz} = 1339.52 KNm > 52.5861 KNm *safe*

Utility ratio = ***0.039***

Moment Capacity about Y axis

P_u = 2535.059 KN ; A_g = 800 x 800 = 640000 mm^2

Assume Strength reduction factor φ = 0.65 *Table 21.2.2 SBC 304* ; f_c' = 35 N/mm^2

ρ_s' = 0.01

$P_n = \frac{P_u}{\phi}$ = 3900.09 KN ; γ_y = 0.81875

$K_n = \frac{P_n}{A_g f_c'}$ = 0.174

Corresponding to the values of K_n ρ_s γ_y $\frac{f_y}{f_c'}$ f_c' the value of $\frac{M_{ny}}{A_g f_c' D_y}$ is interpolated from Interaction diagr. in SP-17M(09)

We get $\frac{M_{ny}}{A_g f_c' D_y}$ = 0.115 *As per graph* *3.3.2* *SP-17M(09)*

M_{ny} = 0.115x640000x35x800/10^6 = 2060.8 KNm

Max moment capacity M_{uy} = ϕM_{ny} = 1339.52 KNm > 685.3371 KNm *safe*

Utility ratio = ***0.512***

Sensitivity: LNT Construction Internal Use

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	C02	D01		
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Outotec / L&T Revision No.:	0	1	2	3
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Combined moment capacity check

M_{noz} = 2060.8 KNm $M_{nz} = M_{uz}/\varphi$ = 80.90169231 KNm

M_{noy} = 2060.8 KNm $M_{ny} = M_{uy}/\varphi$ = 1054.364769 KNm

Assume β = 0.65 log β = -0.187087 log 0.5 = -0.30103

Check $\left(\frac{M_{nz}}{M_{noz}}\right)^{\log 0.5/\log \beta} + \left(\frac{M_{ny}}{M_{noy}}\right)^{\log 0.5/\log \beta} < 1$ *Equation 3-15 SP-17M(09)*

= 0.01 + 0.34 = 0.3456 < 1 *Safe*

Longitudinal reinforcement: 16 nos 20 mm — Required Ast = 6400 mm²

Spread equally on all four side: 4 nos 25 mm

Total Ast = 6990.0437 mm² O.K

Minimim Reinforcement = 0.01Ag = 6400 mm²

Transverse Reinforcement

Minimum diameter of tie = 10 mm

Maximum c/c dist.

a) 16db = 400 mm

b) 48db = 480 mm

c) Smaller column dimension = 800 mm

Provide lateral ties: 10 mm @ 250 c/c

Check for Tension

Tension Forces P_u = 195.44 KN M_{uz} = 96.0519 KNm M_{uy} = 1.6 KNm

Assume Strength reduction factor φ = 0.9 Tension controlled *Table 21.2.2 SBC 304*

$P_n = \frac{P_u}{\emptyset}$ = 217.16 KN A_g = 800 x 800 = 640000 mm²

= 35 N/mm²

$K_n = \frac{P_n}{A_g f_c'}$ = 0.010 ρ_g' = 0.01

γ_z = 0.81875

γ_y = 0.81875

From the tension curve corresponding to the values of γ_y f_y f_c' ρ_s K_n as per interaction diagram generated using clause 3.2.2 of SP-17M(09)

We get,

$\frac{M_{nz}}{A_g f_c' D_z}$ = 0.05 $\frac{M_{ny}}{A_g f_c' D_y}$ = 0.05

M_{nz} = 896 KNm M_{ny} = 896 KNm

= 806.4 KNm >96.0519 = 806.4 KNm

$\emptyset M_{nz}$ *safe* $\emptyset M_{ny}$ >1.5666 **safe**

Utility ratio = ***0.12*** *Utility ratio =* ***0.002***

Sensitivity: LNT Construction Internal Use

Global Project ID	**900707**
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Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
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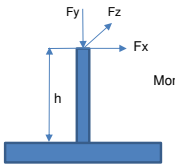
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REFERENCE DOCUMENTS

DOCUMENT NUMBER	DOCUMENT DESCRIPTION
MD-216-1000-EG-ST-CRT-1001	Civil Structural Design Criteria

8.4.3 Design of pedestal P1 at grid 3120/J-3130/12



Moment at pedestal Bottom,
Mx = Fz *h
Mz = Fx *h

Underside of base plate	=	0.500 m
Bottom of foundation	=	-2.500 m
Thickness of footing	=	0.9 m
Height of column	=	2.100 m

Calculation of load in pedestal for all load combinations

L/C	Fx	Fy	Fz	Mx	My	Mz
401	-57.2	71.0	2.6	5.4	0.0	-120.1
402	-30.3	2500.3	6.4	13.5	0.0	-63.6
403	-60.2	304.0	-1.8	-3.8	0.0	-126.4
404	-56.2	387.2	0.9	1.9	0.0	-118.1
405	-33.3	2733.3	2.1	4.3	0.0	-69.9
406	-29.3	2816.5	4.8	10.0	0.0	-61.6
407	85.3	865.4	1.6	3.4	0.0	179.2
408	-182.5	-569.4	1.0	2.1	0.0	-383.2
409	-72.0	-195.9	10.7	22.4	0.0	-151.1
410	-27.0	500.6	-8.0	-16.8	0.0	-56.8
411	108.4	2947.7	4.9	10.4	0.0	227.6
412	-159.4	1512.9	4.3	9.0	0.0	-334.8
413	-48.9	1886.4	14.0	29.3	0.0	-102.7
414	-4.0	2582.9	-4.7	-9.9	0.0	-8.4
415	-49.3	141.6	1.3	2.8	0.0	-103.6
416	-49.3	141.6	1.3	2.8	0.0	-103.6
417	-49.3	141.6	1.3	2.8	0.0	-103.6
418	-49.3	141.6	1.3	2.8	0.0	-103.6
419	-26.3	2223.9	4.6	9.7	0.0	-55.2
420	-26.3	2223.9	4.6	9.7	0.0	-55.2
421	-26.3	2223.9	4.6	9.7	0.0	-55.2
422	-26.3	2223.9	4.6	9.7	0.0	-55.2
423	47.4	498.0	1.8	3.9	0.0	99.5
424	-120.0	-398.7	1.4	3.0	0.0	-252.0
425	-50.9	-165.3	7.5	15.7	0.0	-106.9
426	-22.8	270.0	-4.2	-8.8	0.0	-48.0
427	64.7	2059.7	4.3	9.1	0.0	135.8
428	-102.7	1163.0	3.9	8.2	0.0	-215.7
429	-33.6	1396.4	10.0	20.9	0.0	-70.6
430	-5.6	1831.7	-1.7	-3.6	0.0	-11.7
431	-36.8	45.6	1.6	3.5	0.0	-77.2
432	-36.8	45.6	1.6	3.5	0.0	-77.2
433	-36.8	45.6	1.6	3.5	0.0	-77.2
434	-36.8	45.6	1.6	3.5	0.0	-77.2
435	-19.5	1607.4	4.1	8.6	0.0	-40.9
436	-19.5	1607.4	4.1	8.6	0.0	-40.9
437	-19.5	1607.4	4.1	8.6	0.0	-40.9
438	-19.5	1607.4	4.1	8.6	0.0	-40.9

Summary of load in pedestal

	L/C	Fy(KN)	Mx(KNm)	Mz(KNm)
FY max	411	2947.7	10.4	227.6
Fy min	408	-569.4	2.1	-383.2
Mx max	413	1886.4	29.3	-102.7
Mx min	410	500.6	-16.8	-102.7
Mz max	411	2947.7	10.4	227.6
Mz min	408	-569.4	2.1	-383.2

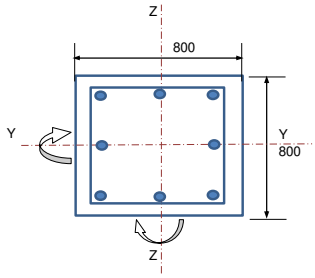
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Dimension of column along Z	Dz	=	800	mm
Dimenson of column along Y	Dy	=	800	mm
Unsupported length of column along Z	Lz	=	2100	mm
Unsupported length of column along Y	Ly	=	2100	mm
Grade of steel	f_y	=	420	MPa
Grade of concrete	f_c'	=	35	MPa
Effective length factor along Z	Kz	=	2	
Effective length factor along Y	Ky	=	2	
Effective Depth in Z dir		=	655	mm
Effective Depth in Y dir		=	655	mm
Cover		=	50	mm
Effective cover		=	72.5	mm
Dia of longitudinal bar		=	25	mm
Dia of link		=	10	mm
Percentage of steel reinforcement		=	1	%



ANALYSIS RESULTS

Analysis result as obtained from staad analysis

Axial load	Fy	=	P_u	=	-569.4	KN
B.M about Z	Mz	=	M_{uz}	=	383.2	KNm
B.M about Y	Mx	=	M_{uy}	=	2.1	KNm

Load Case No.
421 1.2 (DL + SDL2) + 0.5 LL + 1.0 EL(+Z)

CHECK FOR SLENDERNESS

Allowable slenderness for sway column $\frac{kl}{r} \le 22$ where r = 0.3h *Clause 6.2.5 SBC 304*

Slenderness check along Z axis $\frac{k_z l_z}{r_z}$ = 2 x 2100 / 240 r_z = 240

= 17.5 *Non-slender along Z*

Slenderness check along Y axis $\frac{k_y l_y}{r_y}$ = 2 x 2100 / 240 r_z = 240

= 17.5 *Non-slender along Y*

DESIGN FOR CAPACITY OF COLUMN

Moment Capacity about Z axis

P_u	=	-569.399	KN			A_g	=	800 x 800 =	640000 mm^2
Assume Strength reduction factor φ	=	0.65	*Table 21.2.2*	*SBC 304*		f_c'	=	35	N/mm^2
						ρ_s	=	0.01	
$P_n = \frac{P_u}{\phi}$	=	-876.00	KN			γ_z	=	0.81875	
$K_n = \frac{P_n}{A_g f_c'}$	=	-0.039							

Corresponding to the values of K_n ρ_s γ_z f_y f_c' the value of $\frac{M_{nz}}{A_g f_c' D_z}$ is interpolated from Interaction diagr in SP-17M(09)

We get $\frac{M_{nz}}{A_g f_c' D_z}$ = 0.085 *As per graph* *3.3.3* *SP-17M(09)*

M_{nz} = 0.085x640000x35x800/10^6 = 1523.2 KNm

Max moment capacity M_{uz} = ϕM_{nz} = 990.08 KNm > 383.1954 KNm *safe*

Utility ratio = ***0.387***

Moment Capacity about Y axis

P_u	=	-569.399	KN			A_g	=	800 x 800 =	640000 mm^2
Assume Strength reduction factor φ	=	0.65	*Table 21.2.2*	*SBC 304*			=	35	N/mm^2
						ρ_s'	=	0.01	
$P_n = \frac{P_u}{\phi}$	=	-876.00	KN			γ_y	=	0.81875	
$K_n = \frac{P_n}{A_g f_c'}$	=	-0.039							

Corresponding to the values of K_n ρ_s γ_y f_y f_c' the value of $\frac{M_{ny}}{A_g f_c' D_y}$ is interpolated from Interaction diagr in SP-17M(09)

We get $\frac{M_{ny}}{A_g f_c' D_y}$ = 0.085 *As per graph* *3.3.2* *SP-17M(09)*

M_{ny} = 0.085x640000x35x800/10^6 = 1523.2 KNm

Max moment capacity M_{uy} = ϕM_{ny} = 990.08 KNm > 2.0958 KNm *safe*

Utility ratio = ***0.002***

Sensitivity: LNT Construction Internal Use

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	C02	D01		
Date	27-01-2020	12-03-2020	25-04-2022	13/10/2023		
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS DT. 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER SUPPORT STRUCTURE	AS BUILT		
Prepared By	APR	APR	APR	APR		
Checked	SBY	SBY	SBY	SBY		
Approved	AG	AG	AG	AG		

Outotec / L&T Revision No.:	0	1	2	3
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Approved	AG	AG	AG	AG

Combined moment capacity check

M_{noz} = 1523.2 KNm $M_{nz} = M_{uz}/\varphi$ = 589.5313846 KNm

M_{noy} = 1523.2 KNm $M_{ny} = M_{uy}/\varphi$ = 3.224307692 KNm

Assume β = 0.65 log β = -0.187087

log 0.5 = -0.30103

Check $\left(\frac{M_{nz}}{M_{noz}}\right)^{\log 0.5/\log \beta} + \left(\frac{M_{ny}}{M_{noy}}\right)^{\log 0.5/\log \beta} < 1$ *Equation 3-15 SP-17M(09)*

= 0.22 + 0.00 = 0.2172 < 1 *Safe*

Longitudinal reinforcement: 16 nos 20 mm — Required Ast = 6400 mm^2

Spread equally on all four side: 4 nos 25 mm

Total Ast = 6990.0437 mm^2 O.K

Minimim Reinforcement = 0.01Ag = 6400 mm^2

Transverse Reinforcement

Minimum diameter of tie = 10 mm

Maximum c/c dist. a) 16db = 400 mm

b) 48db = 480 mm

c) Smaller column dimension = 800 mm

Provide lateral ties: 10 mm @ 250 c/c

Check for Tension

Tension Forces P_u = 569.40 KN M_{uz} = 383.1954 KNm M_{uy} = 2.1 KNm

Assume Strength reduction factor φ = 0.9 Tension controlled *Table 21.2.2 SBC 304*

$P_n = \frac{P_u}{\emptyset}$ = 632.67 KN A_g = 800 x 800 = 640000 mm^2

f_c' = 35 N/mm^2

$K_n = \frac{P_n}{A_g f_c'}$ = 0.028 ρ_g' = 0.01

γ_z = 0.81875

γ_y = 0.81875

From the tension curve corresponding to the values of γ_y f_y f_c' ρ_s K_n as per interaction diagram generated using clause 3.2.2 of SP-17M(09)

We get,

$\frac{M_{nz}}{A_g f_c' D_z}$ = 0.05 $\frac{M_{ny}}{A_g f_c' D_y}$ = 0.05

M_{nz} = 896 KNm M_{ny} = 896 KNm

$\emptyset M_{nz}$ = 806.4 KNm >383.1954 safe

$\emptyset M_{ny}$ = 806.4 KNm >2.0958 **safe**

Utility ratio = ***0.48*** *Utility ratio =* ***0.003***

Sensitivity: LNT Construction Internal Use

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

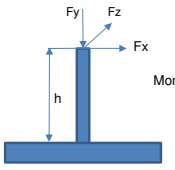
Ma'aden Revision No:	A01	C01	C02	D01
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Approved	AG	AG	AG	AG

REFERENCE DOCUMENTS

DOCUMENT NUMBER	DOCUMENT DESCRIPTION
MD-216-1000-EG-ST-CRT-1001	Civil Structural Design Criteria

8.4.4 Design of pedestal P1 at grid 3120/E-3130/9



Moment at pedestal Bottom,
Mx = Fz *h
Mz = Fx *h

Underside of base plate	=	0.500 m
Bottom of foundation	=	-2.500 m
Thickness of footing	=	0.4 m
Height of column	=	2.600 m

Calculation of load in pedestal for all load combinations

L/C	Fx	Fy	Fz	Mx	My	Mz
401	0.2	382.4	35.5	92.4	0.0	0.5
402	0.2	361.2	40.4	105.2	0.0	0.4
403	-1.5	579.7	123.7	321.5	0.0	-3.9
404	1.9	620.8	-12.0	-31.1	0.0	5.1
405	-1.5	558.5	128.6	334.3	0.0	-4.0
406	1.9	599.6	-7.1	-18.3	0.0	5.0
407	3.0	371.4	38.4	99.9	0.0	7.8
408	-2.7	413.5	34.2	89.0	0.0	-6.9
409	0.2	546.3	86.3	224.3	0.0	0.6
410	0.1	233.5	-14.9	-38.7	0.0	0.3
411	3.0	353.3	42.7	110.9	0.0	7.7
412	-2.7	395.4	38.4	100.0	0.0	-7.0
413	0.2	528.2	90.5	235.3	0.0	0.5
414	0.1	215.4	-10.7	-27.7	0.0	0.2
415	0.2	391.8	36.4	94.7	0.0	0.4
416	0.2	391.8	36.4	94.7	0.0	0.4
417	0.2	391.8	36.4	94.7	0.0	0.4
418	0.2	391.8	36.4	94.7	0.0	0.4
419	0.1	373.7	40.6	105.7	0.0	0.4
420	0.1	373.7	40.6	105.7	0.0	0.4
421	0.1	373.7	40.6	105.7	0.0	0.4
422	0.1	373.7	40.6	105.7	0.0	0.4
423	1.9	233.1	24.1	62.7	0.0	4.9
424	-1.7	259.4	21.5	55.8	0.0	-4.3
425	0.2	342.4	54.0	140.4	0.0	0.4
426	0.1	146.9	-9.2	-24.0	0.0	0.2
427	1.9	219.5	27.3	70.9	0.0	4.9
428	-1.7	245.8	24.6	64.0	0.0	-4.3
429	0.1	328.8	57.2	148.6	0.0	0.4
430	0.1	133.3	-6.1	-15.8	0.0	0.1
431	0.1	245.8	22.8	59.4	0.0	0.3
432	0.1	245.8	22.8	59.4	0.0	0.3
433	0.1	245.8	22.8	59.4	0.0	0.3
434	0.1	245.8	22.8	59.4	0.0	0.3
435	0.1	232.2	26.0	67.6	0.0	0.3
436	0.1	232.2	26.0	67.6	0.0	0.3
437	0.1	232.2	26.0	67.6	0.0	0.3
438	0.1	232.2	26.0	67.6	0.0	0.3

Summary of load in pedestal

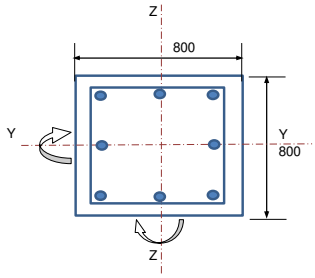
	L/C	Fy(KN)	Mx(KNm)	Mz(KNm)
FY max	404	620.8	-31.1	5.1
Fy min	430	133.3	-15.8	0.1
Mx max	405	558.5	334.3	-4.0
Mx min	410	233.5	-38.7	-4.0
Mz max	407	371.4	99.9	7.8
Mz min	412	395.4	100.0	-7.0

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	C02	D01
Date	27-01-2020	12-03-2020	25-04-2022	13/10/2023
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Dimension of column along Z	Dz	=	800	mm
Dimenson of column along Y	Dy	=	800	mm
Unsupported length of column along Z	Lz	=	2600	mm
Unsupported length of column along Y	Ly	=	2600	mm
Grade of steel	f_y	=	420	MPa
Grade of concrete	f_c'	=	35	MPa
Effective length factor along Z	Kz	=	2	
Effective length factor along Y	Ky	=	2	
Effective Depth in Z dir		=	655	mm
Effective Depth in Y dir		=	655	mm
Cover		=	50	mm
Effective cover		=	72.5	mm
Dia of longitudinal bar		=	25	mm
Dia of link		=	10	mm
Percentage of steel reinforcement		=	1	%



ANALYSIS RESULTS

Analysis result as obtained from staad analysis

Axial load	Fy	=	P_u	=	558.5	KN
B.M about Z	Mz	=	M_{uz}	=	4.0	KNm
B.M about Y	Mx	=	M_{uy}	=	334.3	KNm

Load Case No.
405 1.4 (DL + SDL2 + TL1) + 1.7 LL

CHECK FOR SLENDERNESS

Allowable slenderness for sway column $\frac{kl}{r} \le 22$ where r = 0.3h *Clause 6.2.5 SBC 304*

Slenderness check along Z axis $\frac{k_z l_z}{r_z}$ = 2 x 2600 / 240 r_z = 240

= 21.7 *Non-slender along Z*

Slenderness check along Y axis $\frac{k_y l_y}{r_y}$ = 2 x 2600 / 240 r_z = 240

= 21.7 *Non-slender along Y*

DESIGN FOR CAPACITY OF COLUMN

Moment Capacity about Z axis

P_u = 558.509 KN A_g = 800 x 800 = 640000 mm^2

Assume Strength reduction factor φ = 0.65 *Table 21.2.2 SBC 304* f_c' = 35 N/mm^2

ρ_s = 0.01

$P_n = \frac{P_u}{\phi}$ = 859.24 KN γ_z = 0.81875

$K_n = \frac{P_n}{A_g f_c'}$ = 0.038

Corresponding to the values of K_n ρ_s γ_z f_y f_c' the value of $\frac{M_{nz}}{A_g f_c' D_z}$ is interpolated from Interaction diagr in SP-17M(09)

We get $\frac{M_{nz}}{A_g f_c' D_z}$ = 0.08 *As per graph* *3.3.3* *SP-17M(09)*

M_{nz} = 0.08x640000x35x800/10^6 = 1433.6 KNm

Max moment capacity M_{uz} = ϕM_{nz} = 931.84 KNm > 4.0144 KNm *safe*

Utility ratio = ***0.004***

Moment Capacity about Y axis

P_u = 558.509 KN A_g = 800 x 800 = 640000 mm^2

Assume Strength reduction factor φ = 0.65 *Table 21.2.2 SBC 304* f_c' = 35 N/mm^2

ρ_s = 0.01

$P_n = \frac{P_u}{\phi}$ = 859.24 KN γ_y = 0.81875

$K_n = \frac{P_n}{A_g f_c'}$ = 0.038

Corresponding to the values of K_n ρ_s γ_y f_y f_c' the value of $\frac{M_{ny}}{A_g f_c' D_y}$ is interpolated from Interaction diagr in SP-17M(09)

We get $\frac{M_{ny}}{A_g f_c' D_y}$ = 0.08 *As per graph* *3.3.2* *SP-17M(09)*

M_{ny} = 0.08x640000x35x800/10^6 = 1433.6 KNm

Max moment capacity M_{uy} = ϕM_{ny} = 931.84 KNm > 334.3262 KNm *safe*

Utility ratio = ***0.359***

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	C02	D01		
Date	27-01-2020	12-03-2020	25-04-2022	13/10/2023		
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Combined moment capacity check

M_{noz} = 1433.6 KNm $M_{nz} = M_{uz}/\varphi$ = 6.176 KNm

M_{noy} = 1433.6 KNm $M_{ny} = M_{uy}/\varphi$ = 514.348 KNm

Assume β = 0.65 log β = -0.187087

log 0.5 = -0.30103

Check $\left(\frac{M_{nz}}{M_{noz}}\right)^{\log 0.5/\log \beta} + \left(\frac{M_{ny}}{M_{noy}}\right)^{\log 0.5/\log \beta} < 1$ *Equation 3-15 SP-17M(09)*

= 0.00 + 0.19 = 0.1923 < 1 *Safe*

Longitudinal reinforcement: 16 nos 20 mm — Required Ast = 6400 mm^2

Spread equally on all four side: 4 nos 25 mm

Total Ast = 6990.0437 mm^2 O.K

Minimim Reinforcement = 0.01Ag = 6400 mm^2

Transverse Reinforcement

Minimum diameter of tie = 10 mm

Maximum c/c dist.
a) 16db = 400 mm
b) 48db = 480 mm
c) Smaller column dimension = 800 mm

Provide lateral ties: 10 mm @ 250 c/c

Check for Tension

Tension Forces P_u = 133.27 KN M_{uz} = 0.1482 KNm M_{uy} = 15.8 KNm

Assume Strength reduction factor φ = 0.9 Tension controlled *Table 21.2.2 SBC 304*

$P_n = \frac{P_u}{\emptyset}$ = 148.08 KN A_g = 800 x 800 = 640000 mm^2

f_c' = 35 N/mm^2

$K_n = \frac{P_n}{A_g f_c'}$ = 0.0066 ρ_s = 0.01

γ_z = 0.81875

γ_y = 0.81875

From the tension curve corresponding to the values of γ_y f_y f_c' ρ_s K_n as per interaction diagram generated using clause 3.2.2 of SP-17M(09)

We get,

$\frac{M_{nz}}{A_g f_c' D_z}$ = 0.05 $\frac{M_{ny}}{A_g f_c' D_y}$ = 0.05

M_{nz} = 896 KNm M_{ny} = 896 KNm

$\emptyset M_{nz}$ = 806.4 KNm >0.1482 $\emptyset M_{ny}$ = 806.4 KNm >15.7664

safe **safe**

Utility ratio = ***0.00*** *Utility ratio =* ***0.020***

Global Project ID	**900707**
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Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

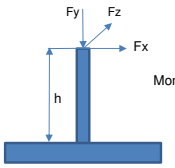
Ma'aden Revision No:	A01	C01	C02	D01
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REFERENCE DOCUMENTS

DOCUMENT NUMBER	DOCUMENT DESCRIPTION
MD-216-1000-EG-ST-CRT-1001	Civil Structural Design Criteria

8.4.5 Design of pedestal P1 at grid 3120/E-3130/10



Moment at pedestal Bottom,
Mx = Fz *h
Mz = Fx *h

Underside of base plate	=	0.500 m
Bottom of foundation	=	-2.500 m
Thickness of footing	=	0.8 m
Height of column	=	2.200 m

Calculation of load in pedestal for all load combinations

L/C	Fx	Fy	Fz	Mx	My	Mz
401	-50.5	1014.6	-8.7	-19.2	0.0	-111.1
402	-20.1	2379.7	-31.4	-69.1	0.0	-44.3
403	-51.2	1472.3	206.6	454.5	0.0	-112.7
404	-47.5	1527.3	-253.5	-557.8	0.0	-104.5
405	-20.8	2837.3	183.9	404.6	0.0	-45.8
406	-17.1	2892.3	-276.2	-607.7	0.0	-37.7
407	114.5	251.7	7.3	16.1	0.0	251.8
408	-199.8	1771.0	-29.9	-65.8	0.0	-439.6
409	-20.2	1066.8	185.0	406.9	0.0	-44.5
410	-65.3	942.0	-213.5	-469.7	0.0	-143.6
411	140.5	1421.7	-12.1	-26.7	0.0	309.1
412	-173.8	2941.0	-49.3	-108.6	0.0	-382.4
413	5.8	2236.8	165.5	364.2	0.0	12.8
414	-39.3	2112.0	-232.9	-512.5	0.0	-86.4
415	-43.0	1012.4	-11.8	-26.0	0.0	-94.5
416	-43.0	1012.4	-11.8	-26.0	0.0	-94.5
417	-43.0	1012.4	-11.8	-26.0	0.0	-94.5
418	-43.0	1012.4	-11.8	-26.0	0.0	-94.5
419	-16.9	2182.4	-31.3	-68.8	0.0	-37.2
420	-16.9	2182.4	-31.3	-68.8	0.0	-37.2
421	-16.9	2182.4	-31.3	-68.8	0.0	-37.2
422	-16.9	2182.4	-31.3	-68.8	0.0	-37.2
423	65.9	176.8	6.3	13.9	0.0	145.0
424	-130.5	1126.4	-16.9	-37.2	0.0	-287.1
425	-18.3	686.3	117.4	258.2	0.0	-40.2
426	-46.4	608.3	-131.7	-289.7	0.0	-102.1
427	85.5	1054.4	-8.2	-18.1	0.0	188.0
428	-111.0	2003.9	-31.5	-69.3	0.0	-244.2
429	1.3	1563.8	102.8	226.2	0.0	2.8
430	-26.9	1485.8	-146.2	-321.7	0.0	-59.2
431	-32.5	652.3	-5.6	-12.4	0.0	-71.4
432	-32.5	652.3	-5.6	-12.4	0.0	-71.4
433	-32.5	652.3	-5.6	-12.4	0.0	-71.4
434	-32.5	652.3	-5.6	-12.4	0.0	-71.4
435	-12.9	1529.8	-20.2	-44.4	0.0	-28.4
436	-12.9	1529.8	-20.2	-44.4	0.0	-28.4
437	-12.9	1529.8	-20.2	-44.4	0.0	-28.4
438	-12.9	1529.8	-20.2	-44.4	0.0	-28.4

Summary of load in pedestal

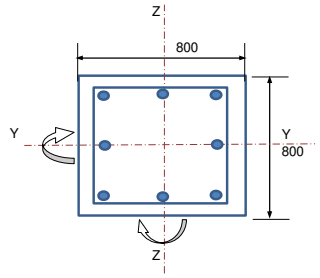
	L/C	Fy(KN)	Mx(KNm)	Mz(KNm)
FY max	412	2941.0	-108.6	-382.35
Fy min	423	176.8	13.9	145.03
Mx max	403	1472.3	454.5	-112.69
Mx min	406	2892.3	-607.7	-112.69
Mz max	411	1421.7	-26.7	309.12
Mz min	408	1771.0	-65.8	-439.65

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	C02	D01
Date	27-01-2020	12-03-2020	25-04-2022	13/10/2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS DT. 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER SUPPORT STRUCTURE	AS BUILT
Prepared By	APR	APR	APR	APR
Checked	SBY	SBY	SBY	SBY
Approved	AG	AG	AG	AG

Outotec / L&T Revision No.:	0	1	2	3
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Dimension of column along Z	Dz	=	800	mm
Dimenson of column along Y	Dy	=	800	mm
Unsupported length of column along Z	Lz	=	2200	mm
Unsupported length of column along Y	Ly	=	2200	mm
Grade of steel	f_y	=	420	MPa
Grade of concrete	f_c'	=	35	MPa
Effective length factor along Z	Kz	=	2	
Effective length factor along Y	Ky	=	2	
Effective Depth in Z dir		=	655	mm
Effective Depth in Y dir		=	655	mm
Cover		=	50	mm
Effective cover		=	72.5	mm
Dia of longitudinal bar		=	25	mm
Dia of link		=	10	mm
Percentage of steel reinforcement		=	1	%



ANALYSIS RESULTS

Analysis result as obtained from staad analysis

Axial load	Fy	=	P_u	=	2892.3 KN	Load Case No.
B.M about Z	Mz	=	M_{uz}	=	112.7 KNm	411 1.2 (DL + SDL2) + 0.5 LL + 1.6 WL(+X)
B.M about Y	Mx	=	M_{uy}	=	607.7 KNm	

CHECK FOR SLENDERNESS

Allowable slenderness for sway column $\frac{kl}{r} \le 22$ where r = 0.3h *Clause 6.2.5 SBC 304*

Slenderness check along Z axis $\frac{k_z l_z}{r_z}$ = 2 x 2200 / 240 r_z = 240

= 18.3 *Non-slender along Z*

Slenderness check along Y axis $\frac{k_y l_y}{r_y}$ = 2 x 2200 / 240 r_z = 240

= 18.3 *Non-slender along Y*

DESIGN FOR CAPACITY OF COLUMN

Moment Capacity about Z axis

P_u = 2892.273 KN A_g = 800 x 800 = 640000 mm^2

Assume Strength reduction factor φ = 0.65 *Table 21.2.2 SBC 304* f_c' = 35 N/mm^2

ρ_s = 0.01

$P_n = \frac{P_u}{\phi}$ = 4449.65 KN γ_z = 0.81875

$K_n = \frac{P_n}{A_g f_c'}$ = 0.199

Corresponding to the values of K_n ρ_s γ_z f_y f_c' the value of $\frac{M_{nz}}{A_g f_c' D_z}$ is interpolated from Interaction diagr in SP-17M(09)

We get $\frac{M_{nz}}{A_g f_c' D_z}$ = 0.12 *As per graph* *3.3.3* *SP-17M(09)*

M_{nz} = 0.12x640000x35x800/10^6 = 2150.4 KNm

Max moment capacity M_{uz} = ϕM_{nz} = 1397.76 KNm > 112.6906 KNm *safe*

Utility ratio = ***0.081***

Moment Capacity about Y axis

P_u = 2892.273 KN A_g = 800 x 800 = 640000 mm^2

Assume Strength reduction factor φ = 0.65 *Table 21.2.2 SBC 304* = 35 N/mm^2

ρ_s' = 0.01

$P_n = \frac{P_u}{\phi}$ = 4449.65 KN γ_y = 0.81875

$K_n = \frac{P_n}{A_g f_c'}$ = 0.199

Corresponding to the values of K_n ρ_s γ_y f_y f_c' the value of $\frac{M_{ny}}{A_g f_c' D_y}$ is interpolated from Interaction diagr in SP-17M(09)

We get $\frac{M_{ny}}{A_g f_c' D_y}$ = 0.12 *As per graph* *3.3.2* *SP-17M(09)*

M_{ny} = 0.12x640000x35x800/10^6 = 2150.4 KNm

Max moment capacity M_{uy} = ϕM_{ny} = 1397.76 KNm > 607.6664 KNm *safe*

Utility ratio = ***0.435***

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	C02	D01		
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Combined moment capacity check

M_{noz} = 2150.4 KNm $M_{nz} = M_{uz}/\varphi$ = 173.3701538 KNm

M_{noy} = 2150.4 KNm $M_{ny} = M_{uy}/\varphi$ = 934.8713846 KNm

Assume β = 0.65 log β = -0.187087

log 0.5 = -0.30103

Check $\left(\frac{M_{nz}}{M_{noz}}\right)^{\log 0.5/\log \beta} + \left(\frac{M_{ny}}{M_{noy}}\right)^{\log 0.5/\log \beta} < 1$ *Equation 3-15 SP-17M(09)*

= 0.02 + 0.26 = 0.2792 < 1 *Safe*

Longitudinal reinforcement: 16 nos 20 mm — Required Ast = 6400 mm^2

Spread equally on all four side: 4 nos 25 mm

Total Ast = 6990.0437 mm^2 O.K

Minimim Reinforcement = 0.01Ag = 6400 mm^2

Transverse Reinforcement

Minimum diameter of tie = 10 mm

Maximum c/c dist. a) 16db = 400 mm

b) 48db = 480 mm

c) Smaller column dimension = 800 mm

Provide lateral ties: 10 mm @ 250 c/c

Check for Tension

Tension Forces P_u = 176.85 KN M_{uz} = 145.0262 KNm M_{uy} = 13.9 KNm

Assume Strength reduction factor φ = 0.9 Tension controlled *Table 21.2.2 SBC 304*

$P_n = \frac{P_u}{\emptyset}$ = 196.49 KN A_g = 800 x 800 = 640000 mm^2

f_c' = 35 N/mm^2

$K_n = \frac{P_n}{A_g f_c'}$ = 0.009 ρ_g' = 0.01

γ_z = 0.81875

γ_y = 0.81875

From the tension curve corresponding to the values of γ_y f_y f_c' ρ_s K_n as per interaction diagram generated using clause 3.2.2 of SP-17M(09)

We get,

$\frac{M_{nz}}{A_g f_c' D_z}$ = 0.05 $\frac{M_{ny}}{A_g f_c' D_y}$ = 0.05

M_{nz} = 896 KNm M_{ny} = 896 KNm

$\emptyset M_{nz}$ = 806.4 KNm >145.0262 $\emptyset M_{ny}$ = 806.4 KNm >13.9436

safe **safe**

Utility ratio = ***0.18*** *Utility ratio =* ***0.017***

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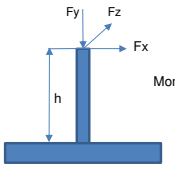
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REFERENCE DOCUMENTS

DOCUMENT NUMBER	DOCUMENT DESCRIPTION
MD-216-1000-EG-ST-CRT-1001	Civil Structural Design Criteria

8.4.6 Design of pedestal P1 at grid 3120/E-3130/12



Moment at pedestal Bottom,
Mx = Fz *h
Mz = Fx *h

Underside of base plate	=	0.500 m
Bottom of foundation	=	-2.500 m
Thickness of footing	=	0.8 m
Height of column	=	2.200 m

Calculation of load in pedestal for all load combinations

L/C	Fx	Fy	Fz	Mx	My	Mz
401	-47.7	-38.5	6.7	14.7	0.0	-104.9
402	-17.5	1891.7	-23.4	-51.4	0.0	-38.5
403	-42.1	277.2	-39.4	-86.7	0.0	-92.6
404	-51.1	262.2	37.9	83.3	0.0	-112.5
405	-11.9	2207.4	-69.5	-152.9	0.0	-26.3
406	-21.0	2192.4	7.8	17.1	0.0	-46.1
407	116.9	798.7	-9.4	-20.8	0.0	257.2
408	-197.5	-677.4	15.8	34.8	0.0	-434.4
409	-18.0	-211.7	90.8	199.9	0.0	-39.5
410	-62.7	356.2	-89.7	-197.4	0.0	-138.0
411	142.8	2453.1	-35.2	-77.5	0.0	314.1
412	-171.6	977.1	-10.0	-21.9	0.0	-377.5
413	7.9	1442.7	65.1	143.1	0.0	17.3
414	-36.9	2010.6	-115.5	-254.1	0.0	-81.2
415	-40.5	57.6	3.5	7.8	0.0	-89.2
416	-40.5	57.6	3.5	7.8	0.0	-89.2
417	-40.5	57.6	3.5	7.8	0.0	-89.2
418	-40.5	57.6	3.5	7.8	0.0	-89.2
419	-14.7	1712.1	-22.2	-48.9	0.0	-32.3
420	-14.7	1712.1	-22.2	-48.9	0.0	-32.3
421	-14.7	1712.1	-22.2	-48.9	0.0	-32.3
422	-14.7	1712.1	-22.2	-48.9	0.0	-32.3
423	67.8	438.4	-3.8	-8.4	0.0	149.1
424	-128.7	-484.1	12.0	26.4	0.0	-283.2
425	-16.5	-193.1	58.9	129.5	0.0	-36.4
426	-44.5	161.8	-54.0	-118.8	0.0	-97.9
427	87.2	1679.2	-23.2	-50.9	0.0	191.7
428	-109.3	756.7	-7.4	-16.2	0.0	-240.5
429	2.9	1047.7	39.5	87.0	0.0	6.3
430	-25.1	1402.7	-73.3	-161.3	0.0	-55.3
431	-30.6	-24.8	4.3	9.5	0.0	-67.4
432	-30.6	-24.8	4.3	9.5	0.0	-67.4
433	-30.6	-24.8	4.3	9.5	0.0	-67.4
434	-30.6	-24.8	4.3	9.5	0.0	-67.4
435	-11.2	1216.1	-15.0	-33.1	0.0	-24.7
436	-11.2	1216.1	-15.0	-33.1	0.0	-24.7
437	-11.2	1216.1	-15.0	-33.1	0.0	-24.7
438	-11.2	1216.1	-15.0	-33.1	0.0	-24.7
439	-14.7	300.8	-45.7	-100.5	0.0	-32.3
440	2.1	185.9	-4.7	-10.3	0.0	4.7
441	2.1	185.9	-4.7	-10.3	0.0	4.7
442	2.1	185.9	-4.7	-10.3	0.0	4.7
443	2.1	185.9	-4.7	-10.3	0.0	4.7

Summary of load in pedestal

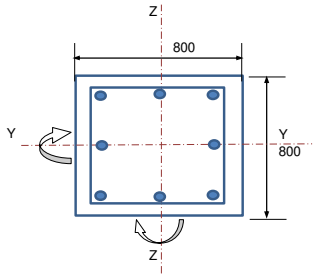
	L/C	Fy(KN)	Mx(KNm)	Mz(KNm)
FY max	411	2453.1	-77.5	314.05
Fy min	408	-677.4	34.8	-434.41
Mx max	409	-211.7	199.9	-39.53
Mx min	414	2010.6	-254.1	-39.53
Mz max	411	2453.1	-77.5	314.05
Mz min	408	-677.4	34.8	-434.41

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Dimension of column along Z	Dz	=	800	mm
Dimenson of column along Y	Dy	=	800	mm
Unsupported length of column along Z	Lz	=	2200	mm
Unsupported length of column along Y	Ly	=	2200	mm
Grade of steel	f_y	=	420	MPa
Grade of concrete	f_c'	=	35	MPa
Effective length factor along Z	Kz	=	2	
Effective length factor along Y	Ky	=	2	
Effective Depth in Z dir		=	655	mm
Effective Depth in Y dir		=	655	mm
Cover		=	50	mm
Effective cover		=	72.5	mm
Dia of longitudinal bar		=	25	mm
Dia of link		=	10	mm
Percentage of steel reinforcement		=	1	%



ANALYSIS RESULTS

Analysis result as obtained from staad analysis

Axial load	Fy	=	P_u	=	-677.4	KN	Load Case No.
B.M about Z	Mz	=	M_{uz}	=	434.4	KNm	421 1.2 (DL + SDL2) + 0.5 LL + 1.0 EL(+Z)
B.M about Y	Mx	=	M_{uy}	=	34.8	KNm	

CHECK FOR SLENDERNESS

Allowable slenderness for sway column $\frac{kl}{r} \le 22$ where r = 0.3h *Clause 6.2.5 SBC 304*

Slenderness check along Z axis $\frac{k_z l_z}{r_z}$ = 2 x 2200 / 240 = 18.3 r_z = 240 *Non-slender along Z*

Slenderness check along Y axis $\frac{k_y l_y}{r_y}$ = 2 x 2200 / 240 = 18.3 r_z = 240 *Non-slender along Y*

DESIGN FOR CAPACITY OF COLUMN

Moment Capacity about Z axis

P_u = -677.367 KN; A_g = 800 x 800 = 640000 mm²

Assume Strength reduction factor φ = 0.65 *Table 21.2.2 SBC 304*; f_c' = 35 N/mm²; ρ_s = 0.01

$P_n = \frac{P_u}{\phi}$ = -1042.10 KN; γ_z = 0.81875

$K_n = \frac{P_n}{A_g f_c'}$ = -0.047

Corresponding to the values of K_n ρ_s γ_z f_y f_c' the value of $\frac{M_{nz}}{A_g f_c' D_z}$ is interpolated from Interaction diagr in SP-17M(09)

We get $\frac{M_{nz}}{A_g f_c' D_z}$ = 0.085 *As per graph* *3.3.3* *SP-17M(09)*

M_{nz} = 0.085x640000x35x800/10^6 = 1523.2 KNm

Max moment capacity M_{uz} = ϕM_{nz} = 990.08 KNm > 434.412 KNm *safe*

Utility ratio = ***0.439***

Moment Capacity about Y axis

P_u = -677.367 KN; A_g = 800 x 800 = 640000 mm²

Assume Strength reduction factor φ = 0.65 *Table 21.2.2 SBC 304*; = 35 N/mm²; ρ_s' = 0.01

$P_n = \frac{P_u}{\phi}$ = -1042.10 KN; γ_y = 0.81875

$K_n = \frac{P_n}{A_g f_c'}$ = -0.047

Corresponding to the values of K_n ρ_s γ_y f_y f_c' the value of $\frac{M_{ny}}{A_g f_c' D_y}$ is interpolated from Interaction diagr in SP-17M(09)

We get $\frac{M_{ny}}{A_g f_c' D_y}$ = 0.085 *As per graph* *3.3.2* *SP-17M(09)*

M_{ny} = 0.085x640000x35x800/10^6 = 1523.2 KNm

Max moment capacity M_{uy} = ϕM_{ny} = 990.08 KNm > 34.8304 KNm *safe*

Utility ratio = ***0.035***

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Combined moment capacity check

M_{noz} = 1523.2 KNm $M_{nz} = M_{uz}/\varphi$ = 668.3261538 KNm

M_{noy} = 1523.2 KNm $M_{ny} = M_{uy}/\varphi$ = 53.58523077 KNm

Assume β = 0.65 log β = -0.187087 log 0.5 = -0.30103

Check $\left(\frac{M_{nz}}{M_{noz}}\right)^{\log 0.5/\log \beta} + \left(\frac{M_{ny}}{M_{noy}}\right)^{\log 0.5/\log \beta} < 1$ *Equation 3-15 SP-17M(09)*

= 0.27 + 0.00 = 0.2702 < 1 *Safe*

Longitudinal reinforcement: 16 nos 20 mm — Required Ast = 6400 mm^2

Spread equally on all four side: 4 nos 25 mm

Total Ast = 6990.0437 mm^2 O.K

Minimim Reinforcement = 0.01Ag = 6400 mm^2

Transverse Reinforcement

Minimum diameter of tie = 10 mm

Maximum c/c dist.
a) 16db = 400 mm
b) 48db = 480 mm
c) Smaller column dimension = 800 mm

Provide lateral ties: 10 mm @ 250 c/c

Check for Tension

Tension Forces P_u = 677.37 KN M_{uz} = 434.412 KNm M_{uy} = 34.8 KNm

Assume Strength reduction factor φ = 0.9 Tension controlled *Table 21.2.2 SBC 304*

P_n = $\frac{P_u}{\emptyset}$ = 752.63 KN A_g = 800 x 800 = 640000 mm^2

K_n = $\frac{P_n}{A_g f_c'}$ = 0.034 f_c' = 35 N/mm^2

ρ_g' = 0.01

γ_z = 0.81875

γ_y = 0.81875

From the tension curve corresponding to the values of γ_y f_y f_c' ρ_s K_n as per interaction diagram generated using clause 3.2.2 of SP-17M(09)

We get,

$\frac{M_{nz}}{A_g f_c' D_z}$ = 0.05 $\frac{M_{ny}}{A_g f_c' D_y}$ = 0.05

M_{nz} = 896 KNm M_{ny} = 896 KNm

$\emptyset M_{nz}$ = 806.4 KNm >434.412 *safe* $\emptyset M_{ny}$ = 806.4 KNm >34.8304 **safe**

Utility ratio = ***0.54*** *Utility ratio =* ***0.043***

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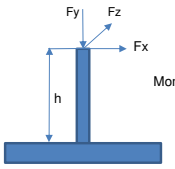
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REFERENCE DOCUMENTS

DOCUMENT NUMBER	DOCUMENT DESCRIPTION
MD-216-1000-EG-ST-CRT-1001	Civil Structural Design Criteria

8.4.7 Design of pedestal P1 at grid 3120/C-3130/10



Moment at pedestal Bottom,
Mx = Fz *h
Mz = Fx *h

Underside of base plate	=	0.500 m
Bottom of foundation	=	-2.500 m
Thickness of footing	=	0.8 m
Height of column	=	2.200 m

Calculation of load in pedestal for all load combinations

L/C	Fx	Fy	Fz	Mx	My	Mz
401	-1.3	84.8	38.5	84.7	0.0	-3.0
402	-1.3	131.3	93.9	206.6	0.0	-2.9
403	-3.0	208.5	155.4	341.9	0.0	-6.7
404	0.3	187.4	-38.4	-84.4	0.0	0.6
405	-3.0	255.1	210.8	463.8	0.0	-6.6
406	0.3	233.9	17.1	37.5	0.0	0.6
407	24.5	-10.1	16.7	36.7	0.0	53.9
408	-26.8	224.1	62.3	137.0	0.0	-59.0
409	-1.2	529.9	285.5	628.1	0.0	-2.7
410	-1.1	-329.8	-214.8	-472.5	0.0	-2.4
411	24.5	29.8	64.2	141.3	0.0	54.0
412	-26.8	264.1	109.8	241.5	0.0	-59.0
413	-1.2	569.8	333.0	732.6	0.0	-2.7
414	-1.1	-289.9	-167.3	-368.0	0.0	-2.4
415	-1.2	105.9	38.9	85.5	0.0	-2.6
416	-1.2	105.9	38.9	85.5	0.0	-2.6
417	-1.2	105.9	38.9	85.5	0.0	-2.6
418	-1.2	105.9	38.9	85.5	0.0	-2.6
419	-1.1	145.9	86.4	190.1	0.0	-2.5
420	-1.1	145.9	86.4	190.1	0.0	-2.5
421	-1.1	145.9	86.4	190.1	0.0	-2.5
422	-1.1	145.9	86.4	190.1	0.0	-2.5
423	15.2	-18.0	10.9	23.9	0.0	33.4
424	-16.9	128.4	39.4	86.6	0.0	-37.2
425	-0.9	319.4	178.9	393.5	0.0	-2.0
426	-0.8	-217.8	-133.8	-294.3	0.0	-1.8
427	15.2	11.9	46.5	102.3	0.0	33.4
428	-16.9	158.3	75.0	165.0	0.0	-37.2
429	-0.9	349.4	214.5	471.9	0.0	-2.0
430	-0.8	-187.9	-98.2	-215.9	0.0	-1.8
431	-0.9	54.5	24.7	54.4	0.0	-1.9
432	-0.9	54.5	24.7	54.4	0.0	-1.9
433	-0.9	54.5	24.7	54.4	0.0	-1.9
434	-0.9	54.5	24.7	54.4	0.0	-1.9
435	-0.8	84.4	60.4	132.8	0.0	-1.9
436	-0.8	84.4	60.4	132.8	0.0	-1.9
437	-0.8	84.4	60.4	132.8	0.0	-1.9
438	-0.8	84.4	60.4	132.8	0.0	-1.9

Summary of load in pedestal

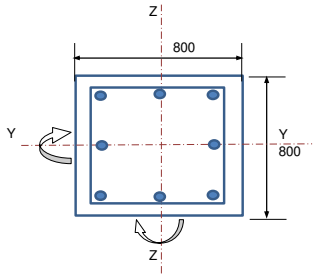
	L/C	Fy(KN)	Mx(KNm)	Mz(KNm)
FY max	413	569.8	732.6	-2.66
Fy min	410	-329.8	-472.5	-2.42
Mx max	413	569.8	732.6	-2.66
Mx min	410	-329.8	-472.5	-2.66
Mz max	411	29.8	141.3	53.96
Mz min	408	224.1	137.0	-59.05

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	C02	D01
Date	27-01-2020	12-03-2020	25-04-2022	13/10/2023
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Checked	SBY	SBY	SBY	SBY
Approved	AG	AG	AG	AG

Outotec / L&T Revision No.:	0	1	2	3
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Dimension of column along Z	Dz	=	800	mm
Dimenson of column along Y	Dy	=	800	mm
Unsupported length of column along Z	Lz	=	2200	mm
Unsupported length of column along Y	Ly	=	2200	mm
Grade of steel	f_y	=	420	MPa
Grade of concrete	f_c'	=	35	MPa
Effective length factor along Z	Kz	=	2	
Effective length factor along Y	Ky	=	2	
Effective Depth in Z dir		=	655	mm
Effective Depth in Y dir		=	655	mm
Cover		=	50	mm
Effective cover		=	72.5	mm
Dia of longitudinal bar		=	25	mm
Dia of link		=	10	mm
Percentage of steel reinforcement		=	1	%



ANALYSIS RESULTS

Analysis result as obtained from staad analysis

Axial load	Fy	=	P_u	=	569.8	KN	Load Case No.
B.M about Z	Mz	=	M_{uz}	=	2.7	KNm	405 1.4 (DL + SDL2 + TL1) + 1.7 LL
B.M about Y	Mx	=	M_{uy}	=	732.6	KNm	

CHECK FOR SLENDERNESS

Allowable slenderness for sway column $\frac{kl}{r} \le 22$ where r = 0.3h *Clause 6.2.5 SBC 304*

Slenderness check along Z axis $\frac{k_z l_z}{r_z}$ = 2 x 2200 / 240 r_z = 240

= 18.3 *Non-slender along Z*

Slenderness check along Y axis $\frac{k_y l_y}{r_y}$ = 2 x 2200 / 240 r_z = 240

= 18.3 *Non-slender along Y*

DESIGN FOR CAPACITY OF COLUMN

Moment Capacity about Z axis

P_u = 569.785 KN A_g = 800 x 800 = 640000 mm^2

Assume Strength reduction factor φ = 0.65 *Table 21.2.2 SBC 304* f_c' = 35 N/mm^2

ρ_s = 0.01

$P_n = \frac{P_u}{\phi}$ = 876.59 KN γ_z = 0.81875

$K_n = \frac{P_n}{A_g f_c'}$ = 0.039

Corresponding to the values of K_n ρ_s γ_z f_y f_c' the value of $\frac{M_{nz}}{A_g f_c' D_z}$ is interpolated from Interaction diagr in SP-17M(09)

We get $\frac{M_{nz}}{A_g f_c' D_z}$ = 0.06 *As per graph* *3.3.3* *SP-17M(09)*

M_{nz} = 0.06x640000x35x800/10^6 = 1075.2 KNm

Max moment capacity M_{uz} = ϕM_{nz} = 698.88 KNm > 2.6554 KNm *safe*

Utility ratio = ***0.004***

Moment Capacity about Y axis

P_u = 569.785 KN A_g = 800 x 800 = 640000 mm^2

Assume Strength reduction factor φ = 0.65 *Table 21.2.2 SBC 304* = 35 N/mm^2

ρ_s' = 0.01

$P_n = \frac{P_u}{\phi}$ = 876.59 KN γ_y = 0.81875

$K_n = \frac{P_n}{A_g f_c'}$ = 0.039

Corresponding to the values of K_n ρ_s γ_y f_y f_c' the value of $\frac{M_{ny}}{A_g f_c' D_y}$ is interpolated from Interaction diagr in SP-17M(09)

We get $\frac{M_{ny}}{A_g f_c' D_y}$ = 0.06 *As per graph* *3.3.2* *SP-17M(09)*

M_{ny} = 0.06x640000x35x800/10^6 = 1075.2 KNm

Max moment capacity M_{uy} = ϕM_{ny} = 698.88 KNm < 732.6242 KNm *UnSafe*

Utility ratio = ***1.048***

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	C02	D01		
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Combined moment capacity check

M_{noz} = 1075.2 KNm $M_{nz} = M_{uz}/\varphi$ = 4.085230769 KNm

M_{noy} = 1075.2 KNm $M_{ny} = M_{uy}/\varphi$ = 1127.114154 KNm

Assume β = 0.65 log β = -0.187087 log 0.5 = -0.30103

Check $\left(\frac{M_{nz}}{M_{noz}}\right)^{\log 0.5/\log \beta} + \left(\frac{M_{ny}}{M_{noy}}\right)^{\log 0.5/\log \beta} < 1$ *Equation 3-15 SP-17M(09)*

= 0.00 + 1.08 = 1.0790 > 1 *Unsafe*

Longitudinal reinforcement: 16 nos 20 mm — Required Ast = 6400 mm^2

Spread equally on all four side: 4 nos 25 mm

Total Ast = 6990.0437 mm^2 O.K

Minimim Reinforcement = 0.01Ag = 6400 mm^2

Transverse Reinforcement

Minimum diameter of tie = 10 mm

Maximum c/c dist. a) 16db = 400 mm

b) 48db = 480 mm

c) Smaller column dimension = 800 mm

Provide lateral ties: 10 mm @ 250 c/c

Check for Tension

Tension Forces P_u = 329.79 KN M_{uz} = 2.4156 KNm M_{uy} = 472.5 KNm

Assume Strength reduction factor φ = 0.9 Tension controlled *Table 21.2.2 SBC 304*

P_n = $\frac{P_u}{\emptyset}$ = 366.43 KN A_g = 800 x 800 = 640000 mm^2

K_n = $\frac{P_n}{A_g f_c'}$ = 0.016 f_c' = 35 N/mm^2

ρ_g' = 0.01

γ_z = 0.81875

γ_y = 0.81875

From the tension curve corresponding to the values of γ_y f_y f_c' ρ_s K_n as per interaction diagram generated using clause 3.2.2 of SP-17M(09)

We get,

$\frac{M_{nz}}{A_g f_c' D_z}$ = 0.05 $\frac{M_{ny}}{A_g f_c' D_y}$ = 0.05

M_{nz} = 896 KNm M_{ny} = 896 KNm

$\emptyset M_{nz}$ = 806.4 KNm >2.4156 *safe* $\emptyset M_{ny}$ = 806.4 KNm >472.4852 **safe**

Utility ratio = ***0.00*** *Utility ratio =* ***0.586***

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

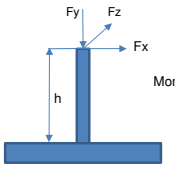
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REFERENCE DOCUMENTS

DOCUMENT NUMBER	DOCUMENT DESCRIPTION
MD-216-1000-EG-ST-CRT-1001	Civil Structural Design Criteria

8.4.8 Design of pedestal P1 at grid 3120/C-3130/12



Moment at pedestal Bottom,
Mx = Fz *h
Mz = Fx *h

Underside of base plate	=	0.500 m
Bottom of foundation	=	-2.500 m
Thickness of footing	=	0.8 m
Height of column	=	2.200 m

Calculation of load in pedestal for all load combinations

L/C	Fx	Fy	Fz	Mx	My	Mz
401	1.3	85.6	12.0	26.4	0.0	2.9
402	1.3	415.4	52.9	116.5	0.0	2.9
403	3.0	231.6	66.7	146.8	0.0	6.6
404	-0.3	196.5	-22.3	-49.1	0.0	-0.6
405	3.0	561.5	107.7	236.8	0.0	6.6
406	-0.3	526.4	18.6	41.0	0.0	-0.7
407	26.9	255.5	25.5	56.2	0.0	59.1
408	-24.6	-39.1	0.0	0.0	0.0	-54.1
409	1.2	719.6	137.0	301.4	0.0	2.6
410	1.1	-548.3	-119.5	-262.8	0.0	2.4
411	26.9	538.2	60.6	133.4	0.0	59.1
412	-24.6	243.6	35.1	77.2	0.0	-54.2
413	1.2	1002.3	172.1	378.6	0.0	2.6
414	1.1	-265.5	-84.4	-185.6	0.0	2.4
415	1.1	111.1	13.3	29.2	0.0	2.5
416	1.1	111.1	13.3	29.2	0.0	2.5
417	1.1	111.1	13.3	29.2	0.0	2.5
418	1.1	111.1	13.3	29.2	0.0	2.5
419	1.1	393.9	48.4	106.4	0.0	2.5
420	1.1	393.9	48.4	106.4	0.0	2.5
421	1.1	393.9	48.4	106.4	0.0	2.5
422	1.1	393.9	48.4	106.4	0.0	2.5
423	16.9	145.2	15.4	33.8	0.0	37.3
424	-15.2	-38.9	-0.6	-1.3	0.0	-33.5
425	0.9	435.3	85.0	187.1	0.0	1.9
426	0.8	-357.1	-75.3	-165.6	0.0	1.8
427	16.9	357.3	41.7	91.7	0.0	37.2
428	-15.3	173.2	25.7	56.6	0.0	-33.6
429	0.9	647.3	111.4	245.0	0.0	1.9
430	0.8	-145.1	-48.9	-107.6	0.0	1.8
431	0.8	55.0	7.7	17.0	0.0	1.9
432	0.8	55.0	7.7	17.0	0.0	1.9
433	0.8	55.0	7.7	17.0	0.0	1.9
434	0.8	55.0	7.7	17.0	0.0	1.9
435	0.8	267.1	34.0	74.9	0.0	1.8
436	0.8	267.1	34.0	74.9	0.0	1.8
437	0.8	267.1	34.0	74.9	0.0	1.8
438	0.8	267.1	34.0	74.9	0.0	1.8

Summary of load in pedestal

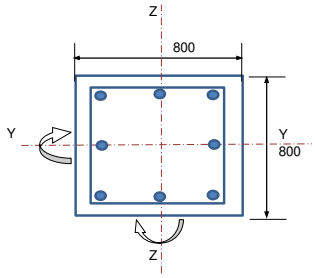
	L/C	Fy(KN)	Mx(KNm)	Mz(KNm)
FY max	413	1002.3	378.6	2.61
Fy min	410	-548.3	-262.8	2.38
Mx max	413	1002.3	378.6	2.61
Mx min	410	-548.3	-262.8	2.61
Mz max	407	255.5	56.2	59.14
Mz min	412	243.6	77.2	-54.15

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
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Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

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Dimension of column along Z	Dz	=	800 mm
Dimenson of column along Y	Dy	=	800 mm
Unsupported length of column along Z	Lz	=	2200 mm
Unsupported length of column along Y	Ly	=	2200 mm
Grade of steel	f_y	=	420 MPa
Grade of concrete	f_c'	=	35 MPa
Effective length factor along Z	Kz	=	2
Effective length factor along Y	Ky	=	2
Effective Depth in Z dir		=	655 mm
Effective Depth in Y dir		=	655 mm
Cover		=	50 mm
Effective cover		=	72.5 mm
Dia of longitudinal bar		=	25 mm
Dia of link		=	10 mm
Percentage of steel reinforcement		=	1 %



ANALYSIS RESULTS

Analysis result as obtained from staad analysis

Axial load	Fy	=	P_u	=	1002.3 KN	Load Case No.
B.M about Z	Mz	=	M_{uz}	=	2.6 KNm	405 1.4 (DL + SDL2 + TL1) + 1.7 LL
B.M about Y	Mx	=	M_{uy}	=	378.6 KNm	

CHECK FOR SLENDERNESS

Allowable slenderness for sway column $\frac{kl}{r} \le 22$ where r = 0.3h *Clause 6.2.5 SBC 304*

Slenderness check along Z axis $\frac{k_z l_z}{r_z}$ = 2 x 2200 / 240 r_z = 240

= 18.3 *Non-slender along Z*

Slenderness check along Y axis $\frac{k_y l_y}{r_y}$ = 2 x 2200 / 240 r_z = 240

= 18.3 *Non-slender along Y*

DESIGN FOR CAPACITY OF COLUMN

Moment Capacity about Z axis

P_u = 1002.308 KN A_g = 800 x 800 = 640000 mm^2

Assume Strength reduction factor φ = 0.65 *Table 21.2.2 SBC 304* f_c' = 35 N/mm^2

ρ_s = 0.01

$P_n = \frac{P_u}{\phi}$ = 1542.01 KN γ_z = 0.81875

$K_n = \frac{P_n}{A_g f_c'}$ = 0.069

Corresponding to the values of K_n ρ_s γ_z f_y f_c' the value of $\frac{M_{nz}}{A_g f_c' D_z}$ is interpolated from Interaction diagr in SP-17M(09)

We get $\frac{M_{nz}}{A_g f_c' D_z}$ = 0.08 *As per graph* *3.3.3* *SP-17M(09)*

M_{nz} = 0.08x640000x35x800/10^6 = 1433.6 KNm

Max moment capacity M_{uz} = ϕM_{nz} = 931.84 KNm > 2.6114 KNm *safe*

Utility ratio = ***0.003***

Moment Capacity about Y axis

P_u = 1002.308 KN A_g = 800 x 800 = 640000 mm^2

Assume Strength reduction factor φ = 0.65 *Table 21.2.2 SBC 304* = 35 N/mm^2

ρ_s' = 0.01

$P_n = \frac{P_u}{\phi}$ = 1542.01 KN γ_y = 0.81875

$K_n = \frac{P_n}{A_g f_c'}$ = 0.069

Corresponding to the values of K_n ρ_s γ_y f_y f_c' the value of $\frac{M_{ny}}{A_g f_c' D_y}$ is interpolated from Interaction diagr in SP-17M(09)

We get $\frac{M_{ny}}{A_g f_c' D_y}$ = 0.08 *As per graph* *3.3.2* *SP-17M(09)*

M_{ny} = 0.08x640000x35x800/10^6 = 1433.6 KNm

Max moment capacity M_{uy} = ϕM_{ny} = 931.84 KNm > 378.6068 KNm *safe*

Utility ratio = ***0.406***

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	C02	D01
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Combined moment capacity check

M_{noz} = 1433.6 KNm $M_{nz} = M_{uz}/\varphi$ = 4.017538462 KNm

M_{noy} = 1433.6 KNm $M_{ny} = M_{uy}/\varphi$ = 582.472 KNm

Assume β = 0.65 log β = -0.187087

log 0.5 = -0.30103

Check $\left(\frac{M_{nz}}{M_{noz}}\right)^{\log 0.5/\log \beta} + \left(\frac{M_{ny}}{M_{noy}}\right)^{\log 0.5/\log \beta} < 1$ *Equation 3-15 SP-17M(09)*

= 0.00 + 0.23 = 0.2348 < 1 *Safe*

Longitudinal reinforcement: 16 nos 20 mm — Required Ast = 6400 mm²

Spread equally on all four side: 4 nos 25 mm

Total Ast = 6990.0437 mm² O.K

Minimim Reinforcement = 0.01Ag = 6400 mm²

Transverse Reinforcement

Minimum diameter of tie = 10 mm

Maximum c/c dist.

a) 16db = 400 mm

b) 48db = 480 mm

c) Smaller column dimension = 800 mm

Provide lateral ties: 10 mm @ 250 c/c

Check for Tension

Tension Forces P_u = 548.25 KN M_{uz} = 2.3804 KNm M_{uy} = 262.8 KNm

Assume Strength reduction factor φ = 0.9 Tension controlled *Table 21.2.2 SBC 304*

$P_n = \frac{P_u}{\emptyset}$ = 609.17 KN A_g = 800 x 800 = 640000 mm²

f_c' = 35 N/mm²

$K_n = \frac{P_n}{A_g f_c'}$ = 0.027 ρ_g' = 0.01

γ_z = 0.81875

γ_y = 0.81875

From the tension curve corresponding to the values of γ_y f_y f_c' ρ_s K_n as per interaction diagram generated using clause 3.2.2 of SP-17M(09)

We get,

$\frac{M_{nz}}{A_g f_c' D_z}$ = 0.05 $\frac{M_{ny}}{A_g f_c' D_y}$ = 0.05

M_{nz} = 896 KNm M_{ny} = 896 KNm

$\emptyset M_{nz}$ = 806.4 KNm >2.3804 $\emptyset M_{ny}$ = 806.4 KNm >262.8208

safe **safe**

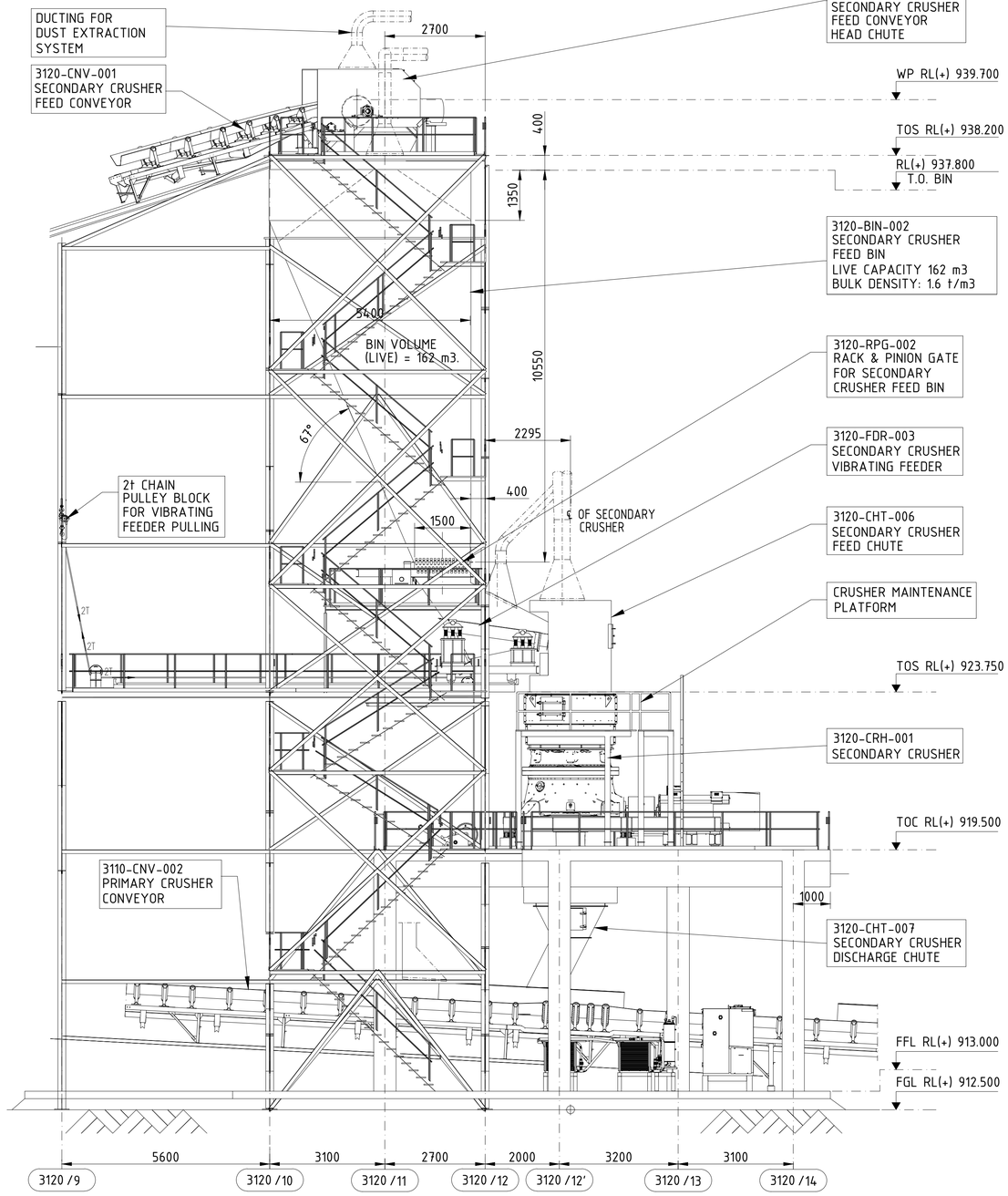
Utility ratio = ***0.00*** *Utility ratio =* ***0.326***

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	C02	D01			Outotec / L&T Revision No.:	0	1	2	3	
Date	27-01-2020	12-03-2020	25-04-2022	13/10/2023			Date	27-01-2020	12-03-2020	25-04-2022	13/10/2023	
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS DT. 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER SUPPORT STRUCTURE	AS BUILT			Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS DT. 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER SUPPORT STRUCTURE	AS BUILT	
Prepared By	APR	APR	APR	APR			Prepared By	APR	APR	APR	APR	
Checked	SBY	SBY	SBY	SBY			Checked	SBY	SBY	SBY	SBY	
Approved	AG	AG	AG	AG			Approved	AG	AG	AG	AG	

ATTACHMENT 1 MECHANICAL GENERAL ARRANGEMENT DRAWING

DRAWING NUMBER: MD-216-3120-EG-ME-ELV-1012 REV A01

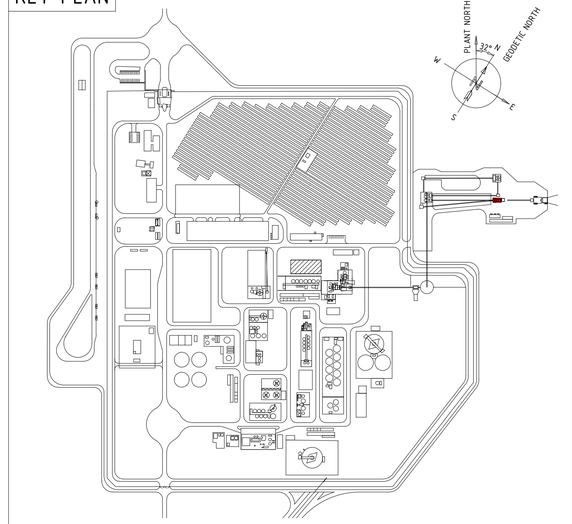


ELEVATION A-A
SCALE 1:75

NOTE:
1. ALL DIMENSIONS ARE IN MILLIMETER, CO-ORDINATES AND ELEVATION ARE IN METER, UNLESS OTHERWISE STATED.
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3. FINISHED GROUND LEVEL (F.G.L) FOR CRUSHING AREA = RL(+) 912.500 m
4. F.F.L, F.G.L, T.O.C, = FINISH FLOOR LEVEL, FINISH GROUND LEVEL, TOP OF CONCRETE, T.O.S, B.O.S, B.O.B= TOP OF STEEL, BOTTOM OF STEEL, BOTTOM OF BASE PLATE
5. ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE ARE SHOWN INDICATIVE.

CONCRETE FLOOR
CHECKERED PLATE
GRATING
BRICK WALL
CONCRETE WALL

KEY PLAN



DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3100-EG-ME-GAD-1013	DUCT - SECONDARY CRUSHER HOUSE								
	DUST EXTRACTION SYSTEM								
MD-216-3120-EG-ME-LPX-1018	SECONDARY CRUSHER HOUSE - PLAN VIEW								
MD-216-3120-EG-ME-LPX-1017	SECONDARY CRUSHER HOUSE - PLAN VIEW								
MD-216-3120-EG-ME-LPX-1016	SECONDARY CRUSHER HOUSE - PLAN VIEW								
MD-216-3120-EG-ME-LPX-1015	SECONDARY CRUSHER HOUSE - PLAN VIEW								
MD-216-3120-EG-ME-LPX-1014	SECONDARY CRUSHER HOUSE - PLAN VIEW								
MD-216-3120-EG-ME-LPX-1013	SECONDARY CRUSHER HOUSE - ELEVATION	A01	02/01/2020	ISSUED FOR APPROVAL	SKH	BRD	BD	BGR	

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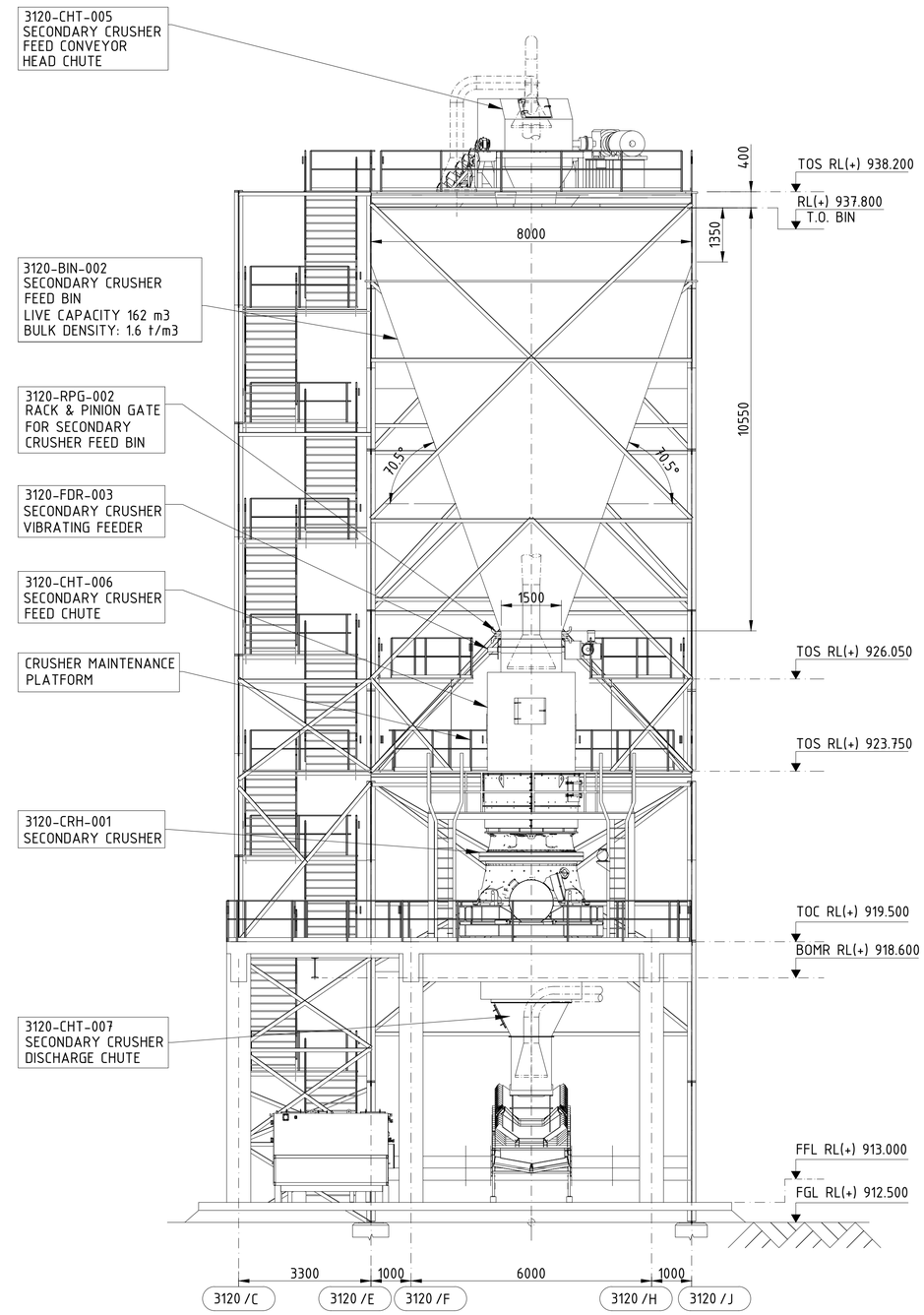
LARSEN & TOUBRO DRAWING NUMBER: O19045_100_H_G_16125_002 — REVISION: 0
OUTOTEC DRAWING NUMBER: 900707__GAU01__ZC01__MTL01__31238 — DOCUMENT ID: OU100669063 — REVISION: 0

DRAWN	SKH	31/12/2019
CHECKED	BRD	02/01/2020
ENGINEER	BD	02/01/2020
APPROVED	BGR	02/01/2020
EMPLOYER		
SCALE	1:75	A1

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
SECONDARY CRUSHING
SECONDARY CRUSHER HOUSE-ELEVATION

DRAWING NUMBER: MD-216-3120-EG-ME-ELV-1012 — REV: A01

DRAWING NUMBER: MD-216-3120-EG-ME-ELV-1012 | REV: A01

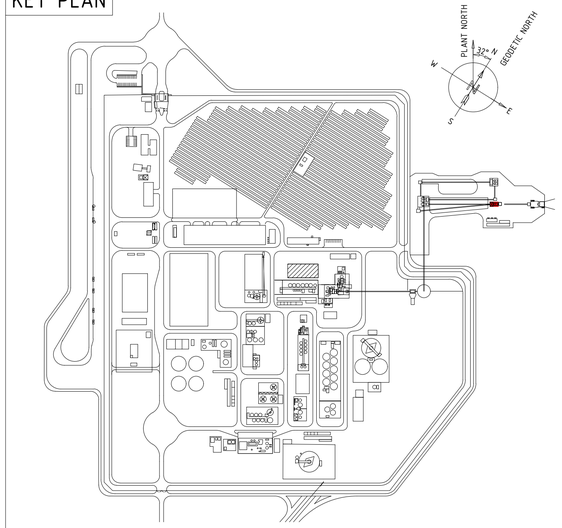


ELEVATION B-B
SCALE 1:75

NOTE:
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3. FINISHED GROUND LEVEL (F.G.L) FOR CRUSHING AREA = RL(+) 912.500 m
4. F.F.L, F.G.L, T.O.C, = FINISH FLOOR LEVEL, FINISH GROUND LEVEL, TOP OF CONCRETE, T.O.S, B.O.S, B.O.B= TOP OF STEEL, BOTTOM OF STEEL, BOTTOM OF BASE PLATE
5. ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE ARE SHOWN INDICATIVE.

CONCRETE FLOOR
CHECKERED PLATE
GRATING
BRICK WALL
CONCRETE WALL

KEY PLAN



DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3100-EG-ME-GAD-1013	DUCT - SECONDARY CRUSHER HOUSE								
	DUST EXTRACTION SYSTEM								
MD-216-3120-EG-ME-LPX-1018	SECONDARY CRUSHER HOUSE - PLAN VIEW								
MD-216-3120-EG-ME-LPX-1017	SECONDARY CRUSHER HOUSE - PLAN VIEW								
MD-216-3120-EG-ME-LPX-1016	SECONDARY CRUSHER HOUSE - PLAN VIEW								
MD-216-3120-EG-ME-LPX-1015	SECONDARY CRUSHER HOUSE - PLAN VIEW								
MD-216-3120-EG-ME-LPX-1014	SECONDARY CRUSHER HOUSE - PLAN VIEW								
MD-216-3120-EG-ME-ELV-1012	SECONDARY CRUSHER HOUSE - ELEVATION	A01	02/01/2020	ISSUED FOR APPROVAL	SKH	BRD	BD	BGR	

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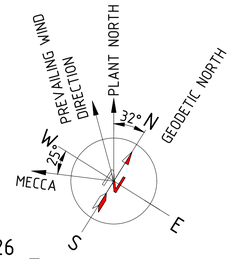
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OUTOTEC DRAWING NUMBER	900707__GAU01__ZC01__MTL01__31239	DOCUMENT ID OU100669064	REVISION 0

DRAWN	SKH	31/12/2019
CHECKED	BRD	02/01/2020
ENGINEER	BD	02/01/2020
APPROVED	BGR	02/01/2020
EMPLOYER		
SCALE	1:75	A1

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
SECONDARY CRUSHING
SECONDARY CRUSHER HOUSE-ELEVATION

DRAWING NUMBER: MD-216-3120-EG-ME-ELV-1013 | REV: A01

DRAWING NUMBER MD-216-3120-EG-ME-LPX-1014 REV A01



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N=2506490.565

3110-CNV-002
PRIMARY CRUSHER
CONVEYOR

3120-CRH-001
SECONDARY CRUSHER

E=264399.362
N=2506500.156

B 002

3120-CNV-001
SECONDARY CRUSHER
FEED CONVEYOR

13.15°

3120 /J
3120 /H
3120 /F
3120 /E
3120 /C

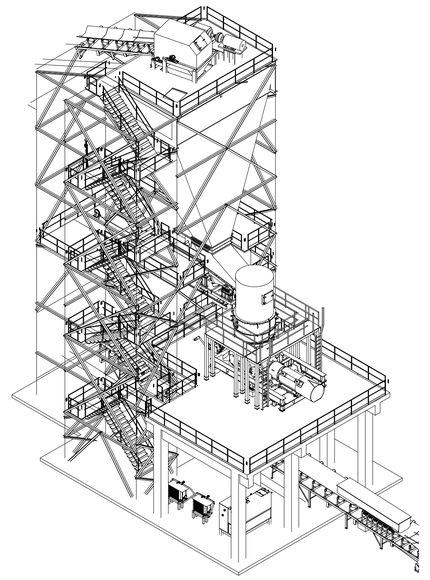
1000
6000
1000
3300

A 002
B 002

3120 /9 | 5600 | 3120 /10 | 3100 | 3120 /11 | 2700 | 3120 /12 | 2000 | 3120 /12' | 3200 | 3120 /13 | 3100 | 3120 /14

PLAN
SCALE 1:50

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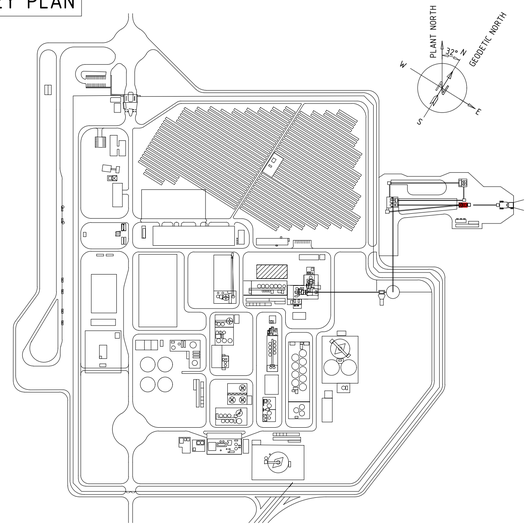


ISO VIEW

NOTE:
1. ALL DIMENSIONS ARE IN MILLIMETER, CO-ORDINATES AND ELEVATION ARE IN METER, UNLESS OTHERWISE STATED.
2. FINISHED FLOOR LEVEL (F.F.L) FOR CRUSHING AREA = RL(+)913.000 m
3. FINISHED GROUND LEVEL (F.G.L) FOR CRUSHING AREA = RL(+)912.500 m
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5. ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE ARE SHOWN INDICATIVE.

CONCRETE FLOOR
CHECKERED PLATE
GRATING
BRICK WALL
CONCRETE WALL

KEY PLAN



DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3120-EG-ME-LPX-1018	SECONDARY CRUSHER HOUSE - PLAN VIEW								
MD-216-3120-EG-ME-LPX-1017	SECONDARY CRUSHER HOUSE - PLAN VIEW								
MD-216-3120-EG-ME-LPX-1016	SECONDARY CRUSHER HOUSE - PLAN VIEW								
MD-216-3120-EG-ME-LPX-1015	SECONDARY CRUSHER HOUSE - PLAN VIEW								
MD-216-3120-EG-ME-LPX-1013	SECONDARY CRUSHER HOUSE - ELEVATION								
MD-216-3120-EG-ME-ELV-1012	SECONDARY CRUSHER HOUSE - ELEVATION	A01	02/01/2020	ISSUED FOR APPROVAL	SKH	BRD	BD	BGR	

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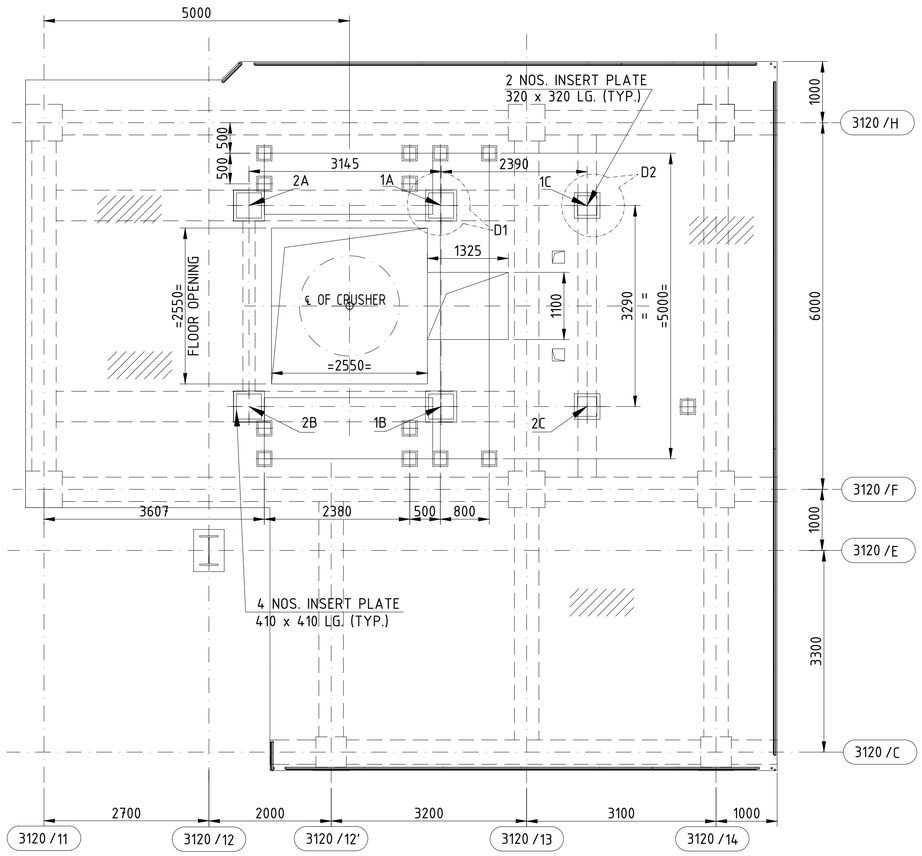
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OUTOTEC DRAWING NUMBER: 900707__GAU01__ZC01__MTL01__31226 — DOCUMENT ID: OU100669045 — REVISION: 0

DRAWN	SKH	31/12/2019	MANSOURAH - MASSARAH GOLD PROJECT
CHECKED	BRD	02/01/2020	GENERAL ARRANGEMENT DRAWING
ENGINEER	BD	02/01/2020	SECONDARY CRUSHING
APPROVED	BGR	02/01/2020	SECONDARY CRUSHER HOUSE-PLAN VIEW
EMPLOYER			
SCALE	1:50	A1	DRAWING NUMBER: MD-216-3120-EG-ME-LPX-1014 — REV: A01

DRAWING NUMBER: MD-216-3120-EG-ME-LPX-1015 | REV: A01

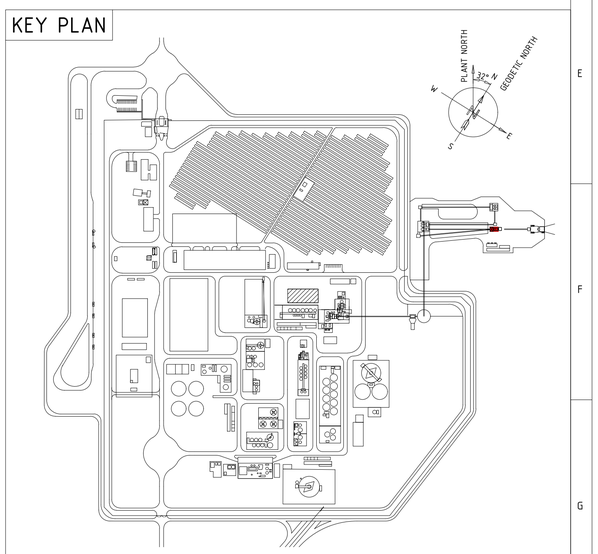


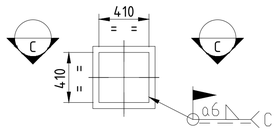
PLAN AT RL(+) 919.500
SCALE 1:50

NOTE:
1. ALL DIMENSIONS ARE IN MILLIMETER, CO-ORDINATES AND ELEVATION ARE IN METER, UNLESS OTHERWISE STATED.
2. FINISHED FLOOR LEVEL (F.F.L) FOR CRUSHING AREA = RL(+)913.000 m
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4. F.F.L, F.G.L, T.O.C, = FINISH FLOOR LEVEL, FINISH GROUND LEVEL, TOP OF CONCRETE, T.O.S, B.O.S, B.O.B= TOP OF STEEL, BOTTOM OF STEEL, BOTTOM OF BASE PLATE
5. ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE ARE SHOWN INDICATIVE.

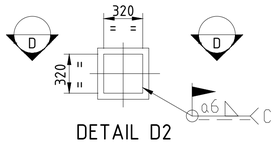
CONCRETE FLOOR
CHECKERED PLATE
GRATING
BRICK WALL
CONCRETE WALL

KEY PLAN

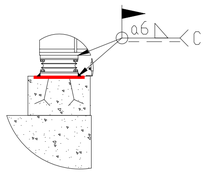




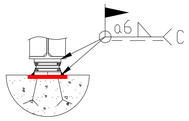
DETAIL D1
SCALE 1:25



DETAIL D2
SCALE 1:25



VIEW C-C
SCALE 1:25



VIEW D-D
SCALE 1:25

TABLE-1: LOAD FOR SECONDARY CRUSHER
3120-CRH-001

	Static Load	Dynamic Load, normal operation f=4,67 Hz		Dynamic Load, start/stop	
Damper position	Vertical	Vertical	Horizontal	Vertical	Horizontal
A [1, 2]	200 kN	± 13 kN	15 kN	± 65 kN	68 kN
B [1, 2]	200 kN	± 14 kN	16 kN	± 70 kN	73 kN
C [1, 2]	45 kN	± 4 kN	2 kN	± 22 kN	12 kN
Vertical load, positive downwards					
Horizontal load, (all horizontal directions)					
Material volume = 6.0 m^3, Density = 1800 kg/m^3					

Drive unit weights	
Component	kg
Sub-frame	1650
Electric motor	7100
Coupling	240

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3120-EG-ME-LPX-1018	SECONDARY CRUSHER HOUSE - PLAN VIEW								
MD-216-3120-EG-ME-LPX-1017	SECONDARY CRUSHER HOUSE - PLAN VIEW								
MD-216-3120-EG-ME-LPX-1016	SECONDARY CRUSHER HOUSE - PLAN VIEW								
MD-216-3120-EG-ME-LPX-1013	SECONDARY CRUSHER HOUSE - ELEVATION								
MD-216-3120-EG-ME-LPX-1012	SECONDARY CRUSHER HOUSE - ELEVATION								
MD-216-3120-EG-ME-LPX-1014	SECONDARY CRUSHER HOUSE - PLAN VIEW	A01	02/01/2020	ISSUED FOR APPROVAL	SKH	BRD	BD	BGR	

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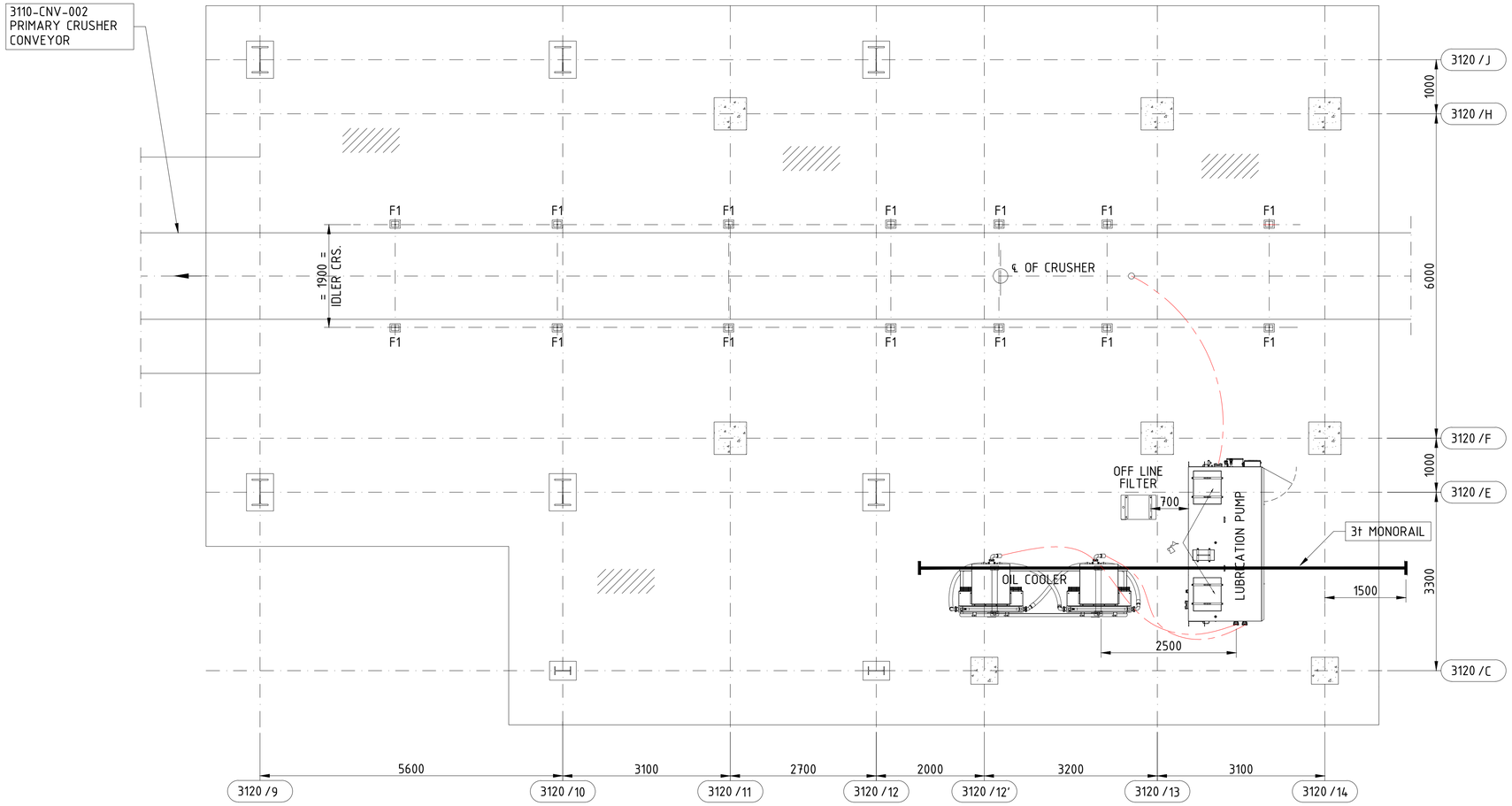
LARSEN & TOUBRO DRAWING NUMBER: 019045_100_H_G_16125_004 | REVISION: 0
OUTOTEC DRAWING NUMBER: 900707__GAU01__ZC01__MTL01__31227 | DOCUMENT ID: OU100669046 | REVISION: 0

DRAWN	SKH	31/12/2019
CHECKED	BRD	02/01/2020
ENGINEER	BD	02/01/2020
APPROVED	BGR	02/01/2020
EMPLOYER		
SCALE	1:50	A1

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
SECONDARY CRUSHING
SECONDARY CRUSHER HOUSE-PLAN VIEW

DRAWING NUMBER: MD-216-3120-EG-ME-LPX-1015 | REV: A01

DRAWING NUMBER MD-216-3120-EG-ME-LPX-1016 REV A01



PLAN AT RL(+) 913.000

SCALE 1:50

NOTE:

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3. FINISHED GROUND LEVEL (F.G.L) FOR CRUSHING AREA = RL(+)912.500 m
4. F.F.L, F.G.L, T.O.C, = FINISH FLOOR LEVEL, FINISH GROUND LEVEL, TOP OF CONCRETE, T.O.S, B.O.S, B.O.B= TOP OF STEEL, BOTTOM OF STEEL, BOTTOM OF BASE PLATE
5. ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE ARE SHOWN INDICATIVE.

CONCRETE FLOOR
CHECKERED PLATE
GRATING
BRICK WALL
CONCRETE WALL

KEY PLAN

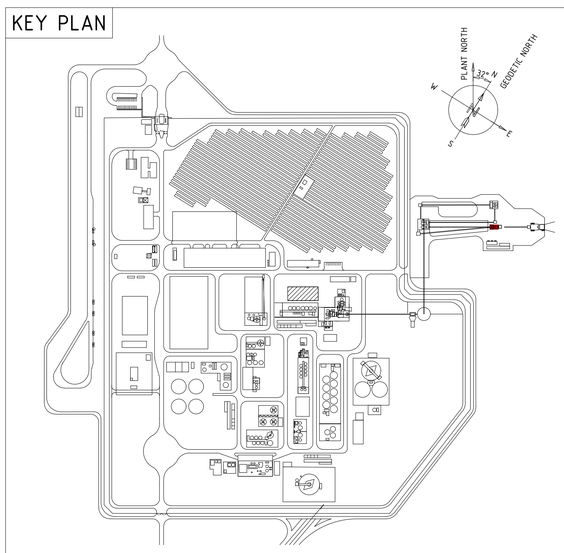


TABLE-2: LOAD FOR CONVEYOR SHORT POST

FOR CONV. 3110-CNV-002

	POINT	ELEVATION IN "m"	VERTICAL LOAD (IN "t")		HORIZONTAL LOAD (IN "t")	
			DEAD LOAD	LIVE LOAD	ALONG	ACROSS
IMPACT ZONE	F1	(+)913.000	(+) 1.09	(+) 0.9	(±) 0.21	(±) 0.11

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3120-EG-ME-LPX-1018	SECONDARY CRUSHER HOUSE - PLAN VIEW								
MD-216-3120-EG-ME-LPX-1017	SECONDARY CRUSHER HOUSE - PLAN VIEW								
MD-216-3120-EG-ME-LPX-1015	SECONDARY CRUSHER HOUSE - PLAN VIEW								
MD-216-3120-EG-ME-LPX-1013	SECONDARY CRUSHER HOUSE - ELEVATION								
MD-216-3120-EG-ME-LPX-1012	SECONDARY CRUSHER HOUSE - ELEVATION								
MD-216-3120-EG-ME-LPX-1014	SECONDARY CRUSHER HOUSE - PLAN VIEW	A01	02/01/2020	ISSUED FOR APPROVAL	SKH	BRD	BD	BGR	

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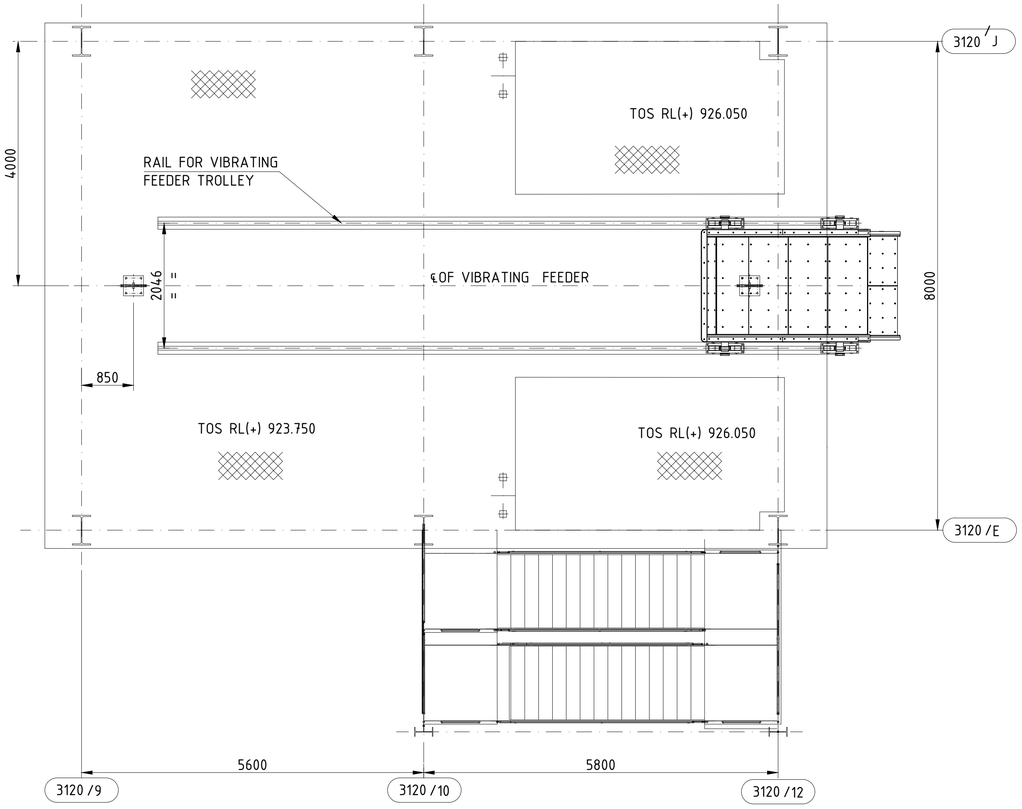
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DRAWN	SKH	31/12/2019
CHECKED	BRD	02/01/2020
ENGINEER	BD	02/01/2020
APPROVED	BGR	02/01/2020
EMPLOYER		
SCALE	1:50	A1

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
SECONDARY CRUSHING
SECONDARY CRUSHER HOUSE-PLAN VIEW

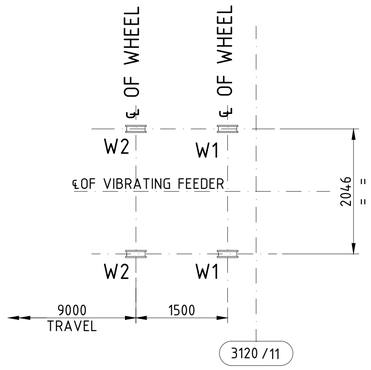
DRAWING NUMBER: MD-216-3120-EG-ME-LPX-1016 — REV: A01

DRAWING NUMBER MD-216-3120-EG-ME-LPX-1017 REV A01



PLAN AT RL(+) 923.750

SCALE 1:50



WHEEL LOAD DIAGRAM

SCALE 1:50

TABLE-3: LOAD FOR WHEEL

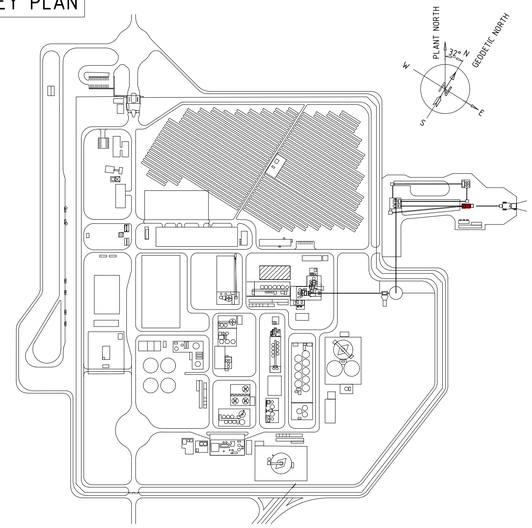
NO.	WHEEL MARKING	LOAD (IN "t")
1	W1	4.0
2	W2	4.0

NOTE:

1. ALL DIMENSIONS ARE IN MILLIMETER, CO-ORDINATES AND ELEVATION ARE IN METER, UNLESS OTHERWISE STATED.
2. FINISHED FLOOR LEVEL (F.F.L) FOR CRUSHING AREA = RL(+)913.000 m
3. FINISHED GROUND LEVEL (F.G.L) FOR CRUSHING AREA = RL(+)912.500 m
4. F.F.L, F.G.L, T.O.C, = FINISH FLOOR LEVEL, FINISH GROUND LEVEL, TOP OF CONCRETE, T.O.S, B.O.S, B.O.B= TOP OF STEEL, BOTTOM OF STEEL, BOTTOM OF BASE PLATE
5. ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE ARE SHOWN INDICATIVE.

CONCRETE FLOOR
CHECKERED PLATE
GRATING
BRICK WALL
CONCRETE WALL

KEY PLAN



DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3120-EG-ME-LPX-1018	SECONDARY CRUSHER HOUSE - PLAN VIEW								
MD-216-3120-EG-ME-LPX-1016	SECONDARY CRUSHER HOUSE - PLAN VIEW								
MD-216-3120-EG-ME-LPX-1015	SECONDARY CRUSHER HOUSE - PLAN VIEW								
MD-216-3120-EG-ME-LPX-1013	SECONDARY CRUSHER HOUSE - ELEVATION								
MD-216-3120-EG-ME-LPX-1012	SECONDARY CRUSHER HOUSE - ELEVATION								
MD-216-3120-EG-ME-LPX-1014	SECONDARY CRUSHER HOUSE - PLAN VIEW	A01	02/01/2020	ISSUED FOR APPROVAL	SKH	BRD	BD	BGR	

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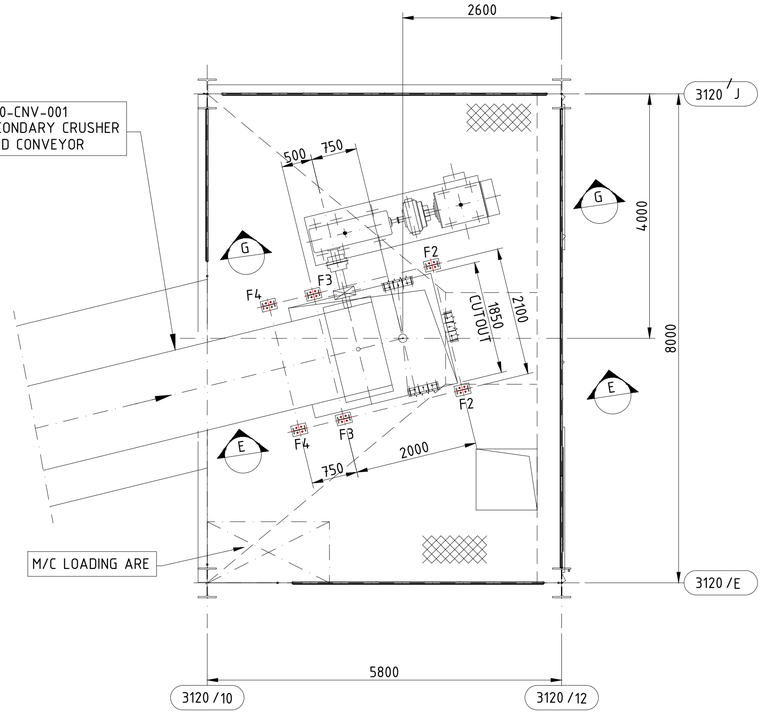
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DRAWN	SKH	31/12/2019
CHECKED	BRD	02/01/2020
ENGINEER	BD	02/01/2020
APPROVED	BGR	02/01/2020
EMPLOYER		
SCALE	1:50	A1

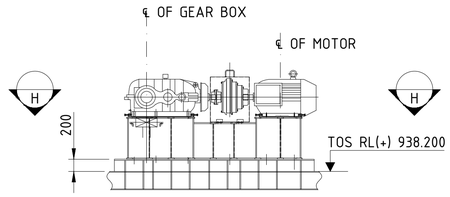
MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
SECONDARY CRUSHING
SECONDARY CRUSHER HOUSE-PLAN VIEW

DRAWING NUMBER: MD-216-3120-EG-ME-LPX-1017 — REV: A01

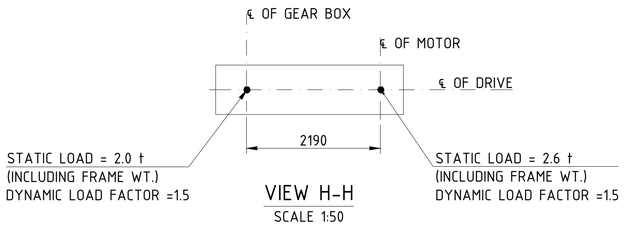
DRAWING NUMBER MD-216-3120-EG-ME-LPX-1018 REV A01



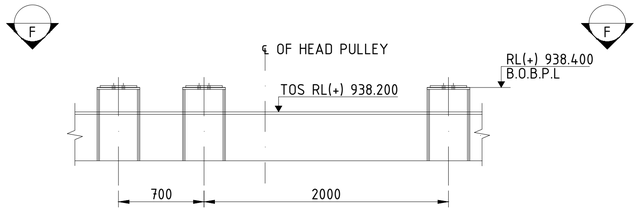
PLAN AT RL(+) 938.200
SCALE 1:50



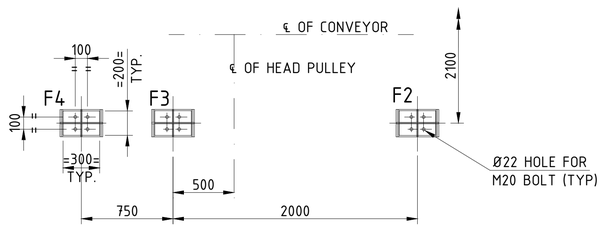
SECTION G-G
SCALE 1:50



VIEW H-H
SCALE 1:50



SECTION E-E
SCALE 1:25

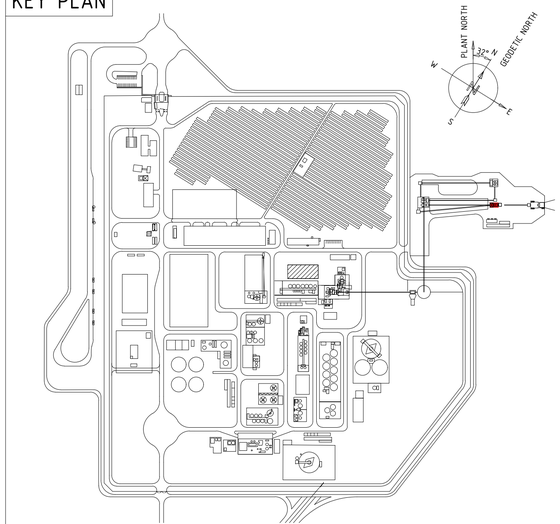


VIEW F-F
SCALE 1:25

NOTE:
1. ALL DIMENSIONS ARE IN MILLIMETER, CO-ORDINATES AND ELEVATION ARE IN METER, UNLESS OTHERWISE STATED.
2. FINISHED FLOOR LEVEL (F.F.L) FOR CRUSHING AREA = RL(+)913.000 m
3. FINISHED GROUND LEVEL (F.G.L) FOR CRUSHING AREA = RL(+)912.500 m
4. F.F.L, F.G.L, T.O.C, = FINISH FLOOR LEVEL, FINISH GROUND LEVEL, TOP OF CONCRETE, T.O.S, B.O.S, B.O.B= TOP OF STEEL, BOTTOM OF STEEL, BOTTOM OF BASE PLATE
5. ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE ARE SHOWN INDICATIVE.

CONCRETE FLOOR
CHECKERED PLATE
GRATING
BRICK WALL
CONCRETE WALL

KEY PLAN



SIGN CONVENTION

(V) VERTICAL LOAD : UPWARD (-)
: DOWNWARD (+)
(H) HORIZONTAL LOAD : TOWARDS HEAD PULLEY=(+)
: TOWARDS TAIL PULLEY=(-)
: REVERSIBLE LOAD=(±)

TABLE-4 : FOR DRIVE PULLEY FRAME LOAD

FOR CONV. 3120-CNV-001

POINTS		ELEVATIONS IN "m"	LOAD CASE I			LOAD CASE II		
			V (IN "t")	H ALONG (IN "t")	H ACROSS (IN "t")	V (IN "t")	H ALONG (IN "t")	H ACROSS (IN "t")
F2		(+)938.200	(+)2.064	(-)0.965	(±)0.21	(-)1.097	(-)2.258	(±)0.11
F3			(+)3.917	(-)2.378	(±)0.40	(+)3.331	(-)2.453	(±)0.33
F4			(+)5.020	(-)3.361	(±)0.50	(+)6.268	(-)4.139	(±)0.63

NOTE:
CASE I : CONVEYOR NORMAL RUNNING + MATERIAL LOAD + DEAD LOAD
CASE II : CONVEYOR STARTING + CHUTE EMPTY + DEAD LOAD

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3120-EG-ME-LPX-1017	SECONDARY CRUSHER HOUSE - PLAN VIEW								
MD-216-3120-EG-ME-LPX-1016	SECONDARY CRUSHER HOUSE - PLAN VIEW								
MD-216-3120-EG-ME-LPX-1015	SECONDARY CRUSHER HOUSE - PLAN VIEW								
MD-216-3120-EG-ME-LPX-1013	SECONDARY CRUSHER HOUSE - ELEVATION								
MD-216-3120-EG-ME-LPX-1012	SECONDARY CRUSHER HOUSE - ELEVATION								
MD-216-3120-EG-ME-LPX-1014	SECONDARY CRUSHER HOUSE - PLAN VIEW	A01	02/01/2020	ISSUED FOR APPROVAL	SKH	BRD	BD	BGR	

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LARSEN & TOUBRO DRAWING NUMBER: O19045_100_H_G_16125_007 — REVISION: 0
OUTOTEC DRAWING NUMBER: 900707__GAU01__ZC01__MTL01__31230 — DOCUMENT ID: OU100669049 — REVISION: 0

DRAWN	SKH	31/12/2019
CHECKED	BRD	02/01/2020
ENGINEER	BD	02/01/2020
APPROVED	BGR	02/01/2020
EMPLOYER		
SCALE	1:50	A1

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
SECONDARY CRUSHING
SECONDARY CRUSHER HOUSE-PLAN VIEW

DRAWING NUMBER: MD-216-3120-EG-ME-LPX-1018 — REV: A01

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	C02	D01			Outotec / L&T Revision No.:	0	1	2	3	
Date	27-01-2020	12-03-2020	25-04-2022	13/10/2023			Date	27-01-2020	12-03-2020	25-04-2022	13/10/2023	
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS DT. 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER SUPPORT STRUCTURE	AS BUILT			Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS DT. 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER SUPPORT STRUCTURE	AS BUILT	
Prepared By	APR	APR	APR	APR			Prepared By	APR	APR	APR	APR	
Checked	SBY	SBY	SBY	SBY			Checked	SBY	SBY	SBY	SBY	
Approved	AG	AG	AG	AG			Approved	AG	AG	AG	AG	

ATTACHMENT 2 STAAD INPUT FILE OF CRUSHER SUPPORT STRUCTURE

```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 08-Jan-20
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
1 3.1 -2.5 2; 2 11 -2.5 2; 3 14.1 -2.5 2; 4 3.1 -2.5 8; 5 11 -2.5 8;
6 14.1 -2.5 8; 7 8.8 -2.5 12.3; 8 14.1 -2.5 12.3; 9 3.1 0 2; 10 11 0 2;
11 14.1 0 2; 12 3.1 6.5 2; 13 11 6.5 2; 14 14.1 6.5 2; 15 3.1 0 8; 16 11 0 8;
17 14.1 0 8; 18 3.1 6.5 8; 19 11 6.5 8; 20 14.1 6.5 8; 21 14.1 0 12.3;
22 14.1 6.5 12.3; 23 8.8 0 12.3; 24 8.8 6.5 12.3; 25 8.8 0 8; 26 3.1 6.5 3.355;
27 3.1 6.5 6.645; 28 11 6.5 3.355; 29 11 6.5 6.645; 30 6.455 6.5 3.355;
31 6.455 6.5 6.645; 32 8.8 6.5 8; 33 11 6.5 12.3; 34 14.1 6.5 3.355;
35 14.1 6.5 6.645; 36 3.1 6.5 1; 37 11 6.5 1; 38 14.1 6.5 1; 39 15.1 6.5 2;
40 15.1 6.5 8; 41 15.1 6.5 12.3; 42 15.1 6.5 1; 43 5.8 0 2; 44 5.8 0 8;
45 8.05 6.5 12.3; 46 8.05 6.5 8; 48 6.455 6.5 1; 50 15.1 6.5 3.355;
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451 3.1 2.5 8; 452 3.1 3 8; 453 3.1 3.5 8; 454 3.1 4 8; 455 3.1 4.5 8;
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MEMBER INCIDENCES
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1450 1350 1351 1358 1357; 1451 1351 1352 1359 1358; 1452 1352 1353 1360 1359;
1453 1353 1354 1361 1360; 1454 1354 1355 1362 1361; 1455 1355 545 564 1362;
1456 1053 1363 1370 1062; 1457 1363 1357 1364 1370; 1458 1357 1358 1365 1364;
1459 1358 1359 1366 1365; 1460 1359 1360 1367 1366; 1461 1360 1361 1368 1367;
1462 1361 1362 1369 1368; 1463 1362 564 583 1369; 1464 1062 1370 1377 1071;
1465 1370 1364 1371 1377; 1466 1364 1365 1372 1371; 1467 1365 1366 1373 1372;
1468 1366 1367 1374 1373; 1469 1367 1368 1375 1374; 1470 1368 1369 1376 1375;
1471 1369 583 602 1376; 1472 1071 1377 1384 1080; 1473 1377 1371 1378 1384;
1474 1371 1372 1379 1378; 1475 1372 1373 1380 1379; 1476 1373 1374 1381 1380;
1477 1374 1375 1382 1381; 1478 1375 1376 1383 1382; 1479 1376 602 621 1383;
1480 1080 1384 1391 1089; 1481 1384 1378 1385 1391; 1482 1378 1379 1386 1385;
1483 1379 1380 1387 1386; 1484 1380 1381 1388 1387; 1485 1381 1382 1389 1388;
1486 1382 1383 1390 1389; 1487 1383 621 640 1390; 1488 1089 1391 1398 1098;
1489 1391 1385 1392 1398; 1490 1385 1386 1393 1392; 1491 1386 1387 1394 1393;
1492 1387 1388 1395 1394; 1493 1388 1389 1396 1395; 1494 1389 1390 1397 1396;
1495 1390 640 659 1397; 1496 1098 1398 1405 1107; 1497 1398 1392 1399 1405;

1498 1392 1393 1400 1399; 1499 1393 1394 1401 1400; 1500 1394 1395 1402 1401;
1501 1395 1396 1403 1402; 1502 1396 1397 1404 1403; 1503 1397 659 678 1404;
1513 23 1406 1415 1017; 1514 1406 1407 1416 1415; 1515 1407 1408 1417 1416;
1516 1408 1409 1418 1417; 1517 1409 1410 1419 1418; 1518 1410 1411 1420 1419;
1519 1411 1412 1421 1420; 1520 1412 1413 1422 1421; 1521 1413 1414 1423 1422;
1522 1414 21 1139 1423; 1523 1017 1415 1424 1026; 1524 1415 1416 1425 1424;
1525 1416 1417 1426 1425; 1526 1417 1418 1427 1426; 1527 1418 1419 1428 1427;
1528 1419 1420 1429 1428; 1529 1420 1421 1430 1429; 1530 1421 1422 1431 1430;
1531 1422 1423 1432 1431; 1532 1423 1139 1148 1432; 1533 1026 1424 1433 1035;
1534 1424 1425 1434 1433; 1535 1425 1426 1435 1434; 1536 1426 1427 1436 1435;
1537 1427 1428 1437 1436; 1538 1428 1429 1438 1437; 1539 1429 1430 1439 1438;
1540 1430 1431 1440 1439; 1541 1431 1432 1441 1440; 1542 1432 1148 1157 1441;
1543 1035 1433 1442 1044; 1544 1433 1434 1443 1442; 1545 1434 1435 1444 1443;
1546 1435 1436 1445 1444; 1547 1436 1437 1446 1445; 1548 1437 1438 1447 1446;
1549 1438 1439 1448 1447; 1550 1439 1440 1449 1448; 1551 1440 1441 1450 1449;
1552 1441 1157 1166 1450; 1553 1044 1442 1451 1053; 1554 1442 1443 1452 1451;
1555 1443 1444 1453 1452; 1556 1444 1445 1454 1453; 1557 1445 1446 1455 1454;
1558 1446 1447 1456 1455; 1559 1447 1448 1457 1456; 1560 1448 1449 1458 1457;
1561 1449 1450 1459 1458; 1562 1450 1166 1175 1459; 1563 1053 1451 1460 1062;
1564 1451 1452 1461 1460; 1565 1452 1453 1462 1461; 1566 1453 1454 1463 1462;
1567 1454 1455 1464 1463; 1568 1455 1456 1465 1464; 1569 1456 1457 1466 1465;
1570 1457 1458 1467 1466; 1571 1458 1459 1468 1467; 1572 1459 1175 1184 1468;
1573 1062 1460 1469 1071; 1574 1460 1461 1470 1469; 1575 1461 1462 1471 1470;
1576 1462 1463 1472 1471; 1577 1463 1464 1473 1472; 1578 1464 1465 1474 1473;
1579 1465 1466 1475 1474; 1580 1466 1467 1476 1475; 1581 1467 1468 1477 1476;
1582 1468 1184 1193 1477; 1583 1071 1469 1478 1080; 1584 1469 1470 1479 1478;
1585 1470 1471 1480 1479; 1586 1471 1472 1481 1480; 1587 1472 1473 1482 1481;
1588 1473 1474 1483 1482; 1589 1474 1475 1484 1483; 1590 1475 1476 1485 1484;
1591 1476 1477 1486 1485; 1592 1477 1193 1202 1486; 1593 1080 1478 1487 1089;
1594 1478 1479 1488 1487; 1595 1479 1480 1489 1488; 1596 1480 1481 1490 1489;
1597 1481 1482 1491 1490; 1598 1482 1483 1492 1491; 1599 1483 1484 1493 1492;
1600 1484 1485 1494 1493; 1601 1485 1486 1495 1494; 1602 1486 1202 1211 1495;
1603 1089 1487 1496 1098; 1604 1487 1488 1497 1496; 1605 1488 1489 1498 1497;
1606 1489 1490 1499 1498; 1607 1490 1491 1500 1499; 1608 1491 1492 1501 1500;
1609 1492 1493 1502 1501; 1610 1493 1494 1503 1502; 1611 1494 1495 1504 1503;
1612 1495 1211 1220 1504; 1613 1098 1496 1505 1107; 1614 1496 1497 1506 1505;
1615 1497 1498 1507 1506; 1616 1498 1499 1508 1507; 1617 1499 1500 1509 1508;
1618 1500 1501 1510 1509; 1619 1501 1502 1511 1510; 1620 1502 1503 1512 1511;
1621 1503 1504 1513 1512; 1622 1504 1220 1229 1513; 1623 452 1524 1535 453;
1624 1524 1523 1534 1535; 1625 1523 1522 1533 1534; 1626 1522 1521 1532 1533;
1627 1521 1520 1531 1532; 1628 1520 1519 1530 1531; 1629 1519 1518 1529 1530;
1630 1518 1517 1528 1529; 1631 1517 1516 1527 1528; 1632 1516 1515 1526 1527;
1633 1515 1514 1525 1526; 1634 1514 737 738 1525; 1635 453 1535 1546 454;
1636 1535 1534 1545 1546; 1637 1534 1533 1544 1545; 1638 1533 1532 1543 1544;
1639 1532 1531 1542 1543; 1640 1531 1530 1541 1542; 1641 1530 1529 1540 1541;
1642 1529 1528 1539 1540; 1643 1528 1527 1538 1539; 1644 1527 1526 1537 1538;
1645 1526 1525 1536 1537; 1646 1525 738 739 1536; 1647 454 1546 1557 455;
1648 1546 1545 1556 1557; 1649 1545 1544 1555 1556; 1650 1544 1543 1554 1555;
1651 1543 1542 1553 1554; 1652 1542 1541 1552 1553; 1653 1541 1540 1551 1552;
1654 1540 1539 1550 1551; 1655 1539 1538 1549 1550; 1656 1538 1537 1548 1549;
1657 1537 1536 1547 1548; 1658 1536 739 740 1547; 1659 455 1557 1568 456;
1660 1557 1556 1567 1568; 1661 1556 1555 1566 1567; 1662 1555 1554 1565 1566;
1663 1554 1553 1564 1565; 1664 1553 1552 1563 1564; 1665 1552 1551 1562 1563;
1666 1551 1550 1561 1562; 1667 1550 1549 1560 1561; 1668 1549 1548 1559 1560;

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1669 1548 1547 1558 1559; 1670 1547 740 741 1558; 1671 456 1568 1579 457;
1672 1568 1567 1578 1579; 1673 1567 1566 1577 1578; 1674 1566 1565 1576 1577;
1675 1565 1564 1575 1576; 1676 1564 1563 1574 1575; 1677 1563 1562 1573 1574;
1678 1562 1561 1572 1573; 1679 1561 1560 1571 1572; 1680 1560 1559 1570 1571;
1681 1559 1558 1569 1570; 1682 1558 741 742 1569;
ELEMENT PROPERTY
469 TO 688 789 TO 1008 1052 TO 1059 1062 TO 1069 1072 TO 1079 1082 TO 1089 -
1092 TO 1099 1102 TO 1109 1112 TO 1119 1122 TO 1127 1132 TO 1137 -
1142 TO 1147 1152 TO 1157 1162 TO 1169 1177 TO 1184 1223 TO 1284 -
1287 TO 1292 1295 TO 1300 1303 TO 1308 1311 TO 1326 1416 TO 1503 -
1513 TO 1622 THICKNESS 0.3
1623 TO 1682 THICKNESS 0.5
MEMBER RELEASE
7 TO 9 17 TO 19 25 TO 55 59 60 67 TO 70 72 1033 TO 1038 1041 1042 -
1683 START MPX 0.99
8 9 18 26 TO 30 33 TO 45 47 TO 55 67 70 TO 72 80 89 440 447 452 757 770 1033 -
1038 1041 1042 1049 1176 1191 1222 1415 1512 1683 END MPX 0.99
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.7806e+07
POISSON 0.17
DENSITY 25
ALPHA 1e-05
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
1 2 11 12 412 TO 423 729 TO 740 PRIS YD 0.7 ZD 0.7
21 TO 24 1050 1060 1070 1080 1090 1100 1110 1120 1130 1140 1150 1160 1192 -
1194 1196 1198 1200 1202 1204 1206 1208 1210 1212 1214 PRIS YD 0.5 ZD 0.5
25 27 TO 30 32 70 1409 TO 1415 1504 TO 1512 PRIS YD 0.5 ZD 0.3
7 8 17 18 31 68 69 436 TO 452 753 TO 770 PRIS YD 0.55 ZD 0.3
40 45 47 49 PRIS YD 0.5 ZD 0.3
9 19 46 50 51 71 PRIS YD 0.7 ZD 0.4
41 TO 44 PRIS YD 0.9 ZD 0.4
MEMBER PROPERTY AMERICAN
33 TO 39 52 54 PRIS YD 1 ZD 0.4
MEMBER PROPERTY AMERICAN
56 TO 62 64 TO 66 72 80 87 89 1683 PRIS YD 0.4 ZD 0.3
MEMBER PROPERTY AMERICAN
53 55 PRIS YD 0.9 ZD 0.4
3 4 13 14 424 TO 435 741 TO 752 PRIS YD 0.7 ZD 0.7
5 6 15 16 PRIS YD 0.5 ZD 0.5
26 48 PRIS YD 0.7 ZD 0.4
MEMBER PROPERTY
1030 1032 1051 1061 1071 1081 1091 1101 1111 1121 1131 1141 1151 1161 1193 -
1195 1197 1199 1201 1203 1205 1207 1209 1211 1213 1215 1331 -
1360 PRIS YD 1 ZD 1
1038 PRIS YD 0.5 ZD 0.5
1034 1035 1043 TO 1049 1185 TO 1191 PRIS YD 0.4 ZD 0.3
63 67 73 1039 TO 1042 PRIS YD 0.2 ZD 0.3
1036 1037 1170 TO 1176 1216 TO 1222 PRIS YD 0.5 ZD 0.3
1033 PRIS YD 1 ZD 0.5
```

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CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
1 TO 8 1257 1302 FIXED
SLAVE ZX MASTER 19 JOINT 12 TO 14 18 TO 20 22 24 26 TO 42
DEFINE IBC 2003
SDS 0.0667 SD1 0.0133 S1 0.02 IE 1 RX 2.5 RZ 2.5 SCLASS 2 PX 0.237 PZ 0.237
SELFWEIGHT 1
*CRUSHER static LOAD + 25% Material Load = 200+2.7/4
JOINT WEIGHT
30 31 WEIGHT 200.7
MEMBER WEIGHT
42 44 CON 200.7 3.145
53 55 CON 45 0.99
*CONVEYOR walway support, LOAD DL + 25% OF LL
70 CON 17 0.7
70 CON 17 4.9
28 CON 9.3 0.7
28 CON 9.3 4.9
*Chute Load
42 44 CON 20 0.45
42 44 CON 20 3.245
*Duct load
41 CON 7.5 1.443
41 CON 7.5 2.573
43 CON 5 1.443
43 CON 5 2.573
6 CON 10 2.6
*Cable Tray Load
22 26 48 1030 1192 TO 1215 UNI 5
9 19 41 43 CON 14 2.95
FLOOR WEIGHT
**Floor DL
YRANGE 6.4 6.6 FLOAD 9.6 XRANGE 3 15.2 ZRANGE 0.9 8
YRANGE 6.4 6.6 FLOAD 5 XRANGE 8 15.2 ZRANGE 8 16.4
**Floor 25% Of LL
YRANGE 6.4 6.6 FLOAD 1.25 XRANGE 3 15.2 ZRANGE 0.9 16.4
CUT OFF MODE SHAPE 50
LOAD 11 SEISMIC (+X)
IBC LOAD X 1
LOAD 12 SEISMIC (+Z)
IBC LOAD Z 1
LOAD 1 DEAD LOAD
SELFWEIGHT Y -1
FLOOR LOAD
YRANGE 6.4 6.6 FLOAD -9.6 XRANGE 3 15.2  ZRANGE 0.9 8  GY
YRANGE 6.4 6.6 FLOAD -5 XRANGE 8 15.2  ZRANGE 8 16.4  GY
*CONVEYOR Walwway Dead Load
MEMBER LOAD
70 CON GY -10 0.7
70 CON GY -10 4.9
28 CON GY -5 0.7
28 CON GY -5 4.9
*******Equipments Dead Load*********
```

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*Crusher Static Load
JOINT LOAD
30 FY -200
31 FY -200
MEMBER LOAD
42 CON GY -200 3.145
44 CON GY -200 3.145
53 55 CON GY -45 0.99
*Crusher Material Load
JOINT LOAD
30 31 FY -2.7
MEMBER LOAD
42 44 CON GY -2.7 3.145
*Chute Load
42 44 CON GY -20 0.45
42 44 CON GY -20 3.245
*Conveyor Support Load
MEMBER LOAD
70 CON GY -55 0.7
70 CON GY -55 4.9
28 CON GY -42 0.7
28 CON GY -42 4.9
*Duct load
41 CON GY -7.5 1.443
41 CON GY -7.5 2.573
43 CON GY -5 1.443
43 CON GY -5 2.573
6 CON GY -10 2.6
6 CMOM GZ -10 2.6
*Cable Tray Load
22 26 48 1192 1194 1196 1198 1200 1202 1204 1206 1208 1210 1212 1214 UNI GY -5
41 43 CON GY -14 2.95
*Monorail Beam Load of 3T
40 49 CON GY -40 2.4
47 CON GY -40 1.9
LOAD 2 LIVE LOAD
FLOOR LOAD
YRANGE 6.4 6.6 FLOAD -5 XRANGE 3 15.2  ZRANGE 0.9 16.4  GY
*CONVEYOR walway Live Load
MEMBER LOAD
70 CON GY -27 0.7
70 CON GY -27 4.9
28 CON GY -17 0.7
28 CON GY -17 4.9
LOAD 3 SDL1_R - FOR CRUSHER DYANMIC LOAD
JOINT LOAD
30 FY -65 FZ 68
31 FY -70 FZ 73
MEMBER LOAD
42 CON GY -65 3.145
44 CON GY -70 3.145
42 CON GZ 68 3.145
44 CON GZ 73 3.145
53 55 CON GY -22 0.99
```

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53 55 CON GZ 12 0.99
LOAD 4 SDL2_R - FOR CRUSHER DYANMIC LOAD
JOINT LOAD
30 FX 68 FY -65
31 FX 73 FY -70
MEMBER LOAD
42 CON GY -65 3.145
44 CON GY -70 3.145
42 CON GX 68 3.145
44 CON GX 73 3.145
53 55 CON GY -22 0.99
53 55 CON GX 12 0.99
LOAD 5 WIND LOAD IN +X DIRN
MEMBER LOAD
2 4 6 12 412 TO 423 729 TO 752 UNI GX 3.28
14 16 424 TO 435 UNI GX 5.64
22 24 1050 1060 1070 1080 1090 1100 1110 1120 1130 1140 1150 1160 1192 1194 -
1196 1198 1200 1202 1204 1206 1208 1210 1212 1214 UNI GX 4.54
1030 1032 1051 1061 1071 1081 1091 1101 1111 1121 1131 1141 1151 1161 1193 -
1195 1197 1199 1201 1203 1205 1207 1209 1211 1213 1215 UNI GX 2.19
*WL due to Crusher
42 44 CON GX 3.93 3.145
42 44 CON GY -5.22 3.145
JOINT LOAD
30 31 FX 3.93 FY 5.22
LOAD 6 WIND LOAD IN -X DIRN
MEMBER LOAD
2 4 6 12 412 TO 423 729 TO 752 UNI GX -3.28
12 14 16 412 TO 435 UNI GX -5.64
22 24 1050 1060 1070 1080 1090 1100 1110 1120 1130 1140 1150 1160 1192 1194 -
1196 1198 1200 1202 1204 1206 1208 1210 1212 1214 UNI GX -4.54
1030 1032 1051 1061 1071 1081 1091 1101 1111 1121 1131 1141 1151 1161 1193 -
1195 1197 1199 1201 1203 1205 1207 1209 1211 1213 1215 UNI GX -2.19
*WL due to Crusher
42 44 CON GX -3.93 3.145
42 44 CON GY 5.22 3.145
JOINT LOAD
30 31 FX -3.93 FY -5.22
LOAD 7 WIND LOAD IN +Z DIRN
MEMBER LOAD
2 12 412 TO 423 729 TO 740 UNI GZ 4.38
4 14 424 TO 435 741 TO 752 UNI GZ 6.1
6 16 UNI GZ 1.72
22 24 1030 1032 1050 1051 1060 1061 1070 1071 1080 1081 1090 1091 1100 1101 -
1110 1111 1120 1121 1130 1131 1140 1141 1150 1151 1160 1161 1192 TO 1214 -
1215 UNI GZ 2.94
*WL due to Crusher
42 44 CON GZ 5.21 3.145
42 CON GY 6.62 3.145
44 CON GY -6.62 3.145
JOINT LOAD
30 FY 6.62 FZ 5.21
31 FY -6.62 FZ 5.21
LOAD 8 WIND LOAD IN -Z DIRN
```

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MEMBER LOAD
2 12 412 TO 423 729 TO 740 UNI GZ -1.7
4 14 24 424 TO 435 741 TO 752 1032 1050 1051 1060 1061 1070 1071 1080 1081 -
1090 1091 1100 1101 1110 1111 1120 1121 1130 1131 1140 1141 1150 1151 1160 -
1161 UNI GZ -2.17
6 16 22 1030 1192 TO 1215 UNI GZ -0.67
*WL due to Crusher
42 44 CON GZ -5.21 3.145
42 CON GY -6.62 3.145
44 CON GY 6.62 3.145
JOINT LOAD
30 FY -6.62 FZ -5.21
31 FY 6.62 FZ -5.21
************************************************
*LOAD COMBINATION FOR BASE PRESSURE & Servicibility
*********************************************
LOAD 301 DL+SDL1_R
REPEAT LOAD
1 1.0 3 1.0
LOAD 302 DL+SDL2_R
REPEAT LOAD
1 1.0 4 1.0
LOAD 303 DL-SDL1_R
REPEAT LOAD
1 1.0 3 -1.0
LOAD 304 DL-SDL2_R
REPEAT LOAD
1 1.0 4 -1.0
**********************************
LOAD 305 DL + SDL1_R + LL
REPEAT LOAD
1 1.0 3 1.0 2 1.0
LOAD 306 DL + SDL2_R + LL
REPEAT LOAD
1 1.0 4 1.0 2 1.0
LOAD 307 DL - SDL1_R + LL
REPEAT LOAD
1 1.0 3 -1.0 2 1.0
LOAD 308 DL - SDL2_R + LL
REPEAT LOAD
1 1.0 4 -1.0 2 1.0
************************************************
LOAD 309 DL + SDL1_R + 0.75 LL + 0.75 WL(+X)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 5 0.75
LOAD 310 DL + SDL1_R + 0.75 LL + 0.75 WL(-X)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 6 0.75
LOAD 311 DL + SDL1_R + 0.75 LL + 0.75 WL(+Z)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 7 0.75
LOAD 312 DL + SDL1_R + 0.75 LL + 0.75 WL(-Z)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 8 0.75
```

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LOAD 313 DL + SDL2_R + 0.75 LL + 0.75 WL(+X)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 5 0.75
LOAD 314 DL + SDL2_R + 0.75 LL + 0.75 WL(-X)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 6 0.75
LOAD 315 DL + SDL2_R + 0.75 LL + 0.75 WL(+Z)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 7 0.75
LOAD 316 DL + SDL2_R + 0.75 LL + 0.75 WL(-Z)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 8 0.75
LOAD 317 DL - SDL1_R + 0.75 LL + 0.75 WL(+X)
REPEAT LOAD
1 1.0 3 -1.0 2 0.75 5 0.75
LOAD 318 DL - SDL1_R + 0.75 LL + 0.75 WL(-X)
REPEAT LOAD
1 1.0 3 -1.0 2 0.75 6 0.75
LOAD 319 DL - SDL1_R + 0.75 LL + 0.75 WL(+Z)
REPEAT LOAD
1 1.0 3 -1.0 2 0.75 7 0.75
LOAD 320 DL - SDL1_R + 0.75 LL + 0.75 WL(-Z)
REPEAT LOAD
1 1.0 3 -1.0 2 0.75 8 0.75
LOAD 321 DL - SDL2_R + 0.75 LL + 0.75 WL(+X)
REPEAT LOAD
1 1.0 4 -1.0 2 0.75 5 0.75
LOAD 322 DL - SDL2_R + 0.75 LL + 0.75 WL(-X)
REPEAT LOAD
1 1.0 4 -1.0 2 0.75 6 0.75
LOAD 323 DL - SDL2_R + 0.75 LL + 0.75 WL(+Z)
REPEAT LOAD
1 1.0 4 -1.0 2 0.75 7 0.75
LOAD 324 DL - SDL2_R + 0.75 LL + 0.75 WL(-Z)
REPEAT LOAD
1 1.0 4 -1.0 2 0.75 8 0.75
*********************************************
LOAD 325 DL + SDL1_R + 0.75 LL + 0.525 EL(+X)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 11 0.525
LOAD 326 DL + SDL1_R + 0.75 LL + 0.525 EL(-X)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 11 -0.525
LOAD 327 DL + SDL1_R + 0.75 LL + 0.525 EL(+Z)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 12 0.525
LOAD 328 DL + SDL1_R + 0.75 LL + 0.525 EL(-Z)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 12 -0.525
LOAD 329 DL + SDL2_R + 0.75 LL + 0.525 EL(+X)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 11 0.525
LOAD 330 DL + SDL2_R + 0.75 LL + 0.525 EL(-X)
REPEAT LOAD
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1 1.0 4 1.0 2 0.75 11 -0.525
LOAD 331 DL + SDL2_R + 0.75 LL + 0.525 EL(+Z)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 12 0.525
LOAD 332 DL + SDL2_R + 0.75 LL + 0.525 EL(-Z)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 12 -0.525
LOAD 333 DL - SDL1_R + 0.75 LL + 0.525 EL(+X)
REPEAT LOAD
1 1.0 3 -1.0 2 0.75 11 0.525
LOAD 334 DL - SDL1_R + 0.75 LL + 0.525 EL(-X)
REPEAT LOAD
1 1.0 3 -1.0 2 0.75 11 -0.525
LOAD 335 DL - SDL1_R + 0.75 LL + 0.525 EL(+Z)
REPEAT LOAD
1 1.0 3 -1.0 2 0.75 12 0.525
LOAD 336 DL - SDL1_R + 0.75 LL + 0.525 EL(-Z)
REPEAT LOAD
1 1.0 3 -1.0 2 0.75 12 -0.525
LOAD 337 DL - SDL2_R + 0.75 LL + 0.525 EL(+X)
REPEAT LOAD
1 1.0 4 -1.0 2 0.75 11 0.525
LOAD 338 DL - SDL2_R + 0.75 LL + 0.525 EL(-X)
REPEAT LOAD
1 1.0 4 -1.0 2 0.75 11 -0.525
LOAD 339 DL - SDL2_R + 0.75 LL + 0.525 EL(+Z)
REPEAT LOAD
1 1.0 4 -1.0 2 0.75 12 0.525
LOAD 340 DL - SDL2_R + 0.75 LL + 0.525 EL(-Z)
REPEAT LOAD
1 1.0 4 -1.0 2 0.75 12 -0.525
*********************************************
LOAD 341 0.6 (DL + SDL1_R) + WL(+X)
REPEAT LOAD
1 0.6 3 0.6 5 1.0
LOAD 342 0.6 (DL + SDL1_R) + WL(-X)
REPEAT LOAD
1 0.6 3 0.6 6 1.0
LOAD 343 0.6 (DL + SDL1_R) + WL(+Z)
REPEAT LOAD
1 0.6 3 0.6 7 1.0
LOAD 344 0.6 (DL + SDL1_R) + WL(-Z)
REPEAT LOAD
1 0.6 3 0.6 8 1.0
LOAD 345 0.6 (DL + SDL2_R) + WL(+X)
REPEAT LOAD
1 0.6 4 0.6 5 1.0
LOAD 346 0.6 (DL + SDL2_R) + WL(-X)
REPEAT LOAD
1 0.6 4 0.6 6 1.0
LOAD 347 0.6 (DL + SDL2_R) + WL(+Z)
REPEAT LOAD
1 0.6 4 0.6 7 1.0
LOAD 348 0.6 (DL + SDL2_R) + WL(-Z)
```

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REPEAT LOAD
1 0.6 4 0.6 8 1.0
LOAD 349 0.6 (DL - SDL1_R) + WL(+X)
REPEAT LOAD
1 0.6 3 -0.6 5 1.0
LOAD 350 0.6 (DL - SDL1_R) + WL(-X)
REPEAT LOAD
1 0.6 3 -0.6 6 1.0
LOAD 351 0.6 (DL - SDL1_R) + WL(+Z)
REPEAT LOAD
1 0.6 3 -0.6 7 1.0
LOAD 352 0.6 (DL - SDL1_R) + WL(-Z)
REPEAT LOAD
1 0.6 3 -0.6 8 1.0
LOAD 353 0.6 (DL - SDL2_R) + WL(+X)
REPEAT LOAD
1 0.6 4 -0.6 5 1.0
LOAD 354 0.6 (DL - SDL2_R) + WL(-X)
REPEAT LOAD
1 0.6 4 -0.6 6 1.0
LOAD 355 0.6 (DL - SDL2_R) + WL(+Z)
REPEAT LOAD
1 0.6 4 -0.6 7 1.0
LOAD 356 0.6 (DL - SDL2_R) + WL(-Z)
REPEAT LOAD
1 0.6 4 -0.6 8 1.0
**********************************************
LOAD 357 0.6 (DL + SDL1_R) + 0.7 EL(+X)
REPEAT LOAD
1 0.6 3 0.6 11 0.7
LOAD 358 0.6 (DL + SDL1_R) + 0.7 EL(-X)
REPEAT LOAD
1 0.6 3 0.6 11 -0.7
LOAD 359 0.6 (DL + SDL1_R) + 0.7 EL(+Z)
REPEAT LOAD
1 0.6 3 0.6 12 0.7
LOAD 360 0.6 (DL + SDL1_R) + 0.7 EL(-Z)
REPEAT LOAD
1 0.6 3 0.6 12 -0.7
LOAD 361 0.6 (DL + SDL2_R) + 0.7 EL(+X)
REPEAT LOAD
1 0.6 4 0.6 11 0.7
LOAD 362 0.6 (DL + SDL2_R) + 0.7 EL(-X)
REPEAT LOAD
1 0.6 4 0.6 11 -0.7
LOAD 363 0.6 (DL + SDL2_R) + 0.7 EL(+Z)
REPEAT LOAD
1 0.6 4 0.6 12 -0.7
LOAD 364 0.6 (DL + SDL2_R) + 0.7 EL(-Z)
REPEAT LOAD
1 0.6 4 0.6 12 -0.7
LOAD 365 0.6 (DL + SDL1_R) + 0.7 EL(+X)
REPEAT LOAD
1 0.6 3 -0.6 11 0.7
```

```
LOAD 366 0.6 (DL + SDL1_R) + 0.7 EL(-X)
REPEAT LOAD
1 0.6 3 -0.6 11 -0.7
LOAD 367 0.6 (DL + SDL1_R) + 0.7 EL(+Z)
REPEAT LOAD
1 0.6 3 -0.6 12 0.7
LOAD 368 0.6 (DL + SDL1_R) + 0.7 EL(-Z)
REPEAT LOAD
1 0.6 3 -0.6 12 -0.7
LOAD 369 0.6 (DL + SDL2_R) + 0.7 EL(+X)
REPEAT LOAD
1 0.6 4 -0.6 11 0.7
LOAD 370 0.6 (DL + SDL2_R) + 0.7 EL(-X)
REPEAT LOAD
1 0.6 4 -0.6 11 -0.7
LOAD 371 0.6 (DL + SDL2_R) + 0.7 EL(+Z)
REPEAT LOAD
1 0.6 4 -0.6 12 -0.7
LOAD 372 0.6 (DL + SDL2_R) + 0.7 EL(-Z)
REPEAT LOAD
1 0.6 4 -0.6 12 -0.7
************************************************
*Load Combination for Concrete design
************************************************
LOAD 401 1.4 (DL+SDL1_R)
REPEAT LOAD
1 1.4 3 1.4
LOAD 402 1.4 (DL+SDL2_R)
REPEAT LOAD
1 1.4 4 1.4
LOAD 403 1.4 (DL-SDL1_R)
REPEAT LOAD
1 1.4 3 -1.4
LOAD 404 1.4 (DL-SDL2_R)
REPEAT LOAD
1 1.4 4 -1.4
************************************************
LOAD 405 1.4 (DL + SDL1_R ) + 1.7 LL
REPEAT LOAD
1 1.4 3 1.4 2 1.7
LOAD 406 1.4 (DL + SDL2_R ) + 1.7 LL
REPEAT LOAD
1 1.4 4 1.4 2 1.7
LOAD 407 1.4 (DL - SDL1_R ) + 1.7 LL
REPEAT LOAD
1 1.4 3 -1.4 2 1.7
LOAD 408 1.4 (DL - SDL2_R ) + 1.7 LL
REPEAT LOAD
1 1.4 4 -1.4 2 1.7
************************************************
LOAD 409 1.2 (DL + SDL1_R) + 0.5 LL + 1.6 WL(+X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 5 1.6
LOAD 410 1.2 (DL + SDL1_R) + 0.5 LL + 1.6 WL(-X)
```

```
REPEAT LOAD
1 1.2 3 1.2 2 0.5 6 1.6
LOAD 411 1.2 (DL + SDL1_R) + 0.5 LL + 1.6 WL(+Z)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 7 1.6
LOAD 412 1.2 (DL + SDL1_R) + 0.5 LL + 1.6 WL(-Z)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 8 1.6
LOAD 413 1.2 (DL + SDL2_R) + 0.5 LL + 1.6 WL(+X)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 5 1.6
LOAD 414 1.2 (DL + SDL2_R) + 0.5 LL + 1.6 WL(-X)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 6 1.6
LOAD 415 1.2 (DL + SDL2_R) + 0.5 LL + 1.6 WL(+Z)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 7 1.6
LOAD 416 1.2 (DL + SDL2_R) + 0.5 LL + 1.6 WL(-Z)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 8 1.6
LOAD 417 1.2 (DL - SDL1_R) + 0.5 LL + 1.6 WL(+X)
REPEAT LOAD
1 1.2 3 -1.2 2 0.5 5 1.6
LOAD 418 1.2 (DL - SDL1_R) + 0.5 LL + 1.6 WL(-X)
REPEAT LOAD
1 1.2 3 -1.2 2 0.5 6 1.6
LOAD 419 1.2 (DL - SDL1_R) + 0.5 LL + 1.6 WL(+Z)
REPEAT LOAD
1 1.2 3 -1.2 2 0.5 7 1.6
LOAD 420 1.2 (DL - SDL1_R) + 0.5 LL + 1.6 WL(-Z)
REPEAT LOAD
1 1.2 3 -1.2 2 0.5 8 1.6
LOAD 421 1.2 (DL - SDL2_R) + 0.5 LL + 1.6 WL(+X)
REPEAT LOAD
1 1.2 4 -1.2 2 0.5 5 1.6
LOAD 422 1.2 (DL - SDL2_R) + 0.5 LL + 1.6 WL(-X)
REPEAT LOAD
1 1.2 4 -1.2 2 0.5 6 1.6
LOAD 423 1.2 (DL - SDL2_R) + 0.5 LL + 1.6 WL(+Z)
REPEAT LOAD
1 1.2 4 -1.2 2 0.5 7 1.6
LOAD 424 1.2 (DL - SDL2_R) + 0.5 LL + 1.6 WL(-Z)
REPEAT LOAD
1 1.2 4 -1.2 2 0.5 8 1.6
*********************************************
LOAD 425 1.2 (DL + SDL1_R) + 0.5 LL + 1.0 EL(+X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 11 1.0
LOAD 426 1.2 (DL + SDL1_R) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 11 -1.0
LOAD 427 1.2 (DL + SDL1_R) + 0.5 LL + 1.0 EL(+Z)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 12 1.0
```

```
LOAD 428 1.2 (DL + SDL1_R) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 12 -1.0
LOAD 429 1.2 (DL + SDL2_R) + 0.5 LL + 1.0 EL(+X)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 11 1.0
LOAD 430 1.2 (DL + SDL2_R) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 11 -1.0
LOAD 431 1.2 (DL + SDL2_R) + 0.5 LL + 1.0 EL(+Z)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 12 1.0
LOAD 432 1.2 (DL + SDL2_R) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 12 -1.0
LOAD 433 1.2 (DL - SDL1_R) + 0.5 LL + 1.0 EL(+X)
REPEAT LOAD
1 1.2 3 -1.2 2 0.5 11 1.0
LOAD 434 1.2 (DL - SDL1_R) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 3 -1.2 2 0.5 11 -1.0
LOAD 435 1.2 (DL - SDL1_R) + 0.5 LL + 1.0 EL(+Z)
REPEAT LOAD
1 1.2 3 -1.2 2 0.5 12 1.0
LOAD 436 1.2 (DL - SDL1_R) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 3 -1.2 2 0.5 12 -1.0
LOAD 437 1.2 (DL - SDL2_R) + 0.5 LL + 1.0 EL(+X)
REPEAT LOAD
1 1.2 4 -1.2 2 0.5 11 1.0
LOAD 438 1.2 (DL - SDL2_R) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 4 -1.2 2 0.5 11 -1.0
LOAD 439 1.2 (DL - SDL2_R) + 0.5 LL + 1.0 EL(+Z)
REPEAT LOAD
1 1.2 4 -1.2 2 0.5 12 1.0
LOAD 440 1.2 (DL - SDL2_R) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 4 -1.2 2 0.5 12 -1.0
*********************************************
LOAD 441 0.9 (DL + SDL1_R) + 1.6 WL(+X)
REPEAT LOAD
1 0.9 3 0.9 5 1.6
LOAD 442 0.9 (DL + SDL1_R) + 1.6 WL(-X)
REPEAT LOAD
1 0.9 3 0.9 6 1.6
LOAD 443 0.9 (DL + SDL1_R) + 1.6 WL(+Z)
REPEAT LOAD
1 0.9 3 0.9 7 1.6
LOAD 444 0.9 (DL + SDL1_R) + 1.6 WL(-Z)
REPEAT LOAD
1 0.9 3 0.9 8 1.6
LOAD 445 0.9 (DL + SDL2_R) + 1.6 WL(+X)
REPEAT LOAD
```

```
1 0.9 4 0.9 5 1.6
LOAD 446 0.9 (DL + SDL2_R) + 1.6 WL(-X)
REPEAT LOAD
1 0.9 4 0.9 6 1.6
LOAD 447 0.9 (DL + SDL2_R) + 1.6 WL(+Z)
REPEAT LOAD
1 0.9 4 0.9 7 1.6
LOAD 448 0.9 (DL + SDL2_R) + 1.6 WL(-Z)
REPEAT LOAD
1 0.9 4 0.9 8 1.6
LOAD 449 0.9 (DL - SDL1_R) + 1.6 WL(+X)
REPEAT LOAD
1 0.9 3 -0.9 5 1.6
LOAD 450 0.9 (DL - SDL1_R) + 1.6 WL(-X)
REPEAT LOAD
1 0.9 3 -0.9 6 1.6
LOAD 451 0.9 (DL - SDL1_R) + 1.6 WL(+Z)
REPEAT LOAD
1 0.9 3 -0.9 7 1.6
LOAD 452 0.9 (DL - SDL1_R) + 1.6 WL(-Z)
REPEAT LOAD
1 0.9 3 -0.9 8 1.6
LOAD 453 0.9 (DL - SDL2_R) + 1.6 WL(+X)
REPEAT LOAD
1 0.9 4 -0.9 5 1.6
LOAD 454 0.9 (DL - SDL2_R) + 1.6 WL(-X)
REPEAT LOAD
1 0.9 4 -0.9 6 1.6
LOAD 455 0.9 (DL - SDL2_R) + 1.6 WL(+Z)
REPEAT LOAD
1 0.9 4 -0.9 7 1.6
LOAD 456 0.9 (DL - SDL2_R) + 1.6 WL(-Z)
REPEAT LOAD
1 0.9 4 -0.9 8 1.6
************************************************
LOAD 457 0.9 (DL + SDL1_R) + 1.0 EL(+X)
REPEAT LOAD
1 0.9 3 0.9 11 1.0
LOAD 458 0.9 (DL + SDL1_R) + 1.0 EL(-X)
REPEAT LOAD
1 0.9 3 0.9 11 -1.0
LOAD 459 0.9 (DL + SDL1_R) + 1.0 EL(+Z)
REPEAT LOAD
1 0.9 3 0.9 12 1.0
LOAD 460 0.9 (DL + SDL1_R) + 1.0 EL(-Z)
REPEAT LOAD
1 0.9 3 0.9 12 -1.0
LOAD 461 0.9 (DL + SDL2_R) + 1.0 EL(+X)
REPEAT LOAD
1 0.9 4 0.9 11 1.0
LOAD 462 0.9 (DL + SDL2_R) + 1.0 EL(-X)
REPEAT LOAD
1 0.9 4 0.9 11 -1.0
LOAD 463 0.9 (DL + SDL2_R) + 1.0 EL(+Z)
```

```
REPEAT LOAD
1 0.9 4 0.9 12 1.0
LOAD 464 0.9 (DL + SDL2_R) + 1.0 EL(-Z)
REPEAT LOAD
1 0.9 4 0.9 12 -1.0
LOAD 465 0.9 (DL - SDL1_R) + 1.0 EL(+X)
REPEAT LOAD
1 0.9 3 -0.9 11 1.0
LOAD 466 0.9 (DL - SDL1_R) + 1.0 EL(-X)
REPEAT LOAD
1 0.9 3 -0.9 11 -1.0
LOAD 467 0.9 (DL - SDL1_R) + 1.0 EL(+Z)
REPEAT LOAD
1 0.9 3 -0.9 12 1.0
LOAD 468 0.9 (DL - SDL1_R) + 1.0 EL(-Z)
REPEAT LOAD
1 0.9 3 -0.9 12 -1.0
LOAD 469 0.9 (DL - SDL2_R) + 1.0 EL(+X)
REPEAT LOAD
1 0.9 4 -0.9 11 1.0
LOAD 470 0.9 (DL - SDL2_R) + 1.0 EL(-X)
REPEAT LOAD
1 0.9 4 -0.9 11 -1.0
LOAD 471 0.9 (DL - SDL2_R) + 1.0 EL(+Z)
REPEAT LOAD
1 0.9 4 -0.9 12 1.0
LOAD 472 0.9 (DL - SDL2_R) + 1.0 EL(-Z)
REPEAT LOAD
1 0.9 4 -0.9 12 -1.0
PERFORM ANALYSIS
LOAD LIST 401 TO 472
START CONCRETE DESIGN
CODE ACI 2008
FYMAIN 420000 MEMB 1 TO 9 11 TO 19 21 TO 73 80 87 89 412 TO 452 469 TO 688 -
729 TO 770 789 TO 1008 1030 1032 TO 1127 1130 TO 1137 1140 TO 1147 -
1150 TO 1157 1160 TO 1284 1287 TO 1292 1295 TO 1300 1303 TO 1308 -
1311 TO 1326 1331 1360 1409 TO 1682
RHOMN 0.01 MEMB 1 TO 9 11 TO 19 21 TO 73 80 87 89 412 TO 452 469 TO 688 729 -
730 TO 770 789 TO 1008 1030 1032 TO 1127 1130 TO 1137 1140 TO 1147 -
1150 TO 1157 1160 TO 1284 1287 TO 1292 1295 TO 1300 1303 TO 1308 -
1311 TO 1326 1331 1360 1409 TO 1682
FC 35000 MEMB 1 TO 9 11 TO 19 21 TO 73 80 87 89 412 TO 452 469 TO 688 729 -
730 TO 770 789 TO 1008 1030 1032 TO 1127 1130 TO 1137 1140 TO 1147 -
1150 TO 1157 1160 TO 1284 1287 TO 1292 1295 TO 1300 1303 TO 1308 -
1311 TO 1326 1331 1360 1409 TO 1682
SKZ 6 MEMB 2 12 412 TO 416 729 TO 733
SLZ 13 MEMB 4 14 424 TO 435 741 TO 752
SKY 13 MEMB 1030 1032 1051 1061 1071 1081 1091 1101 1111 1121 1131 1141 1151 -
1161 1193 1195 1197 1199 1201 1203 1205 1207 1209 1211 1213 1215 1331 1360
DESIGN COLUMN 1 TO 6 11 TO 16 21 TO 24 412 TO 435 729 TO 752 1030 1032 1050 -
1051 1060 1061 1070 1071 1080 1081 1090 1091 1100 1101 1110 1111 1120 1121 -
1130 1131 1140 1141 1150 1151 1160 1161 1192 TO 1215 1331 1360
END CONCRETE DESIGN
FINISH
```

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	C02	D01			Outotec / L&T Revision No.:	0	1	2	3	
Date	27-01-2020	12-03-2020	25-04-2022	13/10/2023			Date	27-01-2020	12-03-2020	25-04-2022	13/10/2023	
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS DT. 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER SUPPORT STRUCTURE	AS BUILT			Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS DT. 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER SUPPORT STRUCTURE	AS BUILT	
Prepared By	APR	APR	APR	APR			Prepared By	APR	APR	APR	APR	
Checked	SBY	SBY	SBY	SBY			Checked	SBY	SBY	SBY	SBY	
Approved	AG	AG	AG	AG			Approved	AG	AG	AG	AG	

ATTACHMENT 3 STAAD OUTPUT FILE OF CRUSHER SUPPORT STRUCTURE

```
**************************************************
*                                                *
*           STAAD.Pro V8i SELECTseries6          *
*           Version  20.07.11.70                 *
*           Proprietary Program of               *
*           Bentley Systems, Inc.                *
*           Date=    APR 26, 2022                *
*           Time=    16:27:48                    *
*                                                *
*      USER ID: LNT ECC DIV                      *
**************************************************
```

```
     1. STAAD SPACE
INPUT FILE: C:\Users\20321581\Desktop\Staad Crusher\Crusher Support Structure R2.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 08-JAN-20
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. UNIT METER KN
     7. JOINT COORDINATES
     8. 1 3.1 -2.5 2; 2 11 -2.5 2; 3 14.1 -2.5 2; 4 3.1 -2.5 8; 5 11 -2.5 8
     9. 6 14.1 -2.5 8; 7 8.8 -2.5 12.3; 8 14.1 -2.5 12.3; 9 3.1 0 2; 10 11 0 2
    10. 11 14.1 0 2; 12 3.1 6.5 2; 13 11 6.5 2; 14 14.1 6.5 2; 15 3.1 0 8; 16 11 0 8
    11. 17 14.1 0 8; 18 3.1 6.5 8; 19 11 6.5 8; 20 14.1 6.5 8; 21 14.1 0 12.3
    12. 22 14.1 6.5 12.3; 23 8.8 0 12.3; 24 8.8 6.5 12.3; 25 8.8 0 8; 26 3.1 6.5 3.355
    13. 27 3.1 6.5 6.645; 28 11 6.5 3.355; 29 11 6.5 6.645; 30 6.455 6.5 3.355
    14. 31 6.455 6.5 6.645; 32 8.8 6.5 8; 33 11 6.5 12.3; 34 14.1 6.5 3.355
    15. 35 14.1 6.5 6.645; 36 3.1 6.5 1; 37 11 6.5 1; 38 14.1 6.5 1; 39 15.1 6.5 2
    16. 40 15.1 6.5 8; 41 15.1 6.5 12.3; 42 15.1 6.5 1; 43 5.8 0 2; 44 5.8 0 8
    17. 45 8.05 6.5 12.3; 46 8.05 6.5 8; 48 6.455 6.5 1; 50 15.1 6.5 3.355
    18. 51 15.1 6.5 6.645; 447 3.1 0.5 8; 448 3.1 1 8; 449 3.1 1.5 8; 450 3.1 2 8
    19. 451 3.1 2.5 8; 452 3.1 3 8; 453 3.1 3.5 8; 454 3.1 4 8; 455 3.1 4.5 8
    20. 456 3.1 4.9 8; 457 3.1 5.3 8; 458 3.1 5.9 8; 459 11 0.5 8; 460 11 1 8
    21. 461 11 1.5 8; 462 11 2 8; 463 11 2.5 8; 464 11 3 8; 465 11 3.5 8; 466 11 4 8
    22. 467 11 4.5 8; 468 11 4.9 8; 469 11 5.3 8; 470 11 5.9 8; 471 3.6 0 8
    23. 472 4.1 0 8; 473 4.6 0 8; 474 5.1 0 8; 475 5.45 0 8; 476 6.15 0 8
    24. 477 6.455 0 8; 478 6.855 0 8; 479 7.255 0 8; 480 7.655 0 8; 481 8.05 0 8
    25. 482 8.45 0 8; 483 9.25 0 8; 484 9.6 0 8; 485 9.95 0 8; 486 10.3 0 8
    26. 487 10.65 0 8; 488 8.8 0.5 8; 489 5.8 0.5 8; 490 3.6 0.5 8; 491 4.1 0.5 8
    27. 492 4.6 0.5 8; 493 5.1 0.5 8; 494 5.45 0.5 8; 495 6.15 0.5 8; 496 6.455 0.5 8
    28. 497 6.855 0.5 8; 498 7.255 0.5 8; 499 7.655 0.5 8; 500 8.05 0.5 8
    29. 501 8.45 0.5 8; 502 9.25 0.5 8; 503 9.6 0.5 8; 504 9.95 0.5 8; 505 10.3 0.5 8
    30. 506 10.65 0.5 8; 507 8.8 1 8; 508 5.8 1 8; 509 3.6 1 8; 510 4.1 1 8
    31. 511 4.6 1 8; 512 5.1 1 8; 513 5.45 1 8; 514 6.15 1 8; 515 6.455 1 8
    32. 516 6.855 1 8; 517 7.255 1 8; 518 7.655 1 8; 519 8.05 1 8; 520 8.45 1 8
    33. 521 9.25 1 8; 522 9.6 1 8; 523 9.95 1 8; 524 10.3 1 8; 525 10.65 1 8
    34. 526 8.8 1.5 8; 527 5.8 1.5 8; 528 3.6 1.5 8; 529 4.1 1.5 8; 530 4.6 1.5 8
    35. 531 5.1 1.5 8; 532 5.45 1.5 8; 533 6.15 1.5 8; 534 6.455 1.5 8
    36. 535 6.855 1.5 8; 536 7.255 1.5 8; 537 7.655 1.5 8; 538 8.05 1.5 8
    37. 539 8.45 1.5 8; 540 9.25 1.5 8; 541 9.6 1.5 8; 542 9.95 1.5 8; 543 10.3 1.5 8
    38. 544 10.65 1.5 8; 545 8.8 2 8; 546 5.8 2 8; 547 3.6 2 8; 548 4.1 2 8
```

39. 549 4.6 2 8; 550 5.1 2 8; 551 5.45 2 8; 552 6.15 2 8; 553 6.455 2 8
40. 554 6.855 2 8; 555 7.255 2 8; 556 7.655 2 8; 557 8.05 2 8; 558 8.45 2 8
41. 559 9.25 2 8; 560 9.6 2 8; 561 9.95 2 8; 562 10.3 2 8; 563 10.65 2 8
42. 564 8.8 2.5 8; 565 5.8 2.5 8; 566 3.6 2.5 8; 567 4.1 2.5 8; 568 4.6 2.5 8
43. 569 5.1 2.5 8; 570 5.45 2.5 8; 571 6.15 2.5 8; 572 6.455 2.5 8
44. 573 6.855 2.5 8; 574 7.255 2.5 8; 575 7.655 2.5 8; 576 8.05 2.5 8
45. 577 8.45 2.5 8; 578 9.25 2.5 8; 579 9.6 2.5 8; 580 9.95 2.5 8; 581 10.3 2.5 8
46. 582 10.65 2.5 8; 583 8.8 3 8; 584 5.8 3 8; 585 3.6 3 8; 586 4.1 3 8
47. 587 4.6 3 8; 588 5.1 3 8; 589 5.45 3 8; 590 6.15 3 8; 591 6.455 3 8
48. 592 6.855 3 8; 593 7.255 3 8; 594 7.655 3 8; 595 8.05 3 8; 596 8.45 3 8
49. 597 9.25 3 8; 598 9.6 3 8; 599 9.95 3 8; 600 10.3 3 8; 601 10.65 3 8
50. 602 8.8 3.5 8; 603 5.8 3.5 8; 604 3.6 3.5 8; 605 4.1 3.5 8; 606 4.6 3.5 8
51. 607 5.1 3.5 8; 608 5.45 3.5 8; 609 6.15 3.5 8; 610 6.455 3.5 8
52. 611 6.855 3.5 8; 612 7.255 3.5 8; 613 7.655 3.5 8; 614 8.05 3.5 8
53. 615 8.45 3.5 8; 616 9.25 3.5 8; 617 9.6 3.5 8; 618 9.95 3.5 8; 619 10.3 3.5 8
54. 620 10.65 3.5 8; 621 8.8 4 8; 622 5.8 4 8; 623 3.6 4 8; 624 4.1 4 8
55. 625 4.6 4 8; 626 5.1 4 8; 627 5.45 4 8; 628 6.15 4 8; 629 6.455 4 8
56. 630 6.855 4 8; 631 7.255 4 8; 632 7.655 4 8; 633 8.05 4 8; 634 8.45 4 8
57. 635 9.25 4 8; 636 9.6 4 8; 637 9.95 4 8; 638 10.3 4 8; 639 10.65 4 8
58. 640 8.8 4.5 8; 641 5.8 4.5 8; 642 3.6 4.5 8; 643 4.1 4.5 8; 644 4.6 4.5 8
59. 645 5.1 4.5 8; 646 5.45 4.5 8; 647 6.15 4.5 8; 648 6.455 4.5 8
60. 649 6.855 4.5 8; 650 7.255 4.5 8; 651 7.655 4.5 8; 652 8.05 4.5 8
61. 653 8.45 4.5 8; 654 9.25 4.5 8; 655 9.6 4.5 8; 656 9.95 4.5 8; 657 10.3 4.5 8
62. 658 10.65 4.5 8; 659 8.8 4.9 8; 660 5.8 4.9 8; 661 3.6 4.9 8; 662 4.1 4.9 8
63. 663 4.6 4.9 8; 664 5.1 4.9 8; 665 5.45 4.9 8; 666 6.15 4.9 8; 667 6.455 4.9 8
64. 668 6.855 4.9 8; 669 7.255 4.9 8; 670 7.655 4.9 8; 671 8.05 4.9 8
65. 672 8.45 4.9 8; 673 9.25 4.9 8; 674 9.6 4.9 8; 675 9.95 4.9 8; 676 10.3 4.9 8
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165. 1177 14.1 2.5 12.8; 1178 14.1 2.5 13.3; 1179 14.1 2.5 13.8; 1180 14.1 2.5 14.3
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193. 1353 8.8 2 9.6125; 1354 8.8 2 9.075; 1355 8.8 2 8.5375; 1356 8.8 2 11.7625
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197. 1367 8.8 3 9.6125; 1368 8.8 3 9.075; 1369 8.8 3 8.5375; 1370 8.8 3 11.7625
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201. 1381 8.8 4 9.6125; 1382 8.8 4 9.075; 1383 8.8 4 8.5375; 1384 8.8 4 11.7625
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487. 1239 1149 1156 1165 1158; 1240 1156 1155 1164 1165; 1241 1155 1154 1163 1164
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495. 1263 1176 1183 1192 1185; 1264 1183 1182 1191 1192; 1265 1182 1181 1190 1191
496. 1266 1181 1180 1189 1190; 1267 1180 1179 1188 1189; 1268 1179 1178 1187 1188
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516. 1423 1328 25 488 1334; 1424 1017 1335 1342 1026; 1425 1335 1329 1336 1342
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534. 1477 1374 1375 1382 1381; 1478 1375 1376 1383 1382; 1479 1376 602 621 1383
535. 1480 1080 1384 1391 1089; 1481 1384 1378 1385 1391; 1482 1378 1379 1386 1385
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543. 1513 23 1406 1415 1017; 1514 1406 1407 1416 1415; 1515 1407 1408 1417 1416
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546. 1522 1414 21 1139 1423; 1523 1017 1415 1424 1026; 1524 1415 1416 1425 1424
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556. 1552 1441 1157 1166 1450; 1553 1044 1442 1451 1053; 1554 1442 1443 1452 1451
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559. 1561 1449 1450 1459 1458; 1562 1450 1166 1175 1459; 1563 1053 1451 1460 1062
560. 1564 1451 1452 1461 1460; 1565 1452 1453 1462 1461; 1566 1453 1454 1463 1462
561. 1567 1454 1455 1464 1463; 1568 1455 1456 1465 1464; 1569 1456 1457 1466 1465
562. 1570 1457 1458 1467 1466; 1571 1458 1459 1468 1467; 1572 1459 1175 1184 1468
563. 1573 1062 1460 1469 1071; 1574 1460 1461 1470 1469; 1575 1461 1462 1471 1470
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572. 1600 1484 1485 1494 1493; 1601 1485 1486 1495 1494; 1602 1486 1202 1211 1495
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576. 1612 1495 1211 1220 1504; 1613 1098 1496 1505 1107; 1614 1496 1497 1506 1505
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591. 1657 1537 1536 1547 1548; 1658 1536 739 740 1547; 1659 455 1557 1568 456
592. 1660 1557 1556 1567 1568; 1661 1556 1555 1566 1567; 1662 1555 1554 1565 1566
593. 1663 1554 1553 1564 1565; 1664 1553 1552 1563 1564; 1665 1552 1551 1562 1563
594. 1666 1551 1550 1561 1562; 1667 1550 1549 1560 1561; 1668 1549 1548 1559 1560
595. 1669 1548 1547 1558 1559; 1670 1547 740 741 1558; 1671 456 1568 1579 457
596. 1672 1568 1567 1578 1579; 1673 1567 1566 1577 1578; 1674 1566 1565 1576 1577
597. 1675 1565 1564 1575 1576; 1676 1564 1563 1574 1575; 1677 1563 1562 1573 1574
598. 1678 1562 1561 1572 1573; 1679 1561 1560 1571 1572; 1680 1560 1559 1570 1571

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599. 1681 1559 1558 1569 1570; 1682 1558 741 742 1569
600. ELEMENT PROPERTY
601. 469 TO 688 789 TO 1008 1052 TO 1059 1062 TO 1069 1072 TO 1079 1082 TO 1089 -
602. 1092 TO 1099 1102 TO 1109 1112 TO 1119 1122 TO 1127 1132 TO 1137 -
603. 1142 TO 1147 1152 TO 1157 1162 TO 1169 1177 TO 1184 1223 TO 1284 -
604. 1287 TO 1292 1295 TO 1300 1303 TO 1308 1311 TO 1326 1416 TO 1503 -
605. 1513 TO 1622 THICKNESS 0.3
606. 1623 TO 1682 THICKNESS 0.5
607. MEMBER RELEASE
608. 7 TO 9 17 TO 19 25 TO 55 59 60 67 TO 70 72 1033 TO 1038 1041 1042 -
609. 1683 START MPX 0.99
610. 8 9 18 26 TO 30 33 TO 45 47 TO 55 67 70 TO 72 80 89 440 447 452 757 770 1033 -
611. 1038 1041 1042 1049 1176 1191 1222 1415 1512 1683 END MPX 0.99
612. DEFINE MATERIAL START
613. ISOTROPIC CONCRETE
614. E 2.7806E+07
615. POISSON 0.17
616. DENSITY 25
617. ALPHA 1E-05
618. DAMP 0.05
619. TYPE CONCRETE
620. STRENGTH FCU 27579
621. END DEFINE MATERIAL
622. MEMBER PROPERTY AMERICAN
623. 1 2 11 12 412 TO 423 729 TO 740 PRIS YD 0.7 ZD 0.7
624. 21 TO 24 1050 1060 1070 1080 1090 1100 1110 1120 1130 1140 1150 1160 1192 -
625. 1194 1196 1198 1200 1202 1204 1206 1208 1210 1212 1214 PRIS YD 0.5 ZD 0.5
626. 25 27 TO 30 32 70 1409 TO 1415 1504 TO 1512 PRIS YD 0.5 ZD 0.3
627. 7 8 17 18 31 68 69 436 TO 452 753 TO 770 PRIS YD 0.55 ZD 0.3
628. 40 45 47 49 PRIS YD 0.5 ZD 0.3
629. 9 19 46 50 51 71 PRIS YD 0.7 ZD 0.4
630. 41 TO 44 PRIS YD 0.9 ZD 0.4
631. MEMBER PROPERTY AMERICAN
632. 33 TO 39 52 54 PRIS YD 1 ZD 0.4
633. MEMBER PROPERTY AMERICAN
634. 56 TO 62 64 TO 66 72 80 87 89 1683 PRIS YD 0.4 ZD 0.3
635. MEMBER PROPERTY AMERICAN
636. 53 55 PRIS YD 0.9 ZD 0.4
637. 3 4 13 14 424 TO 435 741 TO 752 PRIS YD 0.7 ZD 0.7
638. 5 6 15 16 PRIS YD 0.5 ZD 0.5
639. 26 48 PRIS YD 0.7 ZD 0.4
640. MEMBER PROPERTY
641. 1030 1032 1051 1061 1071 1081 1091 1101 1111 1121 1131 1141 1151 1161 1193 -
642. 1195 1197 1199 1201 1203 1205 1207 1209 1211 1213 1215 1331 -
643. 1360 PRIS YD 1 ZD 1
644. 1038 PRIS YD 0.5 ZD 0.5
645. 1034 1035 1043 TO 1049 1185 TO 1191 PRIS YD 0.4 ZD 0.3
646. 63 67 73 1039 TO 1042 PRIS YD 0.2 ZD 0.3
647. 1036 1037 1170 TO 1176 1216 TO 1222 PRIS YD 0.5 ZD 0.3
648. 1033 PRIS YD 1 ZD 0.5
649. CONSTANTS
650. MATERIAL CONCRETE ALL
651. SUPPORTS
652. 1 TO 8 1257 1302 FIXED
653. SLAVE ZX MASTER 19 JOINT 12 TO 14 18 TO 20 22 24 26 TO 42
654. DEFINE IBC 2003
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  655. SDS 0.0667 SD1 0.0133 S1 0.02 IE 1 RX 2.5 RZ 2.5 SCLASS 2 PX 0.237 PZ 0.237
  656. SELFWEIGHT 1
  657. *CRUSHER STATIC LOAD + 25% MATERIAL LOAD = 200+2.7/4
  658. JOINT WEIGHT
  659. 30 31 WEIGHT 200.7
  660. MEMBER WEIGHT
  661. 42 44 CON 200.7 3.145
  662. 53 55 CON 45 0.99
  663. *CONVEYOR WALWAY SUPPORT, LOAD DL + 25% OF LL
  664. 70 CON 17 0.7
  665. 70 CON 17 4.9
  666. 28 CON 9.3 0.7
  667. 28 CON 9.3 4.9
  668. *CHUTE LOAD
  669. 42 44 CON 20 0.45
  670. 42 44 CON 20 3.245
  671. *DUCT LOAD
  672. 41 CON 7.5 1.443
  673. 41 CON 7.5 2.573
  674. 43 CON 5 1.443
  675. 43 CON 5 2.573
  676. 6 CON 10 2.6
  677. *CABLE TRAY LOAD
  678. 22 26 48 1030 1192 TO 1215 UNI 5
  679. 9 19 41 43 CON 14 2.95
  680. FLOOR WEIGHT
**NOTE** about Floor/OneWay Loads/Weights.
  Please note that depending on the shape of the floor you may
  have to break up the FLOOR/ONEWAY LOAD into multiple commands.
  For details please refer to Technical Reference Manual
  Section 5.32.4.2 Note d and/or "5.32.4.3 Note f.

  681. **FLOOR DL
  682. YRANGE 6.4 6.6 FLOAD 9.6 XRANGE 3 15.2 ZRANGE 0.9 8
  683. YRANGE 6.4 6.6 FLOAD 5 XRANGE 8 15.2 ZRANGE 8 16.4
  684. **FLOOR 25% OF LL
  685. YRANGE 6.4 6.6 FLOAD 1.25 XRANGE 3 15.2 ZRANGE 0.9 16.4
  686. CUT OFF MODE SHAPE 50
  687. LOAD 11 SEISMIC (+X)
  688. IBC LOAD X 1
  689. LOAD 12 SEISMIC (+Z)
  690. IBC LOAD Z 1
  691. LOAD 1 DEAD LOAD
  692. SELFWEIGHT Y -1
  693. FLOOR LOAD
  694. YRANGE 6.4 6.6 FLOAD -9.6 XRANGE 3 15.2  ZRANGE 0.9 8  GY
  695. YRANGE 6.4 6.6 FLOAD -5 XRANGE 8 15.2  ZRANGE 8 16.4  GY
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696. *CONVEYOR WALWWAY DEAD LOAD
697. MEMBER LOAD
698. 70 CON GY -10 0.7
699. 70 CON GY -10 4.9
700. 28 CON GY -5 0.7
701. 28 CON GY -5 4.9
702. *******EQUIPMENTS DEAD LOAD*********
703. *CRUSHER STATIC LOAD
704. JOINT LOAD
705. 30 FY -200
706. 31 FY -200
707. MEMBER LOAD
708. 42 CON GY -200 3.145
709. 44 CON GY -200 3.145
710. 53 55 CON GY -45 0.99
711. *CRUSHER MATERIAL LOAD
712. JOINT LOAD
713. 30 31 FY -2.7
714. MEMBER LOAD
715. 42 44 CON GY -2.7 3.145
716. *CHUTE LOAD
717. 42 44 CON GY -20 0.45
718. 42 44 CON GY -20 3.245
719. *CONVEYOR SUPPORT LOAD
720. MEMBER LOAD
721. 70 CON GY -55 0.7
722. 70 CON GY -55 4.9
723. 28 CON GY -42 0.7
724. 28 CON GY -42 4.9
725. *DUCT LOAD
726. 41 CON GY -7.5 1.443
727. 41 CON GY -7.5 2.573
728. 43 CON GY -5 1.443
729. 43 CON GY -5 2.573
730. 6 CON GY -10 2.6
731. 6 CMOM GZ -10 2.6
732. *CABLE TRAY LOAD
733. 22 26 48 1192 1194 1196 1198 1200 1202 1204 1206 1208 1210 1212 1214 UNI GY -5
734. 41 43 CON GY -14 2.95
735. *MONORAIL BEAM LOAD OF 3T
736. 40 49 CON GY -40 2.4
737. 47 CON GY -40 1.9
738. LOAD 2 LIVE LOAD
739. FLOOR LOAD
740. YRANGE 6.4 6.6 FLOAD -5 XRANGE 3 15.2  ZRANGE 0.9 16.4  GY
741. *CONVEYOR WALWAY LIVE LOAD
742. MEMBER LOAD
743. 70 CON GY -27 0.7
744. 70 CON GY -27 4.9
745. 28 CON GY -17 0.7
746. 28 CON GY -17 4.9
747. LOAD 3 SDL1_R - FOR CRUSHER DYANMIC LOAD
748. JOINT LOAD
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749. 30 FY -65 FZ 68
750. 31 FY -70 FZ 73
751. MEMBER LOAD
752. 42 CON GY -65 3.145
753. 44 CON GY -70 3.145
754. 42 CON GZ 68 3.145
755. 44 CON GZ 73 3.145
756. 53 55 CON GY -22 0.99
757. 53 55 CON GZ 12 0.99
758. LOAD 4 SDL2_R - FOR CRUSHER DYANMIC LOAD
759. JOINT LOAD
760. 30 FX 68 FY -65
761. 31 FX 73 FY -70
762. MEMBER LOAD
763. 42 CON GY -65 3.145
764. 44 CON GY -70 3.145
765. 42 CON GX 68 3.145
766. 44 CON GX 73 3.145
767. 53 55 CON GY -22 0.99
768. 53 55 CON GX 12 0.99
769. LOAD 5 WIND LOAD IN +X DIRN
770. MEMBER LOAD
771. 2 4 6 12 412 TO 423 729 TO 752 UNI GX 3.28
772. 14 16 424 TO 435 UNI GX 5.64
773. 22 24 1050 1060 1070 1080 1090 1100 1110 1120 1130 1140 1150 1160 1192 1194 -
774. 1196 1198 1200 1202 1204 1206 1208 1210 1212 1214 UNI GX 4.54
775. 1030 1032 1051 1061 1071 1081 1091 1101 1111 1121 1131 1141 1151 1161 1193 -
776. 1195 1197 1199 1201 1203 1205 1207 1209 1211 1213 1215 UNI GX 2.19
777. *WL DUE TO CRUSHER
778. 42 44 CON GX 3.93 3.145
779. 42 44 CON GY -5.22 3.145
780. JOINT LOAD
781. 30 31 FX 3.93 FY 5.22
782. LOAD 6 WIND LOAD IN -X DIRN
783. MEMBER LOAD
784. 2 4 6 12 412 TO 423 729 TO 752 UNI GX -3.28
785. 12 14 16 412 TO 435 UNI GX -5.64
786. 22 24 1050 1060 1070 1080 1090 1100 1110 1120 1130 1140 1150 1160 1192 1194 -
787. 1196 1198 1200 1202 1204 1206 1208 1210 1212 1214 UNI GX -4.54
788. 1030 1032 1051 1061 1071 1081 1091 1101 1111 1121 1131 1141 1151 1161 1193 -
789. 1195 1197 1199 1201 1203 1205 1207 1209 1211 1213 1215 UNI GX -2.19
790. *WL DUE TO CRUSHER
791. 42 44 CON GX -3.93 3.145
792. 42 44 CON GY 5.22 3.145
793. JOINT LOAD
794. 30 31 FX -3.93 FY -5.22
795. LOAD 7 WIND LOAD IN +Z DIRN
796. MEMBER LOAD
797. 2 12 412 TO 423 729 TO 740 UNI GZ 4.38
798. 4 14 424 TO 435 741 TO 752 UNI GZ 6.1
799. 6 16 UNI GZ 1.72
800. 22 24 1030 1032 1050 1051 1060 1061 1070 1071 1080 1081 1090 1091 1100 1101 -
801. 1110 1111 1120 1121 1130 1131 1140 1141 1150 1151 1160 1161 1192 TO 1214 -
802. 1215 UNI GZ 2.94
803. *WL DUE TO CRUSHER
804. 42 44 CON GZ 5.21 3.145
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805. 42 CON GY 6.62 3.145
806. 44 CON GY -6.62 3.145
807. JOINT LOAD
808. 30 FY 6.62 FZ 5.21
809. 31 FY -6.62 FZ 5.21
810. LOAD 8 WIND LOAD IN -Z DIRN
811. MEMBER LOAD
812. 2 12 412 TO 423 729 TO 740 UNI GZ -1.7
813. 4 14 24 424 TO 435 741 TO 752 1032 1050 1051 1060 1061 1070 1071 1080 1081 -
814. 1090 1091 1100 1101 1110 1111 1120 1121 1130 1131 1140 1141 1150 1151 1160 -
815. 1161 UNI GZ -2.17
816. 6 16 22 1030 1192 TO 1215 UNI GZ -0.67
817. *WL DUE TO CRUSHER
818. 42 44 CON GZ -5.21 3.145
819. 42 CON GY -6.62 3.145
820. 44 CON GY 6.62 3.145
821. JOINT LOAD
822. 30 FY -6.62 FZ -5.21
823. 31 FY 6.62 FZ -5.21
824. ************************************************
825. *LOAD COMBINATION FOR BASE PRESSURE _SERVICIBILITY
826. ********************************************
827. LOAD 301 DL+SDL1_R
828. REPEAT LOAD
829. 1 1.0 3 1.0
830. LOAD 302 DL+SDL2_R
831. REPEAT LOAD
832. 1 1.0 4 1.0
833. LOAD 303 DL-SDL1_R
834. REPEAT LOAD
835. 1 1.0 3 -1.0
836. LOAD 304 DL-SDL2_R
837. REPEAT LOAD
838. 1 1.0 4 -1.0
839. **********************************
840. LOAD 305 DL + SDL1_R + LL
841. REPEAT LOAD
842. 1 1.0 3 1.0 2 1.0
843. LOAD 306 DL + SDL2_R + LL
844. REPEAT LOAD
845. 1 1.0 4 1.0 2 1.0
846. LOAD 307 DL - SDL1_R + LL
847. REPEAT LOAD
848. 1 1.0 3 -1.0 2 1.0
849. LOAD 308 DL - SDL2_R + LL
850. REPEAT LOAD
851. 1 1.0 4 -1.0 2 1.0
852. ************************************************
853. LOAD 309 DL + SDL1_R + 0.75 LL + 0.75 WL(+X)
854. REPEAT LOAD
855. 1 1.0 3 1.0 2 0.75 5 0.75
856. LOAD 310 DL + SDL1_R + 0.75 LL + 0.75 WL(-X)
857. REPEAT LOAD
858. 1 1.0 3 1.0 2 0.75 6 0.75
859. LOAD 311 DL + SDL1_R + 0.75 LL + 0.75 WL(+Z)
860. REPEAT LOAD
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861. 1 1.0 3 1.0 2 0.75 7 0.75
862. LOAD 312 DL + SDL1_R + 0.75 LL + 0.75 WL(-Z)
863. REPEAT LOAD
864. 1 1.0 3 1.0 2 0.75 8 0.75
865. LOAD 313 DL + SDL2_R + 0.75 LL + 0.75 WL(+X)
866. REPEAT LOAD
867. 1 1.0 4 1.0 2 0.75 5 0.75
868. LOAD 314 DL + SDL2_R + 0.75 LL + 0.75 WL(-X)
869. REPEAT LOAD
870. 1 1.0 4 1.0 2 0.75 6 0.75
871. LOAD 315 DL + SDL2_R + 0.75 LL + 0.75 WL(+Z)
872. REPEAT LOAD
873. 1 1.0 4 1.0 2 0.75 7 0.75
874. LOAD 316 DL + SDL2_R + 0.75 LL + 0.75 WL(-Z)
875. REPEAT LOAD
876. 1 1.0 4 1.0 2 0.75 8 0.75
877. LOAD 317 DL - SDL1_R + 0.75 LL + 0.75 WL(+X)
878. REPEAT LOAD
879. 1 1.0 3 -1.0 2 0.75 5 0.75
880. LOAD 318 DL - SDL1_R + 0.75 LL + 0.75 WL(-X)
881. REPEAT LOAD
882. 1 1.0 3 -1.0 2 0.75 6 0.75
883. LOAD 319 DL - SDL1_R + 0.75 LL + 0.75 WL(+Z)
884. REPEAT LOAD
885. 1 1.0 3 -1.0 2 0.75 7 0.75
886. LOAD 320 DL - SDL1_R + 0.75 LL + 0.75 WL(-Z)
887. REPEAT LOAD
888. 1 1.0 3 -1.0 2 0.75 8 0.75
889. LOAD 321 DL - SDL2_R + 0.75 LL + 0.75 WL(+X)
890. REPEAT LOAD
891. 1 1.0 4 -1.0 2 0.75 5 0.75
892. LOAD 322 DL - SDL2_R + 0.75 LL + 0.75 WL(-X)
893. REPEAT LOAD
894. 1 1.0 4 -1.0 2 0.75 6 0.75
895. LOAD 323 DL - SDL2_R + 0.75 LL + 0.75 WL(+Z)
896. REPEAT LOAD
897. 1 1.0 4 -1.0 2 0.75 7 0.75
898. LOAD 324 DL - SDL2_R + 0.75 LL + 0.75 WL(-Z)
899. REPEAT LOAD
900. 1 1.0 4 -1.0 2 0.75 8 0.75
901. **********************************************
902. LOAD 325 DL + SDL1_R + 0.75 LL + 0.525 EL(+X)
903. REPEAT LOAD
904. 1 1.0 3 1.0 2 0.75 11 0.525
905. LOAD 326 DL + SDL1_R + 0.75 LL + 0.525 EL(-X)
906. REPEAT LOAD
907. 1 1.0 3 1.0 2 0.75 11 -0.525
908. LOAD 327 DL + SDL1_R + 0.75 LL + 0.525 EL(+Z)
909. REPEAT LOAD
910. 1 1.0 3 1.0 2 0.75 12 0.525
911. LOAD 328 DL + SDL1_R + 0.75 LL + 0.525 EL(-Z)
912. REPEAT LOAD
913. 1 1.0 3 1.0 2 0.75 12 -0.525
914. LOAD 329 DL + SDL2_R + 0.75 LL + 0.525 EL(+X)
915. REPEAT LOAD
916. 1 1.0 4 1.0 2 0.75 11 0.525
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917. LOAD 330 DL + SDL2_R + 0.75 LL + 0.525 EL(-X)
918. REPEAT LOAD
919. 1 1.0 4 1.0 2 0.75 11 -0.525
920. LOAD 331 DL + SDL2_R + 0.75 LL + 0.525 EL(+Z)
921. REPEAT LOAD
922. 1 1.0 4 1.0 2 0.75 12 0.525
923. LOAD 332 DL + SDL2_R + 0.75 LL + 0.525 EL(-Z)
924. REPEAT LOAD
925. 1 1.0 4 1.0 2 0.75 12 -0.525
926. LOAD 333 DL - SDL1_R + 0.75 LL + 0.525 EL(+X)
927. REPEAT LOAD
928. 1 1.0 3 -1.0 2 0.75 11 0.525
929. LOAD 334 DL - SDL1_R + 0.75 LL + 0.525 EL(-X)
930. REPEAT LOAD
931. 1 1.0 3 -1.0 2 0.75 11 -0.525
932. LOAD 335 DL - SDL1_R + 0.75 LL + 0.525 EL(+Z)
933. REPEAT LOAD
934. 1 1.0 3 -1.0 2 0.75 12 0.525
935. LOAD 336 DL - SDL1_R + 0.75 LL + 0.525 EL(-Z)
936. REPEAT LOAD
937. 1 1.0 3 -1.0 2 0.75 12 -0.525
938. LOAD 337 DL - SDL2_R + 0.75 LL + 0.525 EL(+X)
939. REPEAT LOAD
940. 1 1.0 4 -1.0 2 0.75 11 0.525
941. LOAD 338 DL - SDL2_R + 0.75 LL + 0.525 EL(-X)
942. REPEAT LOAD
943. 1 1.0 4 -1.0 2 0.75 11 -0.525
944. LOAD 339 DL - SDL2_R + 0.75 LL + 0.525 EL(+Z)
945. REPEAT LOAD
946. 1 1.0 4 -1.0 2 0.75 12 0.525
947. LOAD 340 DL - SDL2_R + 0.75 LL + 0.525 EL(-Z)
948. REPEAT LOAD
949. 1 1.0 4 -1.0 2 0.75 12 -0.525
950. ************************************************
951. LOAD 341 0.6 (DL + SDL1_R) + WL(+X)
952. REPEAT LOAD
953. 1 0.6 3 0.6 5 1.0
954. LOAD 342 0.6 (DL + SDL1_R) + WL(-X)
955. REPEAT LOAD
956. 1 0.6 3 0.6 6 1.0
957. LOAD 343 0.6 (DL + SDL1_R) + WL(+Z)
958. REPEAT LOAD
959. 1 0.6 3 0.6 7 1.0
960. LOAD 344 0.6 (DL + SDL1_R) + WL(-Z)
961. REPEAT LOAD
962. 1 0.6 3 0.6 8 1.0
963. LOAD 345 0.6 (DL + SDL2_R) + WL(+X)
964. REPEAT LOAD
965. 1 0.6 4 0.6 5 1.0
966. LOAD 346 0.6 (DL + SDL2_R) + WL(-X)
967. REPEAT LOAD
968. 1 0.6 4 0.6 6 1.0
969. LOAD 347 0.6 (DL + SDL2_R) + WL(+Z)
970. REPEAT LOAD
971. 1 0.6 4 0.6 7 1.0
972. LOAD 348 0.6 (DL + SDL2_R) + WL(-Z)
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 973. REPEAT LOAD
 974. 1 0.6 4 0.6 8 1.0
 975. LOAD 349 0.6 (DL - SDL1_R) + WL(+X)
 976. REPEAT LOAD
 977. 1 0.6 3 -0.6 5 1.0
 978. LOAD 350 0.6 (DL - SDL1_R) + WL(-X)
 979. REPEAT LOAD
 980. 1 0.6 3 -0.6 6 1.0
 981. LOAD 351 0.6 (DL - SDL1_R) + WL(+Z)
 982. REPEAT LOAD
 983. 1 0.6 3 -0.6 7 1.0
 984. LOAD 352 0.6 (DL - SDL1_R) + WL(-Z)
 985. REPEAT LOAD
 986. 1 0.6 3 -0.6 8 1.0
 987. LOAD 353 0.6 (DL - SDL2_R) + WL(+X)
 988. REPEAT LOAD
 989. 1 0.6 4 -0.6 5 1.0
 990. LOAD 354 0.6 (DL - SDL2_R) + WL(-X)
 991. REPEAT LOAD
 992. 1 0.6 4 -0.6 6 1.0
 993. LOAD 355 0.6 (DL - SDL2_R) + WL(+Z)
 994. REPEAT LOAD
 995. 1 0.6 4 -0.6 7 1.0
 996. LOAD 356 0.6 (DL - SDL2_R) + WL(-Z)
 997. REPEAT LOAD
 998. 1 0.6 4 -0.6 8 1.0
 999. ************************************************
1000. LOAD 357 0.6 (DL + SDL1_R) + 0.7 EL(+X)
1001. REPEAT LOAD
1002. 1 0.6 3 0.6 11 0.7
1003. LOAD 358 0.6 (DL + SDL1_R) + 0.7 EL(-X)
1004. REPEAT LOAD
1005. 1 0.6 3 0.6 11 -0.7
1006. LOAD 359 0.6 (DL + SDL1_R) + 0.7 EL(+Z)
1007. REPEAT LOAD
1008. 1 0.6 3 0.6 12 0.7
1009. LOAD 360 0.6 (DL + SDL1_R) + 0.7 EL(-Z)
1010. REPEAT LOAD
1011. 1 0.6 3 0.6 12 -0.7
1012. LOAD 361 0.6 (DL + SDL2_R) + 0.7 EL(+X)
1013. REPEAT LOAD
1014. 1 0.6 4 0.6 11 0.7
1015. LOAD 362 0.6 (DL + SDL2_R) + 0.7 EL(-X)
1016. REPEAT LOAD
1017. 1 0.6 4 0.6 11 -0.7
1018. LOAD 363 0.6 (DL + SDL2_R) + 0.7 EL(+Z)
1019. REPEAT LOAD
1020. 1 0.6 4 0.6 12 -0.7
1021. LOAD 364 0.6 (DL + SDL2_R) + 0.7 EL(-Z)
1022. REPEAT LOAD
1023. 1 0.6 4 0.6 12 -0.7
1024. LOAD 365 0.6 (DL + SDL1_R) + 0.7 EL(+X)
1025. REPEAT LOAD
1026. 1 0.6 3 -0.6 11 0.7
1027. LOAD 366 0.6 (DL + SDL1_R) + 0.7 EL(-X)
1028. REPEAT LOAD
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1029. 1 0.6 3 -0.6 11 -0.7
1030. LOAD 367 0.6 (DL + SDL1_R) + 0.7 EL(+Z)
1031. REPEAT LOAD
1032. 1 0.6 3 -0.6 12 0.7
1033. LOAD 368 0.6 (DL + SDL1_R) + 0.7 EL(-Z)
1034. REPEAT LOAD
1035. 1 0.6 3 -0.6 12 -0.7
1036. LOAD 369 0.6 (DL + SDL2_R) + 0.7 EL(+X)
1037. REPEAT LOAD
1038. 1 0.6 4 -0.6 11 0.7
1039. LOAD 370 0.6 (DL + SDL2_R) + 0.7 EL(-X)
1040. REPEAT LOAD
1041. 1 0.6 4 -0.6 11 -0.7
1042. LOAD 371 0.6 (DL + SDL2_R) + 0.7 EL(+Z)
1043. REPEAT LOAD
1044. 1 0.6 4 -0.6 12 -0.7
1045. LOAD 372 0.6 (DL + SDL2_R) + 0.7 EL(-Z)
1046. REPEAT LOAD
1047. 1 0.6 4 -0.6 12 -0.7
1048. ************************************************
1049. *LOAD COMBINATION FOR CONCRETE DESIGN
1050. ************************************************
1051. LOAD 401 1.4 (DL+SDL1_R)
1052. REPEAT LOAD
1053. 1 1.4 3 1.4
1054. LOAD 402 1.4 (DL+SDL2_R)
1055. REPEAT LOAD
1056. 1 1.4 4 1.4
1057. LOAD 403 1.4 (DL-SDL1_R)
1058. REPEAT LOAD
1059. 1 1.4 3 -1.4
1060. LOAD 404 1.4 (DL-SDL2_R)
1061. REPEAT LOAD
1062. 1 1.4 4 -1.4
1063. ************************************************
1064. LOAD 405 1.4 (DL + SDL1_R ) + 1.7 LL
1065. REPEAT LOAD
1066. 1 1.4 3 1.4 2 1.7
1067. LOAD 406 1.4 (DL + SDL2_R ) + 1.7 LL
1068. REPEAT LOAD
1069. 1 1.4 4 1.4 2 1.7
1070. LOAD 407 1.4 (DL - SDL1_R ) + 1.7 LL
1071. REPEAT LOAD
1072. 1 1.4 3 -1.4 2 1.7
1073. LOAD 408 1.4 (DL - SDL2_R ) + 1.7 LL
1074. REPEAT LOAD
1075. 1 1.4 4 -1.4 2 1.7
1076. ************************************************
1077. LOAD 409 1.2 (DL + SDL1_R) + 0.5 LL + 1.6 WL(+X)
1078. REPEAT LOAD
1079. 1 1.2 3 1.2 2 0.5 5 1.6
1080. LOAD 410 1.2 (DL + SDL1_R) + 0.5 LL + 1.6 WL(-X)
1081. REPEAT LOAD
1082. 1 1.2 3 1.2 2 0.5 6 1.6
1083. LOAD 411 1.2 (DL + SDL1_R) + 0.5 LL + 1.6 WL(+Z)
1084. REPEAT LOAD
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1085. 1 1.2 3 1.2 2 0.5 7 1.6
1086. LOAD 412 1.2 (DL + SDL1_R) + 0.5 LL + 1.6 WL(-Z)
1087. REPEAT LOAD
1088. 1 1.2 3 1.2 2 0.5 8 1.6
1089. LOAD 413 1.2 (DL + SDL2_R) + 0.5 LL + 1.6 WL(+X)
1090. REPEAT LOAD
1091. 1 1.2 4 1.2 2 0.5 5 1.6
1092. LOAD 414 1.2 (DL + SDL2_R) + 0.5 LL + 1.6 WL(-X)
1093. REPEAT LOAD
1094. 1 1.2 4 1.2 2 0.5 6 1.6
1095. LOAD 415 1.2 (DL + SDL2_R) + 0.5 LL + 1.6 WL(+Z)
1096. REPEAT LOAD
1097. 1 1.2 4 1.2 2 0.5 7 1.6
1098. LOAD 416 1.2 (DL + SDL2_R) + 0.5 LL + 1.6 WL(-Z)
1099. REPEAT LOAD
1100. 1 1.2 4 1.2 2 0.5 8 1.6
1101. LOAD 417 1.2 (DL - SDL1_R) + 0.5 LL + 1.6 WL(+X)
1102. REPEAT LOAD
1103. 1 1.2 3 -1.2 2 0.5 5 1.6
1104. LOAD 418 1.2 (DL - SDL1_R) + 0.5 LL + 1.6 WL(-X)
1105. REPEAT LOAD
1106. 1 1.2 3 -1.2 2 0.5 6 1.6
1107. LOAD 419 1.2 (DL - SDL1_R) + 0.5 LL + 1.6 WL(+Z)
1108. REPEAT LOAD
1109. 1 1.2 3 -1.2 2 0.5 7 1.6
1110. LOAD 420 1.2 (DL - SDL1_R) + 0.5 LL + 1.6 WL(-Z)
1111. REPEAT LOAD
1112. 1 1.2 3 -1.2 2 0.5 8 1.6
1113. LOAD 421 1.2 (DL - SDL2_R) + 0.5 LL + 1.6 WL(+X)
1114. REPEAT LOAD
1115. 1 1.2 4 -1.2 2 0.5 5 1.6
1116. LOAD 422 1.2 (DL - SDL2_R) + 0.5 LL + 1.6 WL(-X)
1117. REPEAT LOAD
1118. 1 1.2 4 -1.2 2 0.5 6 1.6
1119. LOAD 423 1.2 (DL - SDL2_R) + 0.5 LL + 1.6 WL(+Z)
1120. REPEAT LOAD
1121. 1 1.2 4 -1.2 2 0.5 7 1.6
1122. LOAD 424 1.2 (DL - SDL2_R) + 0.5 LL + 1.6 WL(-Z)
1123. REPEAT LOAD
1124. 1 1.2 4 -1.2 2 0.5 8 1.6
1125. ************************************************
1126. LOAD 425 1.2 (DL + SDL1_R) + 0.5 LL + 1.0 EL(+X)
1127. REPEAT LOAD
1128. 1 1.2 3 1.2 2 0.5 11 1.0
1129. LOAD 426 1.2 (DL + SDL1_R) + 0.5 LL + 1.0 EL(-X)
1130. REPEAT LOAD
1131. 1 1.2 3 1.2 2 0.5 11 -1.0
1132. LOAD 427 1.2 (DL + SDL1_R) + 0.5 LL + 1.0 EL(+Z)
1133. REPEAT LOAD
1134. 1 1.2 3 1.2 2 0.5 12 1.0
1135. LOAD 428 1.2 (DL + SDL1_R) + 0.5 LL + 1.0 EL(-X)
1136. REPEAT LOAD
1137. 1 1.2 3 1.2 2 0.5 12 -1.0
1138. LOAD 429 1.2 (DL + SDL2_R) + 0.5 LL + 1.0 EL(+X)
1139. REPEAT LOAD
1140. 1 1.2 4 1.2 2 0.5 11 1.0
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1141. LOAD 430 1.2 (DL + SDL2_R) + 0.5 LL + 1.0 EL(-X)
1142. REPEAT LOAD
1143. 1 1.2 4 1.2 2 0.5 11 -1.0
1144. LOAD 431 1.2 (DL + SDL2_R) + 0.5 LL + 1.0 EL(+Z)
1145. REPEAT LOAD
1146. 1 1.2 4 1.2 2 0.5 12 1.0
1147. LOAD 432 1.2 (DL + SDL2_R) + 0.5 LL + 1.0 EL(-X)
1148. REPEAT LOAD
1149. 1 1.2 4 1.2 2 0.5 12 -1.0
1150. LOAD 433 1.2 (DL - SDL1_R) + 0.5 LL + 1.0 EL(+X)
1151. REPEAT LOAD
1152. 1 1.2 3 -1.2 2 0.5 11 1.0
1153. LOAD 434 1.2 (DL - SDL1_R) + 0.5 LL + 1.0 EL(-X)
1154. REPEAT LOAD
1155. 1 1.2 3 -1.2 2 0.5 11 -1.0
1156. LOAD 435 1.2 (DL - SDL1_R) + 0.5 LL + 1.0 EL(+Z)
1157. REPEAT LOAD
1158. 1 1.2 3 -1.2 2 0.5 12 1.0
1159. LOAD 436 1.2 (DL - SDL1_R) + 0.5 LL + 1.0 EL(-X)
1160. REPEAT LOAD
1161. 1 1.2 3 -1.2 2 0.5 12 -1.0
1162. LOAD 437 1.2 (DL - SDL2_R) + 0.5 LL + 1.0 EL(+X)
1163. REPEAT LOAD
1164. 1 1.2 4 -1.2 2 0.5 11 1.0
1165. LOAD 438 1.2 (DL - SDL2_R) + 0.5 LL + 1.0 EL(-X)
1166. REPEAT LOAD
1167. 1 1.2 4 -1.2 2 0.5 11 -1.0
1168. LOAD 439 1.2 (DL - SDL2_R) + 0.5 LL + 1.0 EL(+Z)
1169. REPEAT LOAD
1170. 1 1.2 4 -1.2 2 0.5 12 1.0
1171. LOAD 440 1.2 (DL - SDL2_R) + 0.5 LL + 1.0 EL(-X)
1172. REPEAT LOAD
1173. 1 1.2 4 -1.2 2 0.5 12 -1.0
1174. ************************************************
1175. LOAD 441 0.9 (DL + SDL1_R) + 1.6 WL(+X)
1176. REPEAT LOAD
1177. 1 0.9 3 0.9 5 1.6
1178. LOAD 442 0.9 (DL + SDL1_R) + 1.6 WL(-X)
1179. REPEAT LOAD
1180. 1 0.9 3 0.9 6 1.6
1181. LOAD 443 0.9 (DL + SDL1_R) + 1.6 WL(+Z)
1182. REPEAT LOAD
1183. 1 0.9 3 0.9 7 1.6
1184. LOAD 444 0.9 (DL + SDL1_R) + 1.6 WL(-Z)
1185. REPEAT LOAD
1186. 1 0.9 3 0.9 8 1.6
1187. LOAD 445 0.9 (DL + SDL2_R) + 1.6 WL(+X)
1188. REPEAT LOAD
1189. 1 0.9 4 0.9 5 1.6
1190. LOAD 446 0.9 (DL + SDL2_R) + 1.6 WL(-X)
1191. REPEAT LOAD
1192. 1 0.9 4 0.9 6 1.6
1193. LOAD 447 0.9 (DL + SDL2_R) + 1.6 WL(+Z)
1194. REPEAT LOAD
1195. 1 0.9 4 0.9 7 1.6
1196. LOAD 448 0.9 (DL + SDL2_R) + 1.6 WL(-Z)
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1197. REPEAT LOAD
1198. 1 0.9 4 0.9 8 1.6
1199. LOAD 449 0.9 (DL - SDL1_R) + 1.6 WL(+X)
1200. REPEAT LOAD
1201. 1 0.9 3 -0.9 5 1.6
1202. LOAD 450 0.9 (DL - SDL1_R) + 1.6 WL(-X)
1203. REPEAT LOAD
1204. 1 0.9 3 -0.9 6 1.6
1205. LOAD 451 0.9 (DL - SDL1_R) + 1.6 WL(+Z)
1206. REPEAT LOAD
1207. 1 0.9 3 -0.9 7 1.6
1208. LOAD 452 0.9 (DL - SDL1_R) + 1.6 WL(-Z)
1209. REPEAT LOAD
1210. 1 0.9 3 -0.9 8 1.6
1211. LOAD 453 0.9 (DL - SDL2_R) + 1.6 WL(+X)
1212. REPEAT LOAD
1213. 1 0.9 4 -0.9 5 1.6
1214. LOAD 454 0.9 (DL - SDL2_R) + 1.6 WL(-X)
1215. REPEAT LOAD
1216. 1 0.9 4 -0.9 6 1.6
1217. LOAD 455 0.9 (DL - SDL2_R) + 1.6 WL(+Z)
1218. REPEAT LOAD
1219. 1 0.9 4 -0.9 7 1.6
1220. LOAD 456 0.9 (DL - SDL2_R) + 1.6 WL(-Z)
1221. REPEAT LOAD
1222. 1 0.9 4 -0.9 8 1.6
1223. ************************************************
1224. LOAD 457 0.9 (DL + SDL1_R) + 1.0 EL(+X)
1225. REPEAT LOAD
1226. 1 0.9 3 0.9 11 1.0
1227. LOAD 458 0.9 (DL + SDL1_R) + 1.0 EL(-X)
1228. REPEAT LOAD
1229. 1 0.9 3 0.9 11 -1.0
1230. LOAD 459 0.9 (DL + SDL1_R) + 1.0 EL(+Z)
1231. REPEAT LOAD
1232. 1 0.9 3 0.9 12 1.0
1233. LOAD 460 0.9 (DL + SDL1_R) + 1.0 EL(-Z)
1234. REPEAT LOAD
1235. 1 0.9 3 0.9 12 -1.0
1236. LOAD 461 0.9 (DL + SDL2_R) + 1.0 EL(+X)
1237. REPEAT LOAD
1238. 1 0.9 4 0.9 11 1.0
1239. LOAD 462 0.9 (DL + SDL2_R) + 1.0 EL(-X)
1240. REPEAT LOAD
1241. 1 0.9 4 0.9 11 -1.0
1242. LOAD 463 0.9 (DL + SDL2_R) + 1.0 EL(+Z)
1243. REPEAT LOAD
1244. 1 0.9 4 0.9 12 1.0
1245. LOAD 464 0.9 (DL + SDL2_R) + 1.0 EL(-Z)
1246. REPEAT LOAD
1247. 1 0.9 4 0.9 12 -1.0
1248. LOAD 465 0.9 (DL - SDL1_R) + 1.0 EL(+X)
1249. REPEAT LOAD
1250. 1 0.9 3 -0.9 11 1.0
1251. LOAD 466 0.9 (DL - SDL1_R) + 1.0 EL(-X)
1252. REPEAT LOAD
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1253. 1 0.9 3 -0.9 11 -1.0
1254. LOAD 467 0.9 (DL - SDL1_R) + 1.0 EL(+Z)
1255. REPEAT LOAD
1256. 1 0.9 3 -0.9 12 1.0
1257. LOAD 468 0.9 (DL - SDL1_R) + 1.0 EL(-Z)
1258. REPEAT LOAD
1259. 1 0.9 3 -0.9 12 -1.0
1260. LOAD 469 0.9 (DL - SDL2_R) + 1.0 EL(+X)
1261. REPEAT LOAD
1262. 1 0.9 4 -0.9 11 1.0
1263. LOAD 470 0.9 (DL - SDL2_R) + 1.0 EL(-X)
1264. REPEAT LOAD
1265. 1 0.9 4 -0.9 11 -1.0
1266. LOAD 471 0.9 (DL - SDL2_R) + 1.0 EL(+Z)
1267. REPEAT LOAD
1268. 1 0.9 4 -0.9 12 1.0
1269. LOAD 472 0.9 (DL - SDL2_R) + 1.0 EL(-Z)
1270. REPEAT LOAD
1271. 1 0.9 4 -0.9 12 -1.0
1272. PERFORM ANALYSIS
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P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**1054**	**NUMBER OF MEMBERS**	**264**
NUMBER OF PLATES	**890**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**10**

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 995/ 64/ 965 DOF
TOTAL PRIMARY LOAD CASES = 154, TOTAL DEGREES OF FREEDOM = 6192
TOTAL LOAD COMBINATION CASES = 0 SO FAR.
SIZE OF STIFFNESS MATRIX = 5976 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 92.2/ 184002.5 MB

**WARNING: IF THIS UBC/IBC ANALYSIS HAS TENSION/COMPRESSION
OR REPEAT LOAD OR RE-ANALYSIS OR SELECT OPTIMIZE, THEN EACH
UBC/IBC CASE SHOULD BE FOLLOWED BY PERFORM ANALYSIS _CHANGE.

```
************************************************************
*  IBC 2003 SEISMIC LOAD ALONG X  :                        *
*     CT =  0.020 Cu =  1.700                              *
*  TIME PERIODS :                                          *
*     Ta =  0.253 T =  0.237 Tuser = 0.237                 *
*  TIME PERIOD USED (T) =  0.237                           *
*  LOAD FACTOR          =  1.000                           *
*  DESIGN BASE SHEAR    =  1.000 X  0.022 X      6186.73   *
*                       =       138.88  KN                 *
************************************************************
```

```
************************************************************
*  IBC 2003 SEISMIC LOAD ALONG Z  :                        *
*     CT =  0.020 Cu =  1.700                              *
*  TIME PERIODS :                                          *
*     Ta =  0.253 T =  0.237 Tuser = 0.237                 *
*  TIME PERIOD USED (T) =  0.237                           *
*  LOAD FACTOR          =  1.000                           *
*  DESIGN BASE SHEAR    =  1.000 X  0.022 X      6186.73   *
*                       =       138.88  KN                 *
************************************************************
```

```
1273. LOAD LIST 401 TO 472
1274. START CONCRETE DESIGN
```

```
1275. CODE ACI 2008
1276. FYMAIN 420000 MEMB 1 TO 9 11 TO 19 21 TO 73 80 87 89 412 TO 452 469 TO 688 -
1277. 729 TO 770 789 TO 1008 1030 1032 TO 1127 1130 TO 1137 1140 TO 1147 -
1278. 1150 TO 1157 1160 TO 1284 1287 TO 1292 1295 TO 1300 1303 TO 1308 -
1279. 1311 TO 1326 1331 1360 1409 TO 1682
1280. RHOMN 0.01 MEMB 1 TO 9 11 TO 19 21 TO 73 80 87 89 412 TO 452 469 TO 688 729 -
1281. 730 TO 770 789 TO 1008 1030 1032 TO 1127 1130 TO 1137 1140 TO 1147 -
1282. 1150 TO 1157 1160 TO 1284 1287 TO 1292 1295 TO 1300 1303 TO 1308 -
1283. 1311 TO 1326 1331 1360 1409 TO 1682
1284. FC 35000 MEMB 1 TO 9 11 TO 19 21 TO 73 80 87 89 412 TO 452 469 TO 688 729 -
1285. 730 TO 770 789 TO 1008 1030 1032 TO 1127 1130 TO 1137 1140 TO 1147 -
1286. 1150 TO 1157 1160 TO 1284 1287 TO 1292 1295 TO 1300 1303 TO 1308 -
1287. 1311 TO 1326 1331 1360 1409 TO 1682
1288. SKZ 6 MEMB 2 12 412 TO 416 729 TO 733
1289. SLZ 13 MEMB 4 14 424 TO 435 741 TO 752
1290. SKY 13 MEMB 1030 1032 1051 1061 1071 1081 1091 1101 1111 1121 1131 1141 1151 -
1291. 1161 1193 1195 1197 1199 1201 1203 1205 1207 1209 1211 1213 1215 1331 1360
1292. DESIGN COLUMN 1 TO 6 11 TO 16 21 TO 24 412 TO 435 729 TO 752 1030 1032 1050 -
1293. 1051 1060 1061 1070 1071 1080 1081 1090 1091 1100 1101 1110 1111 1120 1121 -
1294. 1130 1131 1140 1141 1150 1151 1160 1161 1192 TO 1215 1331 1360
```

==========

COLUMN NO. 1 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 700.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 4900.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
44 - 12 MM	1.016	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

==========

COLUMN NO. 2 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 700.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 4900.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
44 - 12 MM	1.016	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

==========

COLUMN NO. 3 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 700.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 4900.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
44 - 12 MM	1.016	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

==========

COLUMN NO. 4 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 700.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 4900.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
44 - 12 MM	1.016	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

==========

COLUMN NO. 5 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 20 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 320.00 MM

==========

COLUMN NO. 6 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 20 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 320.00 MM

==========

COLUMN NO. 11 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 700.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 4900.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
44 - 12 MM	1.016	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

==========

COLUMN NO. 12 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 700.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 4900.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
44 - 12 MM	1.016	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

==========

COLUMN NO. 13 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 700.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 4900.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
44 - 12 MM	1.016	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

==========

COLUMN NO. 14 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 700.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 4900.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
44 - 12 MM	1.016	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

==========

COLUMN NO. 15 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 20 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 320.00 MM

==========

COLUMN NO. 16 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 20 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 320.00 MM

==========

COLUMN NO. 21 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 20 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 320.00 MM

==========

COLUMN NO. 22 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 20 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 320.00 MM

==========

COLUMN NO. 23 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 20 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 320.00 MM

==========

COLUMN NO. 24 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 20 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 320.00 MM

==========

COLUMN NO. 412 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 700.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 4900.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
44 - 12 MM	1.016	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

==========

COLUMN NO. 413 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 700.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 4900.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
44 - 12 MM	1.016	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

==========

COLUMN NO. 414 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 700.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 4900.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
44 - 12 MM	1.016	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

==========

COLUMN NO. 415 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 700.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 4900.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
44 - 12 MM	1.016	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

==========

COLUMN NO. 416 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 700.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 4900.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
44 - 12 MM	1.016	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

==========

COLUMN NO. 417 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 700.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 4900.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
44 - 12 MM	1.016	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

==========

COLUMN NO. 418 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 700.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 4900.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
44 - 12 MM	1.016	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

==========

COLUMN NO. 419 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 700.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 4900.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
44 - 12 MM	1.016	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

==========

COLUMN NO. 420 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 700.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 4900.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
44 - 12 MM	1.016	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

==========

COLUMN NO. 421 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 700.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 4900.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
44 - 12 MM	1.016	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

==========

COLUMN NO. 422 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 700.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 4900.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
44 - 12 MM	1.016	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

==========

COLUMN NO. 423 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 700.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 4900.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
44 - 12 MM	1.016	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

==========

COLUMN NO. 424 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 700.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 4900.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
44 - 12 MM	1.016	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

==========

COLUMN NO. 425 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 700.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 4900.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
44 - 12 MM	1.016	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

==========

COLUMN NO. 426 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 700.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 4900.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
44 - 12 MM	1.016	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

==========

COLUMN NO. 427 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 700.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 4900.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
44 - 12 MM	1.016	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

==========

COLUMN NO. 428 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 700.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 4900.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
44 - 12 MM	1.016	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

==========

COLUMN NO. 429 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 700.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 4900.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
44 - 12 MM	1.016	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

==========

COLUMN NO. 430 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 700.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 4900.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
44 - 12 MM	1.016	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

==========

COLUMN NO. 431 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 700.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 4900.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
44 - 12 MM	1.016	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

==========

COLUMN NO. 432 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 700.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 4900.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
44 - 12 MM	1.016	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

==========

COLUMN NO. 433 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 700.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 4900.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
44 - 12 MM	1.016	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

==========

COLUMN NO. 434 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 700.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 4900.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
44 - 12 MM	1.016	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

==========

COLUMN NO. 435 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 700.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 4900.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
44 - 12 MM	1.016	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

==========

COLUMN NO. 729 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 700.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 4900.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
44 - 12 MM	1.016	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

==========

COLUMN NO. 730 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 700.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 4900.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
44 - 12 MM	1.016	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

==========

COLUMN NO. 731 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 700.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 4900.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
44 - 12 MM	1.016	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

==========

COLUMN NO. 732 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 700.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 4900.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
44 - 12 MM	1.016	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

==========

COLUMN NO. 733 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 700.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 4900.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
44 - 12 MM	1.016	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

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COLUMN NO. 734 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 700.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 4900.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
44 - 12 MM	1.016	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

==========

COLUMN NO. 735 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 700.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 4900.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
44 - 12 MM	1.016	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

==========

COLUMN NO. 736 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 700.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 4900.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
44 - 12 MM	1.016	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

==========

COLUMN NO. 737 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 700.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 4900.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
44 - 12 MM	1.016	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

==========

COLUMN NO. 738 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 700.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 4900.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
44 - 12 MM	1.016	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

==========

COLUMN NO. 739 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 700.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 4900.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
44 - 12 MM	1.016	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

==========

COLUMN NO. 740 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 700.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 4900.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
44 - 12 MM	1.016	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

==========

COLUMN NO. 741 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 700.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 4900.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
44 - 12 MM	1.016	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

==========

COLUMN NO. 742 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 700.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 4900.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
44 - 12 MM	1.016	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

==========

COLUMN NO. 743 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 700.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 4900.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
44 - 12 MM	1.016	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

==========

COLUMN NO. 744 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 700.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 4900.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
44 - 12 MM	1.016	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

==========

COLUMN NO. 745 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 700.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 4900.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
44 - 12 MM	1.016	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

==========

COLUMN NO. 746 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 700.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 4900.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
44 - 12 MM	1.016	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

==========

COLUMN NO. 747 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 700.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 4900.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
44 - 12 MM	1.016	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

==========

COLUMN NO. 748 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 700.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 4900.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
44 - 12 MM	1.016	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

==========

COLUMN NO. 749 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 700.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 4900.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
44 - 12 MM	1.016	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

==========

COLUMN NO. 750 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 700.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 4900.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
44 - 12 MM	1.016	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

==========

COLUMN NO. 751 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 700.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 4900.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
44 - 12 MM	1.016	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

==========

COLUMN NO. 752 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 700.0 X 700.0 MMS, TIED
AREA OF STEEL REQUIRED = 5243.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
48 - 12 MM	1.108	406	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

==========

COLUMN NO. 1030 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE -1000.0 X1000.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 10000.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 40 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 576.00 MM

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COLUMN NO. 1032 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE -1000.0 X1000.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 10000.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 40 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 576.00 MM

==========

COLUMN NO. 1050 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 20 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 320.00 MM

==========

COLUMN NO. 1051 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE -1000.0 X1000.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 10000.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 40 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 576.00 MM

==========

COLUMN NO. 1060 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 20 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 320.00 MM

==========

COLUMN NO. 1061 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE -1000.0 X1000.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 10000.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 40 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 576.00 MM

==========

COLUMN NO. 1070 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 20 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 320.00 MM

==========

COLUMN NO. 1071 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE -1000.0 X1000.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 10000.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 40 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 576.00 MM

==========

COLUMN NO. 1080 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 20 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 320.00 MM

==========

COLUMN NO. 1081 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE -1000.0 X1000.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 10000.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 40 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 576.00 MM

==========

COLUMN NO. 1090 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 20 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 320.00 MM

==========

COLUMN NO. 1091 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE -1000.0 X1000.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 10000.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 40 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 576.00 MM

==========

COLUMN NO. 1100 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 20 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 320.00 MM

==========

COLUMN NO. 1101 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE -1000.0 X1000.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 10000.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 40 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 576.00 MM

==========

COLUMN NO. 1110 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 20 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 320.00 MM

==========

COLUMN NO. 1111 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE -1000.0 X1000.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 10000.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 40 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 576.00 MM

==========

COLUMN NO. 1120 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 20 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 320.00 MM

==========

COLUMN NO. 1121 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE -1000.0 X1000.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 10000.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 40 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 576.00 MM

==========

COLUMN NO. 1130 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 20 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 320.00 MM

==========

COLUMN NO. 1131 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE -1000.0 X1000.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 10000.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 40 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 576.00 MM

==========

COLUMN NO. 1140 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 20 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 320.00 MM

==========

COLUMN NO. 1141 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE -1000.0 X1000.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 10000.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 40 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 576.00 MM

==========

COLUMN NO. 1150 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 20 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 320.00 MM

==========

COLUMN NO. 1151 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE -1000.0 X1000.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 10000.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 40 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 576.00 MM

==========

COLUMN NO. 1160 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 20 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 320.00 MM

==========

COLUMN NO. 1161 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE -1000.0 X1000.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 10000.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 40 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 576.00 MM

==========

COLUMN NO. 1192 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 20 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 320.00 MM

==========

COLUMN NO. 1193 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE -1000.0 X1000.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 10000.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 40 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 576.00 MM

==========

COLUMN NO. 1194 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 20 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 320.00 MM

==========

COLUMN NO. 1195 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE -1000.0 X1000.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 10000.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 40 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 576.00 MM

==========

COLUMN NO. 1196 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 20 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 320.00 MM

==========

COLUMN NO. 1197 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE -1000.0 X1000.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 10000.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 40 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 576.00 MM

==========

COLUMN NO. 1198 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 20 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 320.00 MM

==========

COLUMN NO. 1199 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE -1000.0 X1000.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 10000.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 40 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 576.00 MM

==========

COLUMN NO. 1200 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 20 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 320.00 MM

==========

COLUMN NO. 1201 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE -1000.0 X1000.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 10000.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 40 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 576.00 MM

==========

COLUMN NO. 1202 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 20 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 320.00 MM

==========

COLUMN NO. 1203 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE -1000.0 X1000.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 10000.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 40 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 576.00 MM

==========

COLUMN NO. 1204 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 20 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 320.00 MM

==========

COLUMN NO. 1205 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE -1000.0 X1000.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 10000.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 40 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 576.00 MM

==========

COLUMN NO. 1206 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 20 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 320.00 MM

==========

COLUMN NO. 1207 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE -1000.0 X1000.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 10000.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 40 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 576.00 MM

==========

COLUMN NO. 1208 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 20 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 320.00 MM

==========

COLUMN NO. 1209 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE -1000.0 X1000.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 10000.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 40 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 576.00 MM

==========

COLUMN NO. 1210 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 20 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 320.00 MM

==========

COLUMN NO. 1211 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE -1000.0 X1000.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 10000.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 40 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 576.00 MM

==========

COLUMN NO. 1212 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 20 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 320.00 MM

==========

COLUMN NO. 1213 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE -1000.0 X1000.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 10000.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 40 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 576.00 MM

==========

COLUMN NO. 1214 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 20 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 320.00 MM

==========

COLUMN NO. 1215 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE -1000.0 X1000.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 10000.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 40 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 576.00 MM

==========

COLUMN NO. 1331 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE -1000.0 X1000.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 10000.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 40 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 576.00 MM

==========

COLUMN NO. 1360 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE -1000.0 X1000.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 10000.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 40 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 576.00 MM

**********END OF COLUMN DESIGN RESULTS**********

1295. END CONCRETE DESIGN
1296. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= APR 26,2022 TIME= 16:29: 9 ****

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************************************************************
*   For technical assistance on STAAD.Pro, please visit    *
*   http://selectservices.bentley.com/en-US/               *
*                                                          *
*   Details about additional assistance from               *
*   Bentley and Partners can be found at program menu      *
*   Help->Technical Support                                *
*                                                          *
*      Copyright (c) 1997-2016 Bentley Systems, Inc.       *
*                http://www.bentley.com                    *
************************************************************
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Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	C02	D01			Outotec / L&T Revision No.:	0	1	2	3	
Date	27-01-2020	12-03-2020	25-04-2022	13/10/2023			Date	27-01-2020	12-03-2020	25-04-2022	13/10/2023	
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS DT. 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER SUPPORT STRUCTURE	AS BUILT			Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS DT. 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER SUPPORT STRUCTURE	AS BUILT	
Prepared By	APR	APR	APR	APR			Prepared By	APR	APR	APR	APR	
Checked	SBY	SBY	SBY	SBY			Checked	SBY	SBY	SBY	SBY	
Approved	AG	AG	AG	AG			Approved	AG	AG	AG	AG	

ATTACHMENT 4 STAAD INPUT FILE OF FOUNDATION CF1_1

Job Title:

Client:

Engineer:

```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 17-Jan-20
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
5 0 0.5 0; 7 0 0.5 3.3; 101 0 -2.5 0; 102 0 -2.5 3.3; 103 1.2 -2.5 -1.2;
104 1.2 -2.5 4.5; 105 -1.2 -2.5 -1.2; 106 -1.2 -2.5 4.5; 107 1.2 -2.5 0;
108 -1.2 -2.5 0; 109 1.2 -2.5 3.3; 110 -1.2 -2.5 3.3; 111 0 -2.5 -1.2;
112 0 -2.5 4.5; 113 -1.2 -2.5 -0.8; 114 -0.8 -2.5 -0.8; 115 -0.8 -2.5 -1.2;
116 -1.2 -2.5 -0.4; 117 -0.8 -2.5 -0.4; 118 -0.8 -2.5 0; 119 -0.4 -2.5 -0.8;
120 -0.4 -2.5 -1.2; 121 -0.4 -2.5 -0.4; 122 -0.4 -2.5 0; 123 0 -2.5 -0.8;
124 0 -2.5 -0.4; 125 0.4 -2.5 -0.8; 126 0.4 -2.5 -1.2; 127 0.4 -2.5 -0.4;
128 0.4 -2.5 0; 129 0.8 -2.5 -0.8; 130 0.8 -2.5 -1.2; 131 0.8 -2.5 -0.4;
132 0.8 -2.5 0; 133 1.2 -2.5 -0.8; 134 1.2 -2.5 -0.4; 135 -1.2 -2.5 3.7;
136 -0.8 -2.5 3.7; 137 -0.8 -2.5 3.3; 138 -1.2 -2.5 4.1; 139 -0.8 -2.5 4.1;
140 -0.8 -2.5 4.5; 141 -0.4 -2.5 3.7; 142 -0.4 -2.5 3.3; 143 -0.4 -2.5 4.1;
144 -0.4 -2.5 4.5; 145 0 -2.5 3.7; 146 0 -2.5 4.1; 147 0.4 -2.5 3.7;
148 0.4 -2.5 3.3; 149 0.4 -2.5 4.1; 150 0.4 -2.5 4.5; 151 0.8 -2.5 3.7;
152 0.8 -2.5 3.3; 153 0.8 -2.5 4.1; 154 0.8 -2.5 4.5; 155 1.2 -2.5 3.7;
156 1.2 -2.5 4.1; 157 -1.2 -2.5 0.4; 158 -0.8 -2.5 0.4; 159 -1.2 -2.5 0.66;
160 -0.8 -2.5 0.66; 161 -1.2 -2.5 0.9; 162 -0.8 -2.5 0.9; 163 -1.2 -2.5 1.2;
164 -0.8 -2.5 1.2; 165 -1.2 -2.5 1.65; 166 -0.8 -2.5 1.65; 167 -1.2 -2.5 2.1;
168 -0.8 -2.5 2.1; 169 -1.2 -2.5 2.3; 170 -0.8 -2.5 2.3; 171 -1.2 -2.5 2.65;
172 -0.8 -2.5 2.65; 173 -1.2 -2.5 2.9; 174 -0.8 -2.5 2.9; 175 -0.4 -2.5 0.4;
176 -0.4 -2.5 0.66; 177 -0.4 -2.5 0.9; 178 -0.4 -2.5 1.2; 179 -0.4 -2.5 1.65;
180 -0.4 -2.5 2.1; 181 -0.4 -2.5 2.3; 182 -0.4 -2.5 2.65; 183 -0.4 -2.5 2.9;
184 0 -2.5 0.4; 185 0 -2.5 0.66; 186 0 -2.5 0.9; 187 0 -2.5 1.2;
188 0 -2.5 1.65; 189 0 -2.5 2.1; 190 0 -2.5 2.3; 191 0 -2.5 2.65;
192 0 -2.5 2.9; 193 0.4 -2.5 0.4; 194 0.4 -2.5 0.66; 195 0.4 -2.5 0.9;
196 0.4 -2.5 1.2; 197 0.4 -2.5 1.65; 198 0.4 -2.5 2.1; 199 0.4 -2.5 2.3;
200 0.4 -2.5 2.65; 201 0.4 -2.5 2.9; 202 0.8 -2.5 0.4; 203 0.8 -2.5 0.66;
204 0.8 -2.5 0.9; 205 0.8 -2.5 1.2; 206 0.8 -2.5 1.65; 207 0.8 -2.5 2.1;
208 0.8 -2.5 2.3; 209 0.8 -2.5 2.65; 210 0.8 -2.5 2.9; 211 1.2 -2.5 0.4;
212 1.2 -2.5 0.66; 213 1.2 -2.5 0.9; 214 1.2 -2.5 1.2; 215 1.2 -2.5 1.65;
216 1.2 -2.5 2.1; 217 1.2 -2.5 2.3; 218 1.2 -2.5 2.65; 219 1.2 -2.5 2.9;
220 1.2 -2.5 5.1; 221 -1.2 -2.5 5.1; 222 0 -2.5 5.1; 223 -0.8 -2.5 5.1;
224 -0.4 -2.5 5.1; 225 0.4 -2.5 5.1; 226 0.8 -2.5 5.1; 227 1.2 -2.5 -1.6;
228 -1.2 -2.5 -1.6; 229 0 -2.5 -1.6; 230 -0.8 -2.5 -1.6; 231 -0.4 -2.5 -1.6;
232 0.4 -2.5 -1.6; 233 0.8 -2.5 -1.6; 234 1.2 -2.5 5.7; 235 -1.2 -2.5 5.7;
236 0 -2.5 5.7; 237 -0.8 -2.5 5.7; 238 -0.4 -2.5 5.7; 239 0.4 -2.5 5.7;
240 0.8 -2.5 5.7; 241 -1.6 -2.5 -1.2; 242 -1.6 -2.5 4.5; 243 -1.6 -2.5 0;
244 -1.6 -2.5 3.3; 245 -1.6 -2.5 -0.8; 246 -1.6 -2.5 -0.4; 247 -1.6 -2.5 3.7;
248 -1.6 -2.5 4.1; 249 -1.6 -2.5 0.4; 250 -1.6 -2.5 0.66; 251 -1.6 -2.5 0.9;
252 -1.6 -2.5 1.2; 253 -1.6 -2.5 1.65; 254 -1.6 -2.5 2.1; 255 -1.6 -2.5 2.3;
256 -1.6 -2.5 2.65; 257 -1.6 -2.5 2.9; 258 -1.6 -2.5 5.1; 259 -1.6 -2.5 -1.6;
260 -1.6 -2.5 5.7; 261 1.2 -2.5 -2; 262 -1.2 -2.5 -2; 263 0 -2.5 -2;
264 -0.8 -2.5 -2; 265 -0.4 -2.5 -2; 266 0.4 -2.5 -2; 267 0.8 -2.5 -2;
268 -1.6 -2.5 -2; 269 1.2 -2.5 6.3; 270 -1.2 -2.5 6.3; 271 0 -2.5 6.3;
272 -0.8 -2.5 6.3; 273 -0.4 -2.5 6.3; 274 0.4 -2.5 6.3; 275 0.8 -2.5 6.3;
276 -1.6 -2.5 6.3; 277 -1.9 -2.5 -1.2; 278 -1.9 -2.5 4.5; 279 -1.9 -2.5 0;
280 -1.9 -2.5 3.3; 281 -1.9 -2.5 -0.8; 282 -1.9 -2.5 -0.4; 283 -1.9 -2.5 3.7;
284 -1.9 -2.5 4.1; 285 -1.9 -2.5 0.4; 286 -1.9 -2.5 0.66; 287 -1.9 -2.5 0.9;
```

Job Title:

Client:

Engineer:

288 -1.9 -2.5 1.2; 289 -1.9 -2.5 1.65; 290 -1.9 -2.5 2.1; 291 -1.9 -2.5 2.3;
292 -1.9 -2.5 2.65; 293 -1.9 -2.5 2.9; 294 -1.9 -2.5 5.1; 295 -1.9 -2.5 -1.6;
296 -1.9 -2.5 5.7; 297 -1.9 -2.5 -2; 298 -1.9 -2.5 6.3; 299 1.6 -2.5 -1.2;
300 1.6 -2.5 4.5; 301 1.6 -2.5 0; 302 1.6 -2.5 3.3; 303 1.6 -2.5 -0.8;
304 1.6 -2.5 -0.4; 305 1.6 -2.5 3.7; 306 1.6 -2.5 4.1; 307 1.6 -2.5 0.4;
308 1.6 -2.5 0.66; 309 1.6 -2.5 0.9; 310 1.6 -2.5 1.2; 311 1.6 -2.5 1.65;
312 1.6 -2.5 2.1; 313 1.6 -2.5 2.3; 314 1.6 -2.5 2.65; 315 1.6 -2.5 2.9;
316 1.6 -2.5 5.1; 317 1.6 -2.5 -1.6; 318 1.6 -2.5 5.7; 319 1.6 -2.5 -2;
320 1.6 -2.5 6.3;
MEMBER INCIDENCES
1 101 5; 2 102 7;
ELEMENT INCIDENCES SHELL
12 105 113 114 115; 13 113 116 117 114; 14 116 108 118 117; 15 115 114 119 120;
16 114 117 121 119; 17 117 118 122 121; 18 120 119 123 111; 19 119 121 124 123;
20 121 122 101 124; 21 111 123 125 126; 22 123 124 127 125; 23 124 101 128 127;
24 126 125 129 130; 25 125 127 131 129; 26 127 128 132 131; 27 130 129 133 103;
28 129 131 134 133; 29 131 132 107 134; 30 110 135 136 137; 31 135 138 139 136;
32 138 106 140 139; 33 137 136 141 142; 34 136 139 143 141; 35 139 140 144 143;
36 142 141 145 102; 37 141 143 146 145; 38 143 144 112 146; 39 102 145 147 148;
40 145 146 149 147; 41 146 112 150 149; 42 148 147 151 152; 43 147 149 153 151;
44 149 150 154 153; 45 152 151 155 109; 46 151 153 156 155; 47 153 154 104 156;
48 108 157 158 118; 49 157 159 160 158; 50 159 161 162 160; 51 161 163 164 162;
52 163 165 166 164; 53 165 167 168 166; 54 167 169 170 168; 55 169 171 172 170;
56 171 173 174 172; 57 173 110 137 174; 58 118 158 175 122; 59 158 160 176 175;
60 160 162 177 176; 61 162 164 178 177; 62 164 166 179 178; 63 166 168 180 179;
64 168 170 181 180; 65 170 172 182 181; 66 172 174 183 182; 67 174 137 142 183;
68 122 175 184 101; 69 175 176 185 184; 70 176 177 186 185; 71 177 178 187 186;
72 178 179 188 187; 73 179 180 189 188; 74 180 181 190 189; 75 181 182 191 190;
76 182 183 192 191; 77 183 142 102 192; 78 101 184 193 128; 79 184 185 194 193;
80 185 186 195 194; 81 186 187 196 195; 82 187 188 197 196; 83 188 189 198 197;
84 189 190 199 198; 85 190 191 200 199; 86 191 192 201 200; 87 192 102 148 201;
88 128 193 202 132; 89 193 194 203 202; 90 194 195 204 203; 91 195 196 205 204;
92 196 197 206 205; 93 197 198 207 206; 94 198 199 208 207; 95 199 200 209 208;
96 200 201 210 209; 97 201 148 152 210; 98 132 202 211 107; 99 202 203 212 211;
100 203 204 213 212; 101 204 205 214 213; 102 205 206 215 214;
103 206 207 216 215; 104 207 208 217 216; 105 208 209 218 217;
106 209 210 219 218; 107 210 152 109 219; 108 106 221 223 140;
109 140 223 224 144; 110 144 224 222 112; 111 112 222 225 150;
112 150 225 226 154; 113 154 226 220 104; 114 228 105 115 230;
115 230 115 120 231; 116 231 120 111 229; 117 229 111 126 232;
118 232 126 130 233; 119 233 130 103 227; 120 221 235 237 223;
121 223 237 238 224; 122 224 238 236 222; 123 222 236 239 225;
124 225 239 240 226; 125 226 240 234 220; 126 241 245 113 105;
127 245 246 116 113; 128 246 243 108 116; 129 244 247 135 110;
130 247 248 138 135; 131 248 242 106 138; 132 243 249 157 108;
133 249 250 159 157; 134 250 251 161 159; 135 251 252 163 161;
136 252 253 165 163; 137 253 254 167 165; 138 254 255 169 167;
139 255 256 171 169; 140 256 257 173 171; 141 257 244 110 173;
142 242 258 221 106; 143 259 241 105 228; 144 258 260 235 221;
145 262 228 230 264; 146 264 230 231 265; 147 265 231 229 263;
148 263 229 232 266; 149 266 232 233 267; 150 267 233 227 261;
151 268 259 228 262; 152 235 270 272 237; 153 237 272 273 238;
154 238 273 271 236; 155 236 271 274 239; 156 239 274 275 240;
157 240 275 269 234; 158 260 276 270 235; 159 277 281 245 241;



Job Title:

Client:

Engineer:

```
160 281 282 246 245; 161 282 279 243 246; 162 280 283 247 244;
163 283 284 248 247; 164 284 278 242 248; 165 279 285 249 243;
166 285 286 250 249; 167 286 287 251 250; 168 287 288 252 251;
169 288 289 253 252; 170 289 290 254 253; 171 290 291 255 254;
172 291 292 256 255; 173 292 293 257 256; 174 293 280 244 257;
175 278 294 258 242; 176 295 277 241 259; 177 294 296 260 258;
178 297 295 259 268; 179 296 298 276 260; 180 103 133 303 299;
181 133 134 304 303; 182 134 107 301 304; 183 109 155 305 302;
184 155 156 306 305; 185 156 104 300 306; 186 107 211 307 301;
187 211 212 308 307; 188 212 213 309 308; 189 213 214 310 309;
190 214 215 311 310; 191 215 216 312 311; 192 216 217 313 312;
193 217 218 314 313; 194 218 219 315 314; 195 219 109 302 315;
196 104 220 316 300; 197 227 103 299 317; 198 220 234 318 316;
199 261 227 317 319; 200 234 269 320 318;
START GROUP DEFINITION
ELEMENT
_RAFT_MOMENT 12 TO 19 21 22 24 TO 35 37 38 40 TO 67 69 TO 76 79 TO 86 88 TO 200
_RAFT_SHEAR 52 53 62 63 72 73 82 83 92 93 102 103 108 TO 200
END GROUP DEFINITION
ELEMENT PROPERTY
12 TO 200 THICKNESS 0.8
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 23.5616
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
1 2 PRIS YD 0.8 ZD 0.8
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
101 TO 320 ELASTIC MAT DIRECT Y SUBGRADE 20000 PRINT
LOAD 1 DEAD  LOAD
SELFWEIGHT Y -1 LIST ALL
*Soil Weight
ELEMENT LOAD
12 TO 200 PR GY -28.8
joint load
5   FX  -0.4    FY  -202.9  FZ  -6.3
7   FX  -0.9    FY  -47.0   FZ  8.2
Load 2  LL
joint load
5   FX  0.7 FY  -285.4  FZ  -8.7
7   FX  0.0 FY  -66.6   FZ  11.8
Load 3  SDL1
joint load
5   FX  -35.7   FY  -521.9  FZ  0.1
7   FX  -0.1    FY  -13.6   FZ  19.3
Load 4  SDL2
```



Job Title:

Client:

Engineer:

```
joint load
5   FX  -14.0   FY  -1496.9 FZ  -16.1
7   FX  -0.1    FY  -46.8   FZ  58.9
Load 5  WIND LOAD (+X)
joint load
5   FX  98.384  FY  475.426 FZ  11.954
7   FX  16.046  FY  72.512  FZ  -13.863
Load 6  WIND LOAD (-X)
joint load
5   FX  -98.056 FY  -474.116    FZ  -11.303
7   FX  -16.046 FY  -73.88  FZ  14.616
Load 7  WIND LOAD (+Z)
joint load
5   FX  14.196  FY  -33.994 FZ  122.997
7   FX  -0.04   FY  -264.954    FZ  154.137
Load 8  WIND LOAD (+Z)
joint load
5   FX  -13.966 FY  44.013  FZ  -126.049
7   FX  0.042   FY  272.325 FZ  -158.527
Load 9  TL1
joint load
5   FX  -1.33   FY  19.638  FZ  164.329
7   FX  -1.176  FY  -7.537  FZ  69.202
Load 10 TL2
joint load
5   FX  1.33    FY  -19.638 FZ  -164.329
7   FX  1.176   FY  7.537   FZ  -69.202
Load 11 EL (+X)
joint load
5   FX  5.26    FY  34.499  FZ  0.784
7   FX  0.221   FY  1.501   FZ  -0.827
Load 12 EL (+Z)
joint load
5   FX  0.078   FY  -10.48  FZ  4.237
7   FX  -0.002  FY  -10.017 FZ  5.917


***LOAD COMBINATION FOR BASE PRESSURE
LOAD 301 DL+SDL1
REPEAT LOAD
1 1.0 3 1.0
LOAD 302 DL+SDL2
REPEAT LOAD
1 1.0 4 1.0
LOAD 303 DL + SDL1 + TL1 + LL
REPEAT LOAD
1 1.0 3 1.0 9 1.0 2 1.0
LOAD 304 DL + SDL1 + TL2 + LL
REPEAT LOAD
1 1.0 4 1.0 9 1.0 2 1.0
LOAD 305 DL + SDL2 + TL1 + LL
REPEAT LOAD
1 1.0 3 1.0 10 1.0 2 1.0
LOAD 306 DL + SDL2 + TL2 + LL
```

Job Title:

Client:

Engineer:

```
REPEAT LOAD
1 1.0 4 1.0 10 1.0 2 1.0
LOAD 307 DL + SDL1 + 0.75 LL + 0.75 WL(+X)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 5 0.75
LOAD 308 DL + SDL1 + 0.75 LL + 0.75 WL(-X)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 6 0.75
LOAD 309 DL + SDL1 + 0.75 LL + 0.75 WL(+Z)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 7 0.75
LOAD 310 DL + SDL1 + 0.75 LL + 0.75 WL(-Z)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 8 0.75
LOAD 311 DL + SDL2 + 0.75 LL + 0.75 WL(+X)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 5 0.75
LOAD 312 DL + SDL2 + 0.75 LL + 0.75 WL(-X)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 6 0.75
LOAD 313 DL + SDL2 + 0.75 LL + 0.75 WL(+Z)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 7 0.75
LOAD 314 DL + SDL2 + 0.75 LL + 0.75 WL(-Z)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 8 0.75
LOAD 315 DL + SDL1 + 0.75 LL + 0.525 EL(+X)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 11 0.525
LOAD 316 DL + SDL1 + 0.75 LL + 0.525 EL(-X)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 11 -0.525
LOAD 317 DL + SDL1 + 0.75 LL + 0.525 EL(+Z)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 12 0.525
LOAD 318 DL + SDL1 + 0.75 LL + 0.525 EL(-Z)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 12 -0.525
LOAD 319 DL + SDL2 + 0.75 LL + 0.525 EL(+X)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 11 0.525
LOAD 320 DL + SDL2 + 0.75 LL + 0.525 EL(-X)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 11 -0.525
LOAD 321 DL + SDL2 + 0.75 LL + 0.525 EL(+Z)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 12 0.525
LOAD 322 DL + SDL2 + 0.75 LL + 0.525 EL(-Z)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 12 -0.525
LOAD 323 0.6 (DL + SDL1) + WL(+X)
REPEAT LOAD
1 0.6 3 0.6 5 1.0
LOAD 324 0.6 (DL + SDL1) + WL(-X)
```

Job Title:

Client:

Engineer:

```
REPEAT LOAD
1 0.6 3 0.6 6 1.0
LOAD 325 0.6 (DL + SDL1) + WL(+Z)
REPEAT LOAD
1 0.6 3 0.6 7 1.0
LOAD 326 0.6 (DL + SDL1) + WL(-Z)
REPEAT LOAD
1 0.6 3 0.6 8 1.0
LOAD 327 0.6 (DL + SDL2) + WL(+X)
REPEAT LOAD
1 0.6 4 0.6 5 1.0
LOAD 328 0.6 (DL + SDL2) + WL(-X)
REPEAT LOAD
1 0.6 4 0.6 6 1.0
LOAD 329 0.6 (DL + SDL2) + WL(+Z)
REPEAT LOAD
1 0.6 4 0.6 7 1.0
LOAD 330 0.6 (DL + SDL2) + WL(-Z)
REPEAT LOAD
1 0.6 4 0.6 8 1.0
LOAD 331 0.6 (DL + SDL1) + 0.7 EL(+X)
REPEAT LOAD
1 0.6 3 0.6 11 0.7
LOAD 332 0.6 (DL + SDL1) + 0.7 EL(-X)
REPEAT LOAD
1 0.6 3 0.6 11 -0.7
LOAD 333 0.6 (DL + SDL1) + 0.7 EL(+Z)
REPEAT LOAD
1 0.6 3 0.6 12 0.7
LOAD 334 0.6 (DL + SDL1) + 0.7 EL(-Z)
REPEAT LOAD
1 0.6 3 0.6 12 -0.7
LOAD 335 0.6 (DL + SDL2) + 0.7 EL(+X)
REPEAT LOAD
1 0.6 4 0.6 11 0.7
LOAD 336 0.6 (DL + SDL2) + 0.7 EL(-X)
REPEAT LOAD
1 0.6 4 0.6 11 -0.7
LOAD 337 0.6 (DL + SDL2) + 0.7 EL(+Z)
REPEAT LOAD
1 0.6 4 0.6 12 -0.7
LOAD 338 0.6 (DL + SDL2) + 0.7 EL(-Z)
REPEAT LOAD
1 0.6 4 0.6 12 -0.7
****
*Load Combination for Concrete design
LOAD 401 1.4 (DL+SDL1)
REPEAT LOAD
1 1.4 3 1.4
LOAD 402 1.4 (DL+SDL2)
REPEAT LOAD
1 1.4 4 1.4
LOAD 403 1.4 (DL + SDL1 + TL1) + 1.7 LL
REPEAT LOAD
```



Job Title:

Client:

Engineer:

```
1 1.4 3 1.4 9 1.4 2 1.7
LOAD 404 1.4 (DL + SDL1 + TL2) + 1.7 LL
REPEAT LOAD
1 1.4 3 1.4 10 1.4 2 1.7
LOAD 405 1.4 (DL + SDL2 + TL1) + 1.7 LL
REPEAT LOAD
1 1.4 4 1.4 9 1.4 2 1.7
LOAD 406 1.4 (DL + SDL2 + TL2) + 1.7 LL
REPEAT LOAD
1 1.4 4 1.4 10 1.4 2 1.7
LOAD 407 1.2 (DL + SDL1) + 0.5 LL + 1.6 WL(+X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 5 1.6
LOAD 408 1.2 (DL + SDL1) + 0.5 LL + 1.6 WL(-X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 6 1.6
LOAD 409 1.2 (DL + SDL1) + 0.5 LL + 1.6 WL(+Z)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 7 1.6
LOAD 410 1.2 (DL + SDL1) + 0.5 LL + 1.6 WL(-Z)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 8 1.6
LOAD 411 1.2 (DL + SDL2) + 0.5 LL + 1.6 WL(+X)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 5 1.6
LOAD 412 1.2 (DL + SDL2) + 0.5 LL + 1.6 WL(-X)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 6 1.6
LOAD 413 1.2 (DL + SDL2) + 0.5 LL + 1.6 WL(+Z)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 7 1.6
LOAD 414 1.2 (DL + SDL2) + 0.5 LL + 1.6 WL(-Z)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 8 1.6
LOAD 415 1.2 (DL + SDL1) + 0.5 LL + 1.0 EL(+X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 11 1.0
LOAD 416 1.2 (DL + SDL1) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 11 -1.0
LOAD 417 1.2 (DL + SDL1) + 0.5 LL + 1.0 EL(+Z)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 12 1.0
LOAD 418 1.2 (DL + SDL1) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 12 -1.0
LOAD 419 1.2 (DL + SDL2) + 0.5 LL + 1.0 EL(+X)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 11 1.0
LOAD 420 1.2 (DL + SDL2) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 11 -1.0
LOAD 421 1.2 (DL + SDL2) + 0.5 LL + 1.0 EL(+Z)
REPEAT LOAD
```



Job Title:

Client:

Engineer:

```
1 1.2 4 1.2 2 0.5 12 1.0
LOAD 422 1.2 (DL + SDL2) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 12 -1.0
LOAD 423 0.9 (DL + SDL1) + 1.6 WL(+X)
REPEAT LOAD
1 0.9 3 0.9 5 1.6
LOAD 424 0.9 (DL + SDL1) + 1.6 WL(-X)
REPEAT LOAD
1 0.9 3 0.9 6 1.6
LOAD 425 0.9 (DL + SDL1) + 1.6 WL(+Z)
REPEAT LOAD
1 0.9 3 0.9 7 1.6
LOAD 426 0.9 (DL + SDL1) + 1.6 WL(-Z)
REPEAT LOAD
1 0.9 3 0.9 8 1.6
LOAD 427 0.9 (DL + SDL2) + 1.6 WL(+X)
REPEAT LOAD
1 0.9 4 0.9 5 1.6
LOAD 428 0.9 (DL + SDL2) + 1.6 WL(-X)
REPEAT LOAD
1 0.9 4 0.9 6 1.6
LOAD 429 0.9 (DL + SDL2) + 1.6 WL(+Z)
REPEAT LOAD
1 0.9 4 0.9 7 1.6
LOAD 430 0.9 (DL + SDL2) + 1.6 WL(-Z)
REPEAT LOAD
1 0.9 4 0.9 8 1.6
LOAD 431 0.9 (DL + SDL1) + 1.0 EL(+X)
REPEAT LOAD
1 0.9 3 0.9 11 1.0
LOAD 432 0.9 (DL + SDL1) + 1.0 EL(-X)
REPEAT LOAD
1 0.9 3 0.9 11 -1.0
LOAD 433 0.9 (DL + SDL1) + 1.0 EL(+Z)
REPEAT LOAD
1 0.9 3 0.9 12 1.0
LOAD 434 0.9 (DL + SDL1) + 1.0 EL(-Z)
REPEAT LOAD
1 0.9 3 0.9 12 -1.0
LOAD 435 0.9 (DL + SDL2) + 1.0 EL(+X)
REPEAT LOAD
1 0.9 4 0.9 11 1.0
LOAD 436 0.9 (DL + SDL2) + 1.0 EL(-X)
REPEAT LOAD
1 0.9 4 0.9 11 -1.0
LOAD 437 0.9 (DL + SDL2) + 1.0 EL(+Z)
REPEAT LOAD
1 0.9 4 0.9 12 1.0
LOAD 438 0.9 (DL + SDL2) + 1.0 EL(-Z)
REPEAT LOAD
1 0.9 4 0.9 12 -1.0
PERFORM ANALYSIS
FINISH
```

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	C02	D01			Outotec / L&T Revision No.:	0	1	2	3	
Date	27-01-2020	12-03-2020	25-04-2022	13/10/2023			Date	27-01-2020	12-03-2020	25-04-2022	13/10/2023	
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS DT. 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER SUPPORT STRUCTURE	AS BUILT			Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS DT. 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER SUPPORT STRUCTURE	AS BUILT	
Prepared By	APR	APR	APR	APR			Prepared By	APR	APR	APR	APR	
Checked	SBY	SBY	SBY	SBY			Checked	SBY	SBY	SBY	SBY	
Approved	AG	AG	AG	AG			Approved	AG	AG	AG	AG	

ATTACHMENT 5 STAAD INPUT FILE OF FOUNDATION CF1_2

Job Title:

Client:

Engineer:

```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 17-Jan-20
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
6 0 1 -0.5; 8 0 1 2.8; 101 0 -2 -0.5; 102 0 -2 2.8; 103 1.2 -2 -1.7;
104 1.2 -2 4; 105 -1.2 -2 -1.7; 106 -1.2 -2 4; 107 1.2 -2 -0.5;
108 -1.2 -2 -0.5; 109 1.2 -2 2.8; 110 -1.2 -2 2.8; 111 0 -2 -1.7; 112 0 -2 4;
113 -1.2 -2 -1.3; 114 -0.8 -2 -1.3; 115 -0.8 -2 -1.7; 116 -1.2 -2 -0.9;
117 -0.8 -2 -0.9; 118 -0.8 -2 -0.5; 119 -0.4 -2 -1.3; 120 -0.4 -2 -1.7;
121 -0.4 -2 -0.9; 122 -0.4 -2 -0.5; 123 0 -2 -1.3; 124 0 -2 -0.9;
125 0.4 -2 -1.3; 126 0.4 -2 -1.7; 127 0.4 -2 -0.9; 128 0.4 -2 -0.5;
129 0.8 -2 -1.3; 130 0.8 -2 -1.7; 131 0.8 -2 -0.9; 132 0.8 -2 -0.5;
133 1.2 -2 -1.3; 134 1.2 -2 -0.9; 135 -1.2 -2 3.2; 136 -0.8 -2 3.2;
137 -0.8 -2 2.8; 138 -1.2 -2 3.6; 139 -0.8 -2 3.6; 140 -0.8 -2 4;
141 -0.4 -2 3.2; 142 -0.4 -2 2.8; 143 -0.4 -2 3.6; 144 -0.4 -2 4; 145 0 -2 3.2;
146 0 -2 3.6; 147 0.4 -2 3.2; 148 0.4 -2 2.8; 149 0.4 -2 3.6; 150 0.4 -2 4;
151 0.8 -2 3.2; 152 0.8 -2 2.8; 153 0.8 -2 3.6; 154 0.8 -2 4; 155 1.2 -2 3.2;
156 1.2 -2 3.6; 157 -1.2 -2 -0.1; 158 -0.8 -2 -0.1; 159 -1.2 -2 0.16;
160 -0.8 -2 0.16; 161 -1.2 -2 0.44; 162 -0.8 -2 0.44; 163 -1.2 -2 0.7;
164 -0.8 -2 0.7; 165 -1.2 -2 1.15; 166 -0.8 -2 1.15; 167 -1.2 -2 1.6;
168 -0.8 -2 1.6; 169 -1.2 -2 1.86; 170 -0.8 -2 1.86; 171 -1.2 -2 2.14;
172 -0.8 -2 2.14; 173 -1.2 -2 2.4; 174 -0.8 -2 2.4; 175 -0.4 -2 -0.1;
176 -0.4 -2 0.16; 177 -0.4 -2 0.44; 178 -0.4 -2 0.7; 179 -0.4 -2 1.15;
180 -0.4 -2 1.6; 181 -0.4 -2 1.86; 182 -0.4 -2 2.14; 183 -0.4 -2 2.4;
184 0 -2 -0.1; 185 0 -2 0.16; 186 0 -2 0.44; 187 0 -2 0.7; 188 0 -2 1.15;
189 0 -2 1.6; 190 0 -2 1.86; 191 0 -2 2.14; 192 0 -2 2.4; 193 0.4 -2 -0.1;
194 0.4 -2 0.16; 195 0.4 -2 0.44; 196 0.4 -2 0.7; 197 0.4 -2 1.15;
198 0.4 -2 1.6; 199 0.4 -2 1.86; 200 0.4 -2 2.14; 201 0.4 -2 2.4;
202 0.8 -2 -0.1; 203 0.8 -2 0.16; 204 0.8 -2 0.44; 205 0.8 -2 0.7;
206 0.8 -2 1.15; 207 0.8 -2 1.6; 208 0.8 -2 1.86; 209 0.8 -2 2.14;
210 0.8 -2 2.4; 211 1.2 -2 -0.1; 212 1.2 -2 0.16; 213 1.2 -2 0.44;
214 1.2 -2 0.7; 215 1.2 -2 1.15; 216 1.2 -2 1.6; 217 1.2 -2 1.86;
218 1.2 -2 2.14; 219 1.2 -2 2.4; 220 1.2 -2 4.6; 221 -1.2 -2 4.6; 222 0 -2 4.6;
223 -0.8 -2 4.6; 224 -0.4 -2 4.6; 225 0.4 -2 4.6; 226 0.8 -2 4.6;
227 1.2 -2 -2.1; 228 -1.2 -2 -2.1; 229 0 -2 -2.1; 230 -0.8 -2 -2.1;
231 -0.4 -2 -2.1; 232 0.4 -2 -2.1; 233 0.8 -2 -2.1; 234 1.2 -2 5.2;
235 -1.2 -2 5.2; 236 0 -2 5.2; 237 -0.8 -2 5.2; 238 -0.4 -2 5.2;
239 0.4 -2 5.2; 240 0.8 -2 5.2; 241 1.6 -2 -1.7; 242 1.6 -2 4; 243 1.6 -2 -0.5;
244 1.6 -2 2.8; 245 1.6 -2 -1.3; 246 1.6 -2 -0.9; 247 1.6 -2 3.2;
248 1.6 -2 3.6; 249 1.6 -2 -0.1; 250 1.6 -2 0.16; 251 1.6 -2 0.44;
252 1.6 -2 0.7; 253 1.6 -2 1.15; 254 1.6 -2 1.6; 255 1.6 -2 1.86;
256 1.6 -2 2.14; 257 1.6 -2 2.4; 258 1.6 -2 4.6; 259 1.6 -2 -2.1;
260 1.6 -2 5.2; 261 1.2 -2 -2.5; 262 -1.2 -2 -2.5; 263 0 -2 -2.5;
264 -0.8 -2 -2.5; 265 -0.4 -2 -2.5; 266 0.4 -2 -2.5; 267 0.8 -2 -2.5;
268 1.6 -2 -2.5; 269 1.2 -2 5.8; 270 -1.2 -2 5.8; 271 0 -2 5.8;
272 -0.8 -2 5.8; 273 -0.4 -2 5.8; 274 0.4 -2 5.8; 275 0.8 -2 5.8;
276 1.6 -2 5.8; 277 -1.6 -2 -1.7; 278 -1.6 -2 4; 279 -1.6 -2 -0.5;
280 -1.6 -2 2.8; 281 -1.6 -2 -1.3; 282 -1.6 -2 -0.9; 283 -1.6 -2 3.2;
284 -1.6 -2 3.6; 285 -1.6 -2 -0.1; 286 -1.6 -2 0.16; 287 -1.6 -2 0.44;
288 -1.6 -2 0.7; 289 -1.6 -2 1.15; 290 -1.6 -2 1.6; 291 -1.6 -2 1.86;
292 -1.6 -2 2.14; 293 -1.6 -2 2.4; 294 -1.6 -2 4.6; 295 -1.6 -2 -2.1;
```

Job Title:

Client:

Engineer:

```
296 -1.6 -2 5.2; 297 -1.6 -2 -2.5; 298 -1.6 -2 5.8; 299 1.9 -2 -1.7;
300 1.9 -2 4; 301 1.9 -2 -0.5; 302 1.9 -2 2.8; 303 1.9 -2 -1.3;
304 1.9 -2 -0.9; 305 1.9 -2 3.2; 306 1.9 -2 3.6; 307 1.9 -2 -0.1;
308 1.9 -2 0.16; 309 1.9 -2 0.44; 310 1.9 -2 0.7; 311 1.9 -2 1.15;
312 1.9 -2 1.6; 313 1.9 -2 1.86; 314 1.9 -2 2.14; 315 1.9 -2 2.4;
316 1.9 -2 4.6; 317 1.9 -2 -2.1; 318 1.9 -2 5.2; 319 1.9 -2 -2.5;
320 1.9 -2 5.8;
MEMBER INCIDENCES
1 101 6; 2 102 8;
ELEMENT INCIDENCES SHELL
12 105 113 114 115; 13 113 116 117 114; 14 116 108 118 117; 15 115 114 119 120;
16 114 117 121 119; 17 117 118 122 121; 18 120 119 123 111; 19 119 121 124 123;
20 121 122 101 124; 21 111 123 125 126; 22 123 124 127 125; 23 124 101 128 127;
24 126 125 129 130; 25 125 127 131 129; 26 127 128 132 131; 27 130 129 133 103;
28 129 131 134 133; 29 131 132 107 134; 30 110 135 136 137; 31 135 138 139 136;
32 138 106 140 139; 33 137 136 141 142; 34 136 139 143 141; 35 139 140 144 143;
36 142 141 145 102; 37 141 143 146 145; 38 143 144 112 146; 39 102 145 147 148;
40 145 146 149 147; 41 146 112 150 149; 42 148 147 151 152; 43 147 149 153 151;
44 149 150 154 153; 45 152 151 155 109; 46 151 153 156 155; 47 153 154 104 156;
48 108 157 158 118; 49 157 159 160 158; 50 159 161 162 160; 51 161 163 164 162;
52 163 165 166 164; 53 165 167 168 166; 54 167 169 170 168; 55 169 171 172 170;
56 171 173 174 172; 57 173 110 137 174; 58 118 158 175 122; 59 158 160 176 175;
60 160 162 177 176; 61 162 164 178 177; 62 164 166 179 178; 63 166 168 180 179;
64 168 170 181 180; 65 170 172 182 181; 66 172 174 183 182; 67 174 137 142 183;
68 122 175 184 101; 69 175 176 185 184; 70 176 177 186 185; 71 177 178 187 186;
72 178 179 188 187; 73 179 180 189 188; 74 180 181 190 189; 75 181 182 191 190;
76 182 183 192 191; 77 183 142 102 192; 78 101 184 193 128; 79 184 185 194 193;
80 185 186 195 194; 81 186 187 196 195; 82 187 188 197 196; 83 188 189 198 197;
84 189 190 199 198; 85 190 191 200 199; 86 191 192 201 200; 87 192 102 148 201;
88 128 193 202 132; 89 193 194 203 202; 90 194 195 204 203; 91 195 196 205 204;
92 196 197 206 205; 93 197 198 207 206; 94 198 199 208 207; 95 199 200 209 208;
96 200 201 210 209; 97 201 148 152 210; 98 132 202 211 107; 99 202 203 212 211;
100 203 204 213 212; 101 204 205 214 213; 102 205 206 215 214;
103 206 207 216 215; 104 207 208 217 216; 105 208 209 218 217;
106 209 210 219 218; 107 210 152 109 219; 108 106 221 223 140;
109 140 223 224 144; 110 144 224 222 112; 111 112 222 225 150;
112 150 225 226 154; 113 154 226 220 104; 114 228 105 115 230;
115 230 115 120 231; 116 231 120 111 229; 117 229 111 126 232;
118 232 126 130 233; 119 233 130 103 227; 120 221 235 237 223;
121 223 237 238 224; 122 224 238 236 222; 123 222 236 239 225;
124 225 239 240 226; 125 226 240 234 220; 126 103 133 245 241;
127 133 134 246 245; 128 134 107 243 246; 129 109 155 247 244;
130 155 156 248 247; 131 156 104 242 248; 132 107 211 249 243;
133 211 212 250 249; 134 212 213 251 250; 135 213 214 252 251;
136 214 215 253 252; 137 215 216 254 253; 138 216 217 255 254;
139 217 218 256 255; 140 218 219 257 256; 141 219 109 244 257;
142 104 220 258 242; 143 227 103 241 259; 144 220 234 260 258;
145 262 228 230 264; 146 264 230 231 265; 147 265 231 229 263;
148 263 229 232 266; 149 266 232 233 267; 150 267 233 227 261;
151 261 227 259 268; 152 235 270 272 237; 153 237 272 273 238;
154 238 273 271 236; 155 236 271 274 239; 156 239 274 275 240;
157 240 275 269 234; 158 234 269 276 260; 159 277 281 113 105;
160 281 282 116 113; 161 282 279 108 116; 162 280 283 135 110;
163 283 284 138 135; 164 284 278 106 138; 165 279 285 157 108;
```



Job Title:

Client:

Engineer:

```
166 285 286 159 157; 167 286 287 161 159; 168 287 288 163 161;
169 288 289 165 163; 170 289 290 167 165; 171 290 291 169 167;
172 291 292 171 169; 173 292 293 173 171; 174 293 280 110 173;
175 278 294 221 106; 176 295 277 105 228; 177 294 296 235 221;
178 297 295 228 262; 179 296 298 270 235; 180 241 245 303 299;
181 245 246 304 303; 182 246 243 301 304; 183 244 247 305 302;
184 247 248 306 305; 185 248 242 300 306; 186 243 249 307 301;
187 249 250 308 307; 188 250 251 309 308; 189 251 252 310 309;
190 252 253 311 310; 191 253 254 312 311; 192 254 255 313 312;
193 255 256 314 313; 194 256 257 315 314; 195 257 244 302 315;
196 242 258 316 300; 197 259 241 299 317; 198 258 260 318 316;
199 268 259 317 319; 200 260 276 320 318;
START GROUP DEFINITION
ELEMENT
_RAFT_MOMENT 12 TO 19 21 22 24 TO 35 37 38 40 TO 67 69 TO 76 79 TO 86 88 TO 200
_RAFT_SHEAR 52 53 62 63 72 73 82 83 92 93 102 103 108 TO 200
END GROUP DEFINITION
ELEMENT PROPERTY
12 TO 200 THICKNESS 0.8
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 23.5616
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
1 2 PRIS YD 0.8 ZD 0.8
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
101 TO 320 ELASTIC MAT DIRECT Y SUBGRADE 20000 PRINT
LOAD 1 DEAD  LOAD
SELFWEIGHT Y -1
*Soil Weight
ELEMENT LOAD
12 TO 200 PR GY -28.8
*
joint load
6   FX  1.7 FY  -155.8  FZ  -3.9
8   FX  0.9 FY  -57.5   FZ  4.4
Load 2  LL
joint load
6   FX  0.6 FY  -181.3  FZ  -4.4
8   FX  0.0 FY  -75.6   FZ  6.0
Load 3  SDL1
joint load
6   FX  -35.7   FY  183.3   FZ  8.7
8   FX  0.1 FY  -3.7    FZ  4.2
Load 4  SDL2
joint load
```



Job Title:

Client:

Engineer:

```
6   FX  -14.2   FY  -1195.4 FZ  -12.8
8   FX  0.1 FY  -239.3  FZ  33.4
Load 5  WIND LOAD (+X)
joint load
6   FX  98.4    FY  -463.161    FZ  -8.117
8   FX  16.09   FY  -90.225 FZ  7.668
Load 6  WIND LOAD (-X)
joint load
6   FX  -98.071 FY  459.381 FZ  7.68
8   FX  -16.09  FY  93.9    FZ  -8.294
Load 7  WIND LOAD (+Z)
joint load
6   FX  14.11   FY  168.368 FZ  54.565
8   FX  0.036   FY  -380.275    FZ  77.323
Load 8  WIND LOAD (+Z)
joint load
6   FX  -13.874 FY  -186.591    FZ  -58.299
8   FX  -0.036  FY  412.123 FZ  -82.963
Load 9  TL1
joint load
6   FX  3.229   FY  -5.36   FZ  -27.598
8   FX  1.183   FY  -12.529 FZ  31.798
Load 10 TL2
joint load
6   FX  -3.229  FY  5.36    FZ  27.598
8   FX  -1.183  FY  12.529  FZ  -31.798
Load 11 EL (+X)
joint load
6   FX  5.258   FY  -33.51  FZ  -0.527
8   FX  0.219   FY  -3.822  FZ  0.474
Load 12 EL (+Z)
joint load
6   FX  0.072   FY  15.994  FZ  3.692
8   FX  0.001   FY  -30.922 FZ  5.569
***LOAD COMBINATION FOR BASE PRESSURE
LOAD 301 DL+SDL1
REPEAT LOAD
1 1.0 3 1.0
LOAD 302 DL+SDL2
REPEAT LOAD
1 1.0 4 1.0
LOAD 303 DL + SDL1 + TL1 + LL
REPEAT LOAD
1 1.0 3 1.0 9 1.0 2 1.0
LOAD 304 DL + SDL1 + TL2 + LL
REPEAT LOAD
1 1.0 4 1.0 9 1.0 2 1.0
LOAD 305 DL + SDL2 + TL1 + LL
REPEAT LOAD
1 1.0 3 1.0 10 1.0 2 1.0
LOAD 306 DL + SDL2 + TL2 + LL
REPEAT LOAD
1 1.0 4 1.0 10 1.0 2 1.0
LOAD 307 DL + SDL1 + 0.75 LL + 0.75 WL(+X)
```

Job Title:

Client:

Engineer:

```
REPEAT LOAD
1 1.0 3 1.0 2 0.75 5 0.75
LOAD 308 DL + SDL1 + 0.75 LL + 0.75 WL(-X)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 6 0.75
LOAD 309 DL + SDL1 + 0.75 LL + 0.75 WL(+Z)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 7 0.75
LOAD 310 DL + SDL1 + 0.75 LL + 0.75 WL(-Z)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 8 0.75
LOAD 311 DL + SDL2 + 0.75 LL + 0.75 WL(+X)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 5 0.75
LOAD 312 DL + SDL2 + 0.75 LL + 0.75 WL(-X)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 6 0.75
LOAD 313 DL + SDL2 + 0.75 LL + 0.75 WL(+Z)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 7 0.75
LOAD 314 DL + SDL2 + 0.75 LL + 0.75 WL(-Z)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 8 0.75
LOAD 315 DL + SDL1 + 0.75 LL + 0.525 EL(+X)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 11 0.525
LOAD 316 DL + SDL1 + 0.75 LL + 0.525 EL(-X)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 11 -0.525
LOAD 317 DL + SDL1 + 0.75 LL + 0.525 EL(+Z)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 12 0.525
LOAD 318 DL + SDL1 + 0.75 LL + 0.525 EL(-Z)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 12 -0.525
LOAD 319 DL + SDL2 + 0.75 LL + 0.525 EL(+X)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 11 0.525
LOAD 320 DL + SDL2 + 0.75 LL + 0.525 EL(-X)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 11 -0.525
LOAD 321 DL + SDL2 + 0.75 LL + 0.525 EL(+Z)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 12 0.525
LOAD 322 DL + SDL2 + 0.75 LL + 0.525 EL(-Z)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 12 -0.525
LOAD 323 0.6 (DL + SDL1) + WL(+X)
REPEAT LOAD
1 0.6 3 0.6 5 1.0
LOAD 324 0.6 (DL + SDL1) + WL(-X)
REPEAT LOAD
1 0.6 3 0.6 6 1.0
LOAD 325 0.6 (DL + SDL1) + WL(+Z)
```



Job Title:

Client:

Engineer:

```
REPEAT LOAD
1 0.6 3 0.6 7 1.0
LOAD 326 0.6 (DL + SDL1) + WL(-Z)
REPEAT LOAD
1 0.6 3 0.6 8 1.0
LOAD 327 0.6 (DL + SDL2) + WL(+X)
REPEAT LOAD
1 0.6 4 0.6 5 1.0
LOAD 328 0.6 (DL + SDL2) + WL(-X)
REPEAT LOAD
1 0.6 4 0.6 6 1.0
LOAD 329 0.6 (DL + SDL2) + WL(+Z)
REPEAT LOAD
1 0.6 4 0.6 7 1.0
LOAD 330 0.6 (DL + SDL2) + WL(-Z)
REPEAT LOAD
1 0.6 4 0.6 8 1.0
LOAD 331 0.6 (DL + SDL1) + 0.7 EL(+X)
REPEAT LOAD
1 0.6 3 0.6 11 0.7
LOAD 332 0.6 (DL + SDL1) + 0.7 EL(-X)
REPEAT LOAD
1 0.6 3 0.6 11 -0.7
LOAD 333 0.6 (DL + SDL1) + 0.7 EL(+Z)
REPEAT LOAD
1 0.6 3 0.6 12 0.7
LOAD 334 0.6 (DL + SDL1) + 0.7 EL(-Z)
REPEAT LOAD
1 0.6 3 0.6 12 -0.7
LOAD 335 0.6 (DL + SDL2) + 0.7 EL(+X)
REPEAT LOAD
1 0.6 4 0.6 11 0.7
LOAD 336 0.6 (DL + SDL2) + 0.7 EL(-X)
REPEAT LOAD
1 0.6 4 0.6 11 -0.7
LOAD 337 0.6 (DL + SDL2) + 0.7 EL(+Z)
REPEAT LOAD
1 0.6 4 0.6 12 -0.7
LOAD 338 0.6 (DL + SDL2) + 0.7 EL(-Z)
REPEAT LOAD
1 0.6 4 0.6 12 -0.7
****
*Load Combination for Concrete design
LOAD 401 1.4 (DL+SDL1)
REPEAT LOAD
1 1.4 3 1.4
LOAD 402 1.4 (DL+SDL2)
REPEAT LOAD
1 1.4 4 1.4
LOAD 403 1.4 (DL + SDL1 + TL1) + 1.7 LL
REPEAT LOAD
1 1.4 3 1.4 9 1.4 2 1.7
LOAD 404 1.4 (DL + SDL1 + TL2) + 1.7 LL
REPEAT LOAD
```



Job Title:

Client:

Engineer:

```
1 1.4 3 1.4 10 1.4 2 1.7
LOAD 405 1.4 (DL + SDL2 + TL1) + 1.7 LL
REPEAT LOAD
1 1.4 4 1.4 9 1.4 2 1.7
LOAD 406 1.4 (DL + SDL2 + TL2) + 1.7 LL
REPEAT LOAD
1 1.4 4 1.4 10 1.4 2 1.7
LOAD 407 1.2 (DL + SDL1) + 0.5 LL + 1.6 WL(+X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 5 1.6
LOAD 408 1.2 (DL + SDL1) + 0.5 LL + 1.6 WL(-X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 6 1.6
LOAD 409 1.2 (DL + SDL1) + 0.5 LL + 1.6 WL(+Z)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 7 1.6
LOAD 410 1.2 (DL + SDL1) + 0.5 LL + 1.6 WL(-Z)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 8 1.6
LOAD 411 1.2 (DL + SDL2) + 0.5 LL + 1.6 WL(+X)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 5 1.6
LOAD 412 1.2 (DL + SDL2) + 0.5 LL + 1.6 WL(-X)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 6 1.6
LOAD 413 1.2 (DL + SDL2) + 0.5 LL + 1.6 WL(+Z)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 7 1.6
LOAD 414 1.2 (DL + SDL2) + 0.5 LL + 1.6 WL(-Z)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 8 1.6
LOAD 415 1.2 (DL + SDL1) + 0.5 LL + 1.0 EL(+X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 11 1.0
LOAD 416 1.2 (DL + SDL1) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 11 -1.0
LOAD 417 1.2 (DL + SDL1) + 0.5 LL + 1.0 EL(+Z)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 12 1.0
LOAD 418 1.2 (DL + SDL1) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 12 -1.0
LOAD 419 1.2 (DL + SDL2) + 0.5 LL + 1.0 EL(+X)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 11 1.0
LOAD 420 1.2 (DL + SDL2) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 11 -1.0
LOAD 421 1.2 (DL + SDL2) + 0.5 LL + 1.0 EL(+Z)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 12 1.0
LOAD 422 1.2 (DL + SDL2) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
```



Job Title:

Client:

Engineer:

```
1 1.2 4 1.2 2 0.5 12 -1.0
LOAD 423 0.9 (DL + SDL1) + 1.6 WL(+X)
REPEAT LOAD
1 0.9 3 0.9 5 1.6
LOAD 424 0.9 (DL + SDL1) + 1.6 WL(-X)
REPEAT LOAD
1 0.9 3 0.9 6 1.6
LOAD 425 0.9 (DL + SDL1) + 1.6 WL(+Z)
REPEAT LOAD
1 0.9 3 0.9 7 1.6
LOAD 426 0.9 (DL + SDL1) + 1.6 WL(-Z)
REPEAT LOAD
1 0.9 3 0.9 8 1.6
LOAD 427 0.9 (DL + SDL2) + 1.6 WL(+X)
REPEAT LOAD
1 0.9 4 0.9 5 1.6
LOAD 428 0.9 (DL + SDL2) + 1.6 WL(-X)
REPEAT LOAD
1 0.9 4 0.9 6 1.6
LOAD 429 0.9 (DL + SDL2) + 1.6 WL(+Z)
REPEAT LOAD
1 0.9 4 0.9 7 1.6
LOAD 430 0.9 (DL + SDL2) + 1.6 WL(-Z)
REPEAT LOAD
1 0.9 4 0.9 8 1.6
LOAD 431 0.9 (DL + SDL1) + 1.0 EL(+X)
REPEAT LOAD
1 0.9 3 0.9 11 1.0
LOAD 432 0.9 (DL + SDL1) + 1.0 EL(-X)
REPEAT LOAD
1 0.9 3 0.9 11 -1.0
LOAD 433 0.9 (DL + SDL1) + 1.0 EL(+Z)
REPEAT LOAD
1 0.9 3 0.9 12 1.0
LOAD 434 0.9 (DL + SDL1) + 1.0 EL(-Z)
REPEAT LOAD
1 0.9 3 0.9 12 -1.0
LOAD 435 0.9 (DL + SDL2) + 1.0 EL(+X)
REPEAT LOAD
1 0.9 4 0.9 11 1.0
LOAD 436 0.9 (DL + SDL2) + 1.0 EL(-X)
REPEAT LOAD
1 0.9 4 0.9 11 -1.0
LOAD 437 0.9 (DL + SDL2) + 1.0 EL(+Z)
REPEAT LOAD
1 0.9 4 0.9 12 1.0
LOAD 438 0.9 (DL + SDL2) + 1.0 EL(-Z)
REPEAT LOAD
1 0.9 4 0.9 12 -1.0
PERFORM ANALYSIS
FINISH
```

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	C02	D01			Outotec / L&T Revision No.:	0	1	2	3	
Date	27-01-2020	12-03-2020	25-04-2022	13/10/2023			Date	27-01-2020	12-03-2020	25-04-2022	13/10/2023	
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS DT. 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER SUPPORT STRUCTURE	AS BUILT			Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS DT. 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER SUPPORT STRUCTURE	AS BUILT	
Prepared By	APR	APR	APR	APR			Prepared By	APR	APR	APR	APR	
Checked	SBY	SBY	SBY	SBY			Checked	SBY	SBY	SBY	SBY	
Approved	AG	AG	AG	AG			Approved	AG	AG	AG	AG	

ATTACHMENT 6 STAAD INPUT FILE OF FOUNDATION CF2

```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 17-Jan-20
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
1 0 -2.3 0; 4 0 -2.3 6; 101 0 -2.5 0; 102 0 -2.5 6; 103 1 -2.5 -1.2;
104 1 -2.5 7.2; 105 -1 -2.5 -1.2; 106 -1 -2.5 7.2; 107 1 -2.5 0; 108 -1 -2.5 0;
109 1 -2.5 6; 110 -1 -2.5 6; 111 0 -2.5 -1.2; 112 0 -2.5 7.2;
113 -1 -2.5 -0.85; 114 -0.85 -2.5 -0.85; 115 -0.85 -2.5 -1.2;
116 -1 -2.5 -0.35; 117 -0.85 -2.5 -0.35; 118 -0.85 -2.5 0;
119 -0.35 -2.5 -0.85; 120 -0.35 -2.5 -1.2; 121 -0.35 -2.5 -0.35;
122 -0.35 -2.5 0; 123 0 -2.5 -0.85; 124 0 -2.5 -0.35; 125 0.35 -2.5 -0.85;
126 0.35 -2.5 -1.2; 127 0.35 -2.5 -0.35; 128 0.35 -2.5 0; 129 0.85 -2.5 -0.85;
130 0.85 -2.5 -1.2; 131 0.85 -2.5 -0.35; 132 0.85 -2.5 0; 133 1 -2.5 -0.85;
134 1 -2.5 -0.35; 135 -1 -2.5 6.35; 136 -0.85 -2.5 6.35; 137 -0.85 -2.5 6;
138 -1 -2.5 6.85; 139 -0.85 -2.5 6.85; 140 -0.85 -2.5 7.2; 141 -0.35 -2.5 6.35;
142 -0.35 -2.5 6; 143 -0.35 -2.5 6.85; 144 -0.35 -2.5 7.2; 145 0 -2.5 6.35;
146 0 -2.5 6.85; 147 0.35 -2.5 6.35; 148 0.35 -2.5 6; 149 0.35 -2.5 6.85;
150 0.35 -2.5 7.2; 151 0.85 -2.5 6.35; 152 0.85 -2.5 6; 153 0.85 -2.5 6.85;
154 0.85 -2.5 7.2; 155 1 -2.5 6.35; 156 1 -2.5 6.85; 157 -1 -2.5 0.35;
158 -0.85 -2.5 0.35; 159 -1 -2.5 0.85; 160 -0.85 -2.5 0.85; 161 -1 -2.5 1.55;
162 -0.85 -2.5 1.55; 163 -1 -2.5 2.25; 164 -0.85 -2.5 2.25; 165 -1 -2.5 3;
166 -0.85 -2.5 3; 167 -1 -2.5 3.75; 168 -0.85 -2.5 3.75; 169 -1 -2.5 4.45;
170 -0.85 -2.5 4.45; 171 -1 -2.5 5.15; 172 -0.85 -2.5 5.15; 173 -1 -2.5 5.65;
174 -0.85 -2.5 5.65; 175 -0.35 -2.5 0.35; 176 -0.35 -2.5 0.85;
177 -0.35 -2.5 1.55; 178 -0.35 -2.5 2.25; 179 -0.35 -2.5 3;
180 -0.35 -2.5 3.75; 181 -0.35 -2.5 4.45; 182 -0.35 -2.5 5.15;
183 -0.35 -2.5 5.65; 184 0 -2.5 0.35; 185 0 -2.5 0.85; 186 0 -2.5 1.55;
187 0 -2.5 2.25; 188 0 -2.5 3; 189 0 -2.5 3.75; 190 0 -2.5 4.45;
191 0 -2.5 5.15; 192 0 -2.5 5.65; 193 0.35 -2.5 0.35; 194 0.35 -2.5 0.85;
195 0.35 -2.5 1.55; 196 0.35 -2.5 2.25; 197 0.35 -2.5 3; 198 0.35 -2.5 3.75;
199 0.35 -2.5 4.45; 200 0.35 -2.5 5.15; 201 0.35 -2.5 5.65; 202 0.85 -2.5 0.35;
203 0.85 -2.5 0.85; 204 0.85 -2.5 1.55; 205 0.85 -2.5 2.25; 206 0.85 -2.5 3;
207 0.85 -2.5 3.75; 208 0.85 -2.5 4.45; 209 0.85 -2.5 5.15; 210 0.85 -2.5 5.65;
211 1 -2.5 0.35; 212 1 -2.5 0.85; 213 1 -2.5 1.55; 214 1 -2.5 2.25;
215 1 -2.5 3; 216 1 -2.5 3.75; 217 1 -2.5 4.45; 218 1 -2.5 5.15;
219 1 -2.5 5.65; 220 1 -2.5 7.5; 221 -1 -2.5 7.5; 222 0 -2.5 7.5;
223 -0.85 -2.5 7.5; 224 -0.35 -2.5 7.5; 225 0.35 -2.5 7.5; 226 0.85 -2.5 7.5;
227 1 -2.5 -1.5; 228 -1 -2.5 -1.5; 229 0 -2.5 -1.5; 230 -0.85 -2.5 -1.5;
231 -0.35 -2.5 -1.5; 232 0.35 -2.5 -1.5; 233 0.85 -2.5 -1.5;
MEMBER INCIDENCES
1 101 1; 2 102 4;
ELEMENT INCIDENCES SHELL
12 105 113 114 115; 13 113 116 117 114; 14 116 108 118 117; 15 115 114 119 120;
16 114 117 121 119; 17 117 118 122 121; 18 120 119 123 111; 19 119 121 124 123;
20 121 122 101 124; 21 111 123 125 126; 22 123 124 127 125; 23 124 101 128 127;
24 126 125 129 130; 25 125 127 131 129; 26 127 128 132 131; 27 130 129 133 103;
28 129 131 134 133; 29 131 132 107 134; 30 110 135 136 137; 31 135 138 139 136;
32 138 106 140 139; 33 137 136 141 142; 34 136 139 143 141; 35 139 140 144 143;
36 142 141 145 102; 37 141 143 146 145; 38 143 144 112 146; 39 102 145 147 148;
40 145 146 149 147; 41 146 112 150 149; 42 148 147 151 152; 43 147 149 153 151;
44 149 150 154 153; 45 152 151 155 109; 46 151 153 156 155; 47 153 154 104 156;
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48 108 157 158 118; 49 157 159 160 158; 50 159 161 162 160; 51 161 163 164 162;
52 163 165 166 164; 53 165 167 168 166; 54 167 169 170 168; 55 169 171 172 170;
56 171 173 174 172; 57 173 110 137 174; 58 118 158 175 122; 59 158 160 176 175;
60 160 162 177 176; 61 162 164 178 177; 62 164 166 179 178; 63 166 168 180 179;
64 168 170 181 180; 65 170 172 182 181; 66 172 174 183 182; 67 174 137 142 183;
68 122 175 184 101; 69 175 176 185 184; 70 176 177 186 185; 71 177 178 187 186;
72 178 179 188 187; 73 179 180 189 188; 74 180 181 190 189; 75 181 182 191 190;
76 182 183 192 191; 77 183 142 102 192; 78 101 184 193 128; 79 184 185 194 193;
80 185 186 195 194; 81 186 187 196 195; 82 187 188 197 196; 83 188 189 198 197;
84 189 190 199 198; 85 190 191 200 199; 86 191 192 201 200; 87 192 102 148 201;
88 128 193 202 132; 89 193 194 203 202; 90 194 195 204 203; 91 195 196 205 204;
92 196 197 206 205; 93 197 198 207 206; 94 198 199 208 207; 95 199 200 209 208;
96 200 201 210 209; 97 201 148 152 210; 98 132 202 211 107; 99 202 203 212 211;
100 203 204 213 212; 101 204 205 214 213; 102 205 206 215 214;
103 206 207 216 215; 104 207 208 217 216; 105 208 209 218 217;
106 209 210 219 218; 107 210 152 109 219; 108 106 221 223 140;
109 140 223 224 144; 110 144 224 222 112; 111 112 222 225 150;
112 150 225 226 154; 113 154 226 220 104; 114 228 105 115 230;
115 230 115 120 231; 116 231 120 111 229; 117 229 111 126 232;
118 232 126 130 233; 119 233 130 103 227;
START GROUP DEFINITION
ELEMENT
_RAFT_MOMENT 12 TO 19 21 22 24 TO 35 37 38 40 TO 67 69 TO 76 79 TO 86 88 TO 119
_RAFT_SHEAR 12 TO 15 18 21 24 27 TO 32 35 38 41 44 TO 57 60 TO 65 70 TO 75 -
80 TO 85 90 TO 95 98 TO 119
END GROUP DEFINITION
ELEMENT PROPERTY
12 TO 119 THICKNESS 0.5
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.7806e+07
POISSON 0.17
DENSITY 25
ALPHA 1e-05
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
1 2 PRIS YD 0.8 ZD 0.8
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
101 TO 233 ELASTIC MAT DIRECT Y SUBGRADE 20000 PRINT
LOAD 1 DEAD  LOAD
SELFWEIGHT Y -1
*Soil Weight
ELEMENT LOAD
12 TO 119 PR GY -27.0
JOINT LOAD
1   FX  -34.3   FY  -857.9  FZ  -11.9   MX  -26.0   MZ  35.7
4   FX  -34.7   FY  -791.9  FZ  -4.9    MX  -21.4   MZ  35.7
Load 2  LL
JOINT LOAD
```

```
1   FX  -4.5    FY  -95.9   FZ  -1.1    MX  -2.4    MZ  4.8
4   FX  -3.8    FY  -72.3   FZ  -0.2    MX  -2.0    MZ  3.7
Load 3  SDL1_R
JOINT LOAD
1   FX  -15.6   FY  -41.5   FZ  12.0    MX  30.3    MZ  22.1
4   FX  5.4 FY  -21.5   FZ  12.1    MX  30.2    MZ  -7.9
Load 4  SDL2_R
JOINT LOAD
1   FX  60.5    FY  73.5    FZ  -5.8    MX  -14.8   MZ  -90.5
4   FX  46.2    FY  67.1    FZ  -5.7    MX  -15.0   MZ  -69.3
Load 5  WIND LOAD (+X)
JOINT LOAD
1   FX  35.3    FY  51.2    FZ  1.4 MX  3.5 MZ  -52.2
4   FX  44.2    FY  53.5    FZ  1.4 MX  3.5 MZ  -65.2
Load 6  WIND LOAD (-X)
JOINT LOAD
1   FX  -38.5   FY  -58 FZ  -1.8    MX  -4.4    MZ  56.9
4   FX  -54.5   FY  -60.6   FZ  -1.8    MX  -4.4    MZ  80.4
Load 7  WIND LOAD (+Z)
JOINT LOAD
1   FX  -5.5    FY  -11.4   FZ  11.5    MX  25.7    MZ  8.1
4   FX  3.8 FY  21.9    FZ  11.9    MX  26.1    MZ  -4.9
Load 8  WIND LOAD (+-Z)
JOINT LOAD
1   FX  3.1 FY  3.6 FZ  -5.3    MX  -12.1   MZ  -4.7
4   FX  -1.9    FY  -8.2    FZ  -5.5    MX  -12.3   MZ  2.5
Load 11 EL (+X)
JOINT LOAD
1   FX  21.9    FY  39.2    FZ  -0.2    MX  -0.5    MZ  -32.4
4   FX  22.1    FY  40.2    FZ  -0.2    MX  -0.5    MZ  -32.6
Load 12 EL (+Z)
JOINT LOAD
1   FX  -3.2    FY  -1.1    FZ  5.1 MX  12.4    MZ  4.8
4   FX  2.2 FY  9.9 FZ  5.3 MX  12.6    MZ  -2.8
************************************************
*LOAD COMBINATION FOR BASE PRESSURE
************************************************
LOAD 301 DL+SDL1_R
REPEAT LOAD
1 1.0 3 1.0
LOAD 302 DL+SDL2_R
REPEAT LOAD
1 1.0 4 1.0
LOAD 303 DL-SDL1_R
REPEAT LOAD
1 1.0 3 -1.0
LOAD 304 DL-SDL2_R
REPEAT LOAD
1 1.0 4 -1.0
**********************************
LOAD 305 DL + SDL1_R + LL
REPEAT LOAD
1 1.0 3 1.0 2 1.0
LOAD 306 DL + SDL2_R + LL
```

```
REPEAT LOAD
1 1.0 4 1.0 2 1.0
LOAD 307 DL - SDL1_R + LL
REPEAT LOAD
1 1.0 3 -1.0 2 1.0
LOAD 308 DL - SDL2_R + LL
REPEAT LOAD
1 1.0 4 -1.0 2 1.0
*********************************************
LOAD 309 DL + SDL1_R + 0.75 LL + 0.75 WL(+X)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 5 0.75
LOAD 310 DL + SDL1_R + 0.75 LL + 0.75 WL(-X)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 6 0.75
LOAD 311 DL + SDL1_R + 0.75 LL + 0.75 WL(+Z)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 7 0.75
LOAD 312 DL + SDL1_R + 0.75 LL + 0.75 WL(-Z)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 8 0.75
LOAD 313 DL + SDL2_R + 0.75 LL + 0.75 WL(+X)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 5 0.75
LOAD 314 DL + SDL2_R + 0.75 LL + 0.75 WL(-X)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 6 0.75
LOAD 315 DL + SDL2_R + 0.75 LL + 0.75 WL(+Z)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 7 0.75
LOAD 316 DL + SDL2_R + 0.75 LL + 0.75 WL(-Z)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 8 0.75
LOAD 317 DL - SDL1_R + 0.75 LL + 0.75 WL(+X)
REPEAT LOAD
1 1.0 3 -1.0 2 0.75 5 0.75
LOAD 318 DL - SDL1_R + 0.75 LL + 0.75 WL(-X)
REPEAT LOAD
1 1.0 3 -1.0 2 0.75 6 0.75
LOAD 319 DL - SDL1_R + 0.75 LL + 0.75 WL(+Z)
REPEAT LOAD
1 1.0 3 -1.0 2 0.75 7 0.75
LOAD 320 DL - SDL1_R + 0.75 LL + 0.75 WL(-Z)
REPEAT LOAD
1 1.0 3 -1.0 2 0.75 8 0.75
LOAD 321 DL - SDL2_R + 0.75 LL + 0.75 WL(+X)
REPEAT LOAD
1 1.0 4 -1.0 2 0.75 5 0.75
LOAD 322 DL - SDL2_R + 0.75 LL + 0.75 WL(-X)
REPEAT LOAD
1 1.0 4 -1.0 2 0.75 6 0.75
LOAD 323 DL - SDL2_R + 0.75 LL + 0.75 WL(+Z)
REPEAT LOAD
1 1.0 4 -1.0 2 0.75 7 0.75
```

```
LOAD 324 DL - SDL2_R + 0.75 LL + 0.75 WL(-Z)
REPEAT LOAD
1 1.0 4 -1.0 2 0.75 8 0.75
*********************************************
LOAD 325 DL + SDL1_R + 0.75 LL + 0.525 EL(+X)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 11 0.525
LOAD 326 DL + SDL1_R + 0.75 LL + 0.525 EL(-X)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 11 -0.525
LOAD 327 DL + SDL1_R + 0.75 LL + 0.525 EL(+Z)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 12 0.525
LOAD 328 DL + SDL1_R + 0.75 LL + 0.525 EL(-Z)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 12 -0.525
LOAD 329 DL + SDL2_R + 0.75 LL + 0.525 EL(+X)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 11 0.525
LOAD 330 DL + SDL2_R + 0.75 LL + 0.525 EL(-X)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 11 -0.525
LOAD 331 DL + SDL2_R + 0.75 LL + 0.525 EL(+Z)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 12 0.525
LOAD 332 DL + SDL2_R + 0.75 LL + 0.525 EL(-Z)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 12 -0.525
LOAD 333 DL - SDL1_R + 0.75 LL + 0.525 EL(+X)
REPEAT LOAD
1 1.0 3 -1.0 2 0.75 11 0.525
LOAD 334 DL - SDL1_R + 0.75 LL + 0.525 EL(-X)
REPEAT LOAD
1 1.0 3 -1.0 2 0.75 11 -0.525
LOAD 335 DL - SDL1_R + 0.75 LL + 0.525 EL(+Z)
REPEAT LOAD
1 1.0 3 -1.0 2 0.75 12 0.525
LOAD 336 DL - SDL1_R + 0.75 LL + 0.525 EL(-Z)
REPEAT LOAD
1 1.0 3 -1.0 2 0.75 12 -0.525
LOAD 337 DL - SDL2_R + 0.75 LL + 0.525 EL(+X)
REPEAT LOAD
1 1.0 4 -1.0 2 0.75 11 0.525
LOAD 338 DL - SDL2_R + 0.75 LL + 0.525 EL(-X)
REPEAT LOAD
1 1.0 4 -1.0 2 0.75 11 -0.525
LOAD 339 DL - SDL2_R + 0.75 LL + 0.525 EL(+Z)
REPEAT LOAD
1 1.0 4 -1.0 2 0.75 12 0.525
LOAD 340 DL - SDL2_R + 0.75 LL + 0.525 EL(-Z)
REPEAT LOAD
1 1.0 4 -1.0 2 0.75 12 -0.525
*********************************************
LOAD 341 0.6 (DL + SDL1_R) + WL(+X)
```

```
REPEAT LOAD
1 0.6 3 0.6 5 1.0
LOAD 342 0.6 (DL + SDL1_R) + WL(-X)
REPEAT LOAD
1 0.6 3 0.6 6 1.0
LOAD 343 0.6 (DL + SDL1_R) + WL(+Z)
REPEAT LOAD
1 0.6 3 0.6 7 1.0
LOAD 344 0.6 (DL + SDL1_R) + WL(-Z)
REPEAT LOAD
1 0.6 3 0.6 8 1.0
LOAD 345 0.6 (DL + SDL2_R) + WL(+X)
REPEAT LOAD
1 0.6 4 0.6 5 1.0
LOAD 346 0.6 (DL + SDL2_R) + WL(-X)
REPEAT LOAD
1 0.6 4 0.6 6 1.0
LOAD 347 0.6 (DL + SDL2_R) + WL(+Z)
REPEAT LOAD
1 0.6 4 0.6 7 1.0
LOAD 348 0.6 (DL + SDL2_R) + WL(-Z)
REPEAT LOAD
1 0.6 4 0.6 8 1.0
LOAD 349 0.6 (DL - SDL1_R) + WL(+X)
REPEAT LOAD
1 0.6 3 -0.6 5 1.0
LOAD 350 0.6 (DL - SDL1_R) + WL(-X)
REPEAT LOAD
1 0.6 3 -0.6 6 1.0
LOAD 351 0.6 (DL - SDL1_R) + WL(+Z)
REPEAT LOAD
1 0.6 3 -0.6 7 1.0
LOAD 352 0.6 (DL - SDL1_R) + WL(-Z)
REPEAT LOAD
1 0.6 3 -0.6 8 1.0
LOAD 353 0.6 (DL - SDL2_R) + WL(+X)
REPEAT LOAD
1 0.6 4 -0.6 5 1.0
LOAD 354 0.6 (DL - SDL2_R) + WL(-X)
REPEAT LOAD
1 0.6 4 -0.6 6 1.0
LOAD 355 0.6 (DL - SDL2_R) + WL(+Z)
REPEAT LOAD
1 0.6 4 -0.6 7 1.0
LOAD 356 0.6 (DL - SDL2_R) + WL(-Z)
REPEAT LOAD
1 0.6 4 -0.6 8 1.0
*********************************************
LOAD 357 0.6 (DL + SDL1_R) + 0.7 EL(+X)
REPEAT LOAD
1 0.6 3 0.6 11 0.7
LOAD 358 0.6 (DL + SDL1_R) + 0.7 EL(-X)
REPEAT LOAD
1 0.6 3 0.6 11 -0.7
```

```
LOAD 359 0.6 (DL + SDL1_R) + 0.7 EL(+Z)
REPEAT LOAD
1 0.6 3 0.6 12 0.7
LOAD 360 0.6 (DL + SDL1_R) + 0.7 EL(-Z)
REPEAT LOAD
1 0.6 3 0.6 12 -0.7
LOAD 361 0.6 (DL + SDL2_R) + 0.7 EL(+X)
REPEAT LOAD
1 0.6 4 0.6 11 0.7
LOAD 362 0.6 (DL + SDL2_R) + 0.7 EL(-X)
REPEAT LOAD
1 0.6 4 0.6 11 -0.7
LOAD 363 0.6 (DL + SDL2_R) + 0.7 EL(+Z)
REPEAT LOAD
1 0.6 4 0.6 12 -0.7
LOAD 364 0.6 (DL + SDL2_R) + 0.7 EL(-Z)
REPEAT LOAD
1 0.6 4 0.6 12 -0.7
LOAD 365 0.6 (DL + SDL1_R) + 0.7 EL(+X)
REPEAT LOAD
1 0.6 3 -0.6 11 0.7
LOAD 366 0.6 (DL + SDL1_R) + 0.7 EL(-X)
REPEAT LOAD
1 0.6 3 -0.6 11 -0.7
LOAD 367 0.6 (DL + SDL1_R) + 0.7 EL(+Z)
REPEAT LOAD
1 0.6 3 -0.6 12 0.7
LOAD 368 0.6 (DL + SDL1_R) + 0.7 EL(-Z)
REPEAT LOAD
1 0.6 3 -0.6 12 -0.7
LOAD 369 0.6 (DL + SDL2_R) + 0.7 EL(+X)
REPEAT LOAD
1 0.6 4 -0.6 11 0.7
LOAD 370 0.6 (DL + SDL2_R) + 0.7 EL(-X)
REPEAT LOAD
1 0.6 4 -0.6 11 -0.7
LOAD 371 0.6 (DL + SDL2_R) + 0.7 EL(+Z)
REPEAT LOAD
1 0.6 4 -0.6 12 -0.7
LOAD 372 0.6 (DL + SDL2_R) + 0.7 EL(-Z)
REPEAT LOAD
1 0.6 4 -0.6 12 -0.7
************************************************
*Load Combination for Concrete design
************************************************
LOAD 401 1.4 (DL+SDL1_R)
REPEAT LOAD
1 1.4 3 1.4
LOAD 402 1.4 (DL+SDL2_R)
REPEAT LOAD
1 1.4 4 1.4
LOAD 403 1.4 (DL-SDL1_R)
REPEAT LOAD
1 1.4 3 -1.4
```

```
LOAD 404 1.4 (DL-SDL2_R)
REPEAT LOAD
1 1.4 4 -1.4
************************************************
LOAD 405 1.4 (DL + SDL1_R ) + 1.7 LL
REPEAT LOAD
1 1.4 3 1.4 2 1.7
LOAD 406 1.4 (DL + SDL2_R ) + 1.7 LL
REPEAT LOAD
1 1.4 4 1.4 2 1.7
LOAD 407 1.4 (DL - SDL1_R ) + 1.7 LL
REPEAT LOAD
1 1.4 3 -1.4 2 1.7
LOAD 408 1.4 (DL - SDL2_R ) + 1.7 LL
REPEAT LOAD
1 1.4 4 -1.4 2 1.7
************************************************
LOAD 409 1.2 (DL + SDL1_R) + 0.5 LL + 1.6 WL(+X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 5 1.6
LOAD 410 1.2 (DL + SDL1_R) + 0.5 LL + 1.6 WL(-X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 6 1.6
LOAD 411 1.2 (DL + SDL1_R) + 0.5 LL + 1.6 WL(+Z)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 7 1.6
LOAD 412 1.2 (DL + SDL1_R) + 0.5 LL + 1.6 WL(-Z)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 8 1.6
LOAD 413 1.2 (DL + SDL2_R) + 0.5 LL + 1.6 WL(+X)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 5 1.6
LOAD 414 1.2 (DL + SDL2_R) + 0.5 LL + 1.6 WL(-X)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 6 1.6
LOAD 415 1.2 (DL + SDL2_R) + 0.5 LL + 1.6 WL(+Z)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 7 1.6
LOAD 416 1.2 (DL + SDL2_R) + 0.5 LL + 1.6 WL(-Z)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 8 1.6
LOAD 417 1.2 (DL - SDL1_R) + 0.5 LL + 1.6 WL(+X)
REPEAT LOAD
1 1.2 3 -1.2 2 0.5 5 1.6
LOAD 418 1.2 (DL - SDL1_R) + 0.5 LL + 1.6 WL(-X)
REPEAT LOAD
1 1.2 3 -1.2 2 0.5 6 1.6
LOAD 419 1.2 (DL - SDL1_R) + 0.5 LL + 1.6 WL(+Z)
REPEAT LOAD
1 1.2 3 -1.2 2 0.5 7 1.6
LOAD 420 1.2 (DL - SDL1_R) + 0.5 LL + 1.6 WL(-Z)
REPEAT LOAD
1 1.2 3 -1.2 2 0.5 8 1.6
LOAD 421 1.2 (DL - SDL2_R) + 0.5 LL + 1.6 WL(+X)
```

```
REPEAT LOAD
1 1.2 4 -1.2 2 0.5 5 1.6
LOAD 422 1.2 (DL - SDL2_R) + 0.5 LL + 1.6 WL(-X)
REPEAT LOAD
1 1.2 4 -1.2 2 0.5 6 1.6
LOAD 423 1.2 (DL - SDL2_R) + 0.5 LL + 1.6 WL(+Z)
REPEAT LOAD
1 1.2 4 -1.2 2 0.5 7 1.6
LOAD 424 1.2 (DL - SDL2_R) + 0.5 LL + 1.6 WL(-Z)
REPEAT LOAD
1 1.2 4 -1.2 2 0.5 8 1.6
*********************************************
LOAD 425 1.2 (DL + SDL1_R) + 0.5 LL + 1.0 EL(+X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 11 1.0
LOAD 426 1.2 (DL + SDL1_R) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 11 -1.0
LOAD 427 1.2 (DL + SDL1_R) + 0.5 LL + 1.0 EL(+Z)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 12 1.0
LOAD 428 1.2 (DL + SDL1_R) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 12 -1.0
LOAD 429 1.2 (DL + SDL2_R) + 0.5 LL + 1.0 EL(+X)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 11 1.0
LOAD 430 1.2 (DL + SDL2_R) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 11 -1.0
LOAD 431 1.2 (DL + SDL2_R) + 0.5 LL + 1.0 EL(+Z)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 12 1.0
LOAD 432 1.2 (DL + SDL2_R) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 12 -1.0
LOAD 433 1.2 (DL - SDL1_R) + 0.5 LL + 1.0 EL(+X)
REPEAT LOAD
1 1.2 3 -1.2 2 0.5 11 1.0
LOAD 434 1.2 (DL - SDL1_R) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 3 -1.2 2 0.5 11 -1.0
LOAD 435 1.2 (DL - SDL1_R) + 0.5 LL + 1.0 EL(+Z)
REPEAT LOAD
1 1.2 3 -1.2 2 0.5 12 1.0
LOAD 436 1.2 (DL - SDL1_R) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 3 -1.2 2 0.5 12 -1.0
LOAD 437 1.2 (DL - SDL2_R) + 0.5 LL + 1.0 EL(+X)
REPEAT LOAD
1 1.2 4 -1.2 2 0.5 11 1.0
LOAD 438 1.2 (DL - SDL2_R) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 4 -1.2 2 0.5 11 -1.0
```

```
LOAD 439 1.2 (DL - SDL2_R) + 0.5 LL + 1.0 EL(+Z)
REPEAT LOAD
1 1.2 4 -1.2 2 0.5 12 1.0
LOAD 440 1.2 (DL - SDL2_R) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 4 -1.2 2 0.5 12 -1.0
************************************************
LOAD 441 0.9 (DL + SDL1_R) + 1.6 WL(+X)
REPEAT LOAD
1 0.9 3 0.9 5 1.6
LOAD 442 0.9 (DL + SDL1_R) + 1.6 WL(-X)
REPEAT LOAD
1 0.9 3 0.9 6 1.6
LOAD 443 0.9 (DL + SDL1_R) + 1.6 WL(+Z)
REPEAT LOAD
1 0.9 3 0.9 7 1.6
LOAD 444 0.9 (DL + SDL1_R) + 1.6 WL(-Z)
REPEAT LOAD
1 0.9 3 0.9 8 1.6
LOAD 445 0.9 (DL + SDL2_R) + 1.6 WL(+X)
REPEAT LOAD
1 0.9 4 0.9 5 1.6
LOAD 446 0.9 (DL + SDL2_R) + 1.6 WL(-X)
REPEAT LOAD
1 0.9 4 0.9 6 1.6
LOAD 447 0.9 (DL + SDL2_R) + 1.6 WL(+Z)
REPEAT LOAD
1 0.9 4 0.9 7 1.6
LOAD 448 0.9 (DL + SDL2_R) + 1.6 WL(-Z)
REPEAT LOAD
1 0.9 4 0.9 8 1.6
LOAD 449 0.9 (DL - SDL1_R) + 1.6 WL(+X)
REPEAT LOAD
1 0.9 3 -0.9 5 1.6
LOAD 450 0.9 (DL - SDL1_R) + 1.6 WL(-X)
REPEAT LOAD
1 0.9 3 -0.9 6 1.6
LOAD 451 0.9 (DL - SDL1_R) + 1.6 WL(+Z)
REPEAT LOAD
1 0.9 3 -0.9 7 1.6
LOAD 452 0.9 (DL - SDL1_R) + 1.6 WL(-Z)
REPEAT LOAD
1 0.9 3 -0.9 8 1.6
LOAD 453 0.9 (DL - SDL2_R) + 1.6 WL(+X)
REPEAT LOAD
1 0.9 4 -0.9 5 1.6
LOAD 454 0.9 (DL - SDL2_R) + 1.6 WL(-X)
REPEAT LOAD
1 0.9 4 -0.9 6 1.6
LOAD 455 0.9 (DL - SDL2_R) + 1.6 WL(+Z)
REPEAT LOAD
1 0.9 4 -0.9 7 1.6
LOAD 456 0.9 (DL - SDL2_R) + 1.6 WL(-Z)
REPEAT LOAD
```

```
1 0.9 4 -0.9 8 1.6
************************************************
LOAD 457 0.9 (DL + SDL1_R) + 1.0 EL(+X)
REPEAT LOAD
1 0.9 3 0.9 11 1.0
LOAD 458 0.9 (DL + SDL1_R) + 1.0 EL(-X)
REPEAT LOAD
1 0.9 3 0.9 11 -1.0
LOAD 459 0.9 (DL + SDL1_R) + 1.0 EL(+Z)
REPEAT LOAD
1 0.9 3 0.9 12 1.0
LOAD 460 0.9 (DL + SDL1_R) + 1.0 EL(-Z)
REPEAT LOAD
1 0.9 3 0.9 12 -1.0
LOAD 461 0.9 (DL + SDL2_R) + 1.0 EL(+X)
REPEAT LOAD
1 0.9 4 0.9 11 1.0
LOAD 462 0.9 (DL + SDL2_R) + 1.0 EL(-X)
REPEAT LOAD
1 0.9 4 0.9 11 -1.0
LOAD 463 0.9 (DL + SDL2_R) + 1.0 EL(+Z)
REPEAT LOAD
1 0.9 4 0.9 12 1.0
LOAD 464 0.9 (DL + SDL2_R) + 1.0 EL(-Z)
REPEAT LOAD
1 0.9 4 0.9 12 -1.0
LOAD 465 0.9 (DL - SDL1_R) + 1.0 EL(+X)
REPEAT LOAD
1 0.9 3 -0.9 11 1.0
LOAD 466 0.9 (DL - SDL1_R) + 1.0 EL(-X)
REPEAT LOAD
1 0.9 3 -0.9 11 -1.0
LOAD 467 0.9 (DL - SDL1_R) + 1.0 EL(+Z)
REPEAT LOAD
1 0.9 3 -0.9 12 1.0
LOAD 468 0.9 (DL - SDL1_R) + 1.0 EL(-Z)
REPEAT LOAD
1 0.9 3 -0.9 12 -1.0
LOAD 469 0.9 (DL - SDL2_R) + 1.0 EL(+X)
REPEAT LOAD
1 0.9 4 -0.9 11 1.0
LOAD 470 0.9 (DL - SDL2_R) + 1.0 EL(-X)
REPEAT LOAD
1 0.9 4 -0.9 11 -1.0
LOAD 471 0.9 (DL - SDL2_R) + 1.0 EL(+Z)
REPEAT LOAD
1 0.9 4 -0.9 12 1.0
LOAD 472 0.9 (DL - SDL2_R) + 1.0 EL(-Z)
REPEAT LOAD
1 0.9 4 -0.9 12 -1.0

PERFORM ANALYSIS
FINISH
```

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	C02	D01			Outotec / L&T Revision No.:	0	1	2	3	
Date	27-01-2020	12-03-2020	25-04-2022	13/10/2023			Date	27-01-2020	12-03-2020	25-04-2022	13/10/2023	
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS DT. 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER SUPPORT STRUCTURE	AS BUILT			Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS DT. 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER SUPPORT STRUCTURE	AS BUILT	
Prepared By	APR	APR	APR	APR			Prepared By	APR	APR	APR	APR	
Checked	SBY	SBY	SBY	SBY			Checked	SBY	SBY	SBY	SBY	
Approved	AG	AG	AG	AG			Approved	AG	AG	AG	AG	

ATTACHMENT 7 STAAD INPUT FILE OF FOUNDATION CF3_1

```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 17-Jan-20
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
2 0 -2.3 0; 3 3.1 -2.3 0; 101 0 -2.5 0; 102 3.1 -2.5 0; 103 -1.5 -2.5 -1.5;
104 4.6 -2.5 -1.5; 105 -1.5 -2.5 1.5; 106 4.6 -2.5 1.5; 107 0 -2.5 -1.5;
108 0 -2.5 1.5; 109 3.1 -2.5 -1.5; 110 3.1 -2.5 1.5; 111 -1.5 -2.5 0;
112 4.6 -2.5 0; 113 -0.95 -2.5 1.5; 114 -0.95 -2.5 0.95; 115 -1.5 -2.5 0.95;
116 -0.35 -2.5 1.5; 117 -0.35 -2.5 0.95; 118 0 -2.5 0.95; 119 -0.95 -2.5 0.35;
120 -1.5 -2.5 0.35; 121 -0.35 -2.5 0.35; 122 0 -2.5 0.35; 123 -0.95 -2.5 0;
124 -0.35 -2.5 0; 125 -0.95 -2.5 -0.35; 126 -1.5 -2.5 -0.35;
127 -0.35 -2.5 -0.35; 128 0 -2.5 -0.35; 129 -0.95 -2.5 -0.95;
130 -1.5 -2.5 -0.95; 131 -0.35 -2.5 -0.95; 132 0 -2.5 -0.95;
133 -0.95 -2.5 -1.5; 134 -0.35 -2.5 -1.5; 135 3.35 -2.5 1.5;
136 3.35 -2.5 0.95; 137 3.1 -2.5 0.95; 138 3.95 -2.5 1.5; 139 3.95 -2.5 0.95;
140 4.6 -2.5 0.95; 141 3.35 -2.5 0.35; 142 3.1 -2.5 0.35; 143 3.95 -2.5 0.35;
144 4.6 -2.5 0.35; 145 3.35 -2.5 0; 146 3.95 -2.5 0; 147 3.35 -2.5 -0.35;
148 3.1 -2.5 -0.35; 149 3.95 -2.5 -0.35; 150 4.6 -2.5 -0.35;
151 3.35 -2.5 -0.95; 152 3.1 -2.5 -0.95; 153 3.95 -2.5 -0.95;
154 4.6 -2.5 -0.95; 155 3.35 -2.5 -1.5; 156 3.95 -2.5 -1.5; 157 0.35 -2.5 1.5;
158 0.35 -2.5 0.95; 159 0.95 -2.5 1.5; 160 0.95 -2.5 0.95; 161 1.3 -2.5 1.5;
162 1.3 -2.5 0.95; 163 2.85 -2.5 1.5; 164 2.85 -2.5 0.95; 175 0.35 -2.5 0.35;
176 0.95 -2.5 0.35; 177 1.3 -2.5 0.35; 178 2.85 -2.5 0.35; 184 0.35 -2.5 0;
185 0.95 -2.5 0; 186 1.3 -2.5 0; 187 2.85 -2.5 0; 193 0.35 -2.5 -0.35;
194 0.95 -2.5 -0.35; 195 1.3 -2.5 -0.35; 196 2.85 -2.5 -0.35;
202 0.35 -2.5 -0.95; 203 0.95 -2.5 -0.95; 204 1.3 -2.5 -0.95;
205 2.85 -2.5 -0.95; 211 0.35 -2.5 -1.5; 212 0.95 -2.5 -1.5; 213 1.3 -2.5 -1.5;
214 2.85 -2.5 -1.5; 215 1.8 -2.5 1.5; 216 1.8 -2.5 0.95; 217 1.8 -2.5 0.35;
218 1.8 -2.5 0; 219 1.8 -2.5 -0.35; 220 1.8 -2.5 -0.95; 221 1.8 -2.5 -1.5;
222 2.25 -2.5 1.5; 223 2.25 -2.5 0.95; 224 2.25 -2.5 0.35; 225 2.25 -2.5 0;
226 2.25 -2.5 -0.35; 227 2.25 -2.5 -0.95; 228 2.25 -2.5 -1.5;
MEMBER INCIDENCES
1 101 2; 2 102 3;
ELEMENT INCIDENCES SHELL
12 105 113 114 115; 13 113 116 117 114; 14 116 108 118 117; 15 115 114 119 120;
16 114 117 121 119; 17 117 118 122 121; 18 120 119 123 111; 19 119 121 124 123;
20 121 122 101 124; 21 111 123 125 126; 22 123 124 127 125; 23 124 101 128 127;
24 126 125 129 130; 25 125 127 131 129; 26 127 128 132 131; 27 130 129 133 103;
28 129 131 134 133; 29 131 132 107 134; 30 110 135 136 137; 31 135 138 139 136;
32 138 106 140 139; 33 137 136 141 142; 34 136 139 143 141; 35 139 140 144 143;
36 142 141 145 102; 37 141 143 146 145; 38 143 144 112 146; 39 102 145 147 148;
40 145 146 149 147; 41 146 112 150 149; 42 148 147 151 152; 43 147 149 153 151;
44 149 150 154 153; 45 152 151 155 109; 46 151 153 156 155; 47 153 154 104 156;
48 108 157 158 118; 49 157 159 160 158; 50 159 161 162 160; 52 163 110 137 164;
58 118 158 175 122; 59 158 160 176 175; 60 160 162 177 176; 62 164 137 142 178;
68 122 175 184 101; 69 175 176 185 184; 70 176 177 186 185; 72 178 142 102 187;
78 101 184 193 128; 79 184 185 194 193; 80 185 186 195 194; 82 187 102 148 196;
88 128 193 202 132; 89 193 194 203 202; 90 194 195 204 203; 92 196 148 152 205;
98 132 202 211 107; 99 202 203 212 211; 100 203 204 213 212;
102 205 152 109 214; 103 161 215 216 162; 104 162 216 217 177;
105 177 217 218 186; 106 186 218 219 195; 107 195 219 220 204;
108 204 220 221 213; 109 215 222 223 216; 110 216 223 224 217;
111 217 224 225 218; 112 218 225 226 219; 113 219 226 227 220;
114 220 227 228 221; 115 222 163 164 223; 116 223 164 178 224;
117 224 178 187 225; 118 225 187 196 226; 119 226 196 205 227;
```

```
120 227 205 214 228;
START GROUP DEFINITION
ELEMENT
_RAFT_MOMENT 12 TO 19 21 22 24 TO 35 37 38 40 TO 50 52 58 TO 60 62 69 70 79 -
80 88 TO 90 92 98 TO 100 102 TO 120
_RAFT_SHEAR 12 TO 15 18 21 24 27 TO 32 35 38 41 44 TO 50 52 60 70 80 90 98 -
99 TO 100 102 TO 115 120
END GROUP DEFINITION
ELEMENT PROPERTY
12 TO 50 52 58 TO 60 62 68 TO 70 72 78 TO 80 82 88 TO 90 92 98 TO 100 102 -
103 TO 120 THICKNESS 0.6
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.7806e+07
POISSON 0.17
DENSITY 25
ALPHA 1e-05
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
1 2 PRIS YD 0.8 ZD 0.8
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
101 TO 164 175 TO 178 184 TO 187 193 TO 196 202 TO 205 211 TO 227 -
228 ELASTIC MAT DIRECT Y SUBGRADE 20000 PRINT
LOAD 1 DEAD LOAD
SELFWEIGHT Y -1
*Soil Weight
ELEMENT LOAD
12 TO 50 52 58 TO 60 62 68 TO 70 72 78 TO 80 82 88 TO 90 92 98 TO 100 102 -
103 TO 120 PR GY -25.2
***Surcharge Load
31 32 34 35 37 38 40 41 43 44 46 47 PR GY -50
JOINT LOAD
2   FX  23.5    FY  -1216.4 FZ  18.4    MX  -14.6   MZ  -21.0
3   FX  3.6 FY  -146.4  FZ  -4.4    MX  -4.3    MZ  -3.7
Load 2  LL
JOINT LOAD
2   FX  2.0 FY  -159.0  FZ  -0.6    MX  -3.4    MZ  -1.4
3   FX  0.2 FY  -32.2   FZ  0.4 MX  0.1 MZ  -0.1
Load 3  SDL1_R
JOINT LOAD
2   FX  -11.9   FY  -99.2   FZ  8.7 MX  7.4 MZ  18.3
3   FX  -3.1    FY  19.7    FZ  -0.4    MX  0.5 MZ  4.8
Load 4  SDL2_R
JOINT LOAD
2   FX  66.2    FY  -230.4  FZ  7.1 MX  0.5 MZ  -95.0
3   FX  15.4    FY  -6.7    FZ  -0.2    MX  0.4 MZ  -23.6
Load 5  WIND LOAD (+X)
JOINT LOAD
2   FX  37.9    FY  -42.5   FZ  -0.4    MX  -0.8    MZ  -54.4
3   FX  6.2 FY  -6.2    FZ  0.1 MX  0.0 MZ  -11.9
Load 6  WIND LOAD (-X)
JOINT LOAD
2   FX  -41.1   FY  47.6    FZ  0.3 MX  0.8 MZ  59.0
```

```
3   FX  -7  FY  7.2 FZ  -0.1    MX  0.0 MZ  13.0
Load 7  WIND LOAD (+Z)
JOINT LOAD
2   FX  -5.6    FY  12.1    FZ  11  MX  26.6    MZ  8.1
3   FX  -1.3    FY  -0.7    FZ  0.1 MX  2.0 MZ  2.0
Load 8  WIND LOAD (+-Z)
JOINT LOAD
2   FX  3.4 FY  -11.3   FZ  -4.1    MX  -10.2   MZ  -4.9
3   FX  0.8 FY  0.8 FZ  0   MX  -0.7    MZ  -1.2
Load 11 EL (+X)
JOINT LOAD
2   FX  22.3    FY  -33.1   FZ  -0.2    MX  -0.3    MZ  -32.4
3   FX  5.4 FY  -6.6    FZ  0.1 MX  0.1 MZ  -8.2
Load 12 EL (+Z)
JOINT LOAD
2   FX  -3.4    FY  6.4 FZ  1.5 MX  4.9 MZ  4.9
3   FX  -0.8    FY  1.5 FZ  0.3 MX  0.7 MZ  1.2
************************************************
*LOAD COMBINATION FOR BASE PRESSURE & Servicibility
************************************************
LOAD 301 DL+SDL1_R
REPEAT LOAD
1 1.0 3 1.0
LOAD 302 DL+SDL2_R
REPEAT LOAD
1 1.0 4 1.0
LOAD 303 DL-SDL1_R
REPEAT LOAD
1 1.0 3 -1.0
LOAD 304 DL-SDL2_R
REPEAT LOAD
1 1.0 4 -1.0
*********************************
LOAD 305 DL + SDL1_R + LL
REPEAT LOAD
1 1.0 3 1.0 2 1.0
LOAD 306 DL + SDL2_R + LL
REPEAT LOAD
1 1.0 4 1.0 2 1.0
LOAD 307 DL - SDL1_R + LL
REPEAT LOAD
1 1.0 3 -1.0 2 1.0
LOAD 308 DL - SDL2_R + LL
REPEAT LOAD
1 1.0 4 -1.0 2 1.0
************************************************
LOAD 309 DL + SDL1_R + 0.75 LL + 0.75 WL(+X)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 5 0.75
LOAD 310 DL + SDL1_R + 0.75 LL + 0.75 WL(-X)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 6 0.75
LOAD 311 DL + SDL1_R + 0.75 LL + 0.75 WL(+Z)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 7 0.75
LOAD 312 DL + SDL1_R + 0.75 LL + 0.75 WL(-Z)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 8 0.75
```

```
LOAD 313 DL + SDL2_R + 0.75 LL + 0.75 WL(+X)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 5 0.75
LOAD 314 DL + SDL2_R + 0.75 LL + 0.75 WL(-X)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 6 0.75
LOAD 315 DL + SDL2_R + 0.75 LL + 0.75 WL(+Z)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 7 0.75
LOAD 316 DL + SDL2_R + 0.75 LL + 0.75 WL(-Z)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 8 0.75
LOAD 317 DL - SDL1_R + 0.75 LL + 0.75 WL(+X)
REPEAT LOAD
1 1.0 3 -1.0 2 0.75 5 0.75
LOAD 318 DL - SDL1_R + 0.75 LL + 0.75 WL(-X)
REPEAT LOAD
1 1.0 3 -1.0 2 0.75 6 0.75
LOAD 319 DL - SDL1_R + 0.75 LL + 0.75 WL(+Z)
REPEAT LOAD
1 1.0 3 -1.0 2 0.75 7 0.75
LOAD 320 DL - SDL1_R + 0.75 LL + 0.75 WL(-Z)
REPEAT LOAD
1 1.0 3 -1.0 2 0.75 8 0.75
LOAD 321 DL - SDL2_R + 0.75 LL + 0.75 WL(+X)
REPEAT LOAD
1 1.0 4 -1.0 2 0.75 5 0.75
LOAD 322 DL - SDL2_R + 0.75 LL + 0.75 WL(-X)
REPEAT LOAD
1 1.0 4 -1.0 2 0.75 6 0.75
LOAD 323 DL - SDL2_R + 0.75 LL + 0.75 WL(+Z)
REPEAT LOAD
1 1.0 4 -1.0 2 0.75 7 0.75
LOAD 324 DL - SDL2_R + 0.75 LL + 0.75 WL(-Z)
REPEAT LOAD
1 1.0 4 -1.0 2 0.75 8 0.75
**********************************************
LOAD 325 DL + SDL1_R + 0.75 LL + 0.525 EL(+X)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 11 0.525
LOAD 326 DL + SDL1_R + 0.75 LL + 0.525 EL(-X)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 11 -0.525
LOAD 327 DL + SDL1_R + 0.75 LL + 0.525 EL(+Z)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 12 0.525
LOAD 328 DL + SDL1_R + 0.75 LL + 0.525 EL(-Z)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 12 -0.525
LOAD 329 DL + SDL2_R + 0.75 LL + 0.525 EL(+X)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 11 0.525
LOAD 330 DL + SDL2_R + 0.75 LL + 0.525 EL(-X)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 11 -0.525
LOAD 331 DL + SDL2_R + 0.75 LL + 0.525 EL(+Z)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 12 0.525
```

```
LOAD 332 DL + SDL2_R + 0.75 LL + 0.525 EL(-Z)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 12 -0.525
LOAD 333 DL - SDL1_R + 0.75 LL + 0.525 EL(+X)
REPEAT LOAD
1 1.0 3 -1.0 2 0.75 11 0.525
LOAD 334 DL - SDL1_R + 0.75 LL + 0.525 EL(-X)
REPEAT LOAD
1 1.0 3 -1.0 2 0.75 11 -0.525
LOAD 335 DL - SDL1_R + 0.75 LL + 0.525 EL(+Z)
REPEAT LOAD
1 1.0 3 -1.0 2 0.75 12 0.525
LOAD 336 DL - SDL1_R + 0.75 LL + 0.525 EL(-Z)
REPEAT LOAD
1 1.0 3 -1.0 2 0.75 12 -0.525
LOAD 337 DL - SDL2_R + 0.75 LL + 0.525 EL(+X)
REPEAT LOAD
1 1.0 4 -1.0 2 0.75 11 0.525
LOAD 338 DL - SDL2_R + 0.75 LL + 0.525 EL(-X)
REPEAT LOAD
1 1.0 4 -1.0 2 0.75 11 -0.525
LOAD 339 DL - SDL2_R + 0.75 LL + 0.525 EL(+Z)
REPEAT LOAD
1 1.0 4 -1.0 2 0.75 12 0.525
LOAD 340 DL - SDL2_R + 0.75 LL + 0.525 EL(-Z)
REPEAT LOAD
1 1.0 4 -1.0 2 0.75 12 -0.525
************************************************
LOAD 341 0.6 (DL + SDL1_R) + WL(+X)
REPEAT LOAD
1 0.6 3 0.6 5 1.0
LOAD 342 0.6 (DL + SDL1_R) + WL(-X)
REPEAT LOAD
1 0.6 3 0.6 6 1.0
LOAD 343 0.6 (DL + SDL1_R) + WL(+Z)
REPEAT LOAD
1 0.6 3 0.6 7 1.0
LOAD 344 0.6 (DL + SDL1_R) + WL(-Z)
REPEAT LOAD
1 0.6 3 0.6 8 1.0
LOAD 345 0.6 (DL + SDL2_R) + WL(+X)
REPEAT LOAD
1 0.6 4 0.6 5 1.0
LOAD 346 0.6 (DL + SDL2_R) + WL(-X)
REPEAT LOAD
1 0.6 4 0.6 6 1.0
LOAD 347 0.6 (DL + SDL2_R) + WL(+Z)
REPEAT LOAD
1 0.6 4 0.6 7 1.0
LOAD 348 0.6 (DL + SDL2_R) + WL(-Z)
REPEAT LOAD
1 0.6 4 0.6 8 1.0
LOAD 349 0.6 (DL - SDL1_R) + WL(+X)
REPEAT LOAD
1 0.6 3 -0.6 5 1.0
LOAD 350 0.6 (DL - SDL1_R) + WL(-X)
REPEAT LOAD
1 0.6 3 -0.6 6 1.0
```

```
LOAD 351 0.6 (DL - SDL1_R) + WL(+Z)
REPEAT LOAD
1 0.6 3 -0.6 7 1.0
LOAD 352 0.6 (DL - SDL1_R) + WL(-Z)
REPEAT LOAD
1 0.6 3 -0.6 8 1.0
LOAD 353 0.6 (DL - SDL2_R) + WL(+X)
REPEAT LOAD
1 0.6 4 -0.6 5 1.0
LOAD 354 0.6 (DL - SDL2_R) + WL(-X)
REPEAT LOAD
1 0.6 4 -0.6 6 1.0
LOAD 355 0.6 (DL - SDL2_R) + WL(+Z)
REPEAT LOAD
1 0.6 4 -0.6 7 1.0
LOAD 356 0.6 (DL - SDL2_R) + WL(-Z)
REPEAT LOAD
1 0.6 4 -0.6 8 1.0
*********************************************
LOAD 357 0.6 (DL + SDL1_R) + 0.7 EL(+X)
REPEAT LOAD
1 0.6 3 0.6 11 0.7
LOAD 358 0.6 (DL + SDL1_R) + 0.7 EL(-X)
REPEAT LOAD
1 0.6 3 0.6 11 -0.7
LOAD 359 0.6 (DL + SDL1_R) + 0.7 EL(+Z)
REPEAT LOAD
1 0.6 3 0.6 12 0.7
LOAD 360 0.6 (DL + SDL1_R) + 0.7 EL(-Z)
REPEAT LOAD
1 0.6 3 0.6 12 -0.7
LOAD 361 0.6 (DL + SDL2_R) + 0.7 EL(+X)
REPEAT LOAD
1 0.6 4 0.6 11 0.7
LOAD 362 0.6 (DL + SDL2_R) + 0.7 EL(-X)
REPEAT LOAD
1 0.6 4 0.6 11 -0.7
LOAD 363 0.6 (DL + SDL2_R) + 0.7 EL(+Z)
REPEAT LOAD
1 0.6 4 0.6 12 -0.7
LOAD 364 0.6 (DL + SDL2_R) + 0.7 EL(-Z)
REPEAT LOAD
1 0.6 4 0.6 12 -0.7
LOAD 365 0.6 (DL + SDL1_R) + 0.7 EL(+X)
REPEAT LOAD
1 0.6 3 -0.6 11 0.7
LOAD 366 0.6 (DL + SDL1_R) + 0.7 EL(-X)
REPEAT LOAD
1 0.6 3 -0.6 11 -0.7
LOAD 367 0.6 (DL + SDL1_R) + 0.7 EL(+Z)
REPEAT LOAD
1 0.6 3 -0.6 12 0.7
LOAD 368 0.6 (DL + SDL1_R) + 0.7 EL(-Z)
REPEAT LOAD
1 0.6 3 -0.6 12 -0.7
LOAD 369 0.6 (DL + SDL2_R) + 0.7 EL(+X)
REPEAT LOAD
1 0.6 4 -0.6 11 0.7
```

```
LOAD 370 0.6 (DL + SDL2_R) + 0.7 EL(-X)
REPEAT LOAD
1 0.6 4 -0.6 11 -0.7
LOAD 371 0.6 (DL + SDL2_R) + 0.7 EL(+Z)
REPEAT LOAD
1 0.6 4 -0.6 12 -0.7
LOAD 372 0.6 (DL + SDL2_R) + 0.7 EL(-Z)
REPEAT LOAD
1 0.6 4 -0.6 12 -0.7
************************************************
*Load Combination for Concrete design
************************************************
LOAD 401 1.4 (DL+SDL1_R)
REPEAT LOAD
1 1.4 3 1.4
LOAD 402 1.4 (DL+SDL2_R)
REPEAT LOAD
1 1.4 4 1.4
LOAD 403 1.4 (DL-SDL1_R)
REPEAT LOAD
1 1.4 3 -1.4
LOAD 404 1.4 (DL-SDL2_R)
REPEAT LOAD
1 1.4 4 -1.4
************************************************
LOAD 405 1.4 (DL + SDL1_R ) + 1.7 LL
REPEAT LOAD
1 1.4 3 1.4 2 1.7
LOAD 406 1.4 (DL + SDL2_R ) + 1.7 LL
REPEAT LOAD
1 1.4 4 1.4 2 1.7
LOAD 407 1.4 (DL - SDL1_R ) + 1.7 LL
REPEAT LOAD
1 1.4 3 -1.4 2 1.7
LOAD 408 1.4 (DL - SDL2_R ) + 1.7 LL
REPEAT LOAD
1 1.4 4 -1.4 2 1.7
************************************************
LOAD 409 1.2 (DL + SDL1_R) + 0.5 LL + 1.6 WL(+X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 5 1.6
LOAD 410 1.2 (DL + SDL1_R) + 0.5 LL + 1.6 WL(-X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 6 1.6
LOAD 411 1.2 (DL + SDL1_R) + 0.5 LL + 1.6 WL(+Z)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 7 1.6
LOAD 412 1.2 (DL + SDL1_R) + 0.5 LL + 1.6 WL(-Z)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 8 1.6
LOAD 413 1.2 (DL + SDL2_R) + 0.5 LL + 1.6 WL(+X)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 5 1.6
LOAD 414 1.2 (DL + SDL2_R) + 0.5 LL + 1.6 WL(-X)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 6 1.6
LOAD 415 1.2 (DL + SDL2_R) + 0.5 LL + 1.6 WL(+Z)
REPEAT LOAD
```

```
1 1.2 4 1.2 2 0.5 7 1.6
LOAD 416 1.2 (DL + SDL2_R) + 0.5 LL + 1.6 WL(-Z)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 8 1.6
LOAD 417 1.2 (DL - SDL1_R) + 0.5 LL + 1.6 WL(+X)
REPEAT LOAD
1 1.2 3 -1.2 2 0.5 5 1.6
LOAD 418 1.2 (DL - SDL1_R) + 0.5 LL + 1.6 WL(-X)
REPEAT LOAD
1 1.2 3 -1.2 2 0.5 6 1.6
LOAD 419 1.2 (DL - SDL1_R) + 0.5 LL + 1.6 WL(+Z)
REPEAT LOAD
1 1.2 3 -1.2 2 0.5 7 1.6
LOAD 420 1.2 (DL - SDL1_R) + 0.5 LL + 1.6 WL(-Z)
REPEAT LOAD
1 1.2 3 -1.2 2 0.5 8 1.6
LOAD 421 1.2 (DL - SDL2_R) + 0.5 LL + 1.6 WL(+X)
REPEAT LOAD
1 1.2 4 -1.2 2 0.5 5 1.6
LOAD 422 1.2 (DL - SDL2_R) + 0.5 LL + 1.6 WL(-X)
REPEAT LOAD
1 1.2 4 -1.2 2 0.5 6 1.6
LOAD 423 1.2 (DL - SDL2_R) + 0.5 LL + 1.6 WL(+Z)
REPEAT LOAD
1 1.2 4 -1.2 2 0.5 7 1.6
LOAD 424 1.2 (DL - SDL2_R) + 0.5 LL + 1.6 WL(-Z)
REPEAT LOAD
1 1.2 4 -1.2 2 0.5 8 1.6
************************************************
LOAD 425 1.2 (DL + SDL1_R) + 0.5 LL + 1.0 EL(+X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 11 1.0
LOAD 426 1.2 (DL + SDL1_R) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 11 -1.0
LOAD 427 1.2 (DL + SDL1_R) + 0.5 LL + 1.0 EL(+Z)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 12 1.0
LOAD 428 1.2 (DL + SDL1_R) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 12 -1.0
LOAD 429 1.2 (DL + SDL2_R) + 0.5 LL + 1.0 EL(+X)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 11 1.0
LOAD 430 1.2 (DL + SDL2_R) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 11 -1.0
LOAD 431 1.2 (DL + SDL2_R) + 0.5 LL + 1.0 EL(+Z)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 12 1.0
LOAD 432 1.2 (DL + SDL2_R) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 12 -1.0
LOAD 433 1.2 (DL - SDL1_R) + 0.5 LL + 1.0 EL(+X)
REPEAT LOAD
1 1.2 3 -1.2 2 0.5 11 1.0
LOAD 434 1.2 (DL - SDL1_R) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
```

```
1 1.2 3 -1.2 2 0.5 11 -1.0
LOAD 435 1.2 (DL - SDL1_R) + 0.5 LL + 1.0 EL(+Z)
REPEAT LOAD
1 1.2 3 -1.2 2 0.5 12 1.0
LOAD 436 1.2 (DL - SDL1_R) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 3 -1.2 2 0.5 12 -1.0
LOAD 437 1.2 (DL - SDL2_R) + 0.5 LL + 1.0 EL(+X)
REPEAT LOAD
1 1.2 4 -1.2 2 0.5 11 1.0
LOAD 438 1.2 (DL - SDL2_R) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 4 -1.2 2 0.5 11 -1.0
LOAD 439 1.2 (DL - SDL2_R) + 0.5 LL + 1.0 EL(+Z)
REPEAT LOAD
1 1.2 4 -1.2 2 0.5 12 1.0
LOAD 440 1.2 (DL - SDL2_R) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 4 -1.2 2 0.5 12 -1.0
*********************************************
LOAD 441 0.9 (DL + SDL1_R) + 1.6 WL(+X)
REPEAT LOAD
1 0.9 3 0.9 5 1.6
LOAD 442 0.9 (DL + SDL1_R) + 1.6 WL(-X)
REPEAT LOAD
1 0.9 3 0.9 6 1.6
LOAD 443 0.9 (DL + SDL1_R) + 1.6 WL(+Z)
REPEAT LOAD
1 0.9 3 0.9 7 1.6
LOAD 444 0.9 (DL + SDL1_R) + 1.6 WL(-Z)
REPEAT LOAD
1 0.9 3 0.9 8 1.6
LOAD 445 0.9 (DL + SDL2_R) + 1.6 WL(+X)
REPEAT LOAD
1 0.9 4 0.9 5 1.6
LOAD 446 0.9 (DL + SDL2_R) + 1.6 WL(-X)
REPEAT LOAD
1 0.9 4 0.9 6 1.6
LOAD 447 0.9 (DL + SDL2_R) + 1.6 WL(+Z)
REPEAT LOAD
1 0.9 4 0.9 7 1.6
LOAD 448 0.9 (DL + SDL2_R) + 1.6 WL(-Z)
REPEAT LOAD
1 0.9 4 0.9 8 1.6
LOAD 449 0.9 (DL - SDL1_R) + 1.6 WL(+X)
REPEAT LOAD
1 0.9 3 -0.9 5 1.6
LOAD 450 0.9 (DL - SDL1_R) + 1.6 WL(-X)
REPEAT LOAD
1 0.9 3 -0.9 6 1.6
LOAD 451 0.9 (DL - SDL1_R) + 1.6 WL(+Z)
REPEAT LOAD
1 0.9 3 -0.9 7 1.6
LOAD 452 0.9 (DL - SDL1_R) + 1.6 WL(-Z)
REPEAT LOAD
1 0.9 3 -0.9 8 1.6
LOAD 453 0.9 (DL - SDL2_R) + 1.6 WL(+X)
REPEAT LOAD
```

```
1 0.9 4 -0.9 5 1.6
LOAD 454 0.9 (DL - SDL2_R) + 1.6 WL(-X)
REPEAT LOAD
1 0.9 4 -0.9 6 1.6
LOAD 455 0.9 (DL - SDL2_R) + 1.6 WL(+Z)
REPEAT LOAD
1 0.9 4 -0.9 7 1.6
LOAD 456 0.9 (DL - SDL2_R) + 1.6 WL(-Z)
REPEAT LOAD
1 0.9 4 -0.9 8 1.6
*********************************************
LOAD 457 0.9 (DL + SDL1_R) + 1.0 EL(+X)
REPEAT LOAD
1 0.9 3 0.9 11 1.0
LOAD 458 0.9 (DL + SDL1_R) + 1.0 EL(-X)
REPEAT LOAD
1 0.9 3 0.9 11 -1.0
LOAD 459 0.9 (DL + SDL1_R) + 1.0 EL(+Z)
REPEAT LOAD
1 0.9 3 0.9 12 1.0
LOAD 460 0.9 (DL + SDL1_R) + 1.0 EL(-Z)
REPEAT LOAD
1 0.9 3 0.9 12 -1.0
LOAD 461 0.9 (DL + SDL2_R) + 1.0 EL(+X)
REPEAT LOAD
1 0.9 4 0.9 11 1.0
LOAD 462 0.9 (DL + SDL2_R) + 1.0 EL(-X)
REPEAT LOAD
1 0.9 4 0.9 11 -1.0
LOAD 463 0.9 (DL + SDL2_R) + 1.0 EL(+Z)
REPEAT LOAD
1 0.9 4 0.9 12 1.0
LOAD 464 0.9 (DL + SDL2_R) + 1.0 EL(-Z)
REPEAT LOAD
1 0.9 4 0.9 12 -1.0
LOAD 465 0.9 (DL - SDL1_R) + 1.0 EL(+X)
REPEAT LOAD
1 0.9 3 -0.9 11 1.0
LOAD 466 0.9 (DL - SDL1_R) + 1.0 EL(-X)
REPEAT LOAD
1 0.9 3 -0.9 11 -1.0
LOAD 467 0.9 (DL - SDL1_R) + 1.0 EL(+Z)
REPEAT LOAD
1 0.9 3 -0.9 12 1.0
LOAD 468 0.9 (DL - SDL1_R) + 1.0 EL(-Z)
REPEAT LOAD
1 0.9 3 -0.9 12 -1.0
LOAD 469 0.9 (DL - SDL2_R) + 1.0 EL(+X)
REPEAT LOAD
1 0.9 4 -0.9 11 1.0
LOAD 470 0.9 (DL - SDL2_R) + 1.0 EL(-X)
REPEAT LOAD
1 0.9 4 -0.9 11 -1.0
LOAD 471 0.9 (DL - SDL2_R) + 1.0 EL(+Z)
REPEAT LOAD
1 0.9 4 -0.9 12 1.0
LOAD 472 0.9 (DL - SDL2_R) + 1.0 EL(-Z)
REPEAT LOAD
```

```
1 0.9 4 -0.9 12 -1.0
PERFORM ANALYSIS
FINISH
```

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	C02	D01			Outotec / L&T Revision No.:	0	1	2	3	
Date	27-01-2020	12-03-2020	25-04-2022	13/10/2023			Date	27-01-2020	12-03-2020	25-04-2022	13/10/2023	
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS DT. 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER SUPPORT STRUCTURE	AS BUILT			Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS DT. 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER SUPPORT STRUCTURE	AS BUILT	
Prepared By	APR	APR	APR	APR			Prepared By	APR	APR	APR	APR	
Checked	SBY	SBY	SBY	SBY			Checked	SBY	SBY	SBY	SBY	
Approved	AG	AG	AG	AG			Approved	AG	AG	AG	AG	

ATTACHMENT 8 STAAD INPUT FILE OF FOUNDATION CF3_2

```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 17-Jan-20
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
5 0 -2.3 0; 6 3.1 -2.3 0; 101 0 -2.5 0; 102 3.1 -2.5 0; 103 -1.5 -2.5 -1.5;
104 4.6 -2.5 -1.5; 105 -1.5 -2.5 1.5; 106 4.6 -2.5 1.5; 107 0 -2.5 -1.5;
108 0 -2.5 1.5; 109 3.1 -2.5 -1.5; 110 3.1 -2.5 1.5; 111 -1.5 -2.5 0;
112 4.6 -2.5 0; 113 -0.95 -2.5 1.5; 114 -0.95 -2.5 0.95; 115 -1.5 -2.5 0.95;
116 -0.35 -2.5 1.5; 117 -0.35 -2.5 0.95; 118 0 -2.5 0.95; 119 -0.95 -2.5 0.35;
120 -1.5 -2.5 0.35; 121 -0.35 -2.5 0.35; 122 0 -2.5 0.35; 123 -0.95 -2.5 0;
124 -0.35 -2.5 0; 125 -0.95 -2.5 -0.35; 126 -1.5 -2.5 -0.35;
127 -0.35 -2.5 -0.35; 128 0 -2.5 -0.35; 129 -0.95 -2.5 -0.95;
130 -1.5 -2.5 -0.95; 131 -0.35 -2.5 -0.95; 132 0 -2.5 -0.95;
133 -0.95 -2.5 -1.5; 134 -0.35 -2.5 -1.5; 135 3.35 -2.5 1.5;
136 3.35 -2.5 0.95; 137 3.1 -2.5 0.95; 138 3.95 -2.5 1.5; 139 3.95 -2.5 0.95;
140 4.6 -2.5 0.95; 141 3.35 -2.5 0.35; 142 3.1 -2.5 0.35; 143 3.95 -2.5 0.35;
144 4.6 -2.5 0.35; 145 3.35 -2.5 0; 146 3.95 -2.5 0; 147 3.35 -2.5 -0.35;
148 3.1 -2.5 -0.35; 149 3.95 -2.5 -0.35; 150 4.6 -2.5 -0.35;
151 3.35 -2.5 -0.95; 152 3.1 -2.5 -0.95; 153 3.95 -2.5 -0.95;
154 4.6 -2.5 -0.95; 155 3.35 -2.5 -1.5; 156 3.95 -2.5 -1.5; 157 0.35 -2.5 1.5;
158 0.35 -2.5 0.95; 159 0.95 -2.5 1.5; 160 0.95 -2.5 0.95; 161 1.3 -2.5 1.5;
162 1.3 -2.5 0.95; 163 2.85 -2.5 1.5; 164 2.85 -2.5 0.95; 175 0.35 -2.5 0.35;
176 0.95 -2.5 0.35; 177 1.3 -2.5 0.35; 178 2.85 -2.5 0.35; 184 0.35 -2.5 0;
185 0.95 -2.5 0; 186 1.3 -2.5 0; 187 2.85 -2.5 0; 193 0.35 -2.5 -0.35;
194 0.95 -2.5 -0.35; 195 1.3 -2.5 -0.35; 196 2.85 -2.5 -0.35;
202 0.35 -2.5 -0.95; 203 0.95 -2.5 -0.95; 204 1.3 -2.5 -0.95;
205 2.85 -2.5 -0.95; 211 0.35 -2.5 -1.5; 212 0.95 -2.5 -1.5; 213 1.3 -2.5 -1.5;
214 2.85 -2.5 -1.5; 215 1.8 -2.5 1.5; 216 1.8 -2.5 0.95; 217 1.8 -2.5 0.35;
218 1.8 -2.5 0; 219 1.8 -2.5 -0.35; 220 1.8 -2.5 -0.95; 221 1.8 -2.5 -1.5;
222 2.25 -2.5 1.5; 223 2.25 -2.5 0.95; 224 2.25 -2.5 0.35; 225 2.25 -2.5 0;
226 2.25 -2.5 -0.35; 227 2.25 -2.5 -0.95; 228 2.25 -2.5 -1.5;
MEMBER INCIDENCES
1 101 5; 2 102 6;
ELEMENT INCIDENCES SHELL
12 105 113 114 115; 13 113 116 117 114; 14 116 108 118 117; 15 115 114 119 120;
16 114 117 121 119; 17 117 118 122 121; 18 120 119 123 111; 19 119 121 124 123;
20 121 122 101 124; 21 111 123 125 126; 22 123 124 127 125; 23 124 101 128 127;
24 126 125 129 130; 25 125 127 131 129; 26 127 128 132 131; 27 130 129 133 103;
28 129 131 134 133; 29 131 132 107 134; 30 110 135 136 137; 31 135 138 139 136;
32 138 106 140 139; 33 137 136 141 142; 34 136 139 143 141; 35 139 140 144 143;
36 142 141 145 102; 37 141 143 146 145; 38 143 144 112 146; 39 102 145 147 148;
40 145 146 149 147; 41 146 112 150 149; 42 148 147 151 152; 43 147 149 153 151;
44 149 150 154 153; 45 152 151 155 109; 46 151 153 156 155; 47 153 154 104 156;
48 108 157 158 118; 49 157 159 160 158; 50 159 161 162 160; 52 163 110 137 164;
58 118 158 175 122; 59 158 160 176 175; 60 160 162 177 176; 62 164 137 142 178;
68 122 175 184 101; 69 175 176 185 184; 70 176 177 186 185; 72 178 142 102 187;
78 101 184 193 128; 79 184 185 194 193; 80 185 186 195 194; 82 187 102 148 196;
88 128 193 202 132; 89 193 194 203 202; 90 194 195 204 203; 92 196 148 152 205;
98 132 202 211 107; 99 202 203 212 211; 100 203 204 213 212;
102 205 152 109 214; 103 161 215 216 162; 104 162 216 217 177;
105 177 217 218 186; 106 186 218 219 195; 107 195 219 220 204;
108 204 220 221 213; 109 215 222 223 216; 110 216 223 224 217;
111 217 224 225 218; 112 218 225 226 219; 113 219 226 227 220;
114 220 227 228 221; 115 222 163 164 223; 116 223 164 178 224;
117 224 178 187 225; 118 225 187 196 226; 119 226 196 205 227;
```

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120 227 205 214 228;
START GROUP DEFINITION
ELEMENT
_RAFT_MOMENT 12 TO 19 21 22 24 TO 35 37 38 40 TO 50 52 58 TO 60 62 69 70 79 -
80 88 TO 90 92 98 TO 100 102 TO 120
_RAFT_SHEAR 12 TO 15 18 21 24 27 TO 32 35 38 41 44 TO 50 52 60 70 80 90 98 -
99 TO 100 102 TO 115 120
END GROUP DEFINITION
ELEMENT PROPERTY
12 TO 50 52 58 TO 60 62 68 TO 70 72 78 TO 80 82 88 TO 90 92 98 TO 100 102 -
103 TO 120 THICKNESS 0.6
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.7806e+07
POISSON 0.17
DENSITY 25
ALPHA 1e-05
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
1 2 PRIS YD 0.8 ZD 0.8
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
101 TO 164 175 TO 178 184 TO 187 193 TO 196 202 TO 205 211 TO 227 -
228 ELASTIC MAT DIRECT Y SUBGRADE 20000 PRINT
LOAD 1 DEAD LOAD
SELFWEIGHT Y -1
*Soil Weight
ELEMENT LOAD
12 TO 50 52 58 TO 60 62 68 TO 70 72 78 TO 80 82 88 TO 90 92 98 TO 100 102 -
103 TO 120 PR GY -25.2
*Surcharge Load
31 32 34 35 37 38 40 41 43 44 46 47 PR GY -50
JOINT LOAD
5   FX  26.9    FY  -1303.2 FZ  -27.8   MX  -26.7   MZ  -25.2
6   FX  4.1 FY  -183.1  FZ  2.6 MX  1.0 MZ  -4.5
Load 2  LL
JOINT LOAD
5   FX  2.8 FY  -179.6  FZ  -0.1    MX  -1.2    MZ  -2.7
6   FX  0.4 FY  -54.5   FZ  -0.2    MX  -0.3    MZ  -0.5
Load 3  SDL1_R
JOINT LOAD
5   FX  2.8 FY  -43.8   FZ  -5.4    MX  1.9 MZ  -4.4
6   FX  0.7 FY  18.9    FZ  1.3 MX  1.7 MZ  -1.2
Load 4  SDL2_R
JOINT LOAD
5   FX  51.1    FY  -207.6  FZ  -8.6    MX  -6.1    MZ  -73.2
6   FX  11.8    FY  0.4 FZ  1.2 MX  1.5 MZ  -18.1
Load 5  WIND LOAD (+X)
JOINT LOAD
5   FX  48  FY  -40.2   FZ  -0.3    MX  -0.8    MZ  -68.7
6   FX  6.5 FY  -5.8    FZ  -0.1    MX  -0.1    MZ  -14.1
Load 6  WIND LOAD (-X)
JOINT LOAD
5   FX  -58.1   FY  45.1    FZ  0.2 MX  0.8 MZ  83.4
```

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6   FX  -9.2    FY  6.6 FZ  0.1 MX  0.1 MZ  18.0
Load 7  WIND LOAD (+Z)
JOINT LOAD
5   FX  0   FY  40.6    FZ  13.9    MX  27.4    MZ  -0.9
6   FX  0.2 FY  4.6 FZ  0.4 MX  1.9 MZ  -0.4
Load 8  WIND LOAD (+-Z)
JOINT LOAD
5   FX  -0.2    FY  -15.9   FZ  -5.3    MX  -10.6   MZ  0.7
6   FX  -0.2    FY  -2.8    FZ  -0.1    MX  -0.7    MZ  0.3
Load 11 EL (+X)
JOINT LOAD
5   FX  22.3    FY  -29.8   FZ  -0.1    MX  -0.3    MZ  -32.5
6   FX  5.5 FY  -6.4    FZ  0.1 MX  0.1 MZ  -8.3
Load 12 EL (+Z)
JOINT LOAD
5   FX  -0.4    FY  31.2    FZ  2.5 MX  5.3 MZ  0.0
6   FX  0.1 FY  1.1 FZ  0.5 MX  0.9 MZ  -0.1
************************************************
*LOAD COMBINATION FOR BASE PRESSURE & Servicibility
************************************************
LOAD 301 DL+SDL1_R
REPEAT LOAD
1 1.0 3 1.0
LOAD 302 DL+SDL2_R
REPEAT LOAD
1 1.0 4 1.0
LOAD 303 DL-SDL1_R
REPEAT LOAD
1 1.0 3 -1.0
LOAD 304 DL-SDL2_R
REPEAT LOAD
1 1.0 4 -1.0
*********************************
LOAD 305 DL + SDL1_R + LL
REPEAT LOAD
1 1.0 3 1.0 2 1.0
LOAD 306 DL + SDL2_R + LL
REPEAT LOAD
1 1.0 4 1.0 2 1.0
LOAD 307 DL - SDL1_R + LL
REPEAT LOAD
1 1.0 3 -1.0 2 1.0
LOAD 308 DL - SDL2_R + LL
REPEAT LOAD
1 1.0 4 -1.0 2 1.0
************************************************
LOAD 309 DL + SDL1_R + 0.75 LL + 0.75 WL(+X)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 5 0.75
LOAD 310 DL + SDL1_R + 0.75 LL + 0.75 WL(-X)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 6 0.75
LOAD 311 DL + SDL1_R + 0.75 LL + 0.75 WL(+Z)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 7 0.75
LOAD 312 DL + SDL1_R + 0.75 LL + 0.75 WL(-Z)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 8 0.75
```

```
LOAD 313 DL + SDL2_R + 0.75 LL + 0.75 WL(+X)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 5 0.75
LOAD 314 DL + SDL2_R + 0.75 LL + 0.75 WL(-X)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 6 0.75
LOAD 315 DL + SDL2_R + 0.75 LL + 0.75 WL(+Z)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 7 0.75
LOAD 316 DL + SDL2_R + 0.75 LL + 0.75 WL(-Z)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 8 0.75
LOAD 317 DL - SDL1_R + 0.75 LL + 0.75 WL(+X)
REPEAT LOAD
1 1.0 3 -1.0 2 0.75 5 0.75
LOAD 318 DL - SDL1_R + 0.75 LL + 0.75 WL(-X)
REPEAT LOAD
1 1.0 3 -1.0 2 0.75 6 0.75
LOAD 319 DL - SDL1_R + 0.75 LL + 0.75 WL(+Z)
REPEAT LOAD
1 1.0 3 -1.0 2 0.75 7 0.75
LOAD 320 DL - SDL1_R + 0.75 LL + 0.75 WL(-Z)
REPEAT LOAD
1 1.0 3 -1.0 2 0.75 8 0.75
LOAD 321 DL - SDL2_R + 0.75 LL + 0.75 WL(+X)
REPEAT LOAD
1 1.0 4 -1.0 2 0.75 5 0.75
LOAD 322 DL - SDL2_R + 0.75 LL + 0.75 WL(-X)
REPEAT LOAD
1 1.0 4 -1.0 2 0.75 6 0.75
LOAD 323 DL - SDL2_R + 0.75 LL + 0.75 WL(+Z)
REPEAT LOAD
1 1.0 4 -1.0 2 0.75 7 0.75
LOAD 324 DL - SDL2_R + 0.75 LL + 0.75 WL(-Z)
REPEAT LOAD
1 1.0 4 -1.0 2 0.75 8 0.75
*********************************************
LOAD 325 DL + SDL1_R + 0.75 LL + 0.525 EL(+X)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 11 0.525
LOAD 326 DL + SDL1_R + 0.75 LL + 0.525 EL(-X)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 11 -0.525
LOAD 327 DL + SDL1_R + 0.75 LL + 0.525 EL(+Z)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 12 0.525
LOAD 328 DL + SDL1_R + 0.75 LL + 0.525 EL(-Z)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 12 -0.525
LOAD 329 DL + SDL2_R + 0.75 LL + 0.525 EL(+X)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 11 0.525
LOAD 330 DL + SDL2_R + 0.75 LL + 0.525 EL(-X)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 11 -0.525
LOAD 331 DL + SDL2_R + 0.75 LL + 0.525 EL(+Z)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 12 0.525
```

```
LOAD 332 DL + SDL2_R + 0.75 LL + 0.525 EL(-Z)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 12 -0.525
LOAD 333 DL - SDL1_R + 0.75 LL + 0.525 EL(+X)
REPEAT LOAD
1 1.0 3 -1.0 2 0.75 11 0.525
LOAD 334 DL - SDL1_R + 0.75 LL + 0.525 EL(-X)
REPEAT LOAD
1 1.0 3 -1.0 2 0.75 11 -0.525
LOAD 335 DL - SDL1_R + 0.75 LL + 0.525 EL(+Z)
REPEAT LOAD
1 1.0 3 -1.0 2 0.75 12 0.525
LOAD 336 DL - SDL1_R + 0.75 LL + 0.525 EL(-Z)
REPEAT LOAD
1 1.0 3 -1.0 2 0.75 12 -0.525
LOAD 337 DL - SDL2_R + 0.75 LL + 0.525 EL(+X)
REPEAT LOAD
1 1.0 4 -1.0 2 0.75 11 0.525
LOAD 338 DL - SDL2_R + 0.75 LL + 0.525 EL(-X)
REPEAT LOAD
1 1.0 4 -1.0 2 0.75 11 -0.525
LOAD 339 DL - SDL2_R + 0.75 LL + 0.525 EL(+Z)
REPEAT LOAD
1 1.0 4 -1.0 2 0.75 12 0.525
LOAD 340 DL - SDL2_R + 0.75 LL + 0.525 EL(-Z)
REPEAT LOAD
1 1.0 4 -1.0 2 0.75 12 -0.525
************************************************
LOAD 341 0.6 (DL + SDL1_R) + WL(+X)
REPEAT LOAD
1 0.6 3 0.6 5 1.0
LOAD 342 0.6 (DL + SDL1_R) + WL(-X)
REPEAT LOAD
1 0.6 3 0.6 6 1.0
LOAD 343 0.6 (DL + SDL1_R) + WL(+Z)
REPEAT LOAD
1 0.6 3 0.6 7 1.0
LOAD 344 0.6 (DL + SDL1_R) + WL(-Z)
REPEAT LOAD
1 0.6 3 0.6 8 1.0
LOAD 345 0.6 (DL + SDL2_R) + WL(+X)
REPEAT LOAD
1 0.6 4 0.6 5 1.0
LOAD 346 0.6 (DL + SDL2_R) + WL(-X)
REPEAT LOAD
1 0.6 4 0.6 6 1.0
LOAD 347 0.6 (DL + SDL2_R) + WL(+Z)
REPEAT LOAD
1 0.6 4 0.6 7 1.0
LOAD 348 0.6 (DL + SDL2_R) + WL(-Z)
REPEAT LOAD
1 0.6 4 0.6 8 1.0
LOAD 349 0.6 (DL - SDL1_R) + WL(+X)
REPEAT LOAD
1 0.6 3 -0.6 5 1.0
LOAD 350 0.6 (DL - SDL1_R) + WL(-X)
REPEAT LOAD
1 0.6 3 -0.6 6 1.0
```

```
LOAD 351 0.6 (DL - SDL1_R) + WL(+Z)
REPEAT LOAD
1 0.6 3 -0.6 7 1.0
LOAD 352 0.6 (DL - SDL1_R) + WL(-Z)
REPEAT LOAD
1 0.6 3 -0.6 8 1.0
LOAD 353 0.6 (DL - SDL2_R) + WL(+X)
REPEAT LOAD
1 0.6 4 -0.6 5 1.0
LOAD 354 0.6 (DL - SDL2_R) + WL(-X)
REPEAT LOAD
1 0.6 4 -0.6 6 1.0
LOAD 355 0.6 (DL - SDL2_R) + WL(+Z)
REPEAT LOAD
1 0.6 4 -0.6 7 1.0
LOAD 356 0.6 (DL - SDL2_R) + WL(-Z)
REPEAT LOAD
1 0.6 4 -0.6 8 1.0
**********************************************
LOAD 357 0.6 (DL + SDL1_R) + 0.7 EL(+X)
REPEAT LOAD
1 0.6 3 0.6 11 0.7
LOAD 358 0.6 (DL + SDL1_R) + 0.7 EL(-X)
REPEAT LOAD
1 0.6 3 0.6 11 -0.7
LOAD 359 0.6 (DL + SDL1_R) + 0.7 EL(+Z)
REPEAT LOAD
1 0.6 3 0.6 12 0.7
LOAD 360 0.6 (DL + SDL1_R) + 0.7 EL(-Z)
REPEAT LOAD
1 0.6 3 0.6 12 -0.7
LOAD 361 0.6 (DL + SDL2_R) + 0.7 EL(+X)
REPEAT LOAD
1 0.6 4 0.6 11 0.7
LOAD 362 0.6 (DL + SDL2_R) + 0.7 EL(-X)
REPEAT LOAD
1 0.6 4 0.6 11 -0.7
LOAD 363 0.6 (DL + SDL2_R) + 0.7 EL(+Z)
REPEAT LOAD
1 0.6 4 0.6 12 -0.7
LOAD 364 0.6 (DL + SDL2_R) + 0.7 EL(-Z)
REPEAT LOAD
1 0.6 4 0.6 12 -0.7
LOAD 365 0.6 (DL + SDL1_R) + 0.7 EL(+X)
REPEAT LOAD
1 0.6 3 -0.6 11 0.7
LOAD 366 0.6 (DL + SDL1_R) + 0.7 EL(-X)
REPEAT LOAD
1 0.6 3 -0.6 11 -0.7
LOAD 367 0.6 (DL + SDL1_R) + 0.7 EL(+Z)
REPEAT LOAD
1 0.6 3 -0.6 12 0.7
LOAD 368 0.6 (DL + SDL1_R) + 0.7 EL(-Z)
REPEAT LOAD
1 0.6 3 -0.6 12 -0.7
LOAD 369 0.6 (DL + SDL2_R) + 0.7 EL(+X)
REPEAT LOAD
1 0.6 4 -0.6 11 0.7
```

```
LOAD 370 0.6 (DL + SDL2_R) + 0.7 EL(-X)
REPEAT LOAD
1 0.6 4 -0.6 11 -0.7
LOAD 371 0.6 (DL + SDL2_R) + 0.7 EL(+Z)
REPEAT LOAD
1 0.6 4 -0.6 12 -0.7
LOAD 372 0.6 (DL + SDL2_R) + 0.7 EL(-Z)
REPEAT LOAD
1 0.6 4 -0.6 12 -0.7
************************************************
*Load Combination for Concrete design
************************************************
LOAD 401 1.4 (DL+SDL1_R)
REPEAT LOAD
1 1.4 3 1.4
LOAD 402 1.4 (DL+SDL2_R)
REPEAT LOAD
1 1.4 4 1.4
LOAD 403 1.4 (DL-SDL1_R)
REPEAT LOAD
1 1.4 3 -1.4
LOAD 404 1.4 (DL-SDL2_R)
REPEAT LOAD
1 1.4 4 -1.4
************************************************
LOAD 405 1.4 (DL + SDL1_R ) + 1.7 LL
REPEAT LOAD
1 1.4 3 1.4 2 1.7
LOAD 406 1.4 (DL + SDL2_R ) + 1.7 LL
REPEAT LOAD
1 1.4 4 1.4 2 1.7
LOAD 407 1.4 (DL - SDL1_R ) + 1.7 LL
REPEAT LOAD
1 1.4 3 -1.4 2 1.7
LOAD 408 1.4 (DL - SDL2_R ) + 1.7 LL
REPEAT LOAD
1 1.4 4 -1.4 2 1.7
************************************************
LOAD 409 1.2 (DL + SDL1_R) + 0.5 LL + 1.6 WL(+X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 5 1.6
LOAD 410 1.2 (DL + SDL1_R) + 0.5 LL + 1.6 WL(-X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 6 1.6
LOAD 411 1.2 (DL + SDL1_R) + 0.5 LL + 1.6 WL(+Z)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 7 1.6
LOAD 412 1.2 (DL + SDL1_R) + 0.5 LL + 1.6 WL(-Z)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 8 1.6
LOAD 413 1.2 (DL + SDL2_R) + 0.5 LL + 1.6 WL(+X)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 5 1.6
LOAD 414 1.2 (DL + SDL2_R) + 0.5 LL + 1.6 WL(-X)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 6 1.6
LOAD 415 1.2 (DL + SDL2_R) + 0.5 LL + 1.6 WL(+Z)
REPEAT LOAD
```

```
1 1.2 4 1.2 2 0.5 7 1.6
LOAD 416 1.2 (DL + SDL2_R) + 0.5 LL + 1.6 WL(-Z)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 8 1.6
LOAD 417 1.2 (DL - SDL1_R) + 0.5 LL + 1.6 WL(+X)
REPEAT LOAD
1 1.2 3 -1.2 2 0.5 5 1.6
LOAD 418 1.2 (DL - SDL1_R) + 0.5 LL + 1.6 WL(-X)
REPEAT LOAD
1 1.2 3 -1.2 2 0.5 6 1.6
LOAD 419 1.2 (DL - SDL1_R) + 0.5 LL + 1.6 WL(+Z)
REPEAT LOAD
1 1.2 3 -1.2 2 0.5 7 1.6
LOAD 420 1.2 (DL - SDL1_R) + 0.5 LL + 1.6 WL(-Z)
REPEAT LOAD
1 1.2 3 -1.2 2 0.5 8 1.6
LOAD 421 1.2 (DL - SDL2_R) + 0.5 LL + 1.6 WL(+X)
REPEAT LOAD
1 1.2 4 -1.2 2 0.5 5 1.6
LOAD 422 1.2 (DL - SDL2_R) + 0.5 LL + 1.6 WL(-X)
REPEAT LOAD
1 1.2 4 -1.2 2 0.5 6 1.6
LOAD 423 1.2 (DL - SDL2_R) + 0.5 LL + 1.6 WL(+Z)
REPEAT LOAD
1 1.2 4 -1.2 2 0.5 7 1.6
LOAD 424 1.2 (DL - SDL2_R) + 0.5 LL + 1.6 WL(-Z)
REPEAT LOAD
1 1.2 4 -1.2 2 0.5 8 1.6
************************************************
LOAD 425 1.2 (DL + SDL1_R) + 0.5 LL + 1.0 EL(+X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 11 1.0
LOAD 426 1.2 (DL + SDL1_R) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 11 -1.0
LOAD 427 1.2 (DL + SDL1_R) + 0.5 LL + 1.0 EL(+Z)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 12 1.0
LOAD 428 1.2 (DL + SDL1_R) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 12 -1.0
LOAD 429 1.2 (DL + SDL2_R) + 0.5 LL + 1.0 EL(+X)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 11 1.0
LOAD 430 1.2 (DL + SDL2_R) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 11 -1.0
LOAD 431 1.2 (DL + SDL2_R) + 0.5 LL + 1.0 EL(+Z)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 12 1.0
LOAD 432 1.2 (DL + SDL2_R) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 12 -1.0
LOAD 433 1.2 (DL - SDL1_R) + 0.5 LL + 1.0 EL(+X)
REPEAT LOAD
1 1.2 3 -1.2 2 0.5 11 1.0
LOAD 434 1.2 (DL - SDL1_R) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
```

```
1 1.2 3 -1.2 2 0.5 11 -1.0
LOAD 435 1.2 (DL - SDL1_R) + 0.5 LL + 1.0 EL(+Z)
REPEAT LOAD
1 1.2 3 -1.2 2 0.5 12 1.0
LOAD 436 1.2 (DL - SDL1_R) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 3 -1.2 2 0.5 12 -1.0
LOAD 437 1.2 (DL - SDL2_R) + 0.5 LL + 1.0 EL(+X)
REPEAT LOAD
1 1.2 4 -1.2 2 0.5 11 1.0
LOAD 438 1.2 (DL - SDL2_R) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 4 -1.2 2 0.5 11 -1.0
LOAD 439 1.2 (DL - SDL2_R) + 0.5 LL + 1.0 EL(+Z)
REPEAT LOAD
1 1.2 4 -1.2 2 0.5 12 1.0
LOAD 440 1.2 (DL - SDL2_R) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 4 -1.2 2 0.5 12 -1.0
*********************************************
LOAD 441 0.9 (DL + SDL1_R) + 1.6 WL(+X)
REPEAT LOAD
1 0.9 3 0.9 5 1.6
LOAD 442 0.9 (DL + SDL1_R) + 1.6 WL(-X)
REPEAT LOAD
1 0.9 3 0.9 6 1.6
LOAD 443 0.9 (DL + SDL1_R) + 1.6 WL(+Z)
REPEAT LOAD
1 0.9 3 0.9 7 1.6
LOAD 444 0.9 (DL + SDL1_R) + 1.6 WL(-Z)
REPEAT LOAD
1 0.9 3 0.9 8 1.6
LOAD 445 0.9 (DL + SDL2_R) + 1.6 WL(+X)
REPEAT LOAD
1 0.9 4 0.9 5 1.6
LOAD 446 0.9 (DL + SDL2_R) + 1.6 WL(-X)
REPEAT LOAD
1 0.9 4 0.9 6 1.6
LOAD 447 0.9 (DL + SDL2_R) + 1.6 WL(+Z)
REPEAT LOAD
1 0.9 4 0.9 7 1.6
LOAD 448 0.9 (DL + SDL2_R) + 1.6 WL(-Z)
REPEAT LOAD
1 0.9 4 0.9 8 1.6
LOAD 449 0.9 (DL - SDL1_R) + 1.6 WL(+X)
REPEAT LOAD
1 0.9 3 -0.9 5 1.6
LOAD 450 0.9 (DL - SDL1_R) + 1.6 WL(-X)
REPEAT LOAD
1 0.9 3 -0.9 6 1.6
LOAD 451 0.9 (DL - SDL1_R) + 1.6 WL(+Z)
REPEAT LOAD
1 0.9 3 -0.9 7 1.6
LOAD 452 0.9 (DL - SDL1_R) + 1.6 WL(-Z)
REPEAT LOAD
1 0.9 3 -0.9 8 1.6
LOAD 453 0.9 (DL - SDL2_R) + 1.6 WL(+X)
REPEAT LOAD
```

```
1 0.9 4 -0.9 5 1.6
LOAD 454 0.9 (DL - SDL2_R) + 1.6 WL(-X)
REPEAT LOAD
1 0.9 4 -0.9 6 1.6
LOAD 455 0.9 (DL - SDL2_R) + 1.6 WL(+Z)
REPEAT LOAD
1 0.9 4 -0.9 7 1.6
LOAD 456 0.9 (DL - SDL2_R) + 1.6 WL(-Z)
REPEAT LOAD
1 0.9 4 -0.9 8 1.6
************************************************
LOAD 457 0.9 (DL + SDL1_R) + 1.0 EL(+X)
REPEAT LOAD
1 0.9 3 0.9 11 1.0
LOAD 458 0.9 (DL + SDL1_R) + 1.0 EL(-X)
REPEAT LOAD
1 0.9 3 0.9 11 -1.0
LOAD 459 0.9 (DL + SDL1_R) + 1.0 EL(+Z)
REPEAT LOAD
1 0.9 3 0.9 12 1.0
LOAD 460 0.9 (DL + SDL1_R) + 1.0 EL(-Z)
REPEAT LOAD
1 0.9 3 0.9 12 -1.0
LOAD 461 0.9 (DL + SDL2_R) + 1.0 EL(+X)
REPEAT LOAD
1 0.9 4 0.9 11 1.0
LOAD 462 0.9 (DL + SDL2_R) + 1.0 EL(-X)
REPEAT LOAD
1 0.9 4 0.9 11 -1.0
LOAD 463 0.9 (DL + SDL2_R) + 1.0 EL(+Z)
REPEAT LOAD
1 0.9 4 0.9 12 1.0
LOAD 464 0.9 (DL + SDL2_R) + 1.0 EL(-Z)
REPEAT LOAD
1 0.9 4 0.9 12 -1.0
LOAD 465 0.9 (DL - SDL1_R) + 1.0 EL(+X)
REPEAT LOAD
1 0.9 3 -0.9 11 1.0
LOAD 466 0.9 (DL - SDL1_R) + 1.0 EL(-X)
REPEAT LOAD
1 0.9 3 -0.9 11 -1.0
LOAD 467 0.9 (DL - SDL1_R) + 1.0 EL(+Z)
REPEAT LOAD
1 0.9 3 -0.9 12 1.0
LOAD 468 0.9 (DL - SDL1_R) + 1.0 EL(-Z)
REPEAT LOAD
1 0.9 3 -0.9 12 -1.0
LOAD 469 0.9 (DL - SDL2_R) + 1.0 EL(+X)
REPEAT LOAD
1 0.9 4 -0.9 11 1.0
LOAD 470 0.9 (DL - SDL2_R) + 1.0 EL(-X)
REPEAT LOAD
1 0.9 4 -0.9 11 -1.0
LOAD 471 0.9 (DL - SDL2_R) + 1.0 EL(+Z)
REPEAT LOAD
1 0.9 4 -0.9 12 1.0
LOAD 472 0.9 (DL - SDL2_R) + 1.0 EL(-Z)
REPEAT LOAD
```

```
1 0.9 4 -0.9 12 -1.0
PERFORM ANALYSIS
FINISH
```

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	C02	D01			Outotec / L&T Revision No.:	0	1	2	3	
Date	27-01-2020	12-03-2020	25-04-2022	13/10/2023			Date	27-01-2020	12-03-2020	25-04-2022	13/10/2023	
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS DT. 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER SUPPORT STRUCTURE	AS BUILT			Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS DT. 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER SUPPORT STRUCTURE	AS BUILT	
Prepared By	APR	APR	APR	APR			Prepared By	APR	APR	APR	APR	
Checked	SBY	SBY	SBY	SBY			Checked	SBY	SBY	SBY	SBY	
Approved	AG	AG	AG	AG			Approved	AG	AG	AG	AG	

ATTACHMENT 9 COMMENT RESOLUTION LOG SHEET

```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 17-Jan-20
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
2 0 -2.3 0; 3 5.3 -2.3 0; 101 0 -2.5 0; 102 5.3 -2.5 0; 107 0 -2.5 -1.575;
108 0 -2.5 1.575; 109 5.3 -2.5 -1.575; 110 5.3 -2.5 1.575; 113 -1 -2.5 1.575;
114 -1 -2.5 1.15; 116 -0.5 -2.5 1.575; 117 -0.5 -2.5 1.15; 118 0 -2.5 1.15;
119 -1 -2.5 0.5; 121 -0.5 -2.5 0.5; 122 0 -2.5 0.5; 123 -1 -2.5 0;
124 -0.5 -2.5 0; 125 -1 -2.5 -0.5; 127 -0.5 -2.5 -0.5; 128 0 -2.5 -0.5;
129 -1 -2.5 -1.15; 131 -0.5 -2.5 -1.15; 132 0 -2.5 -1.15; 133 -1 -2.5 -1.575;
134 -0.5 -2.5 -1.575; 135 5.8 -2.5 1.575; 136 5.8 -2.5 1.15; 137 5.3 -2.5 1.15;
138 6.3 -2.5 1.575; 139 6.3 -2.5 1.15; 141 5.8 -2.5 0.5; 142 5.3 -2.5 0.5;
143 6.3 -2.5 0.5; 145 5.8 -2.5 0; 146 6.3 -2.5 0; 147 5.8 -2.5 -0.5;
148 5.3 -2.5 -0.5; 149 6.3 -2.5 -0.5; 151 5.8 -2.5 -1.15; 152 5.3 -2.5 -1.15;
153 6.3 -2.5 -1.15; 155 5.8 -2.5 -1.575; 156 6.3 -2.5 -1.575;
157 0.5 -2.5 1.575; 158 0.5 -2.5 1.15; 159 1.2 -2.5 1.575; 160 1.2 -2.5 1.15;
161 1.65 -2.5 1.575; 162 1.65 -2.5 1.15; 163 4.8 -2.5 1.575; 164 4.8 -2.5 1.15;
175 0.5 -2.5 0.5; 176 1.2 -2.5 0.5; 177 1.65 -2.5 0.5; 178 4.8 -2.5 0.5;
184 0.5 -2.5 0; 185 1.2 -2.5 0; 186 1.65 -2.5 0; 187 4.8 -2.5 0;
193 0.5 -2.5 -0.5; 194 1.2 -2.5 -0.5; 195 1.65 -2.5 -0.5; 196 4.8 -2.5 -0.5;
202 0.5 -2.5 -1.15; 203 1.2 -2.5 -1.15; 204 1.65 -2.5 -1.15;
205 4.8 -2.5 -1.15; 211 0.5 -2.5 -1.575; 212 1.2 -2.5 -1.575;
213 1.65 -2.5 -1.575; 214 4.8 -2.5 -1.575; 215 3.65 -2.5 1.575;
216 3.65 -2.5 1.15; 217 3.65 -2.5 0.5; 218 3.65 -2.5 0; 219 3.65 -2.5 -0.5;
220 3.65 -2.5 -1.15; 221 3.65 -2.5 -1.575; 222 4.1 -2.5 1.575;
223 4.1 -2.5 1.15; 224 4.1 -2.5 0.5; 225 4.1 -2.5 0; 226 4.1 -2.5 -0.5;
227 4.1 -2.5 -1.15; 228 4.1 -2.5 -1.575; 231 0 -2.5 -2; 232 5.3 -2.5 -2;
233 -1 -2.5 -2; 234 -0.5 -2.5 -2; 235 5.8 -2.5 -2; 236 6.3 -2.5 -2;
237 0.5 -2.5 -2; 238 1.2 -2.5 -2; 239 1.65 -2.5 -2; 240 4.8 -2.5 -2;
241 3.65 -2.5 -2; 242 4.1 -2.5 -2; 245 0 -2.5 2; 246 5.3 -2.5 2; 247 -1 -2.5 2;
248 -0.5 -2.5 2; 249 5.8 -2.5 2; 250 6.3 -2.5 2; 251 0.5 -2.5 2;
252 1.2 -2.5 2; 253 1.65 -2.5 2; 254 4.8 -2.5 2; 255 3.65 -2.5 2;
256 4.1 -2.5 2; 257 2.15 -2.5 1.575; 258 2.15 -2.5 1.15; 259 2.15 -2.5 0.5;
260 2.15 -2.5 0; 261 2.15 -2.5 -0.5; 262 2.15 -2.5 -1.15; 263 2.15 -2.5 -1.575;
264 2.15 -2.5 -2; 265 2.15 -2.5 2; 266 2.65 -2.5 1.575; 267 2.65 -2.5 1.15;
268 2.65 -2.5 0.5; 269 2.65 -2.5 0; 270 2.65 -2.5 -0.5; 271 2.65 -2.5 -1.15;
272 2.65 -2.5 -1.575; 273 2.65 -2.5 -2; 274 2.65 -2.5 2; 275 3.15 -2.5 1.575;
276 3.15 -2.5 1.15; 277 3.15 -2.5 0.5; 278 3.15 -2.5 0; 279 3.15 -2.5 -0.5;
280 3.15 -2.5 -1.15; 281 3.15 -2.5 -1.575; 282 3.15 -2.5 -2; 283 3.15 -2.5 2;
MEMBER INCIDENCES
1 101 2; 2 102 3;
ELEMENT INCIDENCES SHELL
13 113 116 117 114; 14 116 108 118 117; 16 114 117 121 119; 17 117 118 122 121;
19 119 121 124 123; 20 121 122 101 124; 22 123 124 127 125; 23 124 101 128 127;
25 125 127 131 129; 26 127 128 132 131; 28 129 131 134 133; 29 131 132 107 134;
30 110 135 136 137; 31 135 138 139 136; 33 137 136 141 142; 34 136 139 143 141;
36 142 141 145 102; 37 141 143 146 145; 39 102 145 147 148; 40 145 146 149 147;
42 148 147 151 152; 43 147 149 153 151; 45 152 151 155 109; 46 151 153 156 155;
48 108 157 158 118; 49 157 159 160 158; 50 159 161 162 160; 52 163 110 137 164;
58 118 158 175 122; 59 158 160 176 175; 60 160 162 177 176; 62 164 137 142 178;
68 122 175 184 101; 69 175 176 185 184; 70 176 177 186 185; 72 178 142 102 187;
78 101 184 193 128; 79 184 185 194 193; 80 185 186 195 194; 82 187 102 148 196;
88 128 193 202 132; 89 193 194 203 202; 90 194 195 204 203; 92 196 148 152 205;
98 132 202 211 107; 99 202 203 212 211; 100 203 204 213 212;
102 205 152 109 214; 109 215 222 223 216; 110 216 223 224 217;
```

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111 217 224 225 218; 112 218 225 226 219; 113 219 226 227 220;
114 220 227 228 221; 115 222 163 164 223; 116 223 164 178 224;
117 224 178 187 225; 118 225 187 196 226; 119 226 196 205 227;
120 227 205 214 228; 122 133 134 234 233; 123 134 107 231 234;
124 109 155 235 232; 125 155 156 236 235; 127 107 211 237 231;
128 211 212 238 237; 129 212 213 239 238; 130 214 109 232 240;
132 221 228 242 241; 133 228 214 240 242; 135 247 248 116 113;
136 248 245 108 116; 137 246 249 135 110; 138 249 250 138 135;
140 245 251 157 108; 141 251 252 159 157; 142 252 253 161 159;
143 254 246 110 163; 145 255 256 222 215; 146 256 254 163 222;
147 161 257 258 162; 148 162 258 259 177; 149 177 259 260 186;
150 186 260 261 195; 151 195 261 262 204; 152 204 262 263 213;
153 213 263 264 239; 154 253 265 257 161; 155 257 266 267 258;
156 258 267 268 259; 157 259 268 269 260; 158 260 269 270 261;
159 261 270 271 262; 160 262 271 272 263; 161 263 272 273 264;
162 265 274 266 257; 163 266 275 276 267; 164 267 276 277 268;
165 268 277 278 269; 166 269 278 279 270; 167 270 279 280 271;
168 271 280 281 272; 169 272 281 282 273; 170 274 283 275 266;
171 275 215 216 276; 172 276 216 217 277; 173 277 217 218 278;
174 278 218 219 279; 175 279 219 220 280; 176 280 220 221 281;
177 281 221 241 282; 178 283 255 215 275;
START GROUP DEFINITION
ELEMENT
_RAFT_MOMENT 13 14 16 17 19 22 25 26 28 TO 31 33 34 37 40 42 43 45 46 -
48 TO 50 52 58 TO 60 62 69 70 79 80 88 TO 90 92 98 TO 100 102 109 TO 120 -
122 TO 125 127 TO 130 132 133 135 TO 138 140 TO 143 145 TO 178
_RAFT_SHEAR 13 14 28 TO 31 45 46 48 TO 50 52 60 70 80 90 98 TO 100 102 109 -
110 TO 115 120 122 TO 125 127 TO 130 132 133 135 TO 138 140 TO 143 -
145 TO 178
END GROUP DEFINITION
ELEMENT PROPERTY
13 14 16 17 19 20 22 23 25 26 28 TO 31 33 34 36 37 39 40 42 43 45 46 -
48 TO 50 52 58 TO 60 62 68 TO 70 72 78 TO 80 82 88 TO 90 92 98 TO 100 102 -
109 TO 120 122 TO 125 127 TO 130 132 133 135 TO 138 140 TO 143 145 TO 177 -
178 THICKNESS 2.35
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+07
POISSON 0.17
DENSITY 25
ALPHA 1e-05
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
1 2 PRIS YD 0.8 ZD 0.8
CONSTANTS
MATERIAL CONCRETE MEMB 1 2 13 14 16 17 19 20 22 23 25 26 28 TO 31 33 34 36 -
37 39 40 42 43 45 46 48 TO 50 52 58 TO 60 62 68 TO 70 72 78 TO 80 82 -
88 TO 90 92 98 TO 100 102 109 TO 120 122 TO 125 127 TO 130 132 133 -
135 TO 138 140 TO 143 145 TO 178
SUPPORTS
101 102 107 TO 110 113 114 116 TO 119 121 TO 125 127 TO 129 131 TO 139 141 -
142 TO 143 145 TO 149 151 TO 153 155 TO 164 175 TO 178 184 TO 187 193 TO 196 -
202 TO 205 211 TO 228 231 TO 242 245 TO 282 -
283 ELASTIC MAT DIRECT Y SUBGRADE 20000 PRINT
LOAD 1 DEAD LOAD
```

```
SELFWEIGHT Y -1
JOINT LOAD
2   FX  -5.2    FY  -399.9  FZ  17.1    MX  -75.9   MZ  -15.7
3   FX  9.8 FY  -408.6  FZ  7.4 MX  -81.4   MZ  1.7
Load 2  LL
JOINT LOAD
2   FX  -0.7    FY  -35.0   FZ  -2.9    MX  -12.6   MZ  3.7
3   FX  2.0 FY  -46.0   FZ  4.5 MX  -10.1   MZ  -9.0
Load 3  SDL1_R
JOINT LOAD
2   FX  5.3 FY  -186.2  FZ  214.8   MX  158.7   MZ  -14.0
3   FX  4.8 FY  -86.8   FZ  55.5    MX  50.8    MZ  -13.2
Load 4  SDL2_R
JOINT LOAD
2   FX  9.6 FY  71.3    FZ  -35.5   MX  -27.5   MZ  -23.7
3   FX  8.9 FY  -75.0   FZ  46.6    MX  34.3    MZ  -22.5
Load 5  WIND LOAD (+X)
JOINT LOAD
2   FX  22.9    FY  18  FZ  -1.7    MX  0.8 MZ  -51.9
3   FX  22.3    FY  -4.7    FZ  -0.8    MX  -7.3    MZ  -50.9
Load 6  WIND LOAD (-X)
JOINT LOAD
2   FX  -24.1   FY  -20.2   FZ  1.7 MX  -1.2    MZ  55.4
3   FX  -23.5   FY  4.1 FZ  1.5 MX  9.3 MZ  54.5
Load 7  WIND LOAD (+Z)
JOINT LOAD
2   FX  2.2 FY  -101.2  FZ  136.5   MX  101.3   MZ  -5.7
3   FX  1.9 FY  -53.7   FZ  65  MX  53.0    MZ  -5.3
Load 8  WIND LOAD (+-Z)
JOINT LOAD
2   FX  -1.3    FY  49.2    FZ  -70.6   MX  -52.0   MZ  3.5
3   FX  -1.2    FY  22.6    FZ  -23.3   MX  -19.3   MZ  3.3
Load 11 EL (+X)
JOINT LOAD
2   FX  10.6    FY  19.4    FZ  -7.7    MX  -4.2    MZ  -28.3
3   FX  10.6    FY  -16.8   FZ  8   MX  3.1 MZ  -28.4
Load 12 EL (+Z)
JOINT LOAD
2   FX  1.2 FY  -74.1   FZ  86.6    MX  64.9    MZ  -3.0
3   FX  1.3 FY  -44 FZ  34  MX  30.9    MZ  -3.3

************************************************
*LOAD COMBINATION FOR BASE PRESSURE & Servicibility
************************************************
LOAD 301 DL+SDL1_R
REPEAT LOAD
1 1.0 3 1.0
LOAD 302 DL+SDL2_R
REPEAT LOAD
1 1.0 4 1.0
LOAD 303 DL-SDL1_R
REPEAT LOAD
1 1.0 3 -1.0
LOAD 304 DL-SDL2_R
REPEAT LOAD
1 1.0 4 -1.0
**********************************
LOAD 305 DL + SDL1_R + LL
```

```
REPEAT LOAD
1 1.0 3 1.0 2 1.0
LOAD 306 DL + SDL2_R + LL
REPEAT LOAD
1 1.0 4 1.0 2 1.0
LOAD 307 DL - SDL1_R + LL
REPEAT LOAD
1 1.0 3 -1.0 2 1.0
LOAD 308 DL - SDL2_R + LL
REPEAT LOAD
1 1.0 4 -1.0 2 1.0
*********************************************
LOAD 309 DL + SDL1_R + 0.75 LL + 0.75 WL(+X)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 5 0.75
LOAD 310 DL + SDL1_R + 0.75 LL + 0.75 WL(-X)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 6 0.75
LOAD 311 DL + SDL1_R + 0.75 LL + 0.75 WL(+Z)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 7 0.75
LOAD 312 DL + SDL1_R + 0.75 LL + 0.75 WL(-Z)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 8 0.75
LOAD 313 DL + SDL2_R + 0.75 LL + 0.75 WL(+X)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 5 0.75
LOAD 314 DL + SDL2_R + 0.75 LL + 0.75 WL(-X)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 6 0.75
LOAD 315 DL + SDL2_R + 0.75 LL + 0.75 WL(+Z)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 7 0.75
LOAD 316 DL + SDL2_R + 0.75 LL + 0.75 WL(-Z)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 8 0.75
LOAD 317 DL - SDL1_R + 0.75 LL + 0.75 WL(+X)
REPEAT LOAD
1 1.0 3 -1.0 2 0.75 5 0.75
LOAD 318 DL - SDL1_R + 0.75 LL + 0.75 WL(-X)
REPEAT LOAD
1 1.0 3 -1.0 2 0.75 6 0.75
LOAD 319 DL - SDL1_R + 0.75 LL + 0.75 WL(+Z)
REPEAT LOAD
1 1.0 3 -1.0 2 0.75 7 0.75
LOAD 320 DL - SDL1_R + 0.75 LL + 0.75 WL(-Z)
REPEAT LOAD
1 1.0 3 -1.0 2 0.75 8 0.75
LOAD 321 DL - SDL2_R + 0.75 LL + 0.75 WL(+X)
REPEAT LOAD
1 1.0 4 -1.0 2 0.75 5 0.75
LOAD 322 DL - SDL2_R + 0.75 LL + 0.75 WL(-X)
REPEAT LOAD
1 1.0 4 -1.0 2 0.75 6 0.75
LOAD 323 DL - SDL2_R + 0.75 LL + 0.75 WL(+Z)
REPEAT LOAD
1 1.0 4 -1.0 2 0.75 7 0.75
LOAD 324 DL - SDL2_R + 0.75 LL + 0.75 WL(-Z)
```

```
REPEAT LOAD
1 1.0 4 -1.0 2 0.75 8 0.75
*********************************************
LOAD 325 DL + SDL1_R + 0.75 LL + 0.525 EL(+X)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 11 0.525
LOAD 326 DL + SDL1_R + 0.75 LL + 0.525 EL(-X)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 11 -0.525
LOAD 327 DL + SDL1_R + 0.75 LL + 0.525 EL(+Z)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 12 0.525
LOAD 328 DL + SDL1_R + 0.75 LL + 0.525 EL(-Z)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 12 -0.525
LOAD 329 DL + SDL2_R + 0.75 LL + 0.525 EL(+X)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 11 0.525
LOAD 330 DL + SDL2_R + 0.75 LL + 0.525 EL(-X)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 11 -0.525
LOAD 331 DL + SDL2_R + 0.75 LL + 0.525 EL(+Z)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 12 0.525
LOAD 332 DL + SDL2_R + 0.75 LL + 0.525 EL(-Z)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 12 -0.525
LOAD 333 DL - SDL1_R + 0.75 LL + 0.525 EL(+X)
REPEAT LOAD
1 1.0 3 -1.0 2 0.75 11 0.525
LOAD 334 DL - SDL1_R + 0.75 LL + 0.525 EL(-X)
REPEAT LOAD
1 1.0 3 -1.0 2 0.75 11 -0.525
LOAD 335 DL - SDL1_R + 0.75 LL + 0.525 EL(+Z)
REPEAT LOAD
1 1.0 3 -1.0 2 0.75 12 0.525
LOAD 336 DL - SDL1_R + 0.75 LL + 0.525 EL(-Z)
REPEAT LOAD
1 1.0 3 -1.0 2 0.75 12 -0.525
LOAD 337 DL - SDL2_R + 0.75 LL + 0.525 EL(+X)
REPEAT LOAD
1 1.0 4 -1.0 2 0.75 11 0.525
LOAD 338 DL - SDL2_R + 0.75 LL + 0.525 EL(-X)
REPEAT LOAD
1 1.0 4 -1.0 2 0.75 11 -0.525
LOAD 339 DL - SDL2_R + 0.75 LL + 0.525 EL(+Z)
REPEAT LOAD
1 1.0 4 -1.0 2 0.75 12 0.525
LOAD 340 DL - SDL2_R + 0.75 LL + 0.525 EL(-Z)
REPEAT LOAD
1 1.0 4 -1.0 2 0.75 12 -0.525
*********************************************
LOAD 341 0.6 (DL + SDL1_R) + WL(+X)
REPEAT LOAD
1 0.6 3 0.6 5 1.0
LOAD 342 0.6 (DL + SDL1_R) + WL(-X)
REPEAT LOAD
1 0.6 3 0.6 6 1.0
```

```
LOAD 343 0.6 (DL + SDL1_R) + WL(+Z)
REPEAT LOAD
1 0.6 3 0.6 7 1.0
LOAD 344 0.6 (DL + SDL1_R) + WL(-Z)
REPEAT LOAD
1 0.6 3 0.6 8 1.0
LOAD 345 0.6 (DL + SDL2_R) + WL(+X)
REPEAT LOAD
1 0.6 4 0.6 5 1.0
LOAD 346 0.6 (DL + SDL2_R) + WL(-X)
REPEAT LOAD
1 0.6 4 0.6 6 1.0
LOAD 347 0.6 (DL + SDL2_R) + WL(+Z)
REPEAT LOAD
1 0.6 4 0.6 7 1.0
LOAD 348 0.6 (DL + SDL2_R) + WL(-Z)
REPEAT LOAD
1 0.6 4 0.6 8 1.0
LOAD 349 0.6 (DL - SDL1_R) + WL(+X)
REPEAT LOAD
1 0.6 3 -0.6 5 1.0
LOAD 350 0.6 (DL - SDL1_R) + WL(-X)
REPEAT LOAD
1 0.6 3 -0.6 6 1.0
LOAD 351 0.6 (DL - SDL1_R) + WL(+Z)
REPEAT LOAD
1 0.6 3 -0.6 7 1.0
LOAD 352 0.6 (DL - SDL1_R) + WL(-Z)
REPEAT LOAD
1 0.6 3 -0.6 8 1.0
LOAD 353 0.6 (DL - SDL2_R) + WL(+X)
REPEAT LOAD
1 0.6 4 -0.6 5 1.0
LOAD 354 0.6 (DL - SDL2_R) + WL(-X)
REPEAT LOAD
1 0.6 4 -0.6 6 1.0
LOAD 355 0.6 (DL - SDL2_R) + WL(+Z)
REPEAT LOAD
1 0.6 4 -0.6 7 1.0
LOAD 356 0.6 (DL - SDL2_R) + WL(-Z)
REPEAT LOAD
1 0.6 4 -0.6 8 1.0
*********************************************
LOAD 357 0.6 (DL + SDL1_R) + 0.7 EL(+X)
REPEAT LOAD
1 0.6 3 0.6 11 0.7
LOAD 358 0.6 (DL + SDL1_R) + 0.7 EL(-X)
REPEAT LOAD
1 0.6 3 0.6 11 -0.7
LOAD 359 0.6 (DL + SDL1_R) + 0.7 EL(+Z)
REPEAT LOAD
1 0.6 3 0.6 12 0.7
LOAD 360 0.6 (DL + SDL1_R) + 0.7 EL(-Z)
REPEAT LOAD
1 0.6 3 0.6 12 -0.7
LOAD 361 0.6 (DL + SDL2_R) + 0.7 EL(+X)
REPEAT LOAD
1 0.6 4 0.6 11 0.7
```

```
LOAD 362 0.6 (DL + SDL2_R) + 0.7 EL(-X)
REPEAT LOAD
1 0.6 4 0.6 11 -0.7
LOAD 363 0.6 (DL + SDL2_R) + 0.7 EL(+Z)
REPEAT LOAD
1 0.6 4 0.6 12 -0.7
LOAD 364 0.6 (DL + SDL2_R) + 0.7 EL(-Z)
REPEAT LOAD
1 0.6 4 0.6 12 -0.7
LOAD 365 0.6 (DL + SDL1_R) + 0.7 EL(+X)
REPEAT LOAD
1 0.6 3 -0.6 11 0.7
LOAD 366 0.6 (DL + SDL1_R) + 0.7 EL(-X)
REPEAT LOAD
1 0.6 3 -0.6 11 -0.7
LOAD 367 0.6 (DL + SDL1_R) + 0.7 EL(+Z)
REPEAT LOAD
1 0.6 3 -0.6 12 0.7
LOAD 368 0.6 (DL + SDL1_R) + 0.7 EL(-Z)
REPEAT LOAD
1 0.6 3 -0.6 12 -0.7
LOAD 369 0.6 (DL + SDL2_R) + 0.7 EL(+X)
REPEAT LOAD
1 0.6 4 -0.6 11 0.7
LOAD 370 0.6 (DL + SDL2_R) + 0.7 EL(-X)
REPEAT LOAD
1 0.6 4 -0.6 11 -0.7
LOAD 371 0.6 (DL + SDL2_R) + 0.7 EL(+Z)
REPEAT LOAD
1 0.6 4 -0.6 12 -0.7
LOAD 372 0.6 (DL + SDL2_R) + 0.7 EL(-Z)
REPEAT LOAD
1 0.6 4 -0.6 12 -0.7
**********************************************
*Load Combination for Concrete design
**********************************************
LOAD 401 1.4 (DL+SDL1_R)
REPEAT LOAD
1 1.4 3 1.4
LOAD 402 1.4 (DL+SDL2_R)
REPEAT LOAD
1 1.4 4 1.4
LOAD 403 1.4 (DL-SDL1_R)
REPEAT LOAD
1 1.4 3 -1.4
LOAD 404 1.4 (DL-SDL2_R)
REPEAT LOAD
1 1.4 4 -1.4
**********************************************
LOAD 405 1.4 (DL + SDL1_R ) + 1.7 LL
REPEAT LOAD
1 1.4 3 1.4 2 1.7
LOAD 406 1.4 (DL + SDL2_R ) + 1.7 LL
REPEAT LOAD
1 1.4 4 1.4 2 1.7
LOAD 407 1.4 (DL - SDL1_R ) + 1.7 LL
REPEAT LOAD
1 1.4 3 -1.4 2 1.7
```

```
LOAD 408 1.4 (DL - SDL2_R ) + 1.7 LL
REPEAT LOAD
1 1.4 4 -1.4 2 1.7
*********************************************
LOAD 409 1.2 (DL + SDL1_R) + 0.5 LL + 1.6 WL(+X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 5 1.6
LOAD 410 1.2 (DL + SDL1_R) + 0.5 LL + 1.6 WL(-X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 6 1.6
LOAD 411 1.2 (DL + SDL1_R) + 0.5 LL + 1.6 WL(+Z)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 7 1.6
LOAD 412 1.2 (DL + SDL1_R) + 0.5 LL + 1.6 WL(-Z)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 8 1.6
LOAD 413 1.2 (DL + SDL2_R) + 0.5 LL + 1.6 WL(+X)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 5 1.6
LOAD 414 1.2 (DL + SDL2_R) + 0.5 LL + 1.6 WL(-X)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 6 1.6
LOAD 415 1.2 (DL + SDL2_R) + 0.5 LL + 1.6 WL(+Z)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 7 1.6
LOAD 416 1.2 (DL + SDL2_R) + 0.5 LL + 1.6 WL(-Z)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 8 1.6
LOAD 417 1.2 (DL - SDL1_R) + 0.5 LL + 1.6 WL(+X)
REPEAT LOAD
1 1.2 3 -1.2 2 0.5 5 1.6
LOAD 418 1.2 (DL - SDL1_R) + 0.5 LL + 1.6 WL(-X)
REPEAT LOAD
1 1.2 3 -1.2 2 0.5 6 1.6
LOAD 419 1.2 (DL - SDL1_R) + 0.5 LL + 1.6 WL(+Z)
REPEAT LOAD
1 1.2 3 -1.2 2 0.5 7 1.6
LOAD 420 1.2 (DL - SDL1_R) + 0.5 LL + 1.6 WL(-Z)
REPEAT LOAD
1 1.2 3 -1.2 2 0.5 8 1.6
LOAD 421 1.2 (DL - SDL2_R) + 0.5 LL + 1.6 WL(+X)
REPEAT LOAD
1 1.2 4 -1.2 2 0.5 5 1.6
LOAD 422 1.2 (DL - SDL2_R) + 0.5 LL + 1.6 WL(-X)
REPEAT LOAD
1 1.2 4 -1.2 2 0.5 6 1.6
LOAD 423 1.2 (DL - SDL2_R) + 0.5 LL + 1.6 WL(+Z)
REPEAT LOAD
1 1.2 4 -1.2 2 0.5 7 1.6
LOAD 424 1.2 (DL - SDL2_R) + 0.5 LL + 1.6 WL(-Z)
REPEAT LOAD
1 1.2 4 -1.2 2 0.5 8 1.6
*********************************************
LOAD 425 1.2 (DL + SDL1_R) + 0.5 LL + 1.0 EL(+X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 11 1.0
LOAD 426 1.2 (DL + SDL1_R) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
```

```
1 1.2 3 1.2 2 0.5 11 -1.0
LOAD 427 1.2 (DL + SDL1_R) + 0.5 LL + 1.0 EL(+Z)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 12 1.0
LOAD 428 1.2 (DL + SDL1_R) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 12 -1.0
LOAD 429 1.2 (DL + SDL2_R) + 0.5 LL + 1.0 EL(+X)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 11 1.0
LOAD 430 1.2 (DL + SDL2_R) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 11 -1.0
LOAD 431 1.2 (DL + SDL2_R) + 0.5 LL + 1.0 EL(+Z)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 12 1.0
LOAD 432 1.2 (DL + SDL2_R) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 12 -1.0
LOAD 433 1.2 (DL - SDL1_R) + 0.5 LL + 1.0 EL(+X)
REPEAT LOAD
1 1.2 3 -1.2 2 0.5 11 1.0
LOAD 434 1.2 (DL - SDL1_R) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 3 -1.2 2 0.5 11 -1.0
LOAD 435 1.2 (DL - SDL1_R) + 0.5 LL + 1.0 EL(+Z)
REPEAT LOAD
1 1.2 3 -1.2 2 0.5 12 1.0
LOAD 436 1.2 (DL - SDL1_R) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 3 -1.2 2 0.5 12 -1.0
LOAD 437 1.2 (DL - SDL2_R) + 0.5 LL + 1.0 EL(+X)
REPEAT LOAD
1 1.2 4 -1.2 2 0.5 11 1.0
LOAD 438 1.2 (DL - SDL2_R) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 4 -1.2 2 0.5 11 -1.0
LOAD 439 1.2 (DL - SDL2_R) + 0.5 LL + 1.0 EL(+Z)
REPEAT LOAD
1 1.2 4 -1.2 2 0.5 12 1.0
LOAD 440 1.2 (DL - SDL2_R) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 4 -1.2 2 0.5 12 -1.0
*********************************************
LOAD 441 0.9 (DL + SDL1_R) + 1.6 WL(+X)
REPEAT LOAD
1 0.9 3 0.9 5 1.6
LOAD 442 0.9 (DL + SDL1_R) + 1.6 WL(-X)
REPEAT LOAD
1 0.9 3 0.9 6 1.6
LOAD 443 0.9 (DL + SDL1_R) + 1.6 WL(+Z)
REPEAT LOAD
1 0.9 3 0.9 7 1.6
LOAD 444 0.9 (DL + SDL1_R) + 1.6 WL(-Z)
REPEAT LOAD
1 0.9 3 0.9 8 1.6
LOAD 445 0.9 (DL + SDL2_R) + 1.6 WL(+X)
REPEAT LOAD
```

```
1 0.9 4 0.9 5 1.6
LOAD 446 0.9 (DL + SDL2_R) + 1.6 WL(-X)
REPEAT LOAD
1 0.9 4 0.9 6 1.6
LOAD 447 0.9 (DL + SDL2_R) + 1.6 WL(+Z)
REPEAT LOAD
1 0.9 4 0.9 7 1.6
LOAD 448 0.9 (DL + SDL2_R) + 1.6 WL(-Z)
REPEAT LOAD
1 0.9 4 0.9 8 1.6
LOAD 449 0.9 (DL - SDL1_R) + 1.6 WL(+X)
REPEAT LOAD
1 0.9 3 -0.9 5 1.6
LOAD 450 0.9 (DL - SDL1_R) + 1.6 WL(-X)
REPEAT LOAD
1 0.9 3 -0.9 6 1.6
LOAD 451 0.9 (DL - SDL1_R) + 1.6 WL(+Z)
REPEAT LOAD
1 0.9 3 -0.9 7 1.6
LOAD 452 0.9 (DL - SDL1_R) + 1.6 WL(-Z)
REPEAT LOAD
1 0.9 3 -0.9 8 1.6
LOAD 453 0.9 (DL - SDL2_R) + 1.6 WL(+X)
REPEAT LOAD
1 0.9 4 -0.9 5 1.6
LOAD 454 0.9 (DL - SDL2_R) + 1.6 WL(-X)
REPEAT LOAD
1 0.9 4 -0.9 6 1.6
LOAD 455 0.9 (DL - SDL2_R) + 1.6 WL(+Z)
REPEAT LOAD
1 0.9 4 -0.9 7 1.6
LOAD 456 0.9 (DL - SDL2_R) + 1.6 WL(-Z)
REPEAT LOAD
1 0.9 4 -0.9 8 1.6
************************************************
LOAD 457 0.9 (DL + SDL1_R) + 1.0 EL(+X)
REPEAT LOAD
1 0.9 3 0.9 11 1.0
LOAD 458 0.9 (DL + SDL1_R) + 1.0 EL(-X)
REPEAT LOAD
1 0.9 3 0.9 11 -1.0
LOAD 459 0.9 (DL + SDL1_R) + 1.0 EL(+Z)
REPEAT LOAD
1 0.9 3 0.9 12 1.0
LOAD 460 0.9 (DL + SDL1_R) + 1.0 EL(-Z)
REPEAT LOAD
1 0.9 3 0.9 12 -1.0
LOAD 461 0.9 (DL + SDL2_R) + 1.0 EL(+X)
REPEAT LOAD
1 0.9 4 0.9 11 1.0
LOAD 462 0.9 (DL + SDL2_R) + 1.0 EL(-X)
REPEAT LOAD
1 0.9 4 0.9 11 -1.0
LOAD 463 0.9 (DL + SDL2_R) + 1.0 EL(+Z)
REPEAT LOAD
1 0.9 4 0.9 12 1.0
LOAD 464 0.9 (DL + SDL2_R) + 1.0 EL(-Z)
REPEAT LOAD
```

```
1 0.9 4 0.9 12 -1.0
LOAD 465 0.9 (DL - SDL1_R) + 1.0 EL(+X)
REPEAT LOAD
1 0.9 3 -0.9 11 1.0
LOAD 466 0.9 (DL - SDL1_R) + 1.0 EL(-X)
REPEAT LOAD
1 0.9 3 -0.9 11 -1.0
LOAD 467 0.9 (DL - SDL1_R) + 1.0 EL(+Z)
REPEAT LOAD
1 0.9 3 -0.9 12 1.0
LOAD 468 0.9 (DL - SDL1_R) + 1.0 EL(-Z)
REPEAT LOAD
1 0.9 3 -0.9 12 -1.0
LOAD 469 0.9 (DL - SDL2_R) + 1.0 EL(+X)
REPEAT LOAD
1 0.9 4 -0.9 11 1.0
LOAD 470 0.9 (DL - SDL2_R) + 1.0 EL(-X)
REPEAT LOAD
1 0.9 4 -0.9 11 -1.0
LOAD 471 0.9 (DL - SDL2_R) + 1.0 EL(+Z)
REPEAT LOAD
1 0.9 4 -0.9 12 1.0
LOAD 472 0.9 (DL - SDL2_R) + 1.0 EL(-Z)
REPEAT LOAD
1 0.9 4 -0.9 12 -1.0
PERFORM ANALYSIS
FINISH
```